

Muscles as a Motor

Maximising mechanical power output of muscles

Edwin D.H.M. Reuvers

The research in this thesis was embedded in Amsterdam Movement Sciences Research Institute, at the department of Human Movement Sciences, Vrije Universiteit Amsterdam, the Netherlands.

Cover: Cover design and layout by the author.
The muscle illustration was generated by ChatGPT.
Printer: ?? printer ??
DOI: <https://doi.org/10.5463/thesis.1933>
ISBN: ?? ISBN ??

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This dissertation was prepared using Quarto and typeset in $\text{\LaTeX}$. The body text is set in Minion Pro, with Myriad Pro used for sans serif.

VRIJE UNIVERSITEIT

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ACADEMISCH PROEFSCHRIFT

ter verkrijging van de graad van Doctor of Philosophy aan
de Vrije Universiteit Amsterdam,
op gezag van de rector magnificus
prof.dr. J.J.G. Geurts,
volgens besluit van de decaan
van de Faculteit der Gedrags- en Bewegingswetenschappen
in het openbaar te verdedigen
op dinsdag 8 december 2026 om 13.45 uur
in de universiteit

door

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geboren te Leiderdorp

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1

General introduction

Movement is fundamental to both animal and human performance. From everyday locomotion to athletic activities, muscles generate the mechanical power required to move the body and interact with the environment. During each movement cycle, muscles alternately lengthen (eccentric contraction) and shorten (concentric contraction), a pattern known as the stretch–shortening cycle (SSC). When muscles lengthen while producing force, they deliver negative mechanical power; when they shorten while producing force, they deliver positive mechanical power.

In locomotor activities such as cycling, rowing, speed skating, and running, mechanical energy is dissipated to the environment, for instance due to frictional forces. Therefore, to sustain motion, muscles must generate net positive mechanical work over their SSCs to offset these energy losses. Some of these dissipative forces, such as aerodynamic drag, increase with movement speed. Consequently, the performance in cyclic motor tasks (e.g. the maximally attainable speed) is largely determined by the average mechanical power output (AMPO) over these SSCs.

Typically, during locomotion activities, movements arise from muscle forces and mechanical interactions with the environment, which together determine the SSCs of the muscles involved. In some situations, muscle length changes can be influenced by the individual, who can adjust — for example — frequency and joint

excursion within certain limits (e.g. by changing step length during running). In other situations, when individuals interact with equipment, muscle length changes can be more substantially influenced. For instance, muscle length excursions can be influenced by adjusting crank length in cycling. Thus, by changing equipment, muscle length changes can be influenced, which provides an opportunity to tailor the resulting SSCs to maximise AMPO. Such a design approach, that capitalises on a thorough understanding of the human musculoskeletal system has proven effective, as demonstrated by the Klapskate for speed skating (Ingen Schenau et al., 1996) and the SuperSeat for rowing (Soest and Hofmijster, 2009; Soest et al., 2016). Yet, such design approaches remain rare due to limited understanding of how to optimise AMPO, not only at the level of individual muscles but also for whole-body movements.

The main aim of this thesis is to examine how SSCs influence the maximally attainable AMPO of muscle. A secondary aim is to investigate how constraints affect the maximally attainable AMPO. Before doing so, I will first briefly outline the historical development of muscle mechanics and energetics that underly our current understanding. Subsequently, I will show how this knowledge, together with modern optimisation techniques, can be used to reach the aims of this thesis.

1.1 Historical foundations

1.1.1 Origins of muscle energetics and the conservation of energy

An important step towards understanding muscle metabolism and heat production in physical terms came from the work of Antoine Lavoisier and Pierre-Simon Laplace (Lavoisier and Laplace, 1780). Using a calorimeter, they measured the heat produced by a guinea pig and compared it with the amount of oxygen the animal consumed. From these experiments they concluded that respiration is essentially a form of slow combustion: the body generates heat through chemical reactions involving oxygen. This work provided one of the first demonstrations that metabolism and heat production could be understood in chemical and physical terms.

In 1835, Antoine César Becquerel and Gilbert Breschet extended these investigations to the tissue level (Becquerel and Breschet, 1835). By using thermo-electric

instruments, they were able to measure temperature directly within living organs and muscles. These experiments revealed subtle thermal gradients that varied across different tissues. By doing so, Becquerel and Breschet provided a methodological bridge between the macroscopic calorimetry of Lavoisier and later quantitative approaches in muscle physiology.

This experimental approach coincided with a conceptual breakthrough in physics: the formulation of the law of conservation of energy, which states that energy cannot be created or destroyed, but only transformed from one form to another. Experiments by Julius Robert Mayer and James Prescott Joule demonstrated the quantitative equivalence between mechanical work and heat, providing crucial empirical support for this principle. The German physiologist and physicist Hermann von Helmholtz brought these ideas together in his influential 1847 essay *Über die Erhaltung der Kraft* (“On the Conservation of Force”). Helmholtz argued that the law of conservation of energy is a universal law of nature that must also apply to living organisms. In particular, he emphasised that muscular work cannot arise from a mysterious life force (in accordance with Mayer, 1845), but must ultimately derive from chemical processes within the body (Helmholtz, 1847, p. 52). In 1848, a year later, he provided experimental support for the idea that muscle contraction itself produces heat (Helmholtz, 1848). By measuring temperature changes in frog muscles while isolating them from blood flow, he showed that the heat originated within the muscle tissue itself and could be interpreted as the conversion of chemical energy within the muscle into heat.

To detect temperature changes in muscle, Helmholtz used a thermopile connected to a galvanometer. The thermopile consisted of three thermocouples arranged so that one sensor was placed in the stimulated muscle and a corresponding reference sensor in the unstimulated contralateral muscle. Because the instrumentation had a slow response time, only the total heat produced over an entire contraction could be measured, rather than the instantaneous heat rate—a limitation that persisted until 1937 (see below). Helmholtz’s thermopile method was subsequently adopted and refined by many later researchers studying muscle energetics.

1.1.2 Nineteenth-century myothermic experiments

Following Helmholtz's pioneering investigations in 1848, several studies addressed the heat production of muscle contraction. The most comprehensive contribution was Rudolf Heidenhain's extensive 1864 treatise on the relationship between mechanical work and heat production (Heidenhain, 1864). In a remarkably broad series of experiments, Heidenhain investigated numerous factors affecting muscle heat production such as load, fatigue, and stimulation frequency. Heidenhain later recalled that he had initially expected that, under a constant stimulus, a muscle lifting a heavier load would convert a greater proportion of its available energy into mechanical work and therefore produce less heat (Heidenhain, 1869, p. 423). His 1864 experiments, however, yielded the opposite result. He found that increasing the load increased both mechanical work and heat production up to a certain limit; beyond that limit, both decreased, with heat production declining earlier than mechanical work (Heidenhain, 1864, p. 168). From these observations, Heidenhain concluded that the total energy liberated (i.e. the sum of heat production and mechanical work) by a muscle is not fixed by the stimulus alone, but is influenced by the mechanical conditions under which the muscle contracts. He therefore argued that muscle metabolism continues throughout muscle contraction and varies with the mechanical conditions under which the muscle operates.

Following Heidenhain's work, Adolf Fick became the central figure in the later nineteenth-century study of muscle heat production. He pursued the same basic problem — how heat production, mechanical work and chemical energy were related. After improving the myothermic methods, Fick believed that he could express muscle heat production in absolute units and thereby estimate the mechanical efficiency of the isolated frog muscle (Fick, 1878). He found that, at maximum, roughly one-third of the energy liberated by the muscle could appear as mechanical work. Fick also examined how this efficiency depended on the mechanical conditions of the muscle and other factors such as muscle temperature. An important question for later work was whether heat production differs between a shortening muscle and a muscle held at constant length. In 1884, Fick compared an isometric contraction with a contraction in which the muscle first developed

full tension and was then released to shorten (Fick, 1884). He found that when an active muscle was allowed to shorten it produced slightly more heat (~10%) than when an active muscle was held isometrically. Fick interpreted this as evidence that shortening itself could evoke additional chemical processes, rather than simply using a pre-existing store of elastic energy.

The interpretation of Fick his 1884 experiments remained controversial. Fritz Schenck later repeated similar experiments and found no clear difference between the heat produced in the two conditions investigated by Fick (Schenck, 1892). Magnus Blix, a student of Adolf Fick, later addressed the issue by arguing that heat production was not governed by the mechanical work performed, but rather depended on the length of the muscle (Blix, 1902). As he put it, the relevant conclusion was not that “work produces heat”, but that “length produces heat”. Thus, by the beginning of the twentieth century, myothermal experiments had established that muscle heat production depended on the mechanical conditions of contraction, but they had not established how. It remained unclear whether the relevant factor was the mechanical work produced by the muscle, muscle force/tension, muscle length, muscle shortening itself, or some artefact of the delicate thermoelectric methods used to measure (small) temperature changes.

1.1.3 A.V. Hill's entry into muscle thermodynamics

In 1905, a young Archibald Vivian Hill entered Trinity College Cambridge to study mathematics. During his first year, however, he began to lose interest in subjects he regarded as somewhat removed from physical reality. A decisive influence at this stage was his tutor, Walter Morley Fletcher, who, together with Frederick Gowland Hopkins, was engaged in pioneering studies on the production of lactic acid in muscle. Hill shared with Fletcher a strong interest in athletics, which provided a natural point of contact. Through their conversations, Fletcher introduced him to the production of lactic acid during muscular activity, its removal in the presence of oxygen, and its relation to fatigue and recovery after exercise. Although these ideas made a lasting impression, Fletcher advised him to continue with mathematics for the time being. Following this advice, Hill continued studying mathematics at

first, but decided to change direction later. He resolved to become a physiologist and he started studying physiology, chemistry, and physics at Cambridge. This interdisciplinary path was not unusual: John Newport Langley, then head of the Physiological Laboratory, had himself first studied mathematics before turning to natural sciences.

In November 1909, Langley suggested that Hill should investigate the efficiency of muscle as a thermodynamic machine. This problem, previously studied by Heidenhain and Fick, was particularly promising in the light of Fletcher and Hopkins' work on muscle metabolism under aerobic and anaerobic conditions. Looking back more than half a century later, Hill remarked how striking it was that Langley, "very far from being a biophysicist or biochemist", had proposed a problem so well suited both to the Cambridge environment and to his own inclinations, and he considered it likely that this decision had been made in consultation with colleagues, among them Fletcher (Hill, 1970a, p. 37). In *Trails and Trials in Physiology* (1965, p. 5), he also emphasised the importance of the apparatus that Langley had acquired in 1904 from Magnus Blix: a highly sensitive combination of thermocouple and galvanometer which, according to Blix, was capable of detecting the heat produced in a single muscle twitch. This marked the beginning of over fifty years of experiments conducted by A.V. Hill and colleagues on muscle heat production.

1.1.4 The (visco-)elastic theory of muscle

Hill was well aware of the earlier investigations of Heidenhain and Fick (whom he often described as "a very wise man"), but he was not entirely convinced by Fick's finding that more heat was produced when an active muscle was allowed to shorten compared to when it was held isometrically. Hill raised several methodological concerns about these earlier experiments, particularly that in Fick's apparatus the muscle, after shortening, could not reliably return to the exact same position on the thermopile, potentially distorting the measurements and contributing to the large variance in the results.

Hill decided to replicate and refine Fick's experiments, using a more controlled setup that would reduce these methodological issues. In his own experiments, Hill

measured the heat produced by a muscle that was allowed to shorten at different time points after stimulation and compared this to heat production during isometric contractions. While later found to be incorrect based on subsequent results from his own laboratory, Hill found that heat production was unaffected by shortening when the muscle was allowed to shorten only after it had reached maximal tension. Based on these results, he concluded that “after the maximum tension has been developed, the muscle is merely a new elastic body, whose tension energy may be used indifferently for the accomplishment of work or the production of heat” (Hill, 1913). This theory — later refined into the idea of muscles as visco-elastic bodies — then prevailed in Hill’s research for the next 25 years.

1.1.5 Fenn’s challenge to the (visco-)elastic theory of muscle

At the suggestion of A.V. Hill, Wallace Fenn performed a series of experiments that were published in his famous 1923 paper. In his first set of experiments, Fenn repeated some of the work done by Heidenhain and Fick. By using an isotonic lever, the muscle remained isometric until it had developed sufficient force to lift an attached mass. Once the mass was lifted, the muscle shortened at approximately constant force until the mass — and thus the muscle — returned to its initial position.

Using this setup, muscle heat and the positive mechanical work performed were measured separately and compared with values obtained during isometric contractions. Fenn’s experiments confirmed the pattern previously observed by Heidenhain: as the load increased, mechanical work and heat production initially increased, but declined again at higher loads. These findings reinforced Heidenhain’s conclusion that the total energy liberated by a muscle depends on the mechanical conditions under which it contracts. The comparison with isometric contractions further demonstrated that additional energy was liberated when the muscle shortened and performed work. As Fenn concluded: “Whenever a muscle shortens upon stimulation and does work in lifting a weight, an extra amount of energy is mobilised which does not appear in an isometric contraction. Hence less energy is liberated in an isometric contraction than in any contraction in which the muscle is allowed to shorten” (Fenn, 1923).

In his Nobel Prize lecture (for the 1922 award) at the end of 1923, Hill recognised the implications of Fenn's findings: "It is clear also that those of us who supposed that a stimulated muscle is simply a new elastic body were wrong. A muscle may possess some elastic properties, but it requires more energy to do more work, a fact which is fundamentally in opposition to the elastic body theory." Despite the recognition of Fenn's findings, Hill "went on thinking in terms of it for a long time after" (Hill, 1970b, p. 2).

1.1.6 The heat of shortening and the dynamic constants of muscle

In the years following Fenn's work, many attempts were made to further clarify his findings. However, the success was limited, except for the consistent observation that additional energy is liberated when muscles shorten and perform mechanical work (Hill, 1965, p. 148). Nevertheless, important methodological developments occurred, particularly in the refinement of thermocouples. The work of A. C. Downing enabled the construction of fast-responding thermocouples, which, by 1937, made it possible to measure heat production during muscle contraction (Hill, 1937). This represented a substantial improvement over earlier instruments, whose slow response times caused heat measurements to be lumped at the end of contractions. These technical advances provided the foundation for a new generation of experiments on muscle energetics.

Using these improved instruments, A.V. Hill explored the heat production of muscle during two distinct experiments in his famous 1938 paper (Hill, 1938). In the first set of experiments, a muscle first isometrically contracted until the muscle force reached a steady plateau. Then, a magnet was released from the lever causing a sudden drop in the force acting on the muscle after which the muscle shortened against a constant load until a fixed shortening distance was reached (see Figure 1.1). While the force acting on the muscle (after the magnet was released) was held constant across trials, the permitted shortening distance varied. Hill observed that the heat produced increased linearly with shortening distance (see Figure 1.1D). Because muscle force was held constant, this meant that that mechanical work

also scaled linearly with shortening. From these observations, Hill formulated the relation:

$$E = (F + a) \cdot \Delta L$$

where F is the muscle force during shortening, ΔL is the shortening distance, and a is a constant representing the so-called shortening heat.

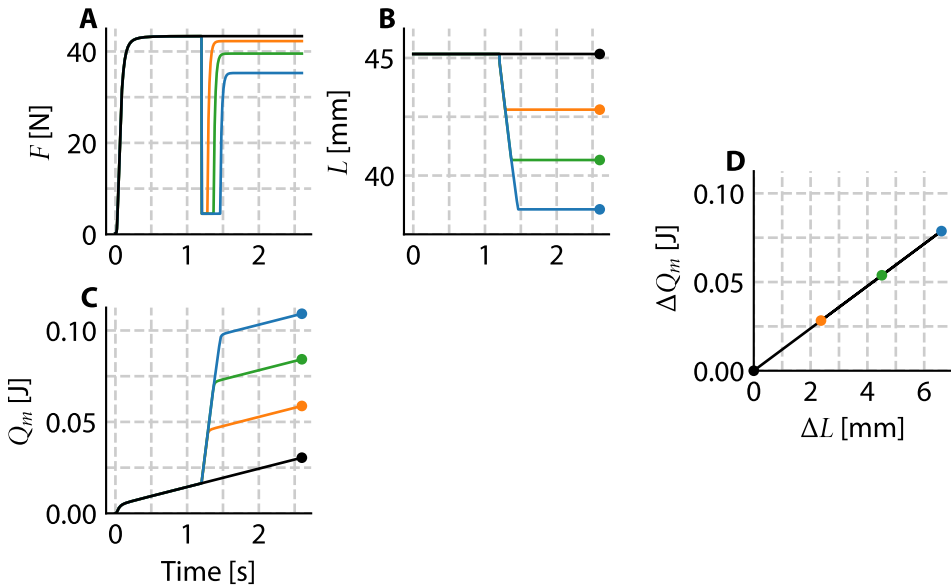


Figure 1.1: Simulation results mimicking Hill (1938) first set of experiments. In these experiments, the muscle was first held isometric and activated until force reached a steady plateau. An external load was then suddenly reduced, allowing the muscle to shorten against a constant force until a prescribed shortening distance was reached. Simulations were repeated for three different shortening distances (indicated by different colours) while maintaining the same post-release force. An isometric contraction, in which no shortening was allowed, is shown in red. A) Muscle force (F) over time. B) Muscle length (L) over time. C) Cumulative measured heat (Q_m) over time. D) Shortening heat (ΔQ_m), defined as the measured heat minus the heat produced during an isometric contraction, plotted against shortening distance (ΔL). The slope of the resulting linear fit corresponds to Hill's parameter a .

In a second set of experiments, Hill performed similar experiments as in the first set, but this time varied the force acting on the muscle while keeping the shortening distance constant (see Figure 1.2). These experiments revealed that muscle heat production was independent of muscle force (see Figure 1.2C). Together, both experiments showed that the maximum rate of heat production plus the

instantaneous mechanical power decreased linearly as the force during shortening approached the maximum isometric force, reaching zero at maximum isometric force (see Figure 1.2E). This led to the second fundamental relation:

$$\dot{E} = b \cdot (F_o - F)$$

where F_o is the maximal isometric muscle force and b is a constant.

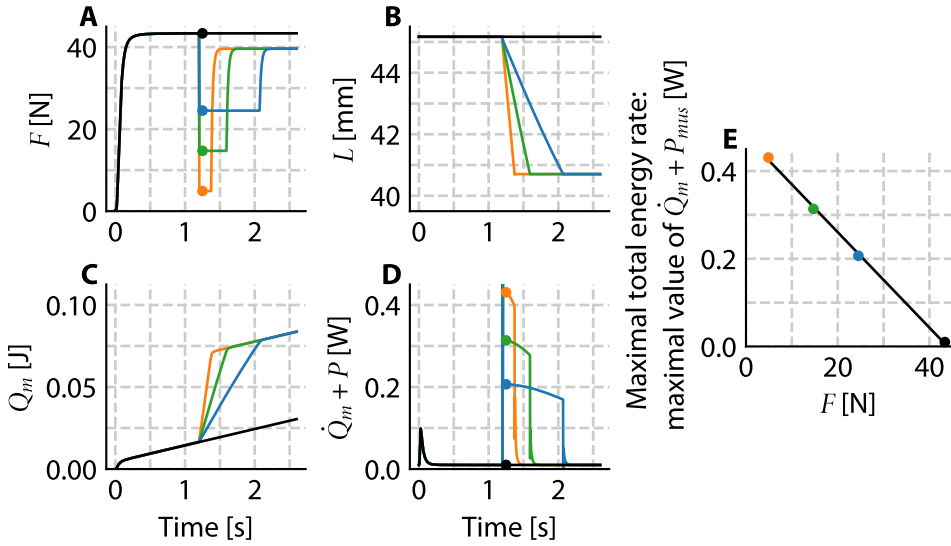


Figure 1.2: Simulation results mimicking Hill (1938) second set of experiments. In these experiments, the muscle was first held isometric and activated until force reached a steady plateau. An external load was then suddenly reduced, allowing the muscle to shorten through a fixed distance against different constant forces. Simulations were repeated for four different post-release forces (indicated by different colours) while maintaining the same shortening distance. An isometric contraction, in which no shortening was permitted, is shown in black. A) Muscle force (F) over time. B) Muscle length (L) over time. C) Cumulative measured heat (Q_m) over time. D) Instantaneous total energy rate, defined as the sum of the rate of heat production and the mechanical power produced by the muscle, $\dot{Q}_m + P_{mus}$, over time. E) Maximum total energy rate attained during shortening, defined as the maximum value of $\dot{Q}_m + P_{mus}$ during the shortening phase, plotted against muscle force during shortening. The (absolute value of the) slope of the resulting linear fit corresponds to Hill's parameter b .

By differentiating the first equation over time and combining it with the second, Hill derived the characteristic relation:

$$\dot{E} = (F + a) \cdot v = b \cdot (F_o - F)$$

from which an explicit relation between shortening velocity v and muscle force F can be obtained:

$$v = b \cdot \frac{F_o - F}{F + a}$$

This formulation provided a quantitative framework linking muscle mechanics and energetics, and marked a decisive shift in the conceptual understanding of muscle function. While the equation was derived from heat measurement, it was subsequently confirmed by directly relating muscle force and velocity measured at the instant immediately after the magnet was released from the lever. On the basis of these results, Hill (finally) rejected the (visco-)elastic theory of muscle. Instead, he proposed that active muscle should be viewed as a two-component model, consisting of an undamped elastic element arranged in series with a contractile element governed by the characteristic force–velocity relationship. Together, these elements form the muscle–tendon complex (MTC), which is often extended to also include an elastic element parallel to the contractile element (see Figure 2.2). This model is now known as the Hill-type MTC model.

1.2 Maximising average mechanical power output

1.2.1 From instantaneous to average mechanical power output

From the groundbreaking work of A.V. Hill, we have learned two fundamental principles: 1) that the power output of the contractile element (CE) depends on CE shortening velocity (with a specific velocity at which instantaneous mechanical power output is maximal); and 2) that CE is in fact the “motor” of the muscle–tendon complex (MTC), with the series elastic element (SEE) and parallel elastic element (PEE) being primarily passive elastic structures. As such, maximising AMPO requires MTC length over time to allow CE to produce maximal AMPO during SSCs.

While it is obvious that an MTC should maximise positive mechanical work and minimise negative mechanical work in order to maximise AMPO, it is far from trivial what MTC length and stimulation over time yield maximal AMPO. To minimise negative mechanical work, stimulation should cease well before lengthening begins, and CE should ideally lengthen at very (s)low velocities because of the effect of the eccentric CE force–velocity relationship. However, such strategies are suboptimal for AMPO: the time spent lengthening cannot be used to produce positive mechanical work, and early deactivation reduces positive mechanical work during shortening. To maximise positive mechanical work, one might argue that CE should shorten at a velocity that maximises instantaneous mechanical power output. However, CE force also depends on CE length, with CE force decreasing at both shorter and longer lengths than the optimal length. Therefore, to optimise AMPO, it may be beneficial for CE to operate over a limited range of the CE force–length relationship. This raises several open questions: over what range should CE operate to maximise AMPO? At what shortening velocity? And how much time should be spent on shortening (i.e., maximising positive mechanical work) versus lengthening (i.e., minimising negative mechanical work) in order to maximise AMPO?

Finding the CE/MTC length and stimulation over time for maximal AMPO is inherently complex. For instance, if one were to change the CE/MTC length excursion, how should other parameters, such as cycle frequency, be adjusted to maximise AMPO? Let alone how stimulation over time should be adjusted. Moreover, even if maximal AMPO could be experimentally identified in a single muscle, it remains unclear how the results translate across different muscles and/or individuals. Muscle properties vary across species and therefore the effect of CE/MTC length and stimulation over time on the maximal attainable AMPO may differ across muscles. As a result, experimental studies alone provide limited insights.

1.2.2 Predicting maximal AMPO of MTC Models via Optimal Control

MTC models are a valuable tool for studying muscle behavior. One of the pioneers in this field, Herbert Hatze, complemented A.V. Hill's work on muscle contraction dynamics with insights into muscle excitation dynamics (Hatzé, 1981). Although modern MTC models combine Hill-based contraction dynamics with excitation dynamics to which Hatze made substantial contributions, they are still commonly referred to as Hill-type MTC models. These models have proven useful not only for understanding isolated muscle behavior (e.g. Curtin et al., 1998; Ingen Schenau et al., 1988; Lemaire et al., 2016; Sandercock and Heckman, 1997; Wakeling and Johnston, 1999; Williams et al., 1998; Zandwijk et al., 1996), but also, when combined with skeletal models into musculoskeletal models, to simulate complex whole-body tasks such as cycling (Raasch et al., 1997; Soest and Casius, 2000), jumping (Anderson and Pandy, 1999; Soest et al., 1993) and the long jump (Hatzé, 1981). Such musculoskeletal models have yielded predictions consistent with experimental observations and have also been applied to study various aspects of animal behavior (e.g. Biewener et al., 2014; Bijlert et al., 2024; Cox et al., 2019; Heers et al., 2018). These studies indicate that Hill-type MTC models are a powerful tool to examine the influence of CE/MTC length and stimulation over time on AMPO.

To find the optimal CE/MTC length and stimulation over time that maximise AMPO (using a Hill-type MTC model), these variables must be specified at each time instance. In discrete form, when the time axis is divided into n points, both CE/MTC length and stimulation must be determined at each point, resulting in a $2n$ -dimensional optimisation problem. In a classical forward simulation framework, one would use the discretised CE/MTC length and stimulation over time as inputs to simulate the dynamics, after which the objective is evaluated. This approach often performs poorly for the situation described above: small changes in inputs (here: CE/MTC length and stimulation over time) can lead to large and nonlinear effects on the objective, resulting in a highly irregular optimisation landscape, making it difficult to find the global optimum.

As an alternative, Optimal Control techniques can be used to find the CE/MTC length and stimulation over time that maximise AMPO. In this approach, both

the inputs (at the n time points) and the states (at the n time points) are treated as optimisation variables, with the system dynamics imposed as constraints. This allows the optimiser to select the full trajectory of states and inputs simultaneously while ensuring that the dynamics are respected. By including the dynamics as constraints, the sensitivities of the states with respect to the inputs—which are implicit in classical forward simulation—are captured directly through the Lagrangian and the resulting KKT system. The Hessian of the Lagrangian incorporates all relevant partial-derivatives, giving the optimiser precise guidance on how changes in inputs propagate through the states to affect the objective. Consequently, the problem becomes well-structured and numerically stable, enabling efficient convergence to the optimal trajectories even in complex, nonlinear systems – such as muscles.

1.3 Thesis outline

In this thesis, we brought together the current detailed understanding of muscle mechanics and energetics with developments in optimisation techniques to examine how various SSCs influence the maximally attainable AMPO. In addition, we investigated the role of constraints on the maximally attainable AMPO.

In **Chapter 2**, we evaluated the accuracy of commonly used experimental methods — such as quick-release and step-ramp experiments — for estimating key muscle properties in experiments. Despite their widespread use, these methods have not been formally evaluated. A major challenge in experimentally addressing the accuracy of parameter estimation is that the true underlying muscle properties are, by definition, unknown. To overcome this limitation, we adopted a modelling approach in which experimental data were simulated using a Hill-type muscle-tendon complex (MTC) model with a given set of muscle properties. This allowed a direct evaluation of the accuracy of the estimated properties against their a priori known muscle properties.

In **Chapter 3**, we combined *in situ* experiments with Hill-type MTC modelling to examine the influence of various SSCs on the maximally attainable AMPO of rat *m. gastrocnemius medialis*. For each specimen, key muscle properties were estimated individually using the methodology developed in Chapter 2. This allowed us to

compare the experimental results with model-based predictions, which showed a near-perfect correlation between measured and predicted AMPO ($r^2 > 0.98$). The strong correspondence then justified the use of a Hill-type MTC model to explore how various SSCs influence the maximally attainable AMPO across a wide range of conditions.

In **Chapter 4**, we used an approach similar to that used in Chapter 3 and applied it to the human knee-extensor muscle group. Using a Hill-type MTC model, we simulated various SSCs to examine their influence on the maximally attainable AMPO at the joint level. From these simulations, we selected nine insightful conditions to be evaluated experimentally. In the experiment, knee joint movements were fully imposed by a knee dynamometer, while participants were instructed to maximise AMPO. Measured AMPO matched the model predictions reasonably well, confirming that the Hill-type MTC model reliably captures the influence of SSCs on the maximally attainable AMPO at the joint level in human participants.

In **Chapter 5**, we investigated the role of radial pedal forces in sprint cycling performance by using an optimal control musculoskeletal model to predict the maximally attainable AMPO under varying constraints on radial pedal forces. Since radial pedal forces do not contribute to mechanical power, it has been (often) suggested that radial pedal forces arise from suboptimal pedaling technique and that limiting these would increase AMPO and efficiency. However, this can never be true from the first principle of optimal control theory: introducing an additional constraint on the controls (here: restricting radial pedal forces) cannot increase the optimal value of the objective (here: maximal AMPO). Using a musculoskeletal model, we showed how individual muscle power output changes across different levels of allowed radial pedal forces. Specifically, we showed that when radial pedal forces are highly constrained, muscles not only produce much less positive mechanical power, but some muscles also substantially increase their negative mechanical power output. These findings are a clear example of the influence of constraints on the maximally attainable performance.

In **Chapter 6**, we discussed the broader implications of the findings presented in this thesis. We reflected on the transition from highly controlled *in situ* experiments

on rat muscle to less controlled *in vivo* experiments on human muscle, and examined the sensitivity of our results. In addition, we discussed how findings for an individual muscle translate to whole-body movements involving multiple muscles. We further related maximally attainable AMPO across different movement conditions to understand the influence of several constraints on performance. Overall, the points addressed in Chapter 6 highlight that maximising AMPO in a motor task requires consideration of both the intrinsic power-producing capacity of muscle and the constraints imposed by the task.

2

Accuracy of experimentally estimated muscle properties: Evaluation and improvement using a Hill-type model and a newly developed toolbox

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Under review at Journal of Experimental Biology

Abstract

The mechanical behaviour of a muscle-tendon complex depends on properties such as the force-length relationships, the force-velocity relationship, and the excitation dynamics. Quick-release and step-ramp experiments are commonly used to estimate these properties. The accuracy of these methods is unclear, as the actual values of these properties are unknown in experiments on real muscle. We conducted a modelling study using a Hill-type muscle-tendon complex model with literature-derived parameter values and simulated quick-release, step-ramp, and isometric experiments. From the simulated experiments, we assessed how accurately the model's parameter values could be retrieved. Using a method traditionally used in literature, the series elastic element stiffness was underestimated by $\sim 35\%$, due to the incorrect assumption that muscle fibres do not shorten during quick-releases. Consequently, this yielded an overestimation of the excitation dynamics activation time constants of $\sim 20\%$. We developed an improved method that accounted for muscle fibre shortening during quick-releases. Using our improved method, all parameter values closely matched their actual values. A sensitivity analysis showed that the most critical parameters were robust to perturbations in experimental data. Lastly, we compared Hill-type MTC model predictions against *in situ* data from three rat *m. gastrocnemius medialis*. Predictions based on parameters from the improved method showed closer agreement than those based on the traditional method — both for quick-release, step-ramp, and isometric experiments, as well as for independent stretch-shortening cycles. In conclusion, the improved method enables more accurate estimates of muscle-tendon complex properties, addressing limitations of the traditionally used method.

2.1 Introduction

In the early 20th century, A.V. Hill laid the foundation of muscle mechanics by recognising that muscles contain not only muscle fibres (the contractile element; CE) but also connective tissue, both parallel to the muscle fibre (the parallel elastic element; PEE) and in series with the muscle fibre (the tendon or serial elastic element; SEE). Together these three elements form the muscle-tendon-complex (MTC). The contraction dynamics describe how CE force emerges from the interplay between the force-length properties of CE, PEE and SEE, and the force-velocity properties of CE. In addition, CE force depends on the active state (the relative amount of Ca^{2+} bound to troponin C, Ebashi and Endo, 1968). Active state, in turn, depends on activation via the excitation dynamics. Consequently, the mechanical behaviour of MTC arises from the intricate interplay between the contraction and the excitation dynamics.

In his famous 1938 paper A.V. Hill performed so-called ‘isotonic quick-release experiments’ using the now-renowned Levin-Wyman lever (Levin and Wyman, 1927) to estimate the CE force-velocity relationship. In these experiments, MTC was attached to a lever and maximally stimulated. When SEE force plateaued (i.e., MTC was isometrically delivering force), a magnet was released from the lever causing a sudden drop in the force acting on the MTC. As SEE force remained constant right after this drop in SEE force, SEE length remained constant and therefore all MTC length changes were attributed to CE length changes. The CE velocity corresponding to the constant SEE force after the drop was then calculated as the maximum rate of change of MTC length over time. Consequently, each isotonic quick-release experiment contributed a single data point of the CE force-velocity relationship. Nowadays, servomotors are typically used in experiments on isolated MTCs to control MTC length (changes). In contrast to the force-controlled experiments with the Levin-Wyman lever, servomotors control the MTC length over time (see Figure 2.S.1). In order to estimate the CE force-velocity relationship using a servomotor, Cecchi et al. (1978) introduced ‘step-ramp experiments’. In these experiments, MTC is kept isometrically under maximal stimulation until SEE force plateaus. Then, a rapid shortening of MTC length is imposed (the ‘step’),

immediately followed by a constant-velocity shortening of the MTC (the ‘ramp’). This experimental protocol results in a brief plateau in SEE force just after the ramp. During this brief period, SEE length remains constant and therefore CE velocity equals MTC velocity. Thus, step-ramp experiments with servomotors serve as a valid alternative for isotonic quick-release experiments to estimate the CE force-velocity relationship.

In addition to estimating the CE force-velocity relationship using isotonic quick-release experiments, the research group of A.V. Hill also used these experiments to approximate SEE stiffness. The underlying idea was that the drop in MTC force was so fast that, due to contraction dynamics, CE had no time to shorten and all MTC shortening could thus be directly attributed to SEE shortening. Consequently, this experiment allowed to directly relate SEE length changes to changes in SEE force and therefore to estimate SEE stiffness. As mentioned above, in isotonic quick-release experiments, a sudden drop in force acting on the MTC causes a rapid change in MTC length. While in these lever-based experiments MTC force is controlled, in experiments using a servomotor MTC length is controlled. To estimate SEE stiffness with a servomotor setup, the opposite approach is taken: a rapid change in MTC length is imposed by the motor, resulting in a rapid change in SEE force. Somewhat confusingly, these servomotor-based experiments are also referred to as ‘quick-release experiments’, even though their approach differs fundamentally from that of an isotonic quick-release experiment. In the remainder of this paper, the term quick-release experiments will exclusively refer to servomotor-based length-controlled quick-release experiments. In quick-release experiments using a servomotor, the drop in MTC length takes several milliseconds (e.g., approximately 10 ms with the commonly used Aurora Scientific’s 300C series motor). As CE shortens within this short time period, the assumption that MTC length changes are exclusively due to SEE length changes may not be entirely valid. This raises the question to what extent SEE stiffness can be accurately estimated based on quick-release experiments using servomotors.

A.V. Hill already acknowledged in (1950) that CE shortens during the brief period in which SEE force decreases – even for his isotonic quick-release experiments

using the Levin-Wyman lever – and introduced a method to correct for this CE shortening. In this method, the CE shortening that occurred due to the quick-release was estimated based on the CE force-velocity relationship of the MTC investigated. Thus, additional experimental work – such as performing step-ramp experiments and estimating the CE force-velocity relationship – are required to correct for CE shortening during the quick-release. Zandwijk et al. (1997) and Lemaire et al. (2016) developed similar methods to correct for CE shortening. As expected, Zandwijk et al. (1997) showed that this method results in higher estimates of SEE stiffness than with the method without correcting for CE shortening. Nevertheless, the accuracy of SEE stiffness estimates remains uncertain, as the actual SEE stiffness is unknown in experiments, making it infeasible to compare the estimated value against the actual one. Consequently, it remains unclear whether the additional complexity and experimental work required for the improved method are truly justified in comparison with the simpler traditional method, especially when the only interest is SEE stiffness.

In experimental studies in which MTC properties are estimated, SEE stiffness is typically estimated first in order to discriminate between SEE stiffness and the other properties, such as those of the CE force-length relationship. In experiments, CE length is unknown and therefore the properties of the CE force-length relationship cannot be directly estimated based on experimental data. Typically, the CE force-length properties are estimated based on the measured MTC force-length relationship (e.g., Blümel et al., 2012; Lemaire et al., 2016). However, different combinations of CE and SEE properties can yield almost identical MTC force-length relationships. Consequently, inaccuracies in SEE stiffness estimates can propagate to errors in the estimation of the CE force-length properties. While predictions under isometric conditions may remain reasonably accurate, these errors can lead to substantially different mechanical behaviour during dynamic contractions such as stretch-shortening cycles (see Figure 2.1). As such, incorrect estimation of SEE stiffness might (partially) explain difference between predictions derived by a Hill-type MTC model and experimental results. In sum, the question arises to which extent the estimates of other MTC properties are affected by inaccurate SEE stiffness

estimates and whether more accurate SEE stiffness estimates improves predictions derived by a Hill-type MTC model.

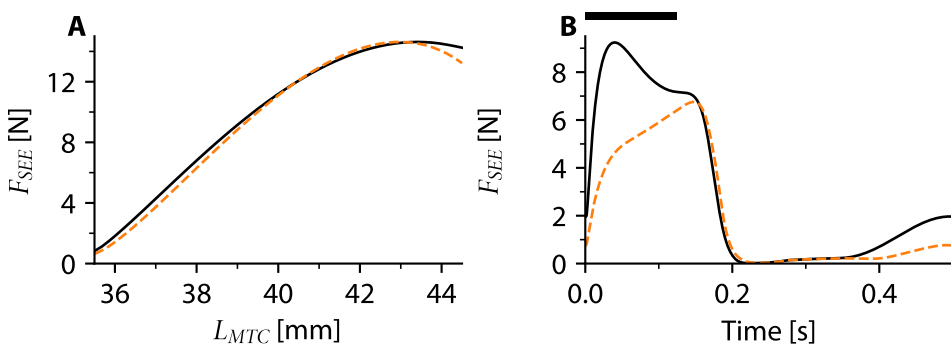


Figure 2.1: Simulated effect of a 2-fold difference in SEE stiffness on predicted mechanical behaviour. Simulated forces are plotted for a muscle model with a stiffer SEE (black lines) and a muscle model with more compliant SEE (dashed orange lines). A) Isometric MTC force-length relationships, showing only marginal differences in behaviour even though differences in SEE stiffness are substantial. B) SEE forces over time for a sinusoidal stretch-shortening cycle with a 2 Hz cycle frequency and a 5 mm MTC length excursion. The thick black bar indicates the where muscle stimulation is maximal, while muscle stimulation is ‘off’ elsewhere. This example shows a substantial difference in predicted forces due to the difference in SEE stiffness.

The potential effect of error propagation also extends to the estimation of the excitation dynamics properties. In the absence of histochemical data, excitation dynamics properties are commonly estimated by fitting predicted SEE force of the Hill-type MTC model to experimental data (e.g., Blümel et al., 2012; Lemaire et al., 2016). As SEE force over time obviously depends on the contraction dynamics properties (e.g., the force-length relationships and the force-velocity relationship), the estimation of the excitation dynamics properties is not independent of the estimation of the contraction dynamics properties. Consequently, inaccuracies in SEE stiffness estimates can propagate to inaccuracies in estimated excitation dynamics properties. This interdependence raises an additional question of what experimental data should be used in order to minimise this potential error propagation.

A major challenge in purely experimentally answering questions above that relating to accuracy of determining muscle properties experimentally is that, obviously, “true” muscle properties are unknown, one is left with comparing predicted mechanical behaviour that observed experimentally. However, as pointed out before

(see Figure 2.1), different combinations of muscle properties yield similar behaviour in some type contracting, yet may yield very different behaviour in others. Thus, relying on the capability to predict mechanical behaviour (in a selected) condition is not a valid way of determining the accuracy estimated muscle properties. To mitigate this, we adopted a modelling approach, in we assessed the accuracy of estimated parameters using a Hill-type MTC model. The rationale is straightforward: if we cannot reliably estimate the a-priori known properties of modelled muscles using simulated experiments, then such estimations are unlikely to be reliable for real muscles in real experiments. This approach allows us to address important methodological questions: can these properties actually be estimated accurately, and if so, what experimental protocols and what is the influence analytical methods on estimated muscle properties?

The primary objective of this study was to systematically evaluate the accuracy of estimating contraction and excitation dynamics properties on the basis of data collected in commonly used experiments. Specifically, we aimed to compare the results of a method that does not correct for CE shortening in quick-release experiments (the ‘traditional method’) with a method that does correct for CE shortening (the ‘improved method’). As explained earlier, the limitation of relying exclusively on an experimental approach is that the actual properties are unknown, which renders a reliable assessment of the accuracy of the estimated properties not feasible. To circumvent this problem, we conducted a modelling study using a Hill-type MTC model. For this purpose, we obtained three different sets of parameter values of a Hill-type MTC model from existing literature. Using these parameter values, we simulated data of quick-release, step-ramp and isometric protocols mimicking experiments on isolated MTCs using servomotors. We then investigated how accurately we could retrieve the model’s parameter values (with respect to their ‘actual’ value). In addition, we employed a comprehensive sensitivity analysis to assess the sensitivity of parameter values against perturbations in experimental data and to examine the interdependency of the estimated properties. The secondary objective of this study was to evaluate if experimental data can be better predicted with parameter values estimated with the improved method vs the traditional

method. To this end, we used *in situ* data from experiments on three rat *m. gastrocnemius medialis*. We estimated the contraction and excitation dynamics properties using both the traditional and the improved method, and compared the resulting model predictions — based on each parameter set — to experimental force data. Overall, this study provides insights into the accuracy of existing commonly used methods for contraction and excitation dynamics property estimation and their influence on predictions of mechanical behaviour derived by a Hill-type MTC model. The approach that we designed in this study for estimating contraction and excitation dynamics properties based on data of quick-release, step-ramp and isometric experiments using servomotors, is made available as an open-source toolbox (<https://github.com/edwinreuvers/mp-estimator>).

2.2 Methods

The primary objective of this study was to assess the accuracy of estimating contraction and excitation dynamics properties. For this purpose, we employed a Hill-type MTC model to simulate MTC mechanical behaviour. Using parameter values obtained from existing literature, we simulated data of quick-release, step-ramp, and isometric experiments, mimicking experiments on isolated MTCs with servomotors. We then estimated the contraction and excitation dynamics parameter values using two methods: one with (the ‘traditional method’) and one without (the ‘improved method’) a correction for CE shortening due to the quick-release. Additionally, we conducted comprehensive sensitivity analyses to assess the accuracy of the improved method. The secondary objective was to evaluate predictions derived by a Hill-type MTC model using two sets of parameter values: one estimated by the traditional method, the other by the improved method. For this purpose, we used *in situ* data from three rat *m. gastrocnemius medialis*. Predictions were derived with experimentally measured MTC length and stimulation over time as model inputs and were compared with experimentally measured data in terms of SEE force. For the remainder of this manuscript, predictions refer to SEE force predictions unless stated otherwise. These comparisons were performed for quick-release, step-ramp, and isometric experiments, as well as independently measured stretch-shortening

cycles. This allowed us to investigate whether the improved method enhanced the predictions derived by a Hill-type MTC model.

2.2.1 Muscle model

2.2.1.1 Contraction dynamics

We employed a Hill-type MTC model consisting of a contractile element (CE) and a parallel elastic element (PEE), which were both in series with a serial elastic element (SEE), as depicted in Figure 2.2. CE represented the contractile element of the muscle fibres, while PEE and SEE represented the tissues arranged in parallel and in series with the muscle fibres, respectively. Given the negligible small mass of MTC compared to the forces typically delivered by CE, PEE, and SEE, we simplified the model by neglecting the second-order dynamics of the system:

$$\begin{aligned} L_{MTC} &= L_{CE} + L_{SEE} \\ F_{SEE} &= F_{CE} + F_{PEE} \end{aligned} \quad (2.1)$$

L_{MTC} , L_{CE} , L_{SEE} and L_{PEE} denote MTC, CE, PEE and SEE length respectively, while F_{SEE} , F_{CE} , and F_{PEE} denote the CE, PEE and SEE force. PEE and SEE were assumed be purely elastic and were modelled as quadratic springs (Zajac, 1989,):

$$F_{EE} = \begin{cases} k_{SEE} \cdot (l_{EE} - L_{EE}^0)^2, & \text{if } l_{EE} \geq L_{EE}^0 \\ 0, & \text{otherwise} \end{cases} \quad (2.2)$$

L_{EE} denotes the length of either PEE or SEE and L_{EE}^0 the slack length of either PEE or SEE. k_{EE} denotes a parameter that scales the stiffness of either PEE or SEE.

Isometric CE force depends on CE length (Blix, 1892; Blix, 1893; Hill, 1925). We simplified the classic sarcomere force-length relationship (Gordon et al., 1966; Walker and Schrodtt, 1974,) to a second order polynomial, which describes the force-length relationship reasonably well (Bobbert et al., 1990; Woittiez et al., 1984):

$$F_{CE}^{isom,rel} = \begin{cases} \frac{-1}{w^2} \cdot (L_{CE}^{rel} - 1)^2 + 1, & \text{if } F_{CE}^{isom,rel} > 0.1 \\ s_{exp} \cdot e^{k_{exp} \cdot L_{CE}^{rel}}, & \text{otherwise} \end{cases} \quad (2.3)$$

$F_{CE}^{isom,rel}$ denotes the CE isometric force normalised by F_{CE}^{max} , w determines the width of the CE force-length relationship (for $F_{CE}^{isom,rel} > 0.1$) and L_{CE}^{rel} denotes CE length normalised by CE optimum length (L_{CE}^{opt} , the CE length at $F_{CE}^{isom,rel} = 1$). Exponential tails were added to the isometric CE force-length relationship such that it had a continuous first derivative with respect to L_{CE}^{rel} (see Figure 2.2). The parameter values of s_{exp} and k_{exp} were determined separately for the ascending and descending limb of the CE force-length relationship.

The concentric CE force-velocity was modelled according to Hill (1938). The original description was adjusted as suggested by Soest and Bobbert (1993) to account for situations where the active state (q , the relative amount of Ca^{2+} bound to troponin C, Ebashi and Endo, 1968) is not maximal:

$$F_{CE} = \frac{q \cdot (F_{CE}^{isom,rel} \cdot F_{CE}^{max} \cdot b + a \cdot v_{CE})}{b - v_{CE}} \quad (2.4)$$

F_{CE} denotes the CE force and v_{CE} denotes the CE velocity. a is a parameter that defines the curvature of the concentric CE force-velocity relationship and defines together with b and F_{CE}^{max} the maximum shortening velocity (v_{CE}^{max} , $v_{CE}^{max} = -b/a \cdot F_{CE}^{max}$). The maximum shortening velocity is scaled by $F_{CE}^{isom,rel}$ due to the formulation of the CE force-velocity relationship (see Equation 2.4). However, the maximum shortening velocity has been reported to be more or less constant above optimum CE length (Gordon et al., 1966; Stern, 1974). Based on this observation, the parameter a was scaled by $F_{CE}^{isom,rel}$ above optimum CE length ($L_{CE}^{rel} > 1$) to make the maximum shortening velocity independent of CE length above optimum CE length. The parameter b was scaled with b_{scale} for low levels of active state in order to make the maximal contraction velocity dependent on the active state (Petrofsky and Phillips, 1981). The original description of this scaling by Soest and Bobbert (1993) was somewhat reformulated, such that b_{scale} was continuously differentiable with respect to q :

$$b_{scale} = \frac{1}{1 + e^{-b_{shape} \cdot (q - q_{sc})}} \quad (2.5)$$

$$\text{with } q_{sc} = \frac{\log\left(\frac{1}{b_{scale}^{min} - 1} + q_o \cdot b_{shape}\right)}{b_{shape}}$$

q_o denotes the minimum value of q , b_{scale}^{min} denotes the minimal scale factor of b (i.e., the minimal value of b_{scale}) and b_{shape} defines the steepness of the relation between b_{scale} and q . The value of b_{shape} was obtained by minimising the least square error between the formulation of Soest and Bobbert (1993) and the reformulated version of the relation between b_{scale} and q .

The eccentric part of the CE force-velocity relationship was modelled as a slanted hyperbola function (Van Soest et al., 2005) of the following form:

$$F_{CE}^{rel} = \frac{c_4 - (c_3 + v_{CE}^{rel} \cdot (c_1 + c_2 \cdot v_{CE}^{rel}))}{c_3 \cdot v_{CE}^{rel}} \quad (2.6)$$

c_1 , c_2 , c_3 and c_4 denote parameters that define the shape of the slanted hyperbola and depend on $F_{CE}^{isom,rel}$ and q . These parameters were chosen such that (1) the concentric and eccentric curves were continuous in the point $v_{CE}^{rel} = 0$; (2) the ratio between the eccentric and concentric derivatives of $\frac{dF_{CE}^{rel}}{dv_{CE}^{rel}}$ in the point $v_{CE}^{rel} = 0$ was defined by r_{slope} ; (3) the oblique asymptote of the eccentric curve was defined by F_{asympt} and (4) the slope of the slanted asymptote was defined by r_{as} .

2.2.1.2 Excitation dynamics

The excitation dynamics were modelled in two steps. The first step, to which we refer as the calcium dynamics, related the rate of change in normalised free Ca^{2+} concentration between the myofilaments (γ) to normalised CE stimulation ($STIM$) and the normalised free Ca^{2+} concentration between the myofilaments itself (Hatze, 1981, pp 31-42):

$$\dot{\gamma} = \begin{cases} \frac{STIM \cdot (1 - \gamma_o) - \gamma + \gamma_o}{\tau_{act}}, & \text{if } STIM \geq \gamma \\ \frac{STIM \cdot (1 - \gamma_o) - \gamma + \gamma_o}{\tau_{deact}}, & \text{otherwise} \end{cases} \quad (2.7)$$

γ_0 denotes the minimum value of γ and had an arbitrary small value such that Equation 2.8 was always solvable. τ_{act} and τ_{deact} are both time constants of the first-order differential equations describing the calcium dynamics. The second step involved the relation between active state (q) and γ , to which we refer as the $q - [Ca^{2+}]$ relation. It is well-known that muscle fibres become increasingly sensitive to $[Ca^{2+}]$ as their length increases (Kistemaker et al., 2005; Rack and Westbury, 1969; Stephenson and Williams, 1982). Consequently, q was also made dependent on L_{CE}^{rel} . The $q - [Ca^{2+}]$ relation was modelled according to Hatze (1981, pp 31-42), but was mathematically reformulated to ensure that the parameter values were physiologically meaningful and to facilitate its application for Optimal Control (e.g., Kistemaker et al., 2023):

$$q = q_0 + \frac{1 - q_0}{1 + (kCa \cdot \gamma)^{A_{act}} \cdot e^{a_{act} B_{act}}} \quad (2.8)$$

with $A_{act} = \log_{10}(e^{-a_{act}})$

and $B_{act} = b_{act,1} + b_{act,2} \cdot L_{CE}^{rel} + b_{act,3} \cdot L_{CE}^{rel^2}$

kCa relates γ to the actual Ca^{2+} concentration between the myofilaments and B_{act} denotes the pCa^{2+} level at which $q = 0.5$. B_{act} depended on L_{CE}^{rel} (Equation 2.8), while parameter a_{act} determined the steepness of the relation between γ and q .

2.2.2 Simulated and *in situ* data

2.2.2.1 Simulated data

To simulate quick-release, step-ramp, and isometric experiments, we used three parameter sets of a Hill-type MTC model from existing literature for three rat *m. gastrocnemius medialis* (GM1, GM2, and GM3; see Table S1). This choice was made between rat *m. gastrocnemius medialis* is a well-studied hind-limb muscle of which an almost complete dataset of muscle properties is available (Zandwijk et al., 1997).

Quick-release experiments Each quick-release experiment consisted of an isometric phase until SEE force plateaued, followed by a rapid (step) change (10 ms) in MTC length and then followed by another isometric phase (see Figure 2.3A). CE stimulation was maximal (i.e., $STIM = 1$) during the first isometric phase and

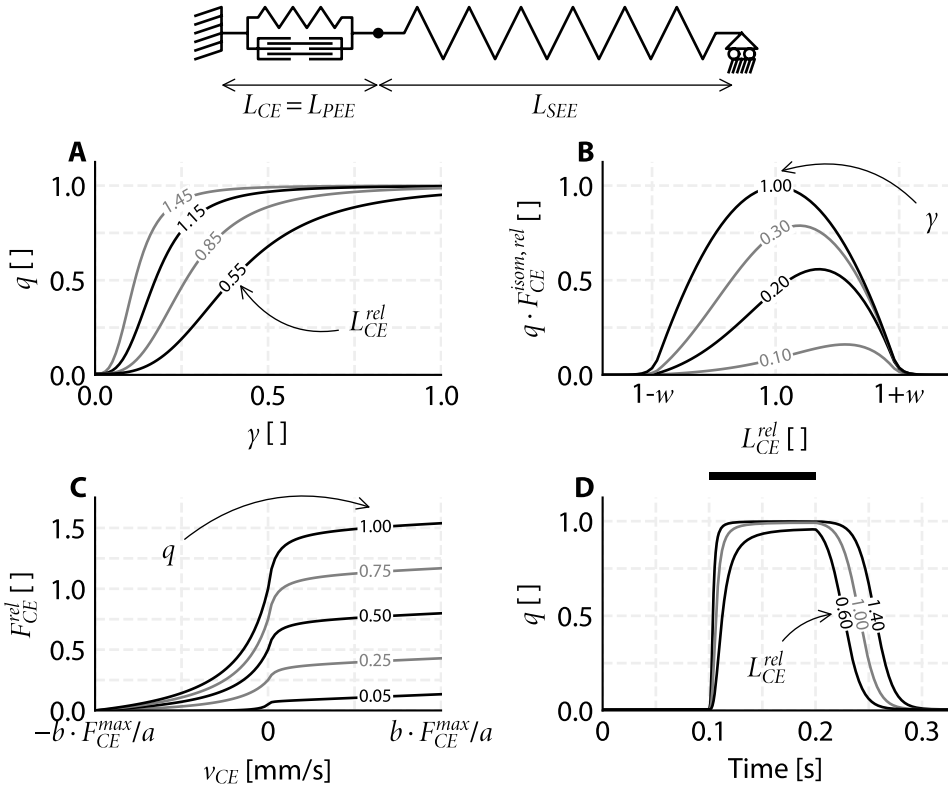


Figure 2.2: Aspects of the Hill-type MTC model used, illustrated at the top. L_{CE} , L_{PEE} and L_{SEE} denote the CE, parallel elastic element (SEE) and serial elastic element (SEE) length. CE represents the contractile part of the muscle fibres, while PEE and SEE represent all elastic tissue in parallel or in series, respectively, with CE. In the Hill-type MTC model, CE force depends on active state (A), CE length (B) and CE velocity (C). The effect of CE stimulation on active state (q) is illustrated in D. A) The relationship between normalised free Ca^{2+} concentration between the myofilaments (γ) and q . q also depends on relative CE length (L_{CE}^{rel}). B) The product of q and the normalised active CE force-length relationship for different values of γ . C) The CE force-velocity relationship for different values of q . D) q over time before, during and after CE stimulation for $L_{CE}^{rel} = 1$. CE stimulation is maximal during the period indicated by the black bar and ‘off’ elsewhere.

continued at this maximal level until it was switched off shortly after the step change in MTC length occurred. The step change in MTC length was 0.2 mm and was chosen such that the resulting change in SEE force was about 5-10% of F_{CE}^{max} . To prevent irreversible damage to muscle fibres, the maximal MTC length used in experiments is typically not far above the MTC length that yields maximal isometric SEE force (L_{MTC}^{opt}). Accordingly, we simulated quick-release experiments at various initial MTC lengths, ranging from a very short MTC length (i.e., a length yielding very low isometric SEE force) and at every 1 mm increment up to a MTC length that was maximal 3 mm above L_{MTC}^{opt} , corresponding to $\sim 85-105\%$ L_{MTC}^{opt} .

Step-ramp experiments Each step-ramp experiment consisted of an isometric phase slightly above (± 0.5 mm) L_{MTC}^{opt} , followed by a rapid (step) change (10 ms) in MTC length and then a constant MTC velocity ramp (see Figure 2.3B). CE stimulation was maximal (i.e., $STIM = 1$) during the first isometric phase and continued at this maximal level until it was switched off 0.1 s after the step change in MTC length occurred. The change in MTC length of this step and the constant MTC velocity of the ramp were chosen such that this resulted in a more or less a constant SEE force (and therefore constant CE force) at the beginning of the ramp. We simulated 9 step-ramp experiments with different combinations of step sizes and constant MTC velocity ramps, to cover a substantial part of the concentric CE force-velocity relationship.

Isometric experiments Isometric experiments were simulated for a combination of different MTC lengths (about 0, -2, -4 and -6 mm below L_{MTC}^{opt}) and stimulation durations (35, 65 and 95 ms). CE stimulation was maximal (i.e., $STIM = 1$) during the indicated stimulation duration and fully off elsewhere (see Figure 2.3C). The combination of four different MTC lengths and three different stimulation durations resulted in twelve different isometric experiments.

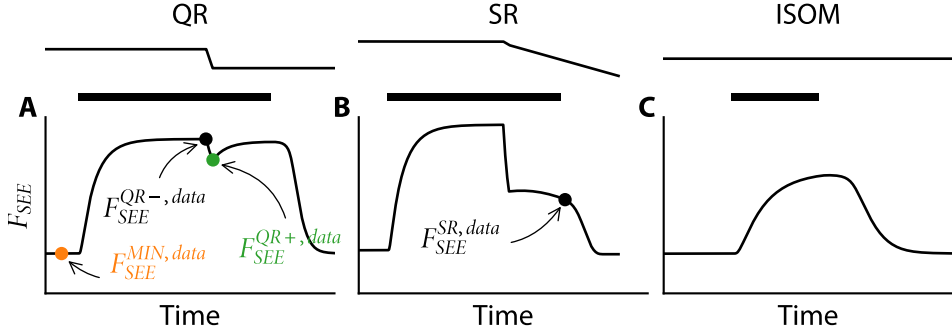


Figure 2.3: Example of simulated data of quick-release (A), step-ramp (B) and isometric (C) experiments. Top: MTC length over time. Bottom: SEE force over time. CE stimulation is maximal during the period indicated by the black bar, and ‘off’ elsewhere. For each experiment, we obtained specific datapoints of SEE force and the corresponding MTC length, which were used to estimate contraction and excitation dynamics parameter values.

2.2.2.2 *In situ* data

We also used data from a previous study in which we conducted *in situ* experiment on isolated rat *m. gastrocnemius medialis*, performed on three male Wistar rats (Reuvers et al., 2025). The original experiment included quick-release, step-ramp, and isometric protocols. In that study, between 11 and 13 quick-release experiments, 10 and 12 step-ramp experiments, and 12 isometric experiments were performed for each rat. Below, we provide a brief description of the experimental procedure, which is fully detailed in Reuvers et al. (2025).

In the experiment, approved by the Committee on the Ethics of Animal Experimentation at the Vrije Universiteit (Permit Number: FBW- AVD11200202114471), rats were first anesthetised with urethane. The hindlimb was then shaved, and the overlying skin and *m. biceps femoris* were removed. The medial and lateral part of *m. gastrocnemius* were carefully separated from their surrounding tissue and exposed as much as possible. The rats were placed in the experimental setup with the hindlimb, femur and foot fully fixed. The distal end of the calcaneal tendon was attached to a servomotor (Aurora 309C, Aurora Scientific, Aurora, Canada) using Kevlar thread and aligned to ensure that *m. gastrocnemius medialis* pulled in its natural direction. All nerves not innervating *m. gastrocnemius medialis* were severed. *M. gastrocnemius medialis* was stimulated with a supramaximal current

via a cuff-electrode placed on the sciatic nerve, with the proximal nerves crushed to prevent spinal reflexes. This experimental setup allowed for precise control of *m. gastrocnemius medialis* length changes and stimulation as well as accurate measurement of *m. gastrocnemius medialis* force.

2.2.3 Parameter value estimation procedure

The general procedure to estimate the parameter values involved minimising the sum of squared differences between the data and the values based on the estimated parameter values. To distinguish between these, we used subscripts: for example, the value of the SEE force before the quick-release of the data is indicated by $F_{SEE}^{QR-,data}$ while the value based on the estimated parameter values is indicated by $F_{SEE}^{QR-,est}$.

The data of the quick-release experiments were used to estimate the parameter values of the CE, PEE and SEE force-length relationships. The data of the step-ramp experiments were used to estimate the parameter values of the CE force-velocity relationship. The data of the isometric experiments were used to estimate the parameter values of the excitation dynamics. Below we provide an overview on the methods used to estimate the parameter values. We also offer an open-source toolbox that automates the estimation of contraction and excitation dynamics parameter values.

2.2.3.1 CE, SEE and PEE force-length parameter value estimation

As explained in the introduction, different combinations of contraction dynamics parameters can yield almost identical mechanical behaviour under isometric conditions (see also Figure 2.1). For this reason, it is important to estimate SEE stiffness first, in order to discriminate between SEE stiffness on the one hand, and L_{CE}^{opt} and L_{SEE}^o on the other. In our approach, the parameter that scales SEE stiffness (k_{SEE}) is therefore estimated first. This is followed by the estimation of the PEE parameters, as k_{SEE} is also required for their estimation. Finally, the parameters F_{CE}^{max} , L_{CE}^{opt} , and L_{SEE}^o are estimated.

Estimation of SEE stiffness. The first step was to estimate SEE stiffness. The model SEE force depends on the parameter values of k_{SEE} and L_{SEE}^o , and obviously

on SEE length. The problem, however, is that SEE length is unknown in experiments. This makes it challenging to estimate the parameter values concerning the SEE force-length relationship. Fortunately, quick-release experiments provide a way out. During quick-release experiments with a servomotor, the motor quickly shortens MTC length such that there is a rapid decline in SEE force (the ‘quick-release’). Due to the shortness of this timeframe, CE shortening is minimal such that *almost all* MTC shortening is taken up by SEE. Often, in experimental studies, it is assumed that *all* MTC shortening can be attributed to SEE shortening. In reality, this assumption leads to an overestimation of SEE shortening, as CE also shortens during this short timeframe. We introduced a method to correct for this overestimation of SEE shortening (see section “Parameter estimation: traditional method & improved method”) and estimated the parameter values considering both methods: without and with correcting for CE shortening during the quick-release.

Obtaining the SEE length change due to the quick-release (ΔL_{SEE}^{QR}) is a crucial step. This is because SEE length after the quick-release (L_{SEE}^{QR+}) can then be expressed as SEE length before the quick-release (L_{SEE}^{QR-}) plus the change in SEE length due to the quick-release: $L_{SEE}^{QR+} = L_{SEE}^{QR-} + \Delta L_{SEE}^{QR}$. This allowed us to rewrite Equation 2.2 to estimate SEE force immediately before ($F_{SEE}^{QR-,est}$) and after ($F_{SEE}^{QR+,est}$) the quick-release. This yielded:

$$\begin{aligned} F_{SEE}^{QR-,est} &= k_{SEE} \cdot (L_{SEE}^{QR-} - L_{SEE}^o)^2 \\ F_{SEE}^{QR+,est} &= k_{SEE} \cdot (L_{SEE}^{QR-} + \Delta L_{SEE}^{QR} - L_{SEE}^o)^2 \end{aligned} \quad (2.9)$$

Now, there are two unknown parameters (i.e., k_{SEE} and L_{SEE}^o), while also SEE length before the quick-release is also unknown for each quick-release experiment. Consequently, there are more unknowns than equations. To address this issue, we replaced $L_{SEE}^{QR-} - L_{SEE}^o$ with a temporary parameter c_{SEE} . This yielded:

$$\begin{aligned} F_{SEE}^{QR-,est} &= k_{SEE} \cdot (c_{SEE})^2 \\ F_{SEE}^{QR+,est} &= k_{SEE} \cdot (c_{SEE} + \Delta L_{SEE}^{QR})^2 \end{aligned} \quad (2.10)$$

As k_{SEE} scales SEE stiffness, its value is constant across all quick-release experiments. In contrast, c_{SEE} depends on the parameter L_{SEE}^o and SEE length before the

quick-release, which differs among each quick-release experiment. Consequently, c_{SEE} should be determined for each quick-release experiment individually. Visually, c_{SEE} determines the shift along the x-axis on the SEE force-length relationship, while k_{SEE} scales SEE stiffness and consequently SEE force (see Figure 2.4A). The value of k_{SEE} (and c_{SEE} for each quick-release experiment) were found by minimising the sum of squared differences between the data and estimated SEE force before (F_{SEE}^{QR-}) and after (F_{SEE}^{QR+}) the quick-release.

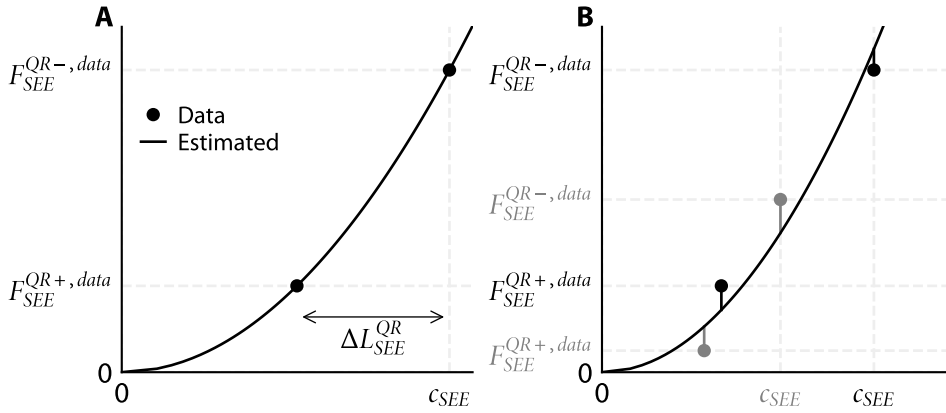


Figure 2.4: Graphical illustration of SEE stiffness parameter estimation. For each quick-release experiment, SEE force before ($F_{SEE}^{QR-,data}$) and after ($F_{SEE}^{QR+,data}$) the quick-release was obtained from the data, as well as the corresponding decrease in SEE length (ΔL_{SEE}^{QR}). Since SEE length prior to the quick-release is typically unknown, a temporary parameter (c_{SEE}) was introduced for each experiment to represent the difference between SEE length before the quick-release and SEE slack length. A) A single quick-release experiment yields two unknowns: c_{SEE} and a parameter that scales SEE stiffness (k_{SEE}). B) Running multiple quick-release experiments yields $n+1$ unknowns: n values of c_{SEE} (one for each experiment) and the parameter scaling SEE stiffness. Here, two quick-release experiments are illustrated (one indicated with black dots, the other with grey dots), while the estimated SEE force-length relationship is depicted with the black line.

Estimation of PEE parameter values. The second step was to estimate PEE stiffness. The model PEE force depends on the parameter values of k_{PEE} and L_{PEE}^o , and obviously on PEE length. In experiments, SEE force is measured, while PEE force is required to estimate PEE stiffness. As such, experimental data should be used in which CE force is negligible such that PEE force is approximately equal to SEE force. At the beginning of the quick-release experiments, the active state is so

low that CE force is negligible. We selected an interval of 10 ms in which the SEE force was minimal (F_{SEE}^{min} , orange dot in Figure 2.3A).

In experiments, PEE length is unknown. This problem can be addressed as follows. First, PEE length equals MTC length minus SEE length: $L_{PEE} = L_{MTC} - L_{SEE}$ (Equation 2.1). Second, SEE length is the sum of SEE slack length and SEE elongation (E_{SEE} : $L_{SEE} = L_{SEE}^o + E_{SEE}$). This allowed us to rewrite Equation 2.2 to estimate PEE force (F_{PEE}^{est}), yielding:

$$F_{PEE}^{est} = k_{PEE} \cdot (L_{MTC} - L_{SEE}^o - E_{SEE} - L_{PEE}^o)^2 \quad (2.11)$$

Now, there are two unknown parameter values (i.e., k_{PEE} and L_{PEE}^o), while also SEE elongation is unknown for every quick-release experiment. Consequently, there are more unknowns than equations. To address this, we replaced $L_{PEE}^o + L_{SEE}^o$ with a temporary parameter c_{PEE} . Subsequently, SEE elongation was computed as $\sqrt{\frac{F_{SEE}^{MIN,data}}{k_{SEE}}}$, given the estimated SEE stiffness scaling factor in the previous step. This yielded:

$$F_{PEE}^{est} = k_{PEE} \cdot (L_{MTC} - \sqrt{\frac{F_{SEE}^{MIN,data}}{k_{SEE}}} - c_{PEE})^2 \quad (2.12)$$

As k_{PEE} only scales PEE stiffness, its value is constant across all quick-release experiments. Similarly, c_{PEE} is the sum of L_{SEE}^o and L_{PEE}^o and should therefore be constant for each quick-release experiment. The values of k_{PEE} and c_{PEE} were then computed by minimising the sum of squared differences between F_{PEE}^{est} and $F_{SEE}^{min,data}$.

Estimation of F_{CE}^{max} , L_{CE}^{opt} and L_{SEE}^o . The third step was to estimate F_{CE}^{max} , L_{CE}^{opt} and L_{SEE}^o . Before the quick-release, MTC is isometrically delivering force. We used the SEE force at this instant ($F_{SEE}^{QR-,data}$, black dot in Figure 2.3A) and the corresponding MTC length to obtain the MTC force-length relationship from the data. We then estimated maximal isometric CE force (F_{CE}^{max}), CE optimum length (L_{CE}^{opt}) and SEE slack length (L_{SEE}^o) by minimising the sum of squared differences between the data and estimated MTC force-length relationship. Lastly, L_{PEE}^o was computed by subtracting L_{SEE}^o from c_{PEE} .

2.2.3.2 CE force-velocity parameter value estimation

The values of the CE force-velocity relationship parameters a and b were estimated from data obtained during the plateau phase of SEE force in step-ramp experiments. This approach was chosen because SEE length is constant when SEE force is constant. Consequently, CE velocity equals MTC velocity under these conditions. Following this argument, we first identified a 10 ms interval in which SEE force changed the least. Second, we derived CE length and CE force as functions of time using Equation 2.1 and Equation 2.2. Third, we calculated the CE velocity as the time-derivative of CE length. Finally, we averaged CE force and CE velocity over the 10 ms interval. This procedure yielded the data of the CE force-velocity relationship.

Now, the model CE force-velocity relationship can be fit to that of the data to obtain value of a and b . However, employing a method that simply minimises the sum of ordinary least squares differences would not be appropriate because in experiments there is uncertainty in both measured CE force and CE velocity. Instead, we used a total least squares method that minimised the distance between the modelled CE force (F_{CE}^{est}) and CE velocity (v_{CE}^{est}) (now called: model values) and those of the data (see also Figure 2.5). To do this, we had to find the nearest model values to the CE force of the data (F_{CE}^{data}) and CE velocity of the data (v_{CE}^{data}) (now called: datapoints). We used the following observations: 1) the model value has to satisfy Equation 2.4 and 2) the derivative of $\frac{dF_{CE}}{dv_{CE}}$ of the model value should be perpendicular to the line from the datapoint to the model value (see Figure 2.5). Hence, the model value could be found by solving Equation 2.4 and Equation 2.13:

$$\frac{dv_{CE}^{est}}{dF_{CE}^{est}} \cdot (v_{CE}^{est} - v_{CE}^{data}) = F_{CE}^{data} - F_{CE}^{est} \quad (2.13)$$

The following cost-function was then minimised to find the parameter values of a and b :

$$J = \sum_{i=1}^n \left(\frac{F_{CE}^{data} - F_{CE}^{est}}{c_1} \right)^2 + \left(\frac{v_{CE}^{data} - v_{CE}^{est}}{c_2} \right)^2$$

c_1 and c_2 denote scaling factors such that both terms of the cost-function are more or less equally weighted, which was done by setting c_1 equal to the maximal range

in CE force of the data and setting c_2 equal to the maximal range of CE velocity of the data.

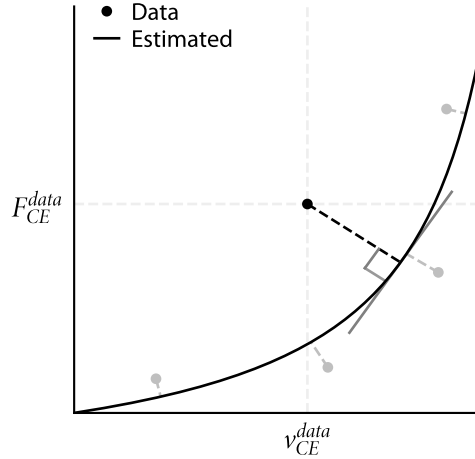


Figure 2.5: Graphical illustration of the CE force-velocity parameter estimation using total least squares. For each step-ramp experiment, CE force and CE velocity was obtained during the interval in which SEE force changed the least. The nearest point on the CE force-velocity relationship was identified based on two criteria: it should satisfy Equation 2.4, and the line from the datapoint to the CE force-velocity relationship should be perpendicular the CE force-velocity relationship. This was done for all step-ramp experiments (other datapoints are depicted in grey), while the estimated CE force-velocity relationship is depicted with the black line.

2.2.3.3 Excitation dynamics parameter value estimation

We estimated the parameters values of τ_{act} and τ_{deact} based on isometric experiments. The choice to estimate only these parameters of the excitation dynamics was made because it is generally challenging to discriminate between parameter values of both the calcium dynamics (Equation 2.7) and the $q - [Ca^{2+}]$ relation (Equation 2.8) based on mechanical measurements outlined above. The parameter values of the $q - [Ca^{2+}]$ relation were set to those used in simulating the isometric experiments. Consequently, these parameters matched their actual values for the simulated data, but obviously were suboptimal for the *in situ* data. To estimate the parameter values of τ_{act} and τ_{deact} , we minimised the sum of squared differences between the SEE force of the data and the estimated SEE force based on the parameter values. For

this, an interval of the data was used starting from maximal CE stimulation to 0.1 s after CE stimulation ceased off.

2.2.3.4 Parameter estimation: traditional method & improved method

To estimate the SEE stiffness it is generally assumed that the quick-release is so fast that all MTC shortening is taken up by SEE. As A.V. Hill already acknowledged in (1950), this is not the case in reality as CE also shortens. Although this CE shortening might be small, SEE is in general stiff and therefore small errors in SEE length may result in large errors in predicted SEE forces and thus in estimated SEE stiffness. We used two different methods to estimate the parameter values: 1) the traditional method and the improved method. In the traditional method, we followed the procedure outlined above. In the improved method, we also followed that procedure after which we incorporated an additional procedure to correct for CE shortening due to the quick-release. This additional procedure was as follows: 1) We calculated CE length just before the (step) change in MTC length occurred using Equation 2.1, 2.2, 2.3 and 2.8, assuming that CE velocity was 0 and that γ equalled 1. 2) We performed a (short) simulation from the time just before the (step) change in MTC length to just after the (step) change in MTC length. 3) We computed the change in CE length over this time interval as the average CE length slightly before and slightly after the (step) change in MTC length. 4) We subtracted the change in CE length from the change in MTC length to obtain the change in SEE length. In addition to correcting for CE length change during the quick-release, we incorporated a step to obtain better estimates of the actual PEE force at the time instance of minimum SEE force ($F_{SEE}^{min,data}$; see Figure 2.3). We computed CE force at this time instance using Equation 2.3 and 2.8, assuming that γ equalled its minimum value. This CE force was then subtracted from $F_{SEE}^{min,data}$.

The improved method leads to a different (i.e., higher) SEE stiffness, which affect the estimation of the parameter values of the CE, PEE and SEE force-length relationships (see Section 2.2.3.1). These estimated parameter values were then used to again estimate the parameter values of the CE force-velocity relationship (see section Section 2.2.3.2). As the parameter values of the CE force-velocity relationship

slightly changed, the estimate of CE length change due to the quick-release also changed and therefore estimated SEE stiffness changed. We found that this process always converged to a stable solution and therefore we used an iterative process until the change in all parameter values was less than 0.1% (see Figure 2.S.2).

2.2.4 Sensitivity analysis

2.2.4.1 Monte Carlo simulations

In an experiment involving isolated rat *m. gastrocnemius medialis* (Reuvers et al., 2025), we observed that the decline in measured GM force at sub-optimal MTC lengths over the 8h experiment was substantially higher than the decline in maximal isometric GM force. In a pilot experiment, we systematically assessed the isometric MTC force-length relationship at the start and end of an ~8h pilot experiment. This revealed a shift in the MTC force-length relationship of ~1 mm, while the maximal isometric force was minimally affected. As such, we deemed it most likely the shift in the MTC force-length was mostly the result of damage of the SEE, similar to the results of Aubert et al. (1951).

To assess the influence of these shifts on the parameter value estimation, we performed Monte Carlo simulations. We assumed that the observed shifts of the MTC force-length relationship were caused solely by a decrease in SEE stiffness. Accordingly, we decreased SEE stiffness to cause random shifts of the MTC force-length relationship between 0 and 1 mm. This way, we simulated data as if they were collected at random intervals throughout an 8-hour period. Importantly, changes in SEE stiffness affect the step size and constant MTC velocity ramp required for a more or less a constant SEE force at the beginning of the ramp in step-ramp experiments. Therefore, we adjusted the step size and constant MTC velocity ramp for each simulated step-ramp trial individually. Parameter values of each MTC were then estimated using the ‘perturbed’ data using the improved method (see Section 2.2.3.4). This process was repeated 50 times for each MTC, allowing us to evaluate both the mean and the spread of the estimated parameter values.

2.2.4.2 Interdependency of parameter values

The parameter estimation procedure detailed above follows a fixed order in which certain parameters are estimated before others. For instance, k_{SEE} is estimated first and therefore affects the estimation of L_{SEE}^o and L_{CE}^{opt} . This sequence creates an interdependency among the parameter values, which has been identified as a potential cause for substantial variation in parameter values among individuals, even within the same muscle type (Lemaire et al., 2016). To investigate this interdependency, we systematically changed parameter values of our model. Specifically, we increased and decreased each parameter value by 5% relative to the values obtained with the improved method. After this, we re-estimated all other parameter values. This approach allowed us to assess the influence of changes in one parameter on the estimation of others.

2.3 Results

2.3.1 Traditional method versus Improved method

The traditional method did not correct for CE shortening during quick-release experiments. Consequently, SEE shortening due to the quick-release was overestimated by 24% on average. This overestimation of SEE shortening caused a decrease in k_{SEE} (and thus SEE stiffness) of 34% on average. The estimation of the activation time constant was most affected by the underestimated SEE stiffness, resulting in an underestimation of 19% on average. All other estimated parameter values were within 8.7% of their actual values (Table 2.1). These findings show the influence of CE shortening — even within a very brief 10 ms interval — on the parameter value estimation of both the contraction dynamics and excitation dynamics.

In contrast, the improved method corrected for CE shortening during the quick-release experiments. The correction substantially reduced the overestimation of SEE shortening, yielding an accurate estimate of k_{SEE} . As a result, all estimated parameter values deviated by no more than 3% from their actual values (Table 2.1). These results demonstrate that accounting for CE shortening — even over a brief 10 ms interval — substantially improves the accuracy of the parameter value estimation.

For interested readers, we discuss below the specific factors contributing to the differences between the actual contraction and excitation dynamics parameter values and those estimated based on the traditional method.

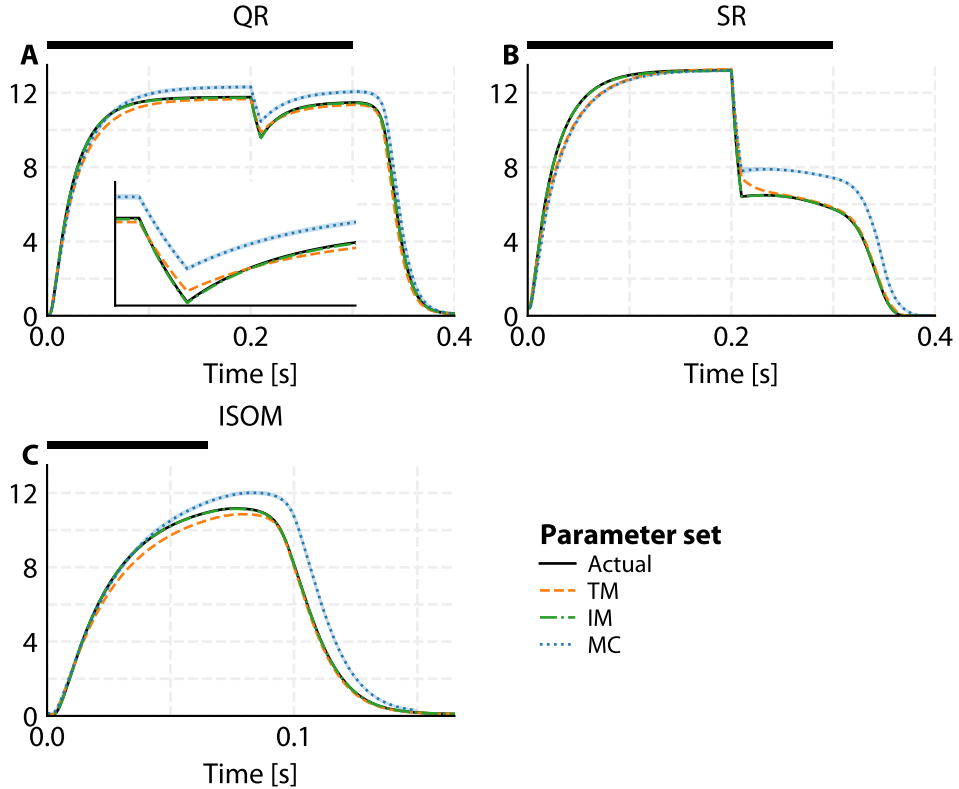


Figure 2.6: Representative example of SEE force over time during a quick-release (A), step-ramp (B) and isometric experiment (C). The inset in (A) depicts the SEE force over time around the quick-release. The SEE force over time is depicted for four sets of parameter values: 1) the actual values (i.e., literature-obtained; black solid line), those obtained with the traditional method (TM; orange dashed line), those obtained with the improved method (IM; green dashed-dotted line) and those resulting from the Monte Carlo Simulations (MC; blue dotted line, with shaded 95% confidence interval).

Table 2.1: Percentage differences between estimated and actual MTC parameter values.

	<i>Traditional method</i>			<i>Improved method</i>			<i>Monte Carlo</i>		
	GM1	GM2	GM3	GM1	GM2	GM3	GM1	GM2	GM3
a	1	4	2	1	3	1	1 ± 2	6 ± 2	1 ± 2
b	-0	2	-1	1	2	1	1 ± 2	5 ± 2	1 ± 2
F_{CE}^{max}	0	-1	0	-0	-1	-0	-0 ± 1	-2 ± 1	-0 ± 1
k_{PEE}	3	-9	9	-0	0	-0	3 ± 17	3 ± 16	6 ± 31
k_{SEE}	-39	-27	-37	0	-1	0	-37 ± 6	-36 ± 6	-36 ± 6
L_{CE}^{opt}	-1	-2	-3	-0	0	-0	0 ± 3	0 ± 2	0 ± 3
L_{PEE}^o	0	-1	-0	-0	0	-0	0 ± 3	1 ± 2	0 ± 2
L_{SEE}^o	-1	-0	-0	0	-0	0	-0 ± 1	-0 ± 1	-0 ± 1
τ_{act}	-21	-16	-21	-0	-1	-0	1 ± 3	-0 ± 4	1 ± 4
τ_{deact}	-5	-4	-5	0	0	0	0 ± 1	0 ± 1	0 ± 1

2.3.1.1 Understanding errors in parameter value estimation

Estimation of SEE stiffness. The SEE stiffness parameter (k_{SEE}) was underestimated by 39%, 27% and 37% for GM1, GM2 and GM3, respectively (see Table 2.1). This underestimation directly resulted from the overestimation of SEE shortening due to the quick-release, which was overestimated by $28 \pm 2\%$, $17 \pm 2\%$ and $26 \pm 2\%$ for GM1, GM2 and GM3, respectively, averaged across all quick-release experiments. In absolute terms, CE shortened only 43 ± 3 , 29 ± 2 and 41 ± 2 μm over the 11 ms interval between the time points at which SEE force and SEE length were sampled (i.e., immediately before and after the quick-release). The amount of CE shortening was smallest in GM2 because GM2 was a slower muscle than GM1 and GM3. Consequently, the overestimation of SEE shortening was also smallest in GM2, which in turn led to the smallest underestimation of k_{SEE} . All in all, this shows that even very small amounts of CE shortening has substantial influence on the estimation of SEE stiffness.

Since SEE was very stiff, the overestimation in SEE elongation at maximal isometric CE force was only 0.50, 0.32 and 0.49 mm for GM1, GM2 and GM3, respectively. This is an important finding because the SEE stiffness is the first parameter in the estimation process and therefore affects all subsequent estimated parameter values. Therefore, since the overestimation of SEE elongation (in mm) at maximal isometric

CE force was only small, the impact on the estimation of the other contraction dynamics parameter values was also small.

Lastly, we note that the correlation coefficient between the data SEE force-length relationship and the one based from the estimated parameters is not an adequate measure of how accurately SEE stiffness is captured. Using the traditional method, we also found high correlation coefficients ($r^2 > 0.99$), yet the SEE stiffness was substantially underestimated. This demonstrated that the correlation coefficient is not sensitive to the overestimation of SEE shortening due to the quick-release. To further investigate this issue, we simulated the quick-release experiments after obtaining all contraction and excitation dynamics parameter values. The simulations clearly showed a slower rise in SEE force after the quick-release in comparison with the experimental data (Figure 2.6A), indicating that SEE stiffness was underestimated. These results suggest that the rise in SEE force is a better measure of SEE stiffness estimation accuracy.

Estimation of PEE stiffness. The PEE stiffness parameter k_{PEE} was 2.8% higher, 8.6% lower and 8.7% than the actual parameter value for GM1, GM2 and GM3, respectively (see Table 2.1). PEE slack length (L_{PEE}^o) values were 0.29% higher, 1.4% lower and 0.05% lower than the actual values for GM1, GM2 and GM3, respectively (see Table 2.1). The available data to estimate k_{PEE} and L_{PEE}^o was limited as in most quick-release experiments PEE length was below L_{PEE}^o . In fact, only 4, 4 and 3 datapoints were used to estimate the PEE force-length relationship for GM1, GM2 and GM3, respectively. Notwithstanding, the estimated PEE parameter values were within 8.7% of their actual value.

Estimation of F_{CE}^{max} , L_{CE}^{opt} and L_{SEE}^o . The estimated values of F_{CE}^{max} , L_{CE}^{opt} and L_{SEE}^o were within 2.6% of their actual values (see Table 2.1). The deviations were all due to the underestimation of k_{SEE} , which can be explained in detail as follows. First, the width of the model's MTC force-length relationship is mainly determined by k_{SEE} and L_{CE}^{opt} . The lower k_{SEE} , the wider the MTC force-length relationship; the shorter L_{CE}^{opt} , the narrower the relationship. Consequently, underestimating k_{SEE} caused an underestimation of L_{CE}^{opt} . Second, a lower L_{CE}^{opt} results in a lower L_{MTC}^{opt} . This led to L_{SEE}^o being overestimated such that L_{MTC}^{opt} of the data and model were aligned.

Third, a lower k_{SEE} , lower L_{CE}^{opt} and higher L_{SEE}^o results in a narrower MTC force-length relationship (see Figure 2.1A). This resulted in an increase F_{CE}^{max} in order to minimise the root mean squared distance between estimated MTC force-length relationship and that of the data. Therefore, the underestimation of SEE stiffness resulted in a small overestimation of F_{CE}^{max} , and L_{SEE}^o , and a small underestimation of L_{CE}^{opt} .

Estimation of CE force-velocity parameter values. The estimated value of a was overestimated by up to 3.8%, while b was within 1.6% of its actual value (see Table 2.1). The slight overestimation of a resulted from the estimation of CE force and CE velocity over a 10 ms interval in which the SEE force changed the least (see Figure 2.3B). At high shortening velocities, CE length decreased so rapidly that — even over a short interval of 10 ms — SEE force changed. Consequently, assuming that CE velocity equalled MTC velocity led to an overestimation of CE velocity. These findings show that it is crucial in experiments to find a combination of step sizes and constant MTC velocity ramps resulting in a nearly constant SEE force. While this is experimentally challenging, achieving this would allow for accurate estimation of the CE force-velocity relationship

Estimation of excitation dynamics parameter values. τ_{act} and τ_{deact} were underestimated across all muscles. For GM1, GM2, and GM3, τ_{act} was underestimated by 21%, 16% and 21%, respectively, while τ_{deact} was underestimated by 4.9%, 3.6% and 5.1%, respectively (see Table 2.1). This underestimation was primarily due to the underestimation of SEE stiffness. In short, the underestimation of SEE stiffness caused an overestimation of SEE lengthening velocity, which in turn lead to an overestimation of CE shortening velocity. This reduced CE force via the CE force-velocity relationship and therefore a slower increase in SEE force. Consequently, τ_{act} was underestimated leading to faster increase in SEE force. The underestimation of τ_{act} also resulted in an overestimation of the relative $[Ca^{2+}]$ at the start of the deactivation, causing a decrease in τ_{deact} (leading to a faster decrease in SEE force). This finding highlights the influence of the contraction dynamics on the estimation of the excitation dynamics parameter values.

To illustrate the influence of the contraction dynamics on the estimation of the excitation dynamics parameter values, we also fitted τ_{act} and τ_{deact} to the data of the quick-release and step-ramp experiments. A similar procedure was used as estimating τ_{act} and τ_{deact} from the isometric experiments. Specifically, we selected an interval of the data starting from maximal CE stimulation to 0.1 s after CE stimulation ceased off. Within this interval, we minimised the root mean sum of squared differences between the SEE force of the data and the estimated SEE force based on the parameter values. Using the quick-release data, τ_{act} was underestimated by 38% on average and based on step-ramp data τ_{act} was underestimated by 64% on average. In turn, τ_{deact} was underestimated by only 1% on average for both data sets. These results show that quick-release and step-ramp experiments are not suitable for accurately estimating τ_{act} due to substantial CE length changes that occur in these experiments. While the effect of contraction dynamics on the estimation of the excitation dynamics parameter values cannot be fully eliminated, it can be minimised by using experimental data in which CE length changes are minimal. Isometric experiments are best suited for this purpose.

2.3.2 Sensitivity analysis

2.3.2.1 Monte Carlo simulations

Monte Carlo simulations were used to examine how shifts in the MTC force-length relationship caused by a decrease in SEE stiffness (e.g., due to irreversible SEE damage in experiments) affect the accuracy of the estimated contraction and excitation dynamics parameter values. The induced shifts of the MTC force-length relationship were between 0–1 mm, and therefore it was no surprise that SEE stiffness decreased by 36% on average — which corresponds to a 0.5 mm shift.

All other estimated contraction dynamics parameter values were within 6.3% on average of their actual value (Table 2.1). The standard deviation in the estimated contraction dynamics parameter values was below 6.1% for all parameters except for the PEE stiffness scaling parameter. The PEE stiffness scaling parameter showed substantially higher variance (with a standard deviation up to 31%), because it was based on four or fewer quick-release trials, making it more sensitive to perturbations

in the data. Regarding the excitation dynamics, the influence on the average time constants was minimal (within 0.7%), but affected the variance in the estimated values of the activation time constant (with a standard deviation of 4% on average). These findings indicate that an average decrease in SEE stiffness of 36% has a much smaller effect on the estimated parameter values of the contraction and excitation dynamics, even at the level of an individual muscle.

2.3.2.2 Interdependency of parameter values

We investigated the interdependency of the estimated parameter values by adjusting each parameter value by 5% and re-estimating all other parameter values. F_{CE}^{max} and L_{SEE}^o were the parameters that had most effect on the estimates of the others.

First, F_{CE}^{max} affected the estimation of parameter a and b of the CE force-velocity relationship. Underestimating F_{CE}^{max} leads to an underestimation of CE force at 0 velocity. Due to the formulation of the CE force-velocity relationship, the curve, by definition, crosses the point at $v_{CE} = 0$ and $F_{CE} = F_{CE}^{max}$. Consequently, the best fit to the data with an underestimated F_{CE}^{max} is a much flatter CE force-velocity relationship, which is realised by an increase in a and a decrease in b . This flatter CE force-velocity relationship affects the estimation of all other contraction dynamics parameter values because the CE force-velocity relationship is used to estimate the CE shortening due to the quick-release with the improved method. As a result, k_{SEE} and L_{SEE}^o decreased due to underestimating F_{CE}^{max} , whereas L_{CE}^{opt} increased. In summary, F_{CE}^{max} substantially influenced the estimation of all contraction dynamics parameter values mainly by its effect on the CE force-velocity relationship.

Second, L_{SEE}^o affected the estimation of L_{CE}^{opt} and F_{CE}^{max} . As explained earlier, when L_{SEE}^o is overestimated, L_{CE}^{opt} is underestimated, such that L_{MTC}^{opt} remains more or less unaffected. Underestimating L_{CE}^{opt} causes a narrower MTC force-length relationship, leading to an overestimation of F_{CE}^{max} to preserve a good fit between the data and the model. As previously discussed, overestimating F_{CE}^{max} , in turn, resulted in an underestimation of parameters a and b . Thus, L_{SEE}^o substantially influenced the estimation of all contraction dynamics parameter values mainly by its effect on the MTC force-length relationship.

Taken together, the interdependence of F_{CE}^{max} and L_{SEE}^o with other parameter values underscores the need for precise estimation of these key parameters. In this regard, it is reassuring that these two parameters were found to be robust for perturbations in the experimental data.

Table 2.2: Interdependency of the estimated MTC parameter values. Each entry shows the percentage change in the row parameter resulting from a 5% change in the column parameter. All values are expressed as percentage changes.

	a	b	F_{CE}^{max}	k_{SEE}	L_{CE}^{opt}	L_{SEE}^o
a	-	10.2 ± 1.1	-14.1 ± 1.7	-0.1 ± 0.0	3.4 ± 0.4	-10.3 ± 1.5
b	2.3 ± 0.2	-	-14.2 ± 2.0	0.1 ± 0.0	3.8 ± 0.5	-10.7 ± 1.3
F_{CE}^{max}	-0.0 ± 0.0	-0.0 ± 0.0	-	-0.1 ± 0.0	-1.2 ± 0.2	3.6 ± 0.8
k_{PEE}	-0.2 ± 0.1	-0.4 ± 0.1	2.2 ± 0.5	-1.3 ± 0.4	-0.5 ± 0.1	1.4 ± 0.3
k_{SEE}	0.6 ± 0.2	1.5 ± 0.3	-6.5 ± 2.1	-	1.6 ± 0.4	-4.2 ± 1.1
L_{CE}^{opt}	0.0 ± 0.0	0.0 ± 0.0	-5.3 ± 0.6	0.2 ± 0.1	-	-14.0 ± 1.3
L_{PEE}^o	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	-10.5 ± 0.7
L_{SEE}^o	0.0 ± 0.0	0.0 ± 0.0	1.8 ± 0.3	0.1 ± 0.0	-1.7 ± 0.1	-
τ_{act}	0.6 ± 0.1	1.7 ± 0.1	-3.3 ± 1.1	2.3 ± 0.2	1.0 ± 0.7	-3.5 ± 2.7
τ_{deact}	0.2 ± 0.0	0.5 ± 0.0	-2.9 ± 0.2	0.6 ± 0.1	1.2 ± 0.1	-2.8 ± 0.3

2.3.3 Evaluating model predictions - *in situ* data

We estimated contraction and excitation dynamics parameter values from *in situ* data of rat *m. gastrocnemius medialis* (Table 2.S.2). The improved method yielded a 67% higher SEE stiffness compared to the traditional method. As explained earlier, higher SEE stiffness also affects the estimation of all other parameter values. The most noticeable changes were longer time constants for both the activation (τ_{act}) and deactivation (τ_{deact}) dynamics, with increases of 61% and 16% on average, respectively. Another noticeable change was in CE optimum length, which was about 11% higher using the improved method compared to the traditional method.

To assess model predictions, we re-ran quick-release, step-ramp and isometric experiments, as well as stretch-shortening cycles with experimentally measured MTC length and CE stimulation over time as inputs to the model. We compared the model predictions using parameter sets obtained from both methods. For all experiments and rats, the average difference between experimental data and

model predictions was 30% smaller on average using parameters from the improved method (Figure 2.8; Table 2.S.3). Overall, the improved method substantially enhanced the predictions derived by the Hill-type MTC model compared to the traditional method.

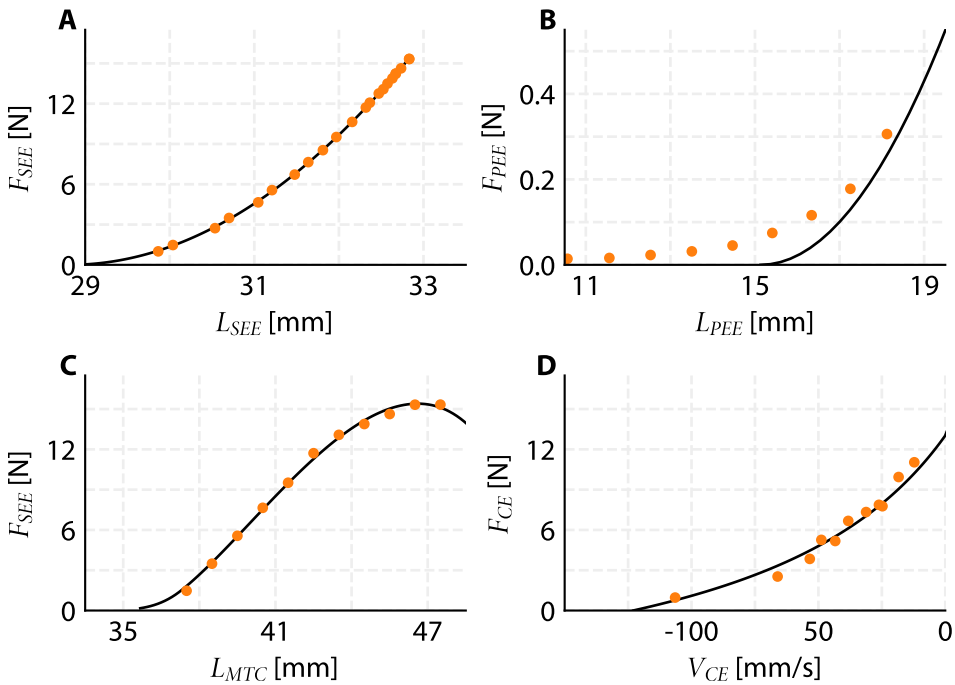


Figure 2.7: Representative example of experimental *in situ* data of rat 1 for the SEE force-length relationship (A), PEE force-length relationship (B), MTC force-length relationship (C) and the CE force-velocity relationship (D). The orange dots depicts the experimental *in situ* data and the solid black line depicts the model fit.

2.4 Discussion

The aim of this study was to evaluate the accuracy of estimating contraction and excitation dynamics parameter values on the basis of data collected in commonly used experimental protocols. In real experiments, the actual parameter values are unknown, making it impossible to directly assess estimation accuracy. In this study, we took a different approach: using a Hill-type MTC model with parameter values obtained from literature we generated synthetic ‘data’ by simulating quick-release, step-ramp, and isometric experiments. We then estimated all important contraction

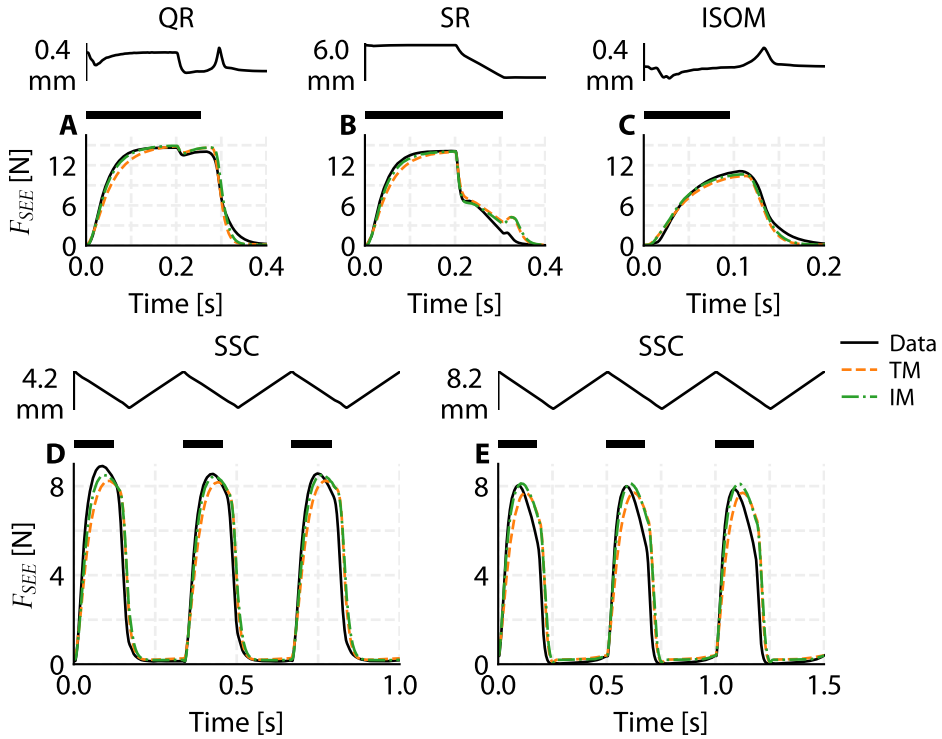


Figure 2.8: Representative example of experimental *in situ* data and simulation results of rat 1 for SEE force over time during a quick-release experiment (A), step-ramp experiment (B), isometric experiment (C) and two stretch-shortening cycles (D & E). The simulation results were obtained by re-simulating the experimental protocol with the MTC length and CE stimulation from the experimental data as input and by using either the parameter set obtained with the traditional method (TM; orange dashed line) or the improved method (IM; green dash-dotted line). For each panel, the top plot represents MTC length over time, with the bar indicating its range in mm. CE stimulation is maximal during the periods indicated by the black bars and is 'off' elsewhere.

and excitation dynamics parameter values by using the synthetic data. Since the actual parameter values were known in this case, we could assess how accurately they were retrieved. Two different estimation methods were compared. The first was the traditional method, commonly used in the literature, which does not account for muscle fibre shortening during quick-release experiments. The second was an improved method that includes a correction for muscle fibre shortening during the quick release. Both methods developed in this study—designed to estimate contraction and excitation dynamics parameters from quick-release, step-ramp, and isometric experiments using servomotors—are made available as an open-source toolbox.

In this study, we focused on the muscle properties that are most commonly estimated in experimental studies and incorporated in musculoskeletal models. Previous work has shown that Hill-type MTC models based on these properties can adequately predict the mechanical behaviour of real muscles with surprising accuracy during relatively complex *in situ* contractions (e.g. Curtin et al., 1998; Ingen Schenau et al., 1988; Lemaire et al., 2016; Sandercock and Heckman, 1997; Wakeling and Johnston, 1999; Williams et al., 1998; Zandwijk et al., 1996). Nevertheless, this set of properties does not provide a complete description of muscle mechanics. Additional muscle properties, such as the potential role of titin in muscle stiffness and history dependent force enhancement, may also importantly influence muscle behaviour. When estimating such properties, the same approach adopted in the present study could be applied: designing dedicated experiments that are sufficiently sensitive to the properties of interest in order to estimate the corresponding muscle properties. In this sense, the framework presented here provides a blueprint for systematically evaluating and estimating muscle properties.

In quick-release experiments using servomotors, muscle fibres shorten, even within the short duration of the release. Substantial muscle fibre shortening during quick-releases occurs because, prior to the quick-release, the muscle fibres are delivering isometric force with all available cross-bridges being attached. As a result, muscle fibres shorten rapidly during the quick-release. Under these conditions, cross-bridge cycling plays only marginally influences the amount of CE shortening.

To demonstrate this, we performed simulations with a Huxley-type MTC model, that explicitly models cross-bridge dynamics (see Lemaire et al., 2016). This model predicted an average CE shortening of 40 μm over the 11 ms interval just before and just after the quick-release for GM1. This CE shortening of 40 μm was only marginally lower than the 43 μm predicted with a Hill-type model. Hence, because nearly all cross-bridges are already attached before and during the quick-release, (the relatively slow) cross-bridge cycling has minimal impact on the magnitude of CE shortening during quick-releases.

Obviously, the extent of muscle fibre shortening depends on the duration of the quick-release. Consequently, the longer the quick-release, the more important it becomes to account for muscle fibre shortening when estimating SEE stiffness (see Figure 2.S.3). For a typical quick-release duration of 10 ms, we found that muscle fibre shortening was nearly 25% of MTC shortening (and thus 33% of SEE shortening), resulting in an underestimation of SEE stiffness by approximately 35%. This finding is important, especially in studies aiming to estimate SEE energy storage from SEE force or length, since energy storage estimates scale linearly with SEE stiffness. Although SEE was substantially underestimated with the traditional method, this had minimal effect on the estimated parameter values of the PEE and CE force-length relationships and the CE force-velocity relationship, but impacted the estimated excitation dynamics parameter values. The improved method, in turn, yielded SEE stiffness values close to the actual ones and substantially enhanced the estimates of the other parameter values.

Our sensitivity analysis revealed substantial interdependencies between certain parameters. Particularly, we observed high sensitivity of estimated parameter values to variations in the maximal isometric CE force (F_{CE}^{max}) and the SEE slack length (L_{SEE}^0) parameter values. Despite this sensitivity, it is important to note that these parameters were accurately estimated even from perturbed experimental data. This was true in general: contraction and excitation dynamics parameter values were quite robust against shifts in the MTC force-length relationship of the simulated data according to the Monte Carlo simulations. These findings indicate the robustness of the improved method.

We applied both the traditional and the improved method to *in situ* data from quick-release, step-ramp, and isometric experiments on rat *m. gastrocnemius medialis*. The improved method yielded substantially higher estimates of SEE stiffness compared to the traditional method (67% on average). Using parameter sets from both methods, we simulated the quick-release, step-ramp, and isometric experiments, as well as independent stretch-shortening cycles, all with experimentally obtained MTC length and CE stimulation over time as input to the model. With parameter estimates obtained with the traditional method, the root mean squared difference between predicted and experimentally measured force was about 5.6% of maximal CE force, averaged across all rats and experiments; with parameter estimates obtained with the improved method this was only 3.9%. The small difference between predicted and experimentally observed force over time with parameters from the improved method suggests that muscle force can be accurately predicted with a Hill-type MTC model across a wide variety of contractions. This is a remarkable finding for two reasons. First, the Hill-type MTC model is obviously a simplification of real muscle, abstracting the muscle belly, aponeurosis, and tendon into the CE, SEE, and PEE components, and thus does not account for certain complexities (e.g., non-constant pennation angle, muscle inhomogeneities, etc.). Second, the Hill-type MTC model used in this study consists of 26 parameters, of which only 10 values were estimated. This means that all other parameters were left suboptimal. Despite these simplifications, the model was able to accurately predict muscle behaviour, demonstrating that a relatively simple model with a limited number of estimated parameter values can still capture important aspects of muscle dynamics.

In conclusion, our results demonstrate that the improved parameter estimation method provides accurate and robust estimates of contraction and excitation dynamics properties, offering a reliable approach for muscle property estimation in both biomechanical and muscle physiology research. The approach that we designed in this study is made available as an open-source toolbox (<https://github.com/edwinreversers/mp-estimator>).

Glossary

Symbol	Description
a	Hill constant
b	Hill constant
b_{min}^{scale}	Minimal scale factor of b (i.e., the minimal value of b_{scale})
b_{shape}	Determines the steepness of the relation between b_{scale} and q
F_{asympt}	Oblique asymptote of the eccentric part of the CE force-velocity relationship
F_{CE}^{max}	Maximum isometric CE force
k_{PEE}	Scales the PEE stiffness
k_{SEE}	Scales the SEE stiffness
L_{CE}^{opt}	CE length at which CE can produce maximal isometric force
L_{PEE}^o	PEE slack length
L_{SEE}^o	SEE slack length
r_{as}	Slope of the slanted asymptote of the eccentric part of the CE force-velocity relationship
r_{slope}	Ratio between the eccentric and concentric derivatives of $\frac{dF_{CE}^{rel}}{dv_{CE}^{rel}}$ @ $v_{CE} = 0$
w	Determines the width of the CE force-length relationship
a_{act}	Determines the steepness of the relation between γ and q
$b_{act,1}$	Determines together with $b_{act,2}$ and $b_{act,3}$ the pCa^{2+} level at which $q = 0.5$
$b_{act,2}$	Determines together with $b_{act,1}$ and $b_{act,3}$ the pCa^{2+} level at which $q = 0.5$
$b_{act,3}$	Determines together with $b_{act,1}$ and $b_{act,2}$ the pCa^{2+} level at which $q = 0.5$
k_{Ca}	Relates γ to the actual Ca^{2+} concentration
γ_o	Minimal value of γ
q_o	Minimal value q
τ_{act}	Activation time constant of the calcium dynamics
τ_{deact}	Deactivation time constant of the calcium dynamics

Acknowledgments

The authors thank Maarten F. Bobbert for helpful comments on a draft of the paper. The authors also thank Koen K. Lemaire for insightful discussions.

Competing interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: E.D.H.M.R.; Formal analysis: E.D.H.M.R.; Investigation: E.D.H.M.R.; Writing - Original Draft: E.D.H.M.R.; Writing - Review & Editing: E.D.H.M.R., D.A.K.; Visualization: E.D.H.M.R.; Supervision: D.A.K.; Funding acquisition: D.A.K.

Funding

This work was funded by The Dutch Research Council (NWO) [21728 to D.A.K.].

Data and resource availability

All data, code, and materials used in this study are openly available:

- **GitHub repository:** All raw data, processed data, and analysis code are hosted on GitHub at <https://github.com/edwinreusers/acc-mp-estimation>.
- **Reproducible analysis website:** Full analysis pipeline — including data, analysis code, and figure/table generation — is available at <https://edwinreusers.github.io/publications/acc-mp-estimation>

2.S Supplementary material

2.S.1 Supplemental figures

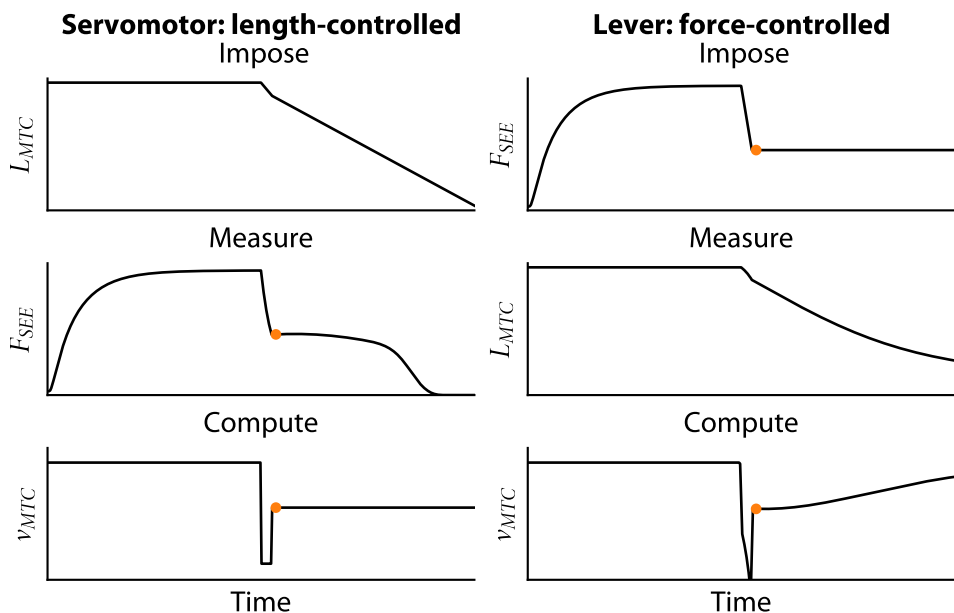


Figure 2.S.1: Comparison between a length-controlled step-ramp experiment and a force-controlled quick-release experiment. In length-controlled step-ramp experiment (left), MTC length over time is imposed using a servomotor, while SEE force is measured. MTC velocity is computed at the time instance at which SEE force is near constant. In the force-controlled quick-release experiment (right), SEE force over time is imposed via a lever, while MTC length is measured. MTC velocity is computed at the time instance where MTC velocity is maximal after the change in SEE force. Both methods yield one datapoint (depicted with the orange dot) of the force-velocity relationship.

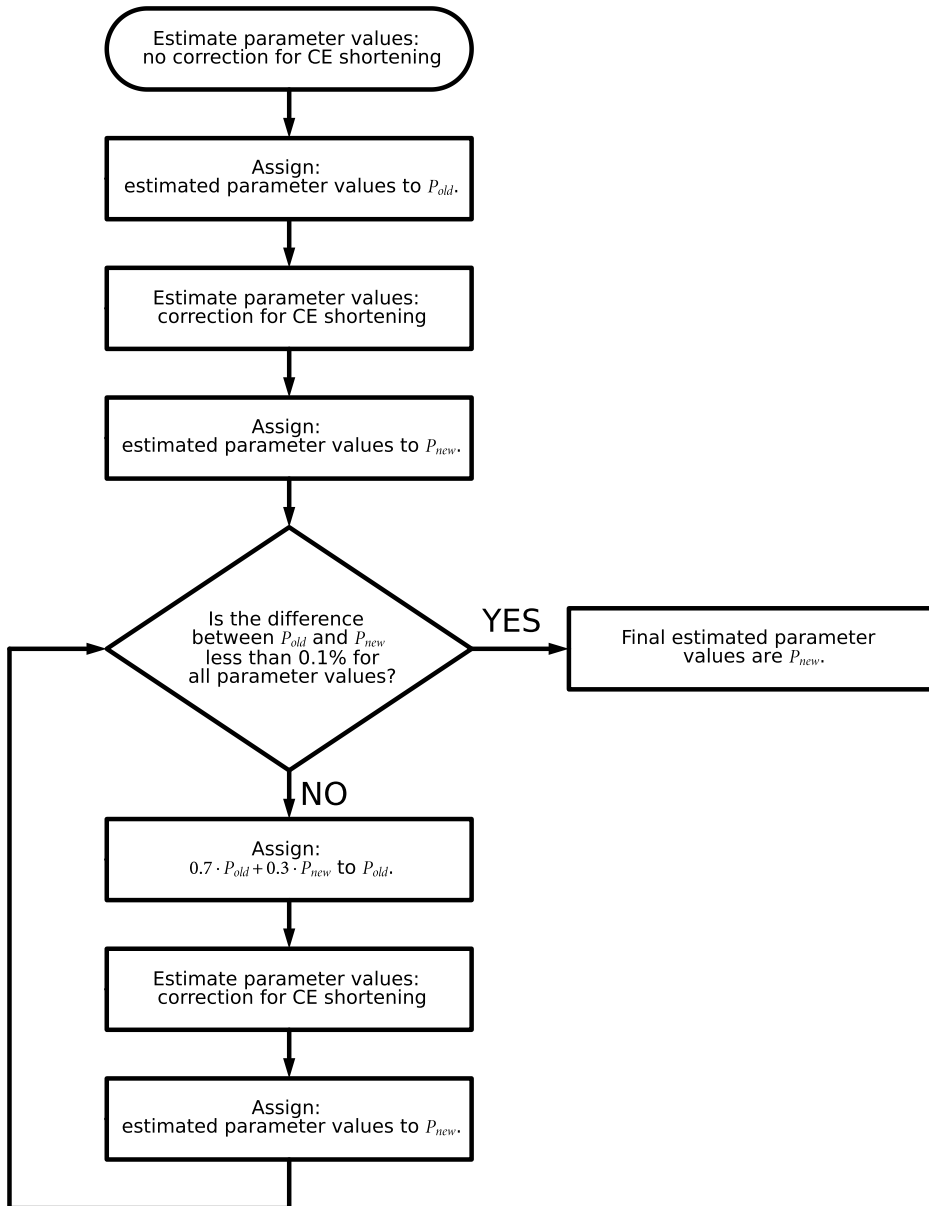


Figure 2.S.2: Flowchart of the improved method. Using the improved method, parameter values were estimated until the change in all parameter values was less than 0.1%.

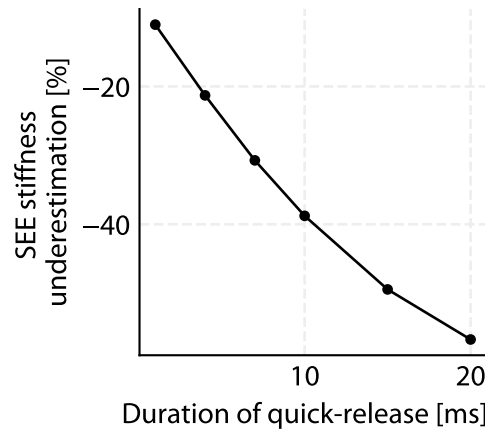


Figure 2.S.3: Relationship between quick-release duration and SEE stiffness underestimation. When the duration of the quick-release increases, the underestimation of SEE stiffness increases.

2.S.2 Supplemental tables

Table 2.S.1: MTC parameter values obtained from literature to simulate data.

Parameter	Unit	GM1	GM2	GM3	Reference
<i>Contraction dynamics</i>					
a	N	2.68	1.80	2.58	Ref. 1
b	mm	41.6	24.8	41.8	Ref. 1
b_{scale}^{min}	-	0.100			Ref. 2
b_{shape}	-	22.0			Ref. 2
F_{asympt}	-	1.50			Ref. 3
F_{CE}^{max}	N	13.4	13.8	12.3	Ref. 1
k_{PEE}	N/mm ²	213	165	511	Ref. 1
k_{SEE}	N/mm ²	4220	3640	3470	Ref. 1
L_{CE}^{opt}	mm	13.2	12.3	11.2	Ref. 1
L_{PEE}^o	mm	13.9	13.2	13.4	Ref. 1
L_{SEE}^o	mm	28.3	30.3	26.5	Ref. 1
r_{as}	-	$3.71 \cdot 10^{-3}$	$5.45 \cdot 10^{-3}$	$3.16 \cdot 10^{-3}$	Ref. 4
<i>Excitation dynamics</i>					
r_{slope}	mm	2.00			Ref. 5
w	-	0.50			Ref. 6
a_{act}	-	-7.37			Ref. 7
$b_{act,1}$	-	5.17			Ref. 7
$b_{act,2}$	-	0.596			Ref. 8
$b_{act,3}$	-	0.00			Ref. 8
kCa	mol/L	$8.00 \cdot 10^{-6}$			Ref. 9
γ_o	-	$1.00 \cdot 10^{-5}$			Ref. 4
q_o	-	$5.00 \cdot 10^{-3}$			Ref. 10
τ_{act}	ms	27.0			Ref. 1
τ_{deact}	ms	27.0			Ref. 1

Ref. 1: van Zandwijk et al. (1996), Ref. 2: van Soest and Bobbert (1993), Ref. 3: Rijkhelhuizen et al. (2003), Ref. 4: Arbitrary small value, Ref. 5: Katz (1939), Ref. 6: Burkholder and Lieber (2001), Ref. 7: Bortolotto et al. (2000), Ref. 8: Stephenson and Williams (1982), Ref. 9: Kistemaker et al. (2005) and Ref. 10: Hatze (1981)

Table 2.S.2: Estimated parameter values of the experimentally measured *in situ* data using both the traditional as well as the improved method.

Parameter	Unit	<i>Traditional method</i>			<i>Improved method</i>		
		Rat 1	Rat 2	Rat 3	Rat 1	Rat 2	Rat 3
a	N	8.90	11.3	10.5	8.82	10.0	11.0
b	mm	76.7	108	84.6	81.9	105	90.6
F_{CE}^{max}	N	15.6	14.1	17.1	15.5	14.1	17.0
k_{PEE}	N/mm ²	9.83	7.25	21.9	28.7	24.1	34.8
k_{SEE}	N/mm ²	608	688	654	952	1260	1050
L_{CE}^{opt}	mm	12.4	12.9	12.3	13.7	14.7	13.3
L_{PEE}^o	mm	12.4	12.4	13.3	15.1	15.6	14.5
L_{SEE}^o	mm	29.0	29.7	25.1	28.8	29.3	25.2
τ_{act}	ms	43.0	32.4	23.6	55.2	57.7	41.6
τ_{deact}	ms	23.8	20.9	19.7	27.1	25.3	22.4

Table 2.S.3: Root mean squared differences between experimentally measured SEE force histories and those predicted by Hill-type MTC model after estimating all contraction and excitation dynamics parameter values.

	QR	SR	ISOM	SSC
<i>Traditional method</i>				
Rat 1	5.6 ± 3.4	5.6 ± 0.8	6.8 ± 3.2	6.2 ± 3.0
Rat 2	4.2 ± 1.5	5.0 ± 0.6	4.3 ± 1.6	4.6 ± 2.2
Rat 3	6.4 ± 2.3	6.8 ± 1.0	6.1 ± 2.1	5.5 ± 2.7
AVG ± SD	5.4 ± 2.6	5.8 ± 1.1	5.7 ± 2.7	5.4 ± 2.7
<i>Improved method</i>				
Rat 1	3.3 ± 1.5	2.8 ± 0.9	5.4 ± 3.0	5.4 ± 3.1
Rat 2	2.7 ± 1.6	2.1 ± 0.5	4.2 ± 2.6	4.2 ± 2.0
Rat 3	4.2 ± 1.8	2.6 ± 0.5	4.9 ± 2.3	5.0 ± 2.7
AVG ± SD	3.4 ± 1.8	2.5 ± 0.7	4.8 ± 2.7	4.9 ± 2.7

Root mean squared differences are expressed as a percentage of the maximal isometric CE force (F_{CE}^{max}). Root mean squared differences were computed over the interval in which CE stimulation was maximal for the quick-release and step-ramp experiments and was computed over the interval from CE stimulation onset to 0.1 s after CE stimulation offset for the isometric experiments and the stretch-shortening cycles.

**Mechanical power output during
stretch–shortening cycles of rat medial
gastrocnemius muscle: influence of
various muscle length trajectories**

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PLOS Computational Biology, 22(8), 1-23

<https://doi.org/10.1371/journal.pcbi.1014587>

Abstract

The average mechanical power output (AMPO) during a stretch–shortening cycle produced by a muscle depends on muscle length and stimulation over time. While the effects of cycle frequency and muscle length excursion on AMPO are well-known, several questions remain about the effects of muscle length and stimulation over time on the maximally attainable AMPO. For example, which precise muscle length and stimulation over time yield maximal AMPO? *In situ* experiments are inherently limited to a finite set of muscle length and stimulation over time. To overcome this limitation, we combined *in situ* experiments on rat *m. gastrocnemius medialis* with Hill-type muscle modelling. We first performed dedicated trials to estimate the muscle-tendon-complex (MTC) properties of each rat. Subsequently, we performed various stretch–shortening cycles with substantial differences in cycle frequency, shortening-to-lengthening time ratio and MTC length excursion. Model-predicted AMPO correlated nearly perfectly with experimentally measured AMPO ($r^2 > 0.98$). This justified further exploration using the Hill-type MTC model. Using the Hill-type MTC model, we predicted that AMPO peaks at a cycle frequency of 3.5 Hz, with a shortening-to-lengthening time ratio of 6:1 and an MTC length excursion of 8 mm. Notably, cycle frequency and MTC length excursion showed a strong interaction: increasing one necessitated a decrease in the other to maximise AMPO. By contrast, the optimal shortening-to-lengthening time ratio remained remarkably constant across all tested combinations of cycle frequency and MTC length excursion. This shows that muscles should spend substantially more time shortening than lengthening to maximise AMPO.

3.1 Introduction

In motor activities such as cycling, rowing, and speed skating, muscles periodically shorten and lengthen, a pattern commonly referred to as the stretch–shortening cycle (SSC). When muscles produce force while shortening, they deliver positive mechanical power, and when they produce force while lengthening, they deliver negative mechanical power. During motor activities, part of the mechanical energy delivered by the muscles is dissipated to the environment, and this energy loss generally increases with movement speed. Consequently, performance critically depends on the average mechanical power output (AMPO) that muscles can produce over these SSCs. This raises several fundamental questions: how does muscle length over time influence the maximally attainable AMPO? More specifically, what are the effects of cycle frequency and muscle length excursion? Which fraction of the cycle should be spent shortening? Do these factors interact, and if so, how? Surprisingly, despite the importance of AMPO for movement performance, many of these questions have not yet been answered.

To study the mechanical output during a stretch–shortening cycle (SSC), researchers use an experimental setup in which a motor imposes cyclical length changes on either isolated muscle fibres or the entire muscle–tendon complex (MTC). While the approach is the same at both levels, this paper focuses on SSCs at the MTC level. During each SSC, the MTC first undergoes lengthening and then shortens back to its initial length. Muscle stimulation is applied during part of the SSC, while MTC force is measured. When MTC force is plotted against MTC length (change), the area enclosed by the resulting loop represents the net mechanical work produced during a full cycle. This net mechanical work is the sum of the positive mechanical work during MTC shortening (Figure 3.1A) and the negative mechanical work during MTC lengthening (Figure 3.1B). This method has been coined the ‘work loop technique’ (Josephson, 1985). The average mechanical power output (AMPO) for each cycle can then be calculated by dividing the net mechanical work per cycle by the duration of the cycle. As such, the work loop technique provides a powerful tool to study the effect of MTC length and stimulation over time on the maximally attainable AMPO of muscles.

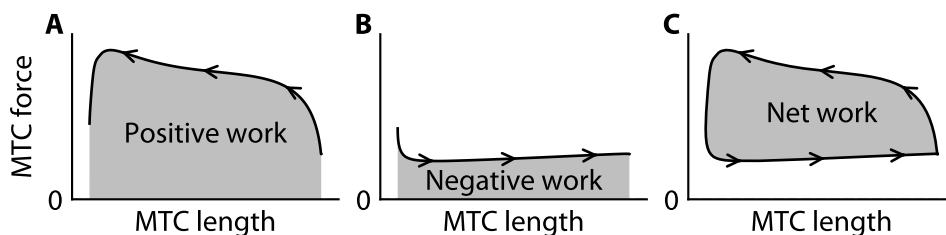


Figure 3.1: An example of a work loop. In a work loop, MTC force is plotted against MTC length (change). The area enclosed by the work loop represents the net mechanical work produced during a full cycle (C), which is the sum of the positive mechanical work during MTC shortening (A) and the negative mechanical work during MTC lengthening (B). The arrows indicate the direction of the work loop over time.

The work loop technique has been extensively employed to examine how MTC (or muscle fibre) length and muscle stimulation over time affect the mechanical behaviour of muscles. In most experiments, SSCs have been studied using sinusoidal length changes (e.g. Josephson, 1985). These SSCs are typically parametrised in terms of cycle frequency and MTC/muscle fibre length excursion, with length changes centred around the length yielding maximal isometric force. Furthermore, muscle stimulation is typically parametrised in terms of stimulation onset time and stimulation duration (i.e. ‘duty cycle’). Since muscle stimulation substantially affects the mechanical behaviour of muscles (e.g. Curtin et al., 2018; Sawicki et al., 2015; Stevens, 1996), stimulation onset time and duration are typically optimised to maximise mechanical work per cycle. This allows isolation of the effects of cycle frequency and MTC/muscle fibre length excursion, and their interaction. Such an approach has been used in experiments to systematically investigate both the individual effects of cycle frequency and MTC/muscle fibre length excursion and their interaction on AMPO. For example, results of such experiments have demonstrated a substantial interaction between cycle frequency and MTC/muscle fibre length excursion, whereby maximising AMPO typically requires larger length excursions at lower frequencies and smaller length excursions at higher frequencies (e.g. Altringham and Johnston, 1990; Caiozzo and Baldwin, 1997; James et al., 1995). Although sinusoidal SSCs have experimentally provided valuable insights, they inherently impose equal shortening and lengthening durations, leaving unexplored how unequal shortening and lengthening durations affect AMPO.

Askew & Marsh (1997; 1998) performed ground-breaking experiments on the effect of unequal shortening and lengthening durations on the maximally attainable AMPO — remaining, to our knowledge, the only researchers to have done so. In their studies, mouse m. soleus and m. extensor digitorum longus were connected at the distal part of the muscle belly to a servomotor for precise control of the imposed SSCs. The SSCs used by Askew & Marsh were parameterised using three variables (see Figure 3.2): cycle frequency, FTS (fraction of the cycle time spent shortening), and muscle fibre length excursion (i.e. maximum minus minimum fibre length), with near-constant shortening and lengthening velocities. Askew & Marsh (1997; 1998) examined three distinct FTS values — 0.25, 0.50, and 0.75 — and heuristically optimised muscle fibre length excursion, muscle stimulation onset time and muscle stimulation duration to achieve the highest AMPO at different combinations of cycle frequency and FTS. Their findings indicated that AMPO was highest for FTS = 0.75, followed by FTS = 0.50, and lowest for FTS = 0.25 at each cycle frequency at the optimal muscle fibre length excursion and muscle stimulation.

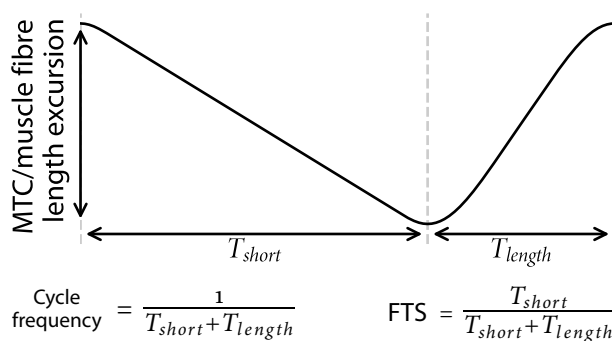


Figure 3.2: Representation of the parameterisation of stretch-shortening cycles. T_{short} and T_{length} denote the shortening and lengthening durations, respectively, of either the muscle-tendon-complex (MTC) or the muscle fibres. FTS denotes the fraction of the cycle time spent shortening. In the example shown, the MTC/muscle fibres shorten 65% of the cycle duration (i.e., FTS = 0.65).

While Askew & Marsh's experimental studies provided valuable insights, several questions remain regarding the effect of SSC parameters on the maximally attainable AMPO. First, Askew & Marsh (1997; 1998) optimised muscle fibre length excursion for each cycle frequency at only three distinct FTS values. Therefore, it remains

unclear how muscle fibre length excursion affects the maximally attainable AMPO at different FTS values. More generally, it remains unclear how sensitive the maximally attainable AMPO is to variations in SSC parameters. For instance, does AMPO substantially decrease with suboptimal SSC parameters, or is there a wide range of SSC parameter combinations that yield near-maximal AMPO? Second, because Askew & Marsh (1997; 1998) only used three distinct FTS values, it remains unclear how cycle frequency and muscle fibre length excursion affect the optimum value of FTS. More generally, it remains unclear how the optimum SSC parameters depend on each other. Third, Askew & Marsh (1997; 1998) imposed near-constant muscle fibre shortening and lengthening velocities. Consequently, it remains unclear how much more AMPO the muscle could produce if the muscle fibre length over time would be optimal for AMPO. For this, the optimal muscle fibre length and stimulation over time should be identified in a parameter-free manner.

While it is obvious that muscle fibres should maximise positive mechanical work and minimise negative mechanical work in order to maximise AMPO, it is far from trivial what SSC and muscle stimulation over time yield maximal AMPO. To minimise negative mechanical work, stimulation should cease well before lengthening begins (because of deactivation time delays) and muscle fibres should ideally lengthen at very low velocities (as muscle fibre force increases with increasing lengthening velocity). However, such strategies are suboptimal for AMPO: the time spent lengthening cannot be used to produce positive mechanical work, and early deactivation reduces the positive mechanical work during shortening. One might argue that muscle fibres should shorten at a velocity that maximises the instantaneous mechanical power output to maximise positive mechanical work. However, muscle fibre force also depends on muscle fibre length, with muscle fibre force decreasing at both shorter and longer lengths than optimum length. Therefore, to maximise AMPO, it may be beneficial for muscle fibres to contract at a lower velocity than the velocity which maximises instantaneous mechanical power output. Doing so limits the negative effects of operating at suboptimal muscle fibre lengths. This raises several open questions: What shortening velocity maximises AMPO? Over what length range should muscle fibres operate? And how much time should

be spent on shortening (to maximise positive mechanical work) versus lengthening (to minimise negative mechanical work) in order to maximise AMPO?

Obviously, it is hard to experimentally find the optimal combination of cycle frequency, FTS, MTC length excursion and muscle stimulation for maximal AMPO. If one would change MTC length excursion, how should the other SSC parameters and muscle stimulation be changed to maximise AMPO? Or, taking it one step further: How should we find the MTC length and stimulation over time that maximises AMPO when the muscle length over time is entirely free to take any form? For this reason, we propose to complement *in situ* experiments with simulations and optimisations using a Hill-type MTC model to study the effects of MTC length and muscle stimulation over time on the maximally attainable AMPO. If the MTC model yields accurate predictions of the influence of SSC parameters and muscle stimulation over time on the maximally attainable AMPO, it would allow us to answer a plethora of questions regarding the maximally attainable AMPO of various SSCs. For instance, we could investigate how the optimum SSC parameters depend on each other, and how sensitive the maximally attainable AMPO is to variations in SSC parameters. Furthermore, using an accurate model, we could optimise MTC length over time for maximal AMPO and quantify how much more AMPO could be produced compared to sinusoidal SSCs or the near-constant velocity SSCs used by Askew & Marsh.

In this study, we aimed to explore the influence of SSC parameters — cycle frequency, FTS and MTC length excursion — on the maximally attainable AMPO of a single isolated MTC. Thus, we considered the maximally attainable AMPO for each combination of SSC parameters. To determine this, muscle stimulation was optimised for every SSC. As discussed earlier, only a limited set of SSCs can be investigated experimentally, so this raises the question: Which SSCs should be selected? To make an informed selection of SSCs to be tested experimentally, we first used a Hill-type MTC model — with parameter values obtained from literature — to simulate a broad range of SSCs. Based on these simulation results, we selected a smaller, representative and informative set of SSCs that captured the most prominent effects, providing a focused basis for evaluating model predictions. This set of SSCs

was then tested in an *in situ* experiment on rat *m. gastrocnemius medialis*. In this experiment, we also performed dedicated trials to estimate the MTC properties of each specimen individually. This allowed us to compare the experimental results with model-based predictions, which showed a nearly perfect correlation between measured and model-predicted AMPO ($r^2 > 0.98$). This strong correspondence then justified the use of the Hill-type MTC model to (finally) explore how SSC parameters — cycle frequency, FTS and MTC length excursion — affect the maximally attainable AMPO and to identify the fully unconstrained optimal MTC length over time for maximal AMPO.

3.2 Methods

We combined *in situ* experiments with simulations and optimisations using a Hill-type MTC model to investigate the influence of SSC parameters — cycle frequency, FTS, and MTC length excursion — on the maximally attainable AMPO of rat *m. gastrocnemius medialis* (GM). We first used a Hill-type MTC model with parameter values from literature to simulate a broad range of SSCs. Based on these simulation results, we selected a representative and informative subset for an *in situ* experiment, in which MTC length and stimulation over time were under full control. In this experiment, we also performed dedicated trials to estimate the MTC properties of each specimen individually. Model-based predictions were then evaluated against the experimental data. The strong correspondence ($r^2 > 0.98$) justified the use of the Hill-type MTC model to explore the influence of SSC parameters on the maximally attainable AMPO across a broad SSC parameter space. First, we derived predictions for SSCs with constant MTC shortening and lengthening velocity. Second, we identified the optimal SSC without any constraints on MTC length over time to investigate how much more AMPO the muscle could produce if the muscle were allowed to shorten-and-lengthen optimally.

3.2.1 *In situ* experiment

3.2.1.1 Animals and ethics

All surgical and experimental procedures conducted in this study were approved by the Committee on the Ethics of Animal Experimentation at the Vrije Universiteit (Permit Number: FBW-AVD11200202114471) and adhered to Dutch law concerning the guidelines and regulations of animal welfare and experimentation. Data was obtained from three male Wistar rats (body mass 365, 410 and 407 gram, respectively). The rats were anaesthetised by intraperitoneally injecting a urethane solution (12.5% urethane) at a dosage of 1.2 mL/100 g body mass. Additional doses of urethane (0.2 mL/g body mass) were administered throughout the experiment as needed to maintain deep anaesthesia. Heart rate, body temperature and withdrawal reflexes to a pain stimulus were monitored every 10 minutes. To prevent dehydration, subcutaneous injections of 1 mL NaCl solution (0.9%) were administered every 1–2 hours. After completion of the experiment, we euthanised the rats by intracardially injecting pentobarbital sodium (Euthasol 20%) followed by a double-sided pneumothorax.

3.2.1.2 Surgical procedure

The hindlimb of the rat was shaved, and the skin was cut and partially removed. Next, *m. biceps femoris* was removed to expose the femur. To minimise mechanical effects of surrounding muscles on GM — in particular myofascial force transmission — the following steps were taken (see Rijkkelijkhuizen et al., 2005): 1) GM was exposed as much as possible from its surrounding tissue. 2) The tendons of *m. plantaris*, *m. gastrocnemius lateralis* and *m. soleus* were cut at their distal ends. 3) The medial and lateral parts of *m. gastrocnemius* were carefully separated. *N. peroneus*, *n. suralis*, and the branch of the *n. tibialis* innervating *m. gastrocnemius lateralis* and *m. soleus* were then cut to further reduce the influence of surrounding tissue on GM. Subsequently, the tissue around the calcaneal tendon was removed such that only a small part of the calcaneus was left. Lastly, a cuff-electrode was placed on *n. ischiadicus*. The part of *n. ischiadicus* proximal to the cuff-electrode was then crushed to prevent muscle excitation via spinal reflexes.

3.2.1.3 Experimental setup

The rats were placed on an electrical heating pad to maintain body temperature at 37 °C. The laboratory temperature was kept constant at 20 °C, with a local humidity around the muscle tissue of about 80%. A servomotor (Aurora 309C, Aurora Scientific, Aurora, Canada) was used to impose MTC length changes and to measure GM force. Before the experiment, we calibrated the position sensor of the servomotor using a dial gauge (2046S, Mitutoyo Nederland BV, Veenendaal, the Netherlands) and calibrated the force sensor using various calibration masses. To relate servomotor position (changes) to MTC length, we measured MTC length at one position with a caliper at the end of the experiment. A direct current stimulator was used to stimulate *n. ischiadicus* with a pulse width of 100 μ s and at a frequency of 100 Hz during all experiments, while the (supramaximal) current was chosen for every rat individually. Before the series of experiments, we ensured that the position, force, and stimulation signals were synchronised in time by applying a physical stimulus that produced a detectable response across all channels. All data were collected at a sample rate of 2000 Hz.

The femur was fixed with a metal clamp and the foot with a plastic plate, such that the hindlimb was rigidly secured to the setup. The distal end of the calcaneal bone was tightly attached to a steel rod with 100% polyester yarn at the leftover part of the calcaneus. The other end of the steel rod was then attached with Kevlar thread to the motor. A similar procedure was applied to *m. gastrocnemius lateralis* (GL) and *m. plantaris* (PL). The distal tendon of GL and PL (GL+PL) was then connected to a second motor to monitor their combined force throughout the experiment in order to verify the absence of mechanical interaction with GM. Care was taken to ensure that the direction of GM force reflected *in vivo* conditions. Lastly, we connected the cuff-electrode to the direct current stimulator (DS2A, Digitimer, Welwyn Garden City, United Kingdom). This setup allowed us to have full control over MTC length and stimulation while measuring GM force (see Figure 3.3).

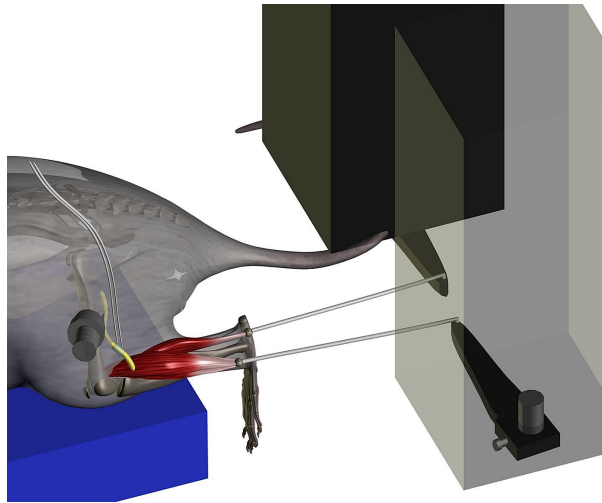


Figure 3.3: Representation of the experimental setup that provided full control of MTC length and stimulation while measuring *m. gastrocnemius medialis* (GM) force. GM was carefully exposed from its surrounding tissue and positioned in the setup such that GM pulled in its natural direction, while the femur and foot were securely fixated. The distal end of the calcaneal tendon was connected to a motor via a steel rod. The distal tendon of *m. gastrocnemius lateralis* and *m. plantaris* was connected to a second motor. A cuff-electrode was placed on n. ischiadicus. N. peroneus, n. suralis and the branch of n. ischiadicus innervating *m. gastrocnemius lateralis* and *m. soleus* were cut such that only GM was innervated.

3.2.1.4 General procedures

After securing the rat in the experimental setup, we conducted a series of tetanic contractions for several purposes. First, we performed isometric contractions to ensure that GM pulled in its natural direction. These isometric contractions were also used to determine the supramaximal stimulation current for GM. The stimulation current was gradually increased until GM force plateaued. The current at which no further increase in GM force was observed was then used consistently throughout the experiment. Additionally, the isometric contractions served as a check to confirm that the surgery was successful, particularly ensuring that no GL+PL force was transmitted to the distal GL+PL tendon, and thus that no myofascial force transmission occurred (Finni et al., 2023). Subsequently, we performed two to three quick-release, step-ramp and isometric experiments along with two to three SSCs. This approach was based on results from pilot experiments, which indicated that imposing these contractions prior to the start of the experiment resulted in more stable GM force throughout the entire experiment.

Following the preparatory procedures, we commenced the main experiment. The main experiment consisted of two parts. In the first part, we performed quick-release, step-ramp and isometric experiments to estimate the MTC properties of each rat. In the second part, we performed various SSCs. To assess if GM was fatigued, two quick-release and two step-ramp experiments were repeated halfway through the second part of the experiment and again at the end of the entire experiment. Additionally, isometric experiments were performed after every six series of SSCs during the second part of the experiment, using a fixed MTC length 2-4 mm below MTC optimum length. Between contractions, GM was returned to slack length, followed by a two-minute rest period. During these rest periods, GM was irrigated with saline to keep the muscle tissue moist. To reduce potentiation effects, two twitches were applied 1.5 seconds before each contraction.

3.2.1.5 Quick-release, step-ramp and isometric experiments to estimate MTC properties

Quick-release, step-ramp, and isometric experiments were performed, with the resulting data being used to estimate MTC properties for each rat individually (see Figure 3 of Reuvers and Kistemaker, 2025). This approach served two purposes. First, it enabled us to predict AMPO for the experimentally tested SSCs for each rat individually. As a result, less rats were required to evaluate predictions against experimental results. Second, it enabled us derive predictions of the influence of a broad range of SSC parameters — cycle frequency, FTS and MTC length excursion — on the maximally attainable AMPO for each rat individually. The full experimental procedure of the quick-release, step-ramp and isometric experiments as well as the MTC property estimation has been described in Reuvers and Kistemaker (2025), where we demonstrated that these properties can be estimated accurately when accounting for muscle fibre shortening quick-releases.

3.2.1.6 stretch–shortening cycles to evaluate predictions

SSCs were parametrised by cycle frequency, FTS, and MTC length excursion (i.e. maximum minus minimum MTC length, see Figure 3.2). The average MTC length was set ~ 3 mm below the MTC length yielding maximal isometric force to prevent irreversible damage to the muscle fibres. MTC velocity was constant during shortening and lengthening phases, except around the transition from MTC lengthening to shortening and vice versa. During these transitioning phases, MTC length changed sinusoidally such that MTC length over time was continuously differentiable to the third order. The maximum MTC acceleration during these transitions was limited to 1500 mm/s^2 to further reduce the risk of irreversible muscle fibre damage. This constraint limited the range of feasible experimental conditions: SSCs with a high cycle frequency, large MTC length excursion, and either a substantially high or low FTS value could not be tested.

Experimental conditions were selected based on preliminary simulations using a Hill-type MTC model with parameter values from literature (see Section 3.S.1). We tested 26 conditions, comprising 13 combinations of cycle frequency and FTS

(see Figure 3.4), each at an MTC length excursion of 4 mm and 8 mm. Full details, including muscle stimulation onset time and duration, are provided in the Supplementary material (Table 3.S.1 & Table 3.S.2).

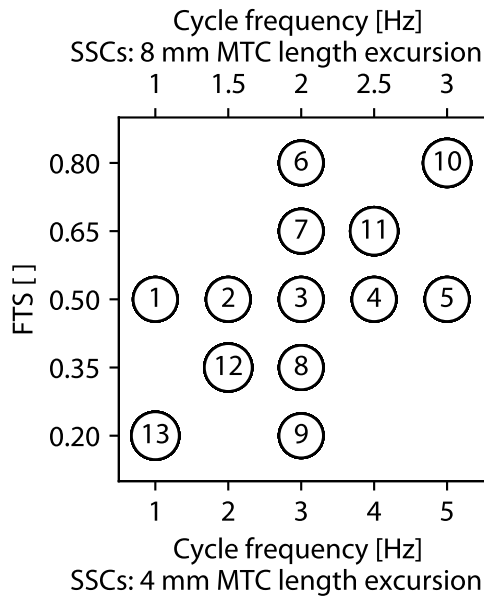


Figure 3.4: Representation of experimentally investigated SSCs. Thirteen combinations of cycle frequency and FTS were tested at two distinct MTC length excursions (4 mm and 8 mm), resulting in a total of 26 experimental SSC conditions.

Each SSC condition consisted of five consecutive cycles, with muscle stimulation applied during the middle three cycles. Muscle stimulation onset time was set at the transition from MTC lengthening to shortening. Muscle stimulation duration was based on preliminary model predictions (see Section 3.S.1), however, we chose a stimulation duration that was slightly shorter (20 ± 10 ms) than that predicted to be optimal for AMPO. This was done to prevent irreversible damage to the muscle fibres as a possible result of substantial GM force during active GM lengthening. After each SSC condition, we inspected GM force over time to heuristically adjust muscle stimulation duration to further enhance AMPO. The SSC condition was then repeated with the adjusted muscle stimulation duration after all other conditions had been performed for that specific MTC length excursion (~40 minutes later in time). This approach aimed to evaluate to what extent we could enhance AMPO

and to evaluate the reliability of measured GM force. Overall, we performed 52 series of SSCs for each rat. For each cycle, AMPO was computed as the product of cycle frequency and the integral of measured GM force over measured MTC length (i.e., the ‘work loop’ area).

Unfortunately, rat 3 passed away near the end of the experiment, which meant that six SSC conditions with an 8 mm MTC length excursion were performed only once instead of twice for this rat, such that 46 instead of 52 experimental SSCs were measured for rat 3.

We tested whether cycle frequency and FTS have a significant effect on the experimentally measured AMPO. This was done on each of the four subsets of the data where one SSC parameter was held constant, which are shown in Figure 3.5. Thus, in total, four statistical tests were performed (two for cycle frequency and two for FTS at the two MTC length excursions). Effects of cycle frequency and FTS (linear and quadratic) were assessed using a linear mixed-effects model, with cycle frequency or FTS as a fixed effect and specimen as a random effect to account for repeated measurements. No correction was made for differences in stimulation duration, so results can be considered a conservative estimate of the statistical effect of cycle frequency/FTS. Formally, the model can be expressed as:

$$y_{ij} = \mu + a \cdot x_{ij} + b \cdot x_{ij}^2 + u_j + \epsilon_{ij}, \quad u_j \sim N(0, \sigma_u^2), \quad \epsilon_{ij} \sim N(0, \sigma^2) \quad (3.1)$$

where y_{ij} is the experimentally measured AMPO for observation i in specimen j , x_{ij} is the corresponding predictor value, a and b are the fixed linear and quadratic effects of x (cycle frequency or FTS), u_j is the random effect for specimen, and ϵ_{ij} is the residual error.

3.2.2 MTC model

A Hill-type MTC model was used to represent GM. The model has been described extensively elsewhere (Reuvers and Kistemaker, 2025; Soest and Bobbert, 1993). In short, the MTC modelled consisted of a contractile element (CE) and a parallel elastic element (PEE) which were both in series with a serial elastic element (SEE). PEE and SEE were modelled as purely passive elastic structures (Zajac, 1989), with

their force non-linearly dependent on their length. The behaviour of CE was more complex: CE force depended on CE length, CE velocity and active state (q), which was defined as the relative amount of Ca^{2+} bound to troponin C (Ebashi and Endo, 1968). The excitation dynamics were modelled according to Hatze (1981, pp 31-32). Here, active state depended on the normalised concentration free Ca^{2+} between the filaments (γ) and CE length, with the sensitivity of active state q to Ca^{2+} increasing with CE length (e.g. Rack and Westbury, 1969; Stephenson and Williams, 1982). The normalised concentration of free Ca^{2+} , in turn, was related to the normalised muscle stimulation ($STIM$) through a first-order differential equation. The independent inputs of the model were $STIM$ and MTC length over time, while CE length and γ were the states.

3.2.2.1 Evaluation of measured versus predicted AMPO

Experimental measurements were compared with predictions derived using a Hill-type MTC model. To ensure a fair comparison, MTC properties were not tuned to the experimental SSC data; instead, they were obtained from dedicated trials (see Reuvers and Kistemaker, 2025). Moreover, all model inputs were identical to those measured experimentally. Specifically, we simulated all experimental SSC conditions for each rat individually based on their estimated MTC properties, using two inputs: the experimentally measured MTC length trajectory and $STIM(t)$ derived from the measured direct current stimulator signal. The latter was done because the measured direct current stimulator signal consisted of a train of pulses. $STIM$ was computed as follows: $STIM$ was maximal (i.e., a value of 1) between two stimulation pulses and was zero exactly halfway through inverse of the stimulation frequency (i.e., 5 ms) after the last stimulation pulse. Model-predicted AMPO was then compared with measured AMPO to evaluate predictive accuracy using the coefficient of determination (r^2) derived from Pearson's correlation (Pearson, 1896).

3.2.2.2 Imposed stretch–shortening cycles — constant MTC shortening/lengthening velocity

To systematically explore the influence of SSC parameters on the maximally attainable AMPO, we derived predictions with a Hill-type MTC model based on estimated MTC properties specific to each rat. Specifically, we examined: 1) how the effect of MTC length excursion on the maximally attainable AMPO varied with FTS; 2) how sensitive the maximally attainable AMPO was to changes in SSC parameters and 3) how optimal values of cycle frequency, FTS, and MTC length excursion co-varied in order to maximise AMPO.

During the imposed SSC, MTC shortening and lengthening velocities were set to constant values throughout their respective phases — thus imposing no constraint on MTC acceleration — in order to explore a broad range of cycle frequency, FTS, and length excursion combinations. The maximally attainable AMPO was predicted for 2280 combinations of cycle frequency (0.5–6.0 Hz in 0.5 Hz steps), FTS (0.05–0.95 in 0.05 steps), and MTC length excursion (2–11 mm in 1 mm steps). For each combination and each rat, we optimised the muscle stimulation onset time (i.e., the time instance at which stimulation switches from 0 to its maximal value) and muscle stimulation duration (i.e., the time period during which stimulation remains at its maximal value before returning to 0) to maximise AMPO using a Nelder-Mead simplex method (Gao and Han, 2012).

To ensure periodic behaviour, simulations were run for at least five cycles and continued thereafter until the difference in GM force at the start and end of the cycle was less than 20 mN. To verify that the global optimum had been found, we systematically varied stimulation onset time and duration. No increase in maximally attainable AMPO was observed across any SSC following changes in muscle stimulation.

Finally, as the simulations also served as a comparison with the experimental results of Askew & Marsh (1997; 1998), we reduced the SEE slack length in our simulations. In their setup, a clip was attached to the distal tendon as closely as possible to the muscle fibres, while the proximal tendon was left intact. Based on this, we simulated the SSCs with a reduced SEE slack length of 3 mm for all rats.

3.2.2.3 Imposed stretch–shortening cycles — optimal MTC shortening/lengthening velocity

The last aim in this study was to identify the MTC length over time for maximal AMPO, when MTC length over time was entirely free to take any form. This allowed us to quantify how much more AMPO could be produced when the SSC shape was completely unconstrained, compared to SSCs with constant MTC shortening and lengthening velocities. In the Hill-type MTC model used, SEE and PEE were assumed to be purely elastic (e.g., Anderson and Pandy, 1999; Soest and Bobbert, 1993) and therefore did not contribute to AMPO. Hence, only CE behaviour affected AMPO. The optimisation problem was thus reduced to finding the periodic CE length and stimulation over time in order to maximise AMPO. These optimisation problems were solved with Optimal Control techniques (see Section 3.S.2) for imposed cycle frequencies ranging from 0.5 to 6.0 Hz. Out of 615 optimisations, 604 converged successfully. Among these, the difference between maximum AMPO and the submaximal solutions was $0.3 \pm 1.1\%$.

3.3 Results

3.3.1 *In situ* experiment

3.3.1.1 Reliability of experimental force measurements

To investigate the influence of SSC parameters on the maximally attainable AMPO of only GM, it was essential to have minimal mechanical interaction between GM and GL+PL. Our setup proved effective: GL+PL force remained below 0.2 N in all rats and conditions, with a maximum within-trial standard deviation of 15 mN. For reference, typical values of maximal isometric force of GL+PL are about 14 N (Bernabei et al., 2015), while the maximal isometric force of GM in our study was about 16 N on average. We therefore concluded that the effect of GL+PL on GM was negligible.

Several checks were used to assess the stability of GM force throughout the experiment. During the first part of the experiment (i.e., the quick-release, step-ramp, and isometric protocols), the maximum isometric GM force in the step-ramp experiments exhibited a coefficient of variation below 1%. This same low coefficient

of variation ($<1\%$) was also observed for the maximum isometric GM force during the isometric contractions in the second part of the experiment (i.e., during the SSCs). The repeated quick-release experiments revealed that GM force was $\sim 11 \pm 8\%$ lower midway compared to the start of the experiment, with no change midway compared to the end of the experiment. Maximal isometric GM force did, however, slightly decrease over the entire experiment. This suggests, in line with literature (e.g. Aubert et al., 1951), that the decrease in GM force was more likely caused by a shift in the MTC force–length relationship, potentially due to a decrease in SEE stiffness. Based on the repeated quick-release experiments, we estimated the shift of the MTC force–length relationship to be $\sim 0.5 \pm 0.2$ mm midway compared to the start of the experiment, with no change midway compared to the end of the experiment. GM force during the ramp was $\sim 5 \pm 3\%$ lower midway compared to the start of the experiment, but remained stable in the second part of the experiment. With regard to individual SSC trials, we found that the second- and third-highest experimentally measured AMPO values among the three stimulated cycles were on average only $4 \pm 3\%$ lower than the highest value across all conditions and rats. Additionally, GM force development was similar in the first and second trial, which had similar muscle stimulation onset times. As intended, the second trial yielded a higher AMPO ($13 \pm 11\%$ across all SSC conditions and rats) due to an adjusted stimulation duration. Overall, GM behaviour was stable and reproducible throughout the whole experiment.

3.3.1.2 MTC property estimation

The estimated MTC properties showed remarkable consistency across all rats (Table 3.1). Based on the estimated MTC properties, we derived the maximum CE shortening velocity (v_{CE}^{max} ; see Table 3.1), the maximum instantaneous CE power output (P_{CE}^{max} ; see Table 3.1) and the CE velocity at which P_{CE}^{max} occurs (v_{CE}^{opt}) (see Table 3.1). Notably, the time to reach 50% of maximal isometric CE force was relatively long — approximately 34 ms at MTC optimum length in all rats. In addition to delays caused by excitation dynamics, the relatively compliant SEE substantially influenced GM force development. To exclude the influence of SEE,

we calculated the time to reach 50% of maximal isometric force while holding CE length constant at its optimal value. The resulting value was considerably shorter (~12 ms vs ~34 ms; see Table 3.1) and provides a better comparison to the half-rise times at constant CE length reported in literature.

Table 3.1: Estimated MTC properties of rat *m. gastorcnemius medialis* as well as metrics derived from these properties.

Description	Symbol	Unit	Rat 1	Rat 2	Rat 3
<i>MTC properties</i>					
Hill constant	a^{rel}	-	0.569	0.712	0.649
Hill constant	b^{rel}	1/s	5.97	7.15	6.80
maximal isometric CE force	F_{CE}^{max}	N	15.5	14.1	17.0
PEE stiffness scaling factor	k_{PEE}	N/mm ²	28.7	24.1	34.8
SEE stiffness scaling factor	k_{SEE}	N/mm ²	952	1255	1050
CE optimum length	L_{CE}^{opt}	mm	13.7	14.7	13.3
PEE slack length	L_{PEE}^o	mm	15.1	15.6	14.5
SEE slack length	L_{SEE}^o	mm	28.8	29.3	25.2
Activation time constant	τ_{act}	ms	55.2	57.7	41.6
Deactivation time constant	τ_{deact}	ms	27.1	25.3	22.4
<i>Derived metrics</i>					
MTC length at maximal isometric SEE force	L_{MTC}^{opt}	mm	46.6	47.3	42.5
maximal instantaneous CE power	P_{CE}^{max}	mW	316	319	352
'half-rise time'	t_{HRT}	ms	13.2	13.8	9.97
maximal CE shortening velocity	v_{CE}^{max}	mm/s	144	147	140
CE velocity at P_{CE}^{max}	v_{CE}^{opt}	mm/s	54.1	57.8	53.8

3.3.1.3 stretch–shortening cycles

The experimentally measured AMPO of the SSC conditions are depicted in Figure 3.5 and detailed in Table 3.S.1 and Table 3.S.2. Both cycle frequency and FTS had a strong effect on the experimentally measured AMPO at each of the two MTC length excursions, with both the linear and quadratic components being highly significant ($p \ll 0.001$). For brevity, we refer to an SSC condition with a cycle frequency of 2 Hz and an MTC length excursion of 8 mm as 8 mm@2 Hz.

In the SSC conditions with FTS set to 0.5, experimentally measured AMPO peaked for ~4 mm@4 Hz and for ~8 mm@2.5 Hz (Figure 3.5). This indicates that the cycle frequency that maximises AMPO depends on the MTC length excursion. Average MTC velocity was 32 mm/s for 4 mm@4 Hz, and 40 mm/s for 8 mm@2.5 Hz. Since GM force was almost zero at the start and end of shortening, average CE shortening equalled average MTC shortening velocity. This shows that average CE velocity that maximises AMPO is substantially lower than the CE velocity at which instantaneous mechanical power output is maximal (~55 mm/s; Table 3.1).

In the SSC conditions with fixed cycle frequency, experimentally measured AMPO increased with FTS, reaching its highest measured value at the maximum imposed FTS of 0.8 (Figure 3.5). This relationship was consistent across both 4 mm and 8 mm MTC length excursions, suggesting that the FTS yielding maximal AMPO is largely independent of MTC length excursion. The effect of FTS on AMPO was most pronounced at low cycle frequencies. For example, raising FTS from 0.2 to 0.5 increased AMPO by ~205% for 4 mm@3 Hz and ~185% for 8 mm@2 Hz, while the increase in AMPO from FTS 0.5 to 0.8 was substantially smaller with 20% for both 4 mm@3 Hz and 8 mm@2 Hz (Figure 3.5; Table 3.S.1; Table 3.S.2). Since the effect of increasing FTS from 0.65 to 0.80 was small — 8% for 4 mm@3 Hz and 4% for 8 mm@2 Hz — we conclude that the FTS yielding maximal AMPO for these combinations of cycle frequency and MTC length excursion was close to 0.8. Lastly, the effect of FTS on AMPO became more pronounced at higher cycle frequencies. For example, increasing FTS from 0.5 to 0.8 increased AMPO by ~50% for 4 mm@5 Hz and ~40% for 8 mm@3 Hz, which is substantially more than the 8% and 4% increase for 4 mm@3 Hz and 8 mm@2 Hz (see Table 3.S.1; Table 3.S.2).

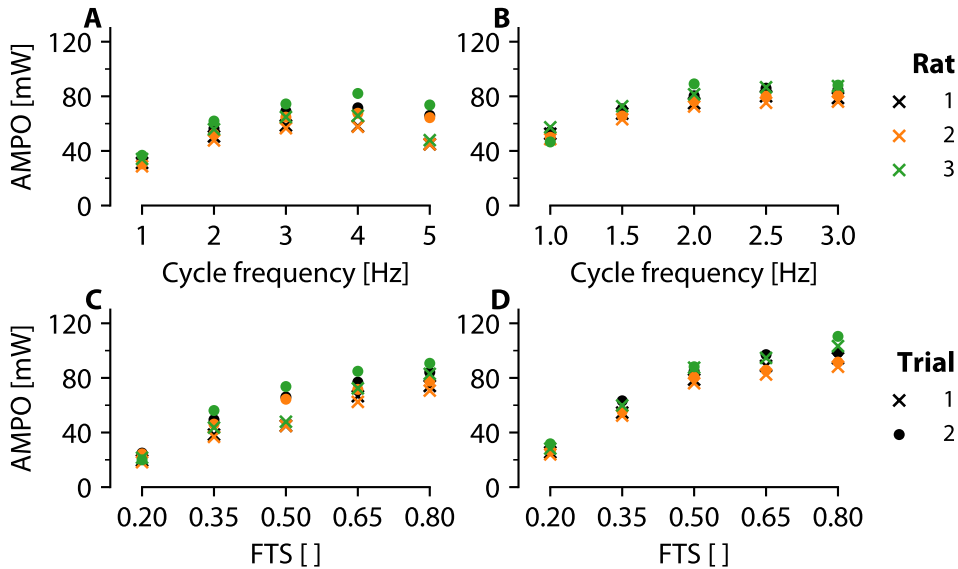


Figure 3.5: Experimentally measured influence of cycle frequency and FTS on AMPO. AMPO as a function of cycle frequency, with a fixed FTS of 0.5 at an MTC length excursion of 4 mm (A) and 8 mm (B). AMPO as a function of FTS, with a fixed cycle frequency of 3 Hz at an MTC length excursion of 4 mm (C) and 8 mm (D). Each combination of cycle frequency, FTS and MTC length excursion was performed twice. The first trial with suboptimal muscle stimulation duration and the second trial with an improved muscle stimulation duration.

3.3.2 MTC model

3.3.2.1 Evaluation of measured versus predicted maximally attainable AMPO

We aimed to explore the influence of SSC parameters – cycle frequency, FTS and MTC length excursion – on the maximally attainable AMPO across a broad SSC parameter space. As discussed earlier, such a comprehensive investigation is not feasible experimentally because only a limited set of conditions can be tested. Therefore, we used a Hill-type MTC model to investigate a broad SSC parameter space. For this modelling approach to be meaningful, it was crucial that the model could accurately predict the influence of MTC length and stimulation over time on AMPO.

The model-predicted GM force closely matched the measured GM force during activation across all rats and SSC conditions (Figure 3.6). During relaxation, measured

GM force was slightly lower than model-predicted GM force in most conditions. As a result, measured AMPO was on average $6 \pm 13\%$ lower than predicted. Nonetheless, the correlations between measured and model-predicted AMPO were nearly perfect ($r^2 = 0.99$, $r^2 = 0.99$ and $r^2 = 0.98$ for rat 1, 2 and 3, respectively; Figure 3.7). This was a remarkable result, as neither the model inputs nor the estimated MTC properties were tuned to experimentally measured SSCs. These findings demonstrate that a Hill-type MTC model can accurately predict the influence of MTC length and stimulation over time on AMPO.

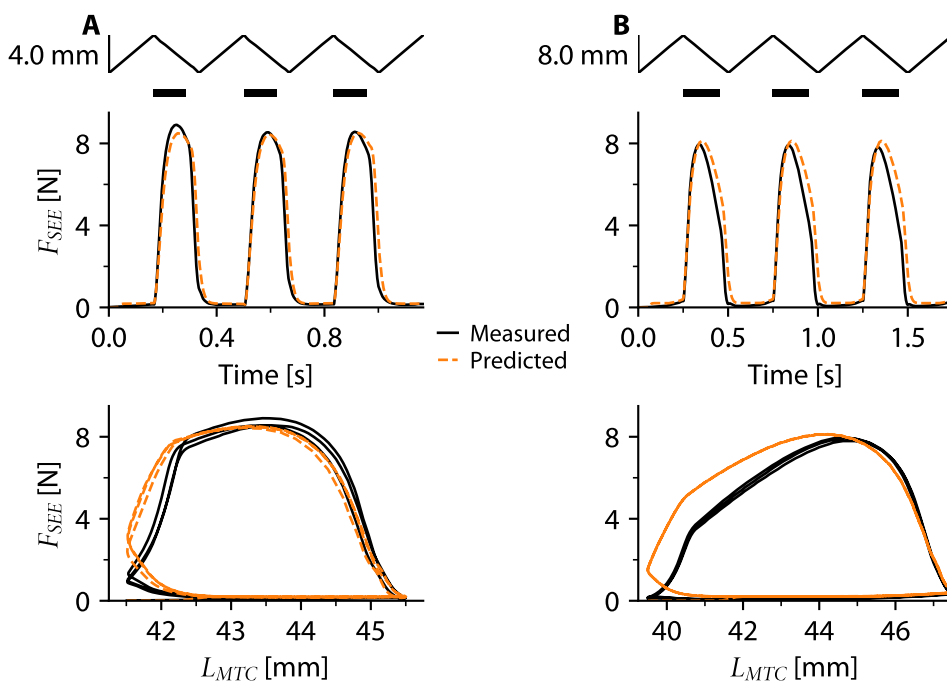


Figure 3.6: Comparison of experimentally measured and predicted GM force over time. Predicted GM force over time was derived using a Hill MTC-type model, with experimentally measured MTC length and stimulation over time as inputs. Predicted SEE force closely matched experimentally measured GM force during activation but was slightly higher than experimentally measured SEE force in most conditions. Top: MTC length over time. Second: Muscle stimulation over time, with maximal stimulation during the periods indicated by the black bars and no stimulation elsewhere. Third: SEE force over time. Bottom: SEE force as a function of MTC length ('the work loop'). A) SSC with a cycle frequency of 3 Hz, an FTS of 0.5 and an MTC length excursion of 4 mm. B) SSC with a cycle frequency of 2 Hz, an FTS of 0.5 and an MTC length excursion of 8 mm.

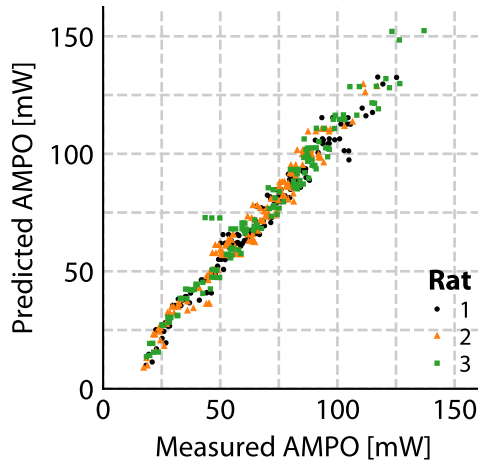


Figure 3.7: Comparison of experimentally measured and predicted AMPO. Measured AMPO was derived from experimentally observed MTC length and GM force. Predicted AMPO was derived using a Hill MTC-type model, with experimentally measured MTC length and stimulation over time as inputs to predict GM force. Each dot represents AMPO of one full cycle where muscle stimulation was present, such that there are three dots for each experimental SSC condition. The nearly perfect correlation demonstrates that a Hill-type MTC model can accurately predict the influence of MTC length and stimulation over time on AMPO.

3.3.2.2 Imposed stretch–shortening cycles — constant MTC shortening/lengthening velocity

Using the Hill-type MTC model, we predicated that AMPO peaks at a value of 169 mW (averaged across the three rats) at cycle frequency of 3.5 Hz, an FTS of 0.85 and an MTC length excursion of 8 mm (Figure 3.8 & Figure 3.9). The optimal FTS remained remarkably constant across all tested combinations of cycle frequency and MTC length excursion (Figure 3.9). By contrast, cycle frequency and MTC length excursion showed a strong interaction, such that when cycle frequency was increased, the optimal MTC length excursion decreased, and vice versa (Figure 3.9). Notably, a broad range of combinations of SSC parameters achieved AMPO values close to its peak value (Figure 3.8 and Figure 3.9). For instance, cycle frequency ranging between 2.9 and 4.5 Hz produced an AMPO within 95% of maximum AMPO, when adjusting FTS and MTC length excursion accordingly. For FTS this was a range between 0.74 and 0.91, when adjusting cycle frequency and MTC length excursion accordingly, while it was between 6.2 and 10.2 mm for the MTC length

excursion when adjusting cycle frequency and FTS accordingly. These findings demonstrate that while a clear optimum exists for maximal AMPO, there is a wide variety of combinations of SSC parameters that yield near-maximal AMPO values.

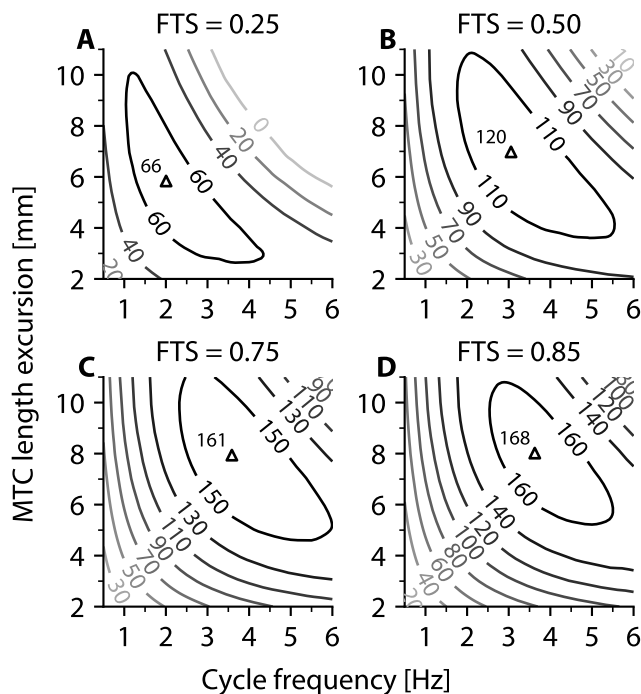


Figure 3.8: Predicted maximally attainable AMPO as a function of cycle frequency and MTC length excursion, shown for four distinct FTS values. The maximally attainable AMPO (in mW), averaged across three rats, is depicted as contour lines. The open triangles indicate the location of peak AMPO at each FTS corresponding to the optimal combination of cycle frequency and MTC length excursion for each FTS, with the corresponding peak AMPO labelled at the top-left of the triangle.

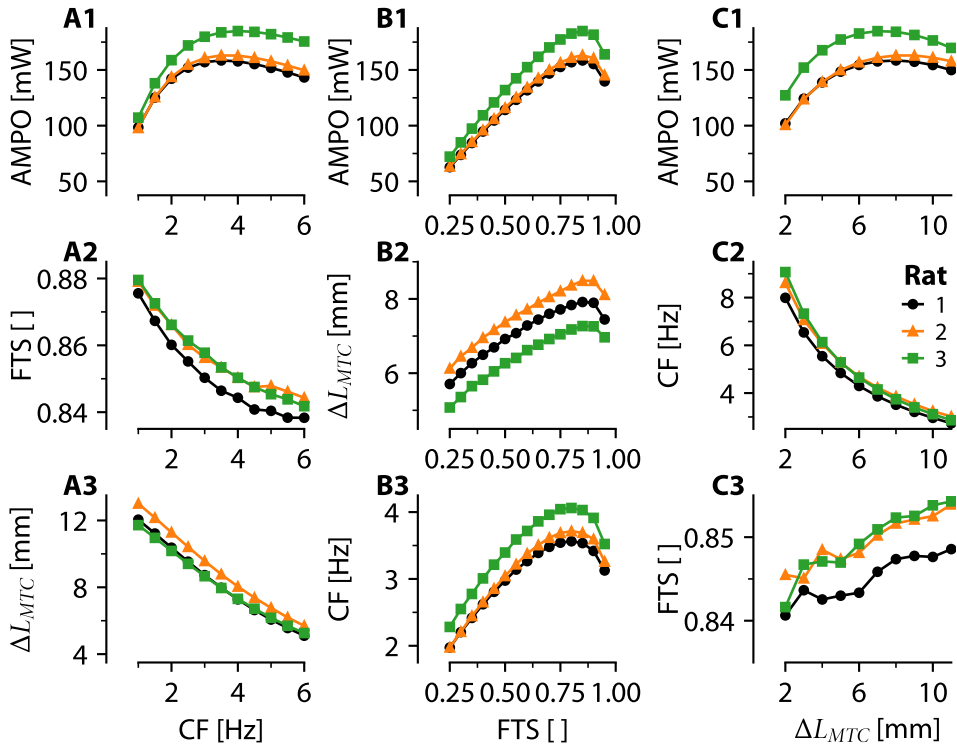


Figure 3.9: Predicted interrelationships between cycle frequency (CF), FTS, MTC length excursion (ΔL_{MTC}) and the maximally attainable AMPO. These plots collectively illustrate how cycle frequency, FTS, and MTC length excursion interact to maximise AMPO under different conditions. This figure presents a 3x3 grid of plots created by imposing one SSC parameter at a time while optimising the two other SSC parameters to maximise AMPO. In the first column, cycle frequency was imposed, and AMPO (A1) was maximised by identifying the optimal FTS (A2) and MTC length excursion (A3) at each cycle frequency. In the second column, FTS was imposed, and AMPO (A1) was maximised by identifying the optimal MTC length excursion (B2) and cycle frequency (B3) at FTS. In the third column, MTC length excursion was imposed, and AMPO (C1) was maximised by identifying the optimal cycle frequency (C2) and FTS (C3) at each MTC length excursion. For all columns, muscle stimulation onset time and duration were optimised to maximise AMPO.

3.3.2.3 Imposed stretch–shortening cycles — optimal MTC shortening/lengthening velocity

Using the Hill-type MTC model, we identified the optimal CE length over time for maximal AMPO without imposing any constraints on MTC length over time. We found that AMPO was maximised when CE shortened at an approximately constant velocity during the shortening phase and lengthened with increasing velocity during

the lengthening phase (see Figure 3.10), across all imposed cycle frequencies. Given CE length and CE force over time (and PEE and SEE properties), the corresponding MTC length over time can be calculated. Thus, MTC length over time is inherently dependent on the properties of SEE. Because SEE force is not constant during CE shortening, the MTC velocity that maximises AMPO is generally not constant during the shortening phase; even if CE shortening velocity is constant. The size of this effect depends on the SEE slack length: with a typical slack length (e.g., 28 mm), the MTC velocity is clearly not constant (see Figure 3.10E), whereas with a short slack length (e.g., 3 mm), the MTC velocity closely resembles the CE velocity and is therefore nearly constant during shortening (see Figure 3.10F).

For a short SEE slack length, the MTC length over time that maximises AMPO has a near-constant shortening and lengthening velocity. As a result, the maximally attainable AMPO using the optimised CE length over time was only $2 \pm 1\%$ higher than that achieved with constant MTC shortening and lengthening velocities with a short SEE slack length. It should be noted, however, that this small increase in AMPO is specific to imposing cycle frequency. When imposing FTS or MTC length excursion, the optimal CE length over time was substantially different from that with constant MTC velocities during shortening and lengthening – especially at suboptimal values. For the interested reader, we refer to the Supplementary Material (Section 3.S.3) for the effect of FTS or MTC length excursion on the optimal MTC behaviour for maximising AMPO.

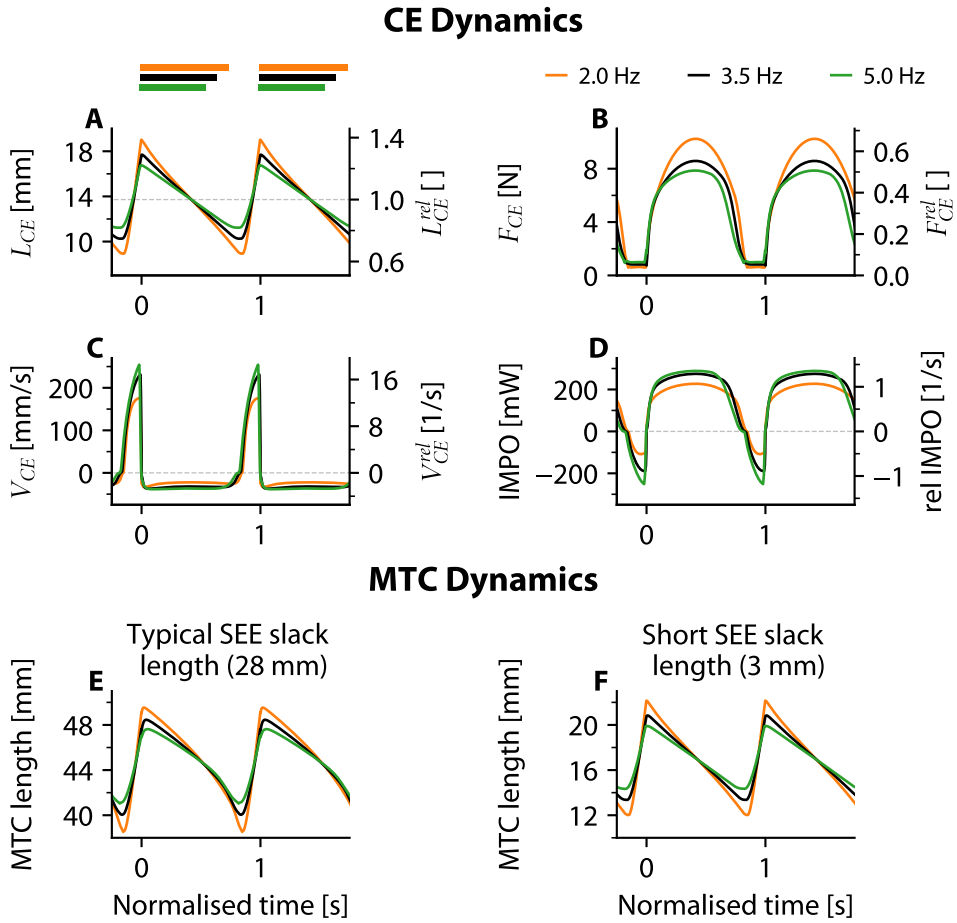


Figure 3.10: Predicted CE behaviour for maximally attainable AMPO, shown at three distinct cycle frequencies for rat 1. CE length (A), CE force (B), CE velocity (C) and instantaneous mechanical power output (IMPO) (D) as a function of normalised time (i.e. time divided by the cycle duration). CE stimulation was maximal during the period indicated by the coloured bars and fully off elsewhere. The right labels depict the normalised values, where CE length and CE velocity are normalised to optimum CE length, CE force is normalised to maximal isometric CE force and IMPO is normalised to the product of optimum CE length and maximal isometric CE force. Based on CE length and force over time, and the SEE properties, the MTC length over time can be calculated. The resulting MTC length over time substantially differs between a typical SEE slack length of 28 mm (E) and a short SEE slack length of 3 mm (F).

3.4 Discussion

The aim of this study was to explore the influence of SSC parameters — cycle frequency, FTS and MTC length excursion — on the maximally attainable AMPO of rat GM. To this end, we combined *in situ* experiments with simulations and optimisations using a Hill-type MTC model. Model parameter values were estimated individually for each specimen based on independent data from quick-release, step-ramp and isometric experiments. Additionally, the model inputs (i.e., MTC length and stimulation over time) were directly derived from the experimental SSC data. The correlation coefficients between model-predicted and measured AMPO were nearly perfect ($r^2 > 0.98$). This is a remarkable result, particularly given that the predictions were not tuned to any of the experimental SSCs. This strong correspondence enabled us to leverage both experimental findings and model-based predictions to investigate how SSC parameters influence the maximally attainable AMPO.

Both experimental data and model-based predictions showed that the MTC should spend substantially more time shortening than lengthening in order to maximise AMPO — at an FTS of approximately 0.85. This optimal FTS value remained remarkably constant across all tested combinations of cycle frequency and MTC length excursion. By contrast, cycle frequency and MTC length excursion showed a strong interaction, such that when cycle frequency increased, the optimal MTC length excursion decreased, and vice versa. Lastly, we found that AMPO peaked at a cycle frequency of 3.5 Hz, an FTS of 0.85 and at an MTC length excursion of 8 mm, with a relatively constant CE velocity throughout the shortening phase and an increasing CE lengthening velocity throughout the lengthening phase.

3.4.1 Robustness and reliability of experimental results

In previous studies, it has been well established that cycle frequency and MTC length excursion significantly affect the maximally attainable AMPO (e.g. Altringham and Johnston, 1990; Askew and Marsh, 1997; Askew and Marsh, 1998; Caiozzo and Baldwin, 1997; James et al., 1995; James et al., 1996), whereas the effect of FTS has only been demonstrated by Askew & Marsh (1997; 1998). Our experimental results also

showed significant effects of cycle frequency and FTS on the maximally attainable AMPO, confirming the sensitivity of AMPO to SSC parameters. Understanding how these SSC parameters interact to influence maximally attainable AMPO requires exploring a larger range of muscle lengths and stimulation timings than is experimentally feasible. Therefore, we complemented experiments with simulations and optimisations using a Hill-type MTC model. For this extensive modelling approach into the influence of SSC parameters on maximally attainable AMPO, it was crucial that the model could accurately predict experimental outcomes. Accordingly, our experimental design focused on selecting a representative and informative set of SSCs to evaluate model-based predictions. To assess the reliability of our experimental outcomes, we repeated experimental conditions with similar stimulation timing (three cycles) and with two different timings. The experimental results were highly consistent across the tested specimens. Both the effects of SSC parameters on maximally attainable AMPO (see Figure 3.5) and the estimated muscle properties (see Table 3.1) showed minimal inter-individual variation. This aligns with previous findings showing highly reproducible mechanical behaviour in muscles of genetically similar animals, making it unlikely that adding additional animals would change the study outcomes. Furthermore, the effects of SSC parameters on maximally attainable AMPO in our study are in strong agreement with prior findings (see sections below). Taken together, we conclude that our experimental data are robust and provide a reliable basis to evaluate model predictions.

3.4.2 Optimal FTS for AMPO was relatively constant across all SSC conditions

Our results show that the maximally attainable AMPO substantially increased when FTS increased from 0.25 to 0.50, regardless of cycle frequency and/or MTC length excursion. For example, at the optimal combination of cycle frequency and MTC length excursion, increasing FTS from 0.25 to 0.50 caused AMPO to increase by about 95%. This finding is consistent with the results of Askew and Marsh (1997), who reported increases of ~90% for *m. soleus* and ~70% for *m. extensor digitorum longus* when FTS increased from 0.25 to 0.50. Further increasing FTS from 0.50 to

0.75 yielded an additional 30% increase in AMPO, aligning with Askew & Marsh's observations of a ~40% increase in AMPO for both muscles. Moreover, Askew and Marsh (1997) observed that AMPO increased even further, and peaked at an FTS between 0.80 and 0.90 at the optimal combination of cycle frequency and MTC length excursion. Our results not only corroborate this finding, but also demonstrate that the optimal FTS was relatively constant (ranging 0.84–0.88) across all tested cycle frequencies (0.5–6 Hz) and MTC length excursions (2–11 mm).

The optimal FTS reflects mainly a trade-off between the CE concentric force–velocity relationship and excitation dynamics. The negative effects of both the CE force–velocity relationship and excitation dynamics on net mechanical work per cycle increase with cycle frequency. Yet, the fact that the optimal FTS remains relatively constant across all cycle frequencies studied suggests that these negative effects increase to a similar extent with cycle frequency (Figure 3.9). Moreover, because the negative effect of both the CE force–velocity relationship and excitation dynamics on AMPO are strongly and comparably linked to muscle fibre type (Close, 1972), their combined influence is likely relatively similar across muscles with different fibre type compositions. Consequently, it is perhaps not surprising that the optimal FTS for AMPO remains relatively consistent across different muscles (~0.85 for rat *m. gastrocnemius medialis* in this study; ~0.85 for mouse *m. soleus* and *m. extensor digitorum longus* (see Askew and Marsh, 1997) and ~0.80 for human *m. quadriceps femoris* (see Reuvers et al., 2026)). Together, these findings suggest that the optimal FTS is not only relatively constant across all SSC conditions, but also across different species and muscles (with varying fibre types).

3.4.3 Effect of FTS on optimal cycle frequency and MTC length excursion for maximising AMPO

While the optimal FTS remained relatively constant across all cycle frequencies and MTC length excursions, FTS itself strongly influenced the optimal values of both cycle frequency and MTC length excursion (Figure 3.9). With respect to cycle frequency, we observed that the optimal value increased with FTS, although the magnitude of this increase diminished at higher FTS values. This observation

aligns well with findings by Askew and Marsh (1997), who reported comparable increases across FTS values for both m. soleus and m. extensor digitorum longus. For MTC length excursion, we found that the optimal value increased with 25% and 10% when raising FTS from 0.25 to 0.50 and from 0.50 to 0.75, respectively. A direct comparison with Askew and Marsh (1997) is difficult, as their results differed between muscles: for m. soleus, these increases were 35% and 40%, while they were 0% and 35% for m. extensor digitorum longus. Lastly, Askew and Marsh (1997) also examined how average muscle shortening velocity varied with FTS. For m. extensor digitorum longus, they found a consistent decrease in optimal shortening velocity as FTS increased. By contrast, m. soleus showed a slight increase in optimal shortening velocity, although this discrepancy may stem from missing data near the optimal frequency at FTS = 0.75 (see their Figure 4). A follow-up study by the same authors (Askew and Marsh, 1998) later confirmed that the optimal average shortening velocity for m. soleus indeed decreases with increasing FTS, independently of cycle frequency. We observed the same for our model-predictions. Overall, our findings regarding the influence of FTS on optimal cycle frequency, MTC length excursion, and average shortening velocity are in line with previous literature. These findings are not surprising, given the effect of the CE force-velocity relationship. After all, at higher FTS, there is relatively more time for shortening than at lower FTS values. This enables an increase in cycle frequency and allows muscles to shorten and lengthen over a greater distance, without substantial negative effects due to the effect of shortening velocity on muscle force.

3.4.4 Optimal cycle frequency and optimal MTC length excursion for maximal AMPO

While the optimal FTS for maximal AMPO remained relatively constant across all SSCs studied, we found that cycle frequency and MTC length excursion strongly interacted for maximal AMPO. Specifically, as cycle frequency increased, the optimal MTC length excursion for the maximally attainable AMPO decreased, and vice versa (see Figure 3.8 & Figure 3.9). This interaction has previously been observed in sinusoidal SSCs (e.g., James et al., 1996) – where shortening and lengthening

durations are equal. A similar interaction between cycle frequency (i.e. cadence) and crank length—which influences MTC length excursion—has also been observed in human sprint cycling (e.g. Martin and Spirduso, 2001), where shortening and lengthening durations of individual MTCs are not necessarily equal. However, to our knowledge, this interaction has not been systematically examined in controlled SSCs with systematically varied unequal shortening and lengthening durations; although Askew and Marsh (1997) briefly noted this interaction, they did not present the relevant data.

The interaction between cycle frequency and MTC length excursion for maximally attainable AMPO did not come as a surprise, given the negative effect of the CE force–velocity relationship on AMPO. In line with Askew and Marsh (1997), Caiozzo and Baldwin (1997) and James et al. (1996), we observed that the maximally attainable AMPO was less sensitive to increases in cycle frequency and MTC length excursion above their respective optimal values than to decreases below these optimal values. This is not only true at optimal FTS (see Figure 3.9), but also at suboptimal FTS. The latter can be observed in Figure 3.8, where the contour lines are more closely spaced below than above the optimum cycle frequency and MTC length excursion. This indicates that, for a given change in cycle frequency or MTC length excursion, AMPO decreases more rapidly below the optimal value than above it. The finding that AMPO was less sensitive to increases in cycle frequency and MTC length excursion above their respective optimal values than to decreases below these optimal values might be counter-intuitive, as one may expect the maximally attainable AMPO to decline rapidly due to a higher shortening velocity (at higher cycle frequencies and MTC length excursion). However, at low cycle frequencies, the MTC must shorten and lengthen over a substantial distance, which reduces the maximally attainable AMPO primarily due the effect of the CE force–length relationship. Conversely, at low MTC length excursions, very high cycle frequencies are required to achieve substantial power output, which in turn compromises AMPO due to the effect of the CE force–velocity relationship (i.e., muscle contracts at lengths at which CE force is substantially lower than the maximal isometric force). In sum, our findings indicate that deviations above the optimal

cycle frequency and MTC length excursion have a less detrimental effect on the maximally attainable AMPO than deviations below optimal values.

3.4.5 Effect of SEE properties on the MTC behaviour for maximal AMPO

To maximise AMPO, CE should shorten at an approximately constant velocity during the shortening phase and lengthen with increasing velocity during the lengthening phase (see Figure 3.10). The resulting MTC length over time for maximal AMPO is determined not only by CE length over time, but also by the SEE properties (i.e. SEE compliance and SEE slack length). Due to SEE elasticity, MTC velocity should be higher (i.e. less negative or even positive) than that of CE during force development, and lower (i.e. more negative) during force relaxation to maximise AMPO (Figure 3.10E).

The same principle applies the other way around: when velocity of MTC is constant, that of CE varies during force development and relaxation. Specifically, changes in CE velocity increase with increasing SEE compliance and SEE slack length. In the experimentally observed SSCs, the estimated SEE slack length was substantially longer (~28 mm) than that used in the simulated SSCs (3 mm), resulting in more variable CE velocity during shortening. This variable CE velocity compromises AMPO: for a given shortening distance, CE should ideally shorten at a constant velocity to maximise AMPO. This more variable CE velocity during the shortening phase of the experimental SSC was the main reason why AMPO in the experimental SSCs was lower than in the simulated SSCs. For instance, at a 2 Hz cycle frequency, FTS of 0.5 and an 8 mm MTC length excursion, AMPO in the simulated SSC was ~40% higher (~80 mW vs. ~110 mW; Figure 3.5 vs. Figure 3.8).

We found that – especially at a low FTS, a high cycle frequency and a small MTC length excursions – muscle stimulation should occur before MTC shortening starts in order to maximise AMPO. Consequently, CE shortening started before MTC shortening, such that CE was shortening over a longer time interval resulting in a lower average CE shortening velocity than MTC. Thus, the behaviour of SEE does

not only cause a discrepancy between instantaneous CE and MTC velocity, it also affects their average shortening velocities.

Askew and Marsh (1998) reported that at low FTS values and high cycle frequencies, the average MTC shortening velocity exceeded the CE velocity that maximises instantaneous power output based on the force–velocity relationship (v_{CE}^{opt}). They acknowledged that this was a unexpected finding, since CE shortening velocity should theoretically be at or below v_{CE}^{opt} in order to maximise AMPO. Similar to our SSCs with a constant MTC shortening and lengthening velocity, Askew and Marsh (1998) left a small portion of the tendon attached to the muscle fibres. Since Askew & Marsh reported that muscle stimulation onset occurred before MTC shortening, it is evident that average CE shortening velocity was lower than that of MTC. Therefore, their finding that optimal CE shortening velocity should exceeds v_{CE}^{opt} can, at least, partly be attributed to an overestimation of CE shortening velocity due to the series elastic structures.

The discrepancy between CE and MTC shortening velocity is most pronounced at low FTS, high cycle frequencies, and small MTC length excursion — exactly where Askew and Marsh (1998) observed CE shortening velocities exceeding v_{CE}^{opt} . At high FTS values, however, this discrepancy is lower. Askew and Marsh (1998) found that at high FTS values (i.e., 0.75), the optimal average shortening velocity was consistently below v_{CE}^{opt} . This aligns with our findings, even when muscle stimulation onset preceded MTC shortening. In conclusion, while other factors may play a role, we are confident that such factors have little influence on the effect of SSC parameters on the maximally attainable AMPO near the optimum.

3.4.6 Applications of findings

In most real-world situations, SSCs of individual muscles emerge from the interaction between the musculoskeletal system and the environment. Consequently, the extent to which SSC parameters can be influenced depends on the mechanical context of the task. During flying, for example, birds can and do actively adjust wing kinematics (e.g. by folding and unfolding their wings), thereby enabling a longer shortening than lengthening duration of *m. pectoralis major* (e.g., Askew and Marsh,

2001; Biewener et al., 1998; Earls, 2000; Ellerby and Askew, 2007; Jackson and Dial, 2011; Spedding, 1987; Williamson et al., 2001). Our findings help to understand this, as a longer shortening duration is favourable for maximising AMPO. In addition, our results may also shed light on (the evolution of) the “design” of musculoskeletal systems, for example the role of moment-arms which affect MTC length excursion thereby influencing AMPO (e.g. Polet and Labonte, 2024)

SSC parameters can be substantially influenced by modifying the interaction between the musculoskeletal system and the environment. One way to achieve this is through the use of external equipment, which can change or constrain movement kinematics. A prime example of such equipment is the bicycle, where crank length affects joint excursions and thus MTC length excursions. Also, non-circular chain-rings can be used to vary the gear ratio throughout the crank revolution. This allows the gear ratio to be increased during the downstroke and to be decreased during the upstroke, thereby changing the relative time spent in each phase. In a clever experiment, (Martin et al., 2002) manipulated chainring shape to alter the relative duration of the downstroke and upstroke during single-leg sprint cycling. AMPO increased with a greater proportion of time spent in the downstroke: an 18% increase was observed when the downstroke occupied 50% rather than 42% of the cycle duration, and a further 4% increase was observed at 58% compared to 50%. Another example is rowing, where the resistance during the stroke and recovery phases could be adjusted in order to alter muscle shortening and lengthening durations. More broadly, such (re)design principles could be applied to any equipment that changes or constrains movement kinematics, thereby enabling the SSCs of the involved muscles to operate closer to their optimum for AMPO.

It should be noted that in motor activities involving multiple muscles (e.g., bicycling, handcycling, rowing, speed skating, etc.), the objective is not to maximise the AMPO of a single muscle. Rather, the goal is to maximise the total AMPO of all muscles combined, which introduces an additional trade-off. In this context, when considering a joint, it should be clear that the optimum FTS for one muscle (e.g. an FTS of ~ 0.85) cannot be realised in both agonists and antagonists. In this situation, the optimal solution arises from a trade-off that depends on properties of both

muscles (e.g. maximal isometric force and muscle fibre length), and is therefore expected to shift in both muscles towards more intermediate values (i.e. closer to an FTS of 0.5). Nevertheless, it stands to reason that larger muscles should operate closer to their individual optimum than smaller muscles in order to maximise performance. Thus, when (re-)designing equipment to maximise total AMPO, configurations should enable larger muscles to operate closer to their individual optimum. Our study provides a starting point for guiding future equipment design aimed at improving total AMPO.

Acknowledgments

The authors thank Guus Baan for creating the illustration used in Figure 3.

Competing interests

The authors declare no competing or financial interests.

Funding

This work was funded by The Dutch Research Council (NWO) [21728 to D.A.K.].

Data and resource availability

All data, code, and materials used in this study are openly available:

- **GitHub repository:** All raw data, processed data, and analysis code are hosted on GitHub at <https://github.com/edwinreuvers/rat-gm-ampo>.
- **Reproducible analysis website:** Full analysis pipeline — including data, analysis code, and figure/table generation — is available at <https://edwinreuvers.github.io/publications/rat-gm-ampo>.

3.S Supplementary material

3.S.1 Selection of experimental SSC conditions

In the experiment, we aimed to measure an informative and representative subset of stretch–shortening cycles (SSCs) to investigate the effect of the SSC parameters — cycle frequency, FTS and MTC length excursion — on the maximally attainable AMPO. Yet, a key question was: Which SSCs should be selected? To make an informed decision, we performed preliminary simulations using a Hill-type MTC model with parameter values from literature for three rat *m. gastrocnemius medialis* (Zandwijk et al., 1996).

We predicted maximal AMPO across various combinations of cycle frequency and FTS at five distinct MTC length excursions (2, 4, 6, 8 and 10 mm), each centred at an average MTC length ~ 3 mm below the MTC length yielding maximal isometric GM force. Muscle stimulation onset time (i.e., the instance at which stimulation switches from 0 to its maximal value) was set to the start of MTC shortening. Muscle stimulation duration (i.e., the time period during which stimulation remains at its maximal value before returning to 0) was optimized using the Nelder-Mead simplex method (Gao and Han, 2012). To ensure periodic behaviour, optimisations were run for at least five cycles and until the muscle force at the beginning of the cycle deviated less than 20 mN from its value at the end of the cycle. Maximal AMPO was calculated for last cycle.

We then made contour plots for every MTC length excursion (Figure 3.S.1), showing the influence of cycle frequency and FTS on maximal AMPO at each MTC length excursion. For the experimental SSC conditions, we limited the range of FTS between 0.2 and 0.8 because values outside this range will yield very short shortening or lengthening durations. Within this FTS range, we observed that AMPO peaked at about 5 and 3 Hz for an MTC length excursion of 4 and 8 mm respectively. Again, to prevent short shortening and lengthening duration, we set the maximum cycle frequencies accordingly. Based on these findings, we selected three subsets of SSC conditions: 1) constant FTS, varying cycle frequency; 2) constant cycle frequency, varying FTS; 3) varying both cycle frequency and FTS. Because there was overlap in the SSC conditions, this yielded 13 unique SSC conditions for

each MTC length excursion, such that there were 26 different SSC conditions in total.

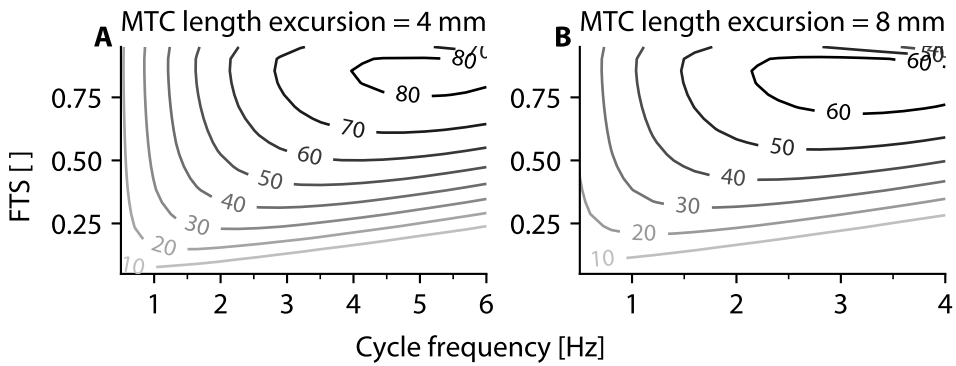


Figure 3.S.1: Preliminary predictions of maximal AMPO as a function of cycle frequency and FTS. The contour lines depict the maximally attainable AMPO (in mW) averaged over three parameter sets. These preliminary predictions were derived with a Hill-type MTC model with parameters obtained from literature (Zandwijk et al., 1996) and were used to inform the selection of experimental SSC conditions.

3.S.2 Optimisation protocol to identify optimal MTC length over time

One of the aims of this study was to identify the MTC length over time for maximal AMPO, when MTC length over time is entirely free to take any form. To predict this, we used a Hill-type MTC model. In our Hill-type MTC model used, we assumed that both SEE and PEE were purely elastic elements (e.g., Anderson and Pandy, 1999; Soest and Bobbert, 1993). Therefore, neither PEE nor SEE contributed to AMPO of the MTC during fully periodic contractions. Hence, only CE behaviour influenced AMPO and therefore only CE was modelled to simplify the optimisation problem.

The optimisation problem was thus to find the periodic CE length and *STIM* over time for maximal AMPO. Optimisations were performed for 1) imposed cycle frequencies (1.0–6.0 Hz in 0.5 Hz steps); 2) imposed FTS values (0.05–0.95 in 0.05 steps) and 3) imposed MTC length excursions (2–11 mm in 1 mm steps). For all optimisations, the following cost function was minimised:

$$J = \int_{t_0}^{t_0+T_c} F_{CE}^{rel} \cdot v_{CE}^{rel} dt \quad (3.2)$$

t_0 denotes the initial time and T_c denotes the cycle time.

The solution had to satisfy the differential equation describing the contraction and excitation dynamics:

$$\dot{x} = f(x(t), u(t)) \quad (3.3)$$

$x(t)$ denotes that states of the system ($L_{CE}^{rel}(t)$, $\gamma(t)$) and $u(t)$ denotes the inputs of the system ($F_{CE}^{rel}(t)$ and $STIM(t)$). Relative CE velocity was obtained by satisfying Equation 3.3.

The following task constraint was added to achieve periodic behaviour:

$$x(t_0) = x(t_0 + T_c) \quad (3.4)$$

Moreover, inequality constraints were imposed on $\gamma(t)$, $L_{CE}^{rel}(t)$ and $STIM(t)$:

$$\begin{aligned}\gamma_0 &\leq \gamma(t) \leq 1 \\ (1 - w) &\leq L_{CE}^{rel}(t) \leq (1 + w) \\ 0 &\leq STIM(t) \leq 1\end{aligned}\tag{3.5}$$

We implemented our problem as a Multiphase Optimal Control Problem to assure that CE shortening and lengthening occurred only once per full periodic cycle, which was done by inequality constraints on $v_{CE}^{rel}(t)$:

$$\begin{aligned}v_{CE}^{rel}(t_0 \dots t_0 + T_c \cdot FTS) &\leq 0 \\ v_{CE}^{rel}(t_0 + T_c \cdot FTS \dots t_0 + T_c) &\geq 0\end{aligned}\tag{3.6}$$

FTS was a numerical value (between 0.05 and 0.95) when FTS were imposed (see above). When FTS was not imposed, FTS was a control variable optimised for maximal AMPO.

When MTC length excursion was imposed, an extra inequality constraint was added:

$$L_{MTC}(t_0) - L_{MTC}(t_0 + T_c \cdot FTS) = \text{MTC length excursion}\tag{3.7}$$

The dynamic, task and (in)equality constraints were implemented in CasADi (Andersson et al., 2019) and the cost function was minimised using a Direct Collocation method using a nonlinear interior point method (IPOPT, Wächter and Biegler, 2006). The optimal control solution was checked by running forward simulations of the model with a linear-interpolated $F_{CE}^{rel}(t)$ and $STIM(t)$ as inputs to confirm that no relevant constraints were violated between the collocation points. This was done using SciPy's ODE integrator with an Implicit Runge-Kutta method of Radau IIA family of order 5 (Hairer and Wanner, 1996) and an absolute and relative tolerance of $1e-6$ and a maximum timestep of $1e-3$. This yielded only very small differences between the states at the collocation points and those at the same time instance of the forward simulation for all optimisations.

3.S.3 Optimal CE length over time: influence of FTS and MTC length excursion

Using the Hill-type MTC model, we identified the optimal CE length over time for maximal AMPO without imposing any constraints on MTC length over time. When imposing cycle frequency, the resulting MTC length over time was reasonably similar to those when with constant MTC shortening and lengthening velocity (see Section 3.3.2.3 and Figure 3.10). However, when imposing FTS or MTC length excursion the shape of CE and MTC length over time dramatically changed.

For imposed FTS values below the optimal value, CE lengthening velocity was found to be almost zero at the onset of CE lengthening (Figure 3.S.2A). This infinitely low velocity at the start of CE lengthening was optimal for AMPO because it allowed the active state to decrease fully before lengthening occurred, thereby reducing the amount of negative mechanical work performed. On top of that, it allowed the active state to remain high throughout most part of CE shortening to enable substantial positive mechanical work. Towards the end of CE lengthening, CE velocity again approached zero, which made it possible to initiate muscle stimulation before CE shortening began. This resulted to a high active state at the onset of CE shortening to enhance positive mechanical work. For imposed FTS values above the optimal value, CE shortening velocity decreased over time at the end of CE shortening (Figure 3.S.2A). To avoid substantial negative mechanical work during the subsequent CE lengthening, muscle stimulation ceased well before the end of CE shortening. While CE could theoretically maintain a constant shortening velocity throughout the shortening phase, any additional positive mechanical work gained near the end was outweighed by the increased negative mechanical work. Thus, an almost isometric phase preceding CE lengthening is optimal for maximising AMPO. In sum, the CE length over time for imposed FTS values mostly results from the effect of the excitation dynamics.

For imposed MTC length excursions above the optimal value, CE shortening velocity was found to be higher at the beginning and end of the shortening phase compared to the middle part (Figure 3.S.2B). At the start and end of the shortening phase, only limited positive mechanical work could be produced due to the low

active state and the suboptimal CE length (that is due to the CE force–length relationship). While CE could theoretically shorten at a constant velocity throughout the entire shortening phase, this would reduce the positive mechanical work during the middle part due to the effect of the force–velocity relationship. At the same time, little additional positive mechanical work could be achieved at the beginning and end of the shortening phase (due to the low active state and suboptimal CE length). Consequently, a non-constant CE shortening velocity — with CE shortening first decreasing and then increasing during the shortening phase — is optimal for maximising AMPO at imposed MTC length excursions above the optimal value. For imposed MTC length excursions below the optimal value, the opposite pattern was found: CE shortening velocity was found to be slower at the beginning and end of the shortening phase compared to the middle part (Figure 3.S.2B). As explained, only limited positive mechanical work could be produced at the beginning and end of the shortening phase due to the low active state. If CE would shorten with a constant velocity throughout the entire shortening phase, it would decrease the shortening distance during the middle part (at which the active state is high) and thereby reducing the positive mechanical work. As a result, a non-constant CE shortening velocity — with a slow shortening velocity at the end and beginning of the shortening phase — is optimal for maximising AMPO at imposed MTC length excursions above the optimal value. In sum, the optimal CE length over time for imposed MTC length excursions shows the intricate trade-off between the CE force–length relationship, CE force–velocity relationship and the and excitation dynamics.

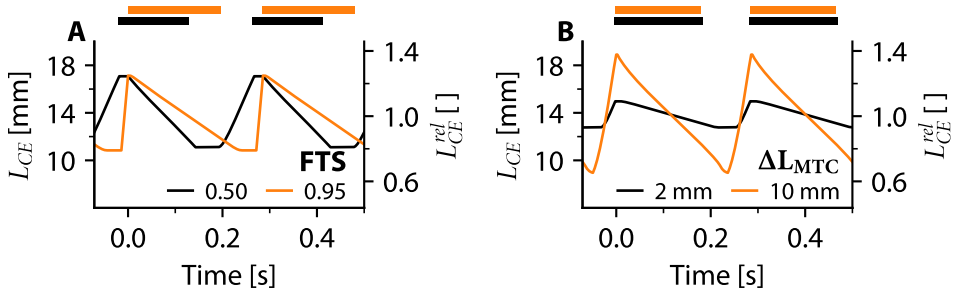


Figure 3.S.2: Predicted CE length over time for maximally attainable AMPO, shown for two imposed FTS values (A) and two imposed MTC length excursions (B), at a cycle frequency of 3.5 Hz for rat 1. In panel A, only cycle frequency and FTS were imposed, while in panel B only cycle frequency and MTC length excursions were imposed. Apart from these two constraints, CE and MTC length over time were completely unconstrained and thus the shape of CE length over time could be different between every combination of imposed cycle frequency and FTS (A) or imposed cycle frequency and MTC length excursion (B). CE stimulation was maximal during the period indicated by the coloured bars and off elsewhere.

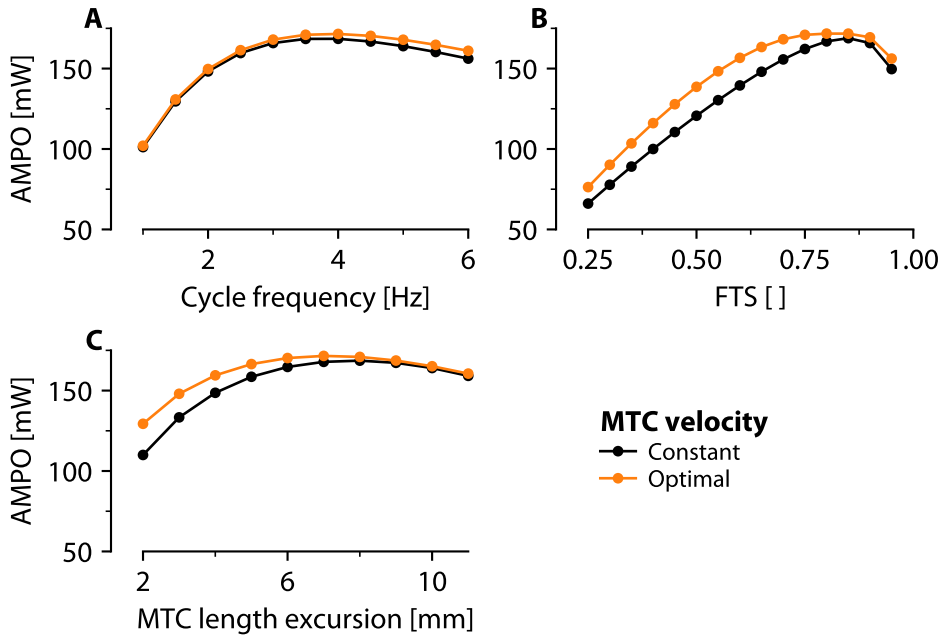


Figure 3.S.3: Predicted influence of cycle frequency (A), FTS (B) and MTC length excursion (C) on the maximum attainable AMPO. The maximally attainable AMPO — averaged across the three rats — slightly increased when there was no constraint on MTC length over time (orange lines), compared to SSCs with a constant MTC shortening and lengthening velocity (black lines).

Table 3-S.1: SSC parameters, stimulation durations, and measured AMPO of experimental stretch-shortening cycles with a 4 mm MTC length excursion. Stimulation onset was set at the start of MTC shortening in all conditions

		Condition												
	Unit	1	2	3	4	5	6	7	8	9	10	11	12	13
SSC parameters														
Cycle frequency	Hz	1.0	2.0	3.0	4.0	5.0	3.0	3.0	3.0	3.0	5.0	4.0	2.0	1.0
FTS	-	0.50	0.50	0.50	0.50	0.50	0.80	0.65	0.35	0.20	0.80	0.65	0.35	0.20
MTC shortening and lengthening times														
MTC shortening time	ms	500	250	167	125	100	267	217	117	66.7	160	162	175	200
MTC lengthening time	ms	500	250	167	125	100	66.7	117	217	267	40.0	87.5	325	800
Stimulation durations														
Trial 1	ms	405	175	103	65	35	183	133	53	23	85	95	115	135
Trial 2	ms	445 ¹	205	123	85	55	213	163	73	33 ²	115	115	135	155 ³
Measured AMPO of Rat 1														
Trial 1	mW	31	51	59	58	45	74	67	38	20	82	74	43	23
Trial 2	mW	34	56	69	72	66	84	77	49	25	104	87	47	27
Measured AMPO of Rat 2														
Trial 1	mW	29	48	57	58	45	71	62	37	18	79	70	40	22
Trial 2	mW	30	52	65	68	64	78	72	46	24	93	79	45	24
Measured AMPO of Rat 3														
Trial 1	mW	34	56	65	66	48	83	72	44	22	91	82	47	25
Trial 2	mW	37	62	74	82	74	91	85	56	20	116	96	54	28

¹ For condition 1: stimulation duration of rat 3 was 455 ms. ² For condition 9: stimulation duration of rat 3 was 23 ms. ³ For condition 13: stimulation duration of rat 1 was 165 ms.

Table 3.S.2: SSC parameters, stimulation durations, and measured AMPO of experimental stretch-shortening cycles with a 8 mm MTC length excursion. Stimulation onset was set at the start of MTC shortening in all conditions

		Condition												
	Unit	1	2	3	4	5	6	7	8	9	10	11	12	13
SSC parameters														
Cycle frequency	Hz	1.0	1.5	2.0	2.5	3.0	2.0	2.0	2.0	2.0	3.0	2.5	1.5	1.0
FTS	-	0.50	0.50	0.50	0.50	0.50	0.80	0.65	0.35	0.20	0.80	0.65	0.35	0.20
MTC shortening and lengthening times														
MTC shortening time	ms	500	333	250	200	167	400	325	175	100	267	260	233	200
MTC lengthening time	ms	500	333	250	200	167	100.0	175	325	400	66.7	140	433	800
Stimulation durations														
Trial 1	ms	405	253	175	135	103	295	245	115	55	173	175	163	145
Trial 2	ms	455	293	205	165	123	345	275	135	75 ¹	213 ²	225	193	165 ³
Measured AMPO of Rat 1														
Trial 1	mW	53	67	75	80	79	94	89	55	26	113	95	54	34
Trial 2	mW	54	47	80	86	85	99	97	63	28	121	104	62	35
Measured AMPO of Rat 2														
Trial 1	mW	49	63	73	75	76	88	83	52	24	104	88	50	32
Trial 2	mW	50	66	76	80	81	92	86	55	27	113	95	54	33
Measured AMPO of Rat 3														
Trial 1	mW	57	73	81	87	88	103	95	60	28	123	101	60	37
Trial 2	mW	47	-	89	-	88	110	-	-	32	129	-	-	40

¹ For condition 9: stimulation duration of rat 3 was 65 ms. ² For condition 10: stimulation duration of rat 3 was 223 ms. ³ For condition 13: stimulation duration of rat 1 was 175 ms.

4

What is the influence of knee joint movement on the maximal average mechanical power output of human quadriceps femoris muscle?

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Journal of Applied Physiology, 141(1), 4-18
<https://doi.org/10.1152/jappphysiol.01127.2025>

Abstract

Performance during motor activities such as wheelchair riding, cycling, rowing and speed skating, depends critically on the average mechanical power output (AMPO) produced by the muscles. To maximise short-duration performance, limb movements should allow muscles to deliver maximal AMPO. However, it is unclear which movement maximises AMPO of human muscle. In this study, we employed a Hill-type muscle-tendon-complex (MTC) model to predict the maximally attainable AMPO of human *m. quadriceps femoris* for various imposed periodic knee joint movements. Based on these predictions, we selected one set of conditions predicted to yield identical maximally attainable AMPO despite substantial variations in knee joint movements and another set of conditions predicted to yield substantial variations in maximally attainable AMPO. In the experiment, periodic knee joint movements were fully imposed by a knee dynamometer. Participants were instructed to maximise AMPO and, to this end, received visual feedback on their cumulative mechanical work throughout each cycle. Experimental data closely matched predictions derived from the Hill-type MTC model, confirming the validity of the model. Model predictions showed a substantial influence of knee joint movement on the maximally attainable AMPO. Specifically, predictions revealed a strong interaction between cycle frequency and knee joint excursion: increasing one necessitates a decrease in the other to maximise AMPO. Even more interestingly, *m. quadriceps femoris* should spend about 80% of the cycle duration while shortening, independent of cycle frequency and/or knee joint excursion.

4.1 Introduction

Basic motor activities such as walking, running, swimming, and activities involving man-made appliances like wheelchair riding, cycling, rowing, and speed skating are characterised by repetitive limb movements that result in periodic stretch-shortening cycles (SSCs) of muscles. In each SSC, muscles alternate between concentric and eccentric phases: they deliver positive mechanical power when generating force while shortening, and they deliver negative mechanical power when generating force while lengthening. In many motor activities, performance largely depends on the average mechanical power output (AMPO) over a full periodic cycle of all muscles involved. Maximising performance therefore requires the muscles to maximise average positive mechanical power during shortening and to minimise average negative mechanical power during lengthening. This raises the question: what periodic muscle length/velocity over time should be imposed in order to maximise AMPO?

A muscle consists of contractile fibres (the contractile element; CE) and connective tissue, both parallel to the fibres (the parallel elastic element; PEE) and in series with the fibres (the series elastic element; SEE). These three elements together form the muscle-tendon complex (MTC). Because connective tissue behaves predominantly elastically, AMPO of the MTC is determined almost exclusively by AMPO of the CE. However, CE length depends on MTC length and deformation of the elastic elements, which in turn depends on MTC force. Consequently, and ultimately, MTC AMPO depends not only on stimulation over time but also on MTC length over time.

During limb movement, MTC length varies with the joint(s) it spans. Understanding how periodic MTC length changes affect AMPO thus provides insight into how joints should move to maximise AMPO. These insights are not only interesting from a fundamental scientific perspective, but also from an applied sciences perspective, as they can inform the (re-)design of equipment. Such a design approach, that capitalises on a thorough understanding of the human musculoskeletal system has proven effective, as demonstrated by the Klapskate for speed skating (Ingen Schenau et al., 1996) and the SuperSeat for rowing (Soest and Hofmijster, 2009). Yet, such

design approaches remain rare due to limited understanding of how changes in MTC length over time affect AMPO, or other measures of performance in human muscle.

Much of the existing knowledge of the influence of CE/MTC length over time on AMPO is inferred from experiments on isolated animal muscle. In these experiments, CE or MTC length and muscle stimulation are imposed and resulting AMPO is measured. This method is often referred to as the ‘workloop technique’ (Josephson, 1985). These experiments have often been performed for sinusoidal SSCs, showing that the effects of cycle frequency and CE/MTC length excursion interact: at lower cycle frequencies, length excursions should be larger in order to maximise AMPO. However, sinusoidal SSCs by definition result in equal shortening and lengthening durations, leaving unanswered how unequal shortening and lengthening duration affects AMPO.

Askew & Marsh (1997; 1998) were the first, and until very recently (see Reuvers et al., 2025), the only researchers to perform experiments on the influence of unequal shortening and lengthening duration on the maximally attainable AMPO of muscle during short-duration, all-out (sprint) conditions. In their study, Askew & Marsh (1997; 1998) imposed SSCs with near-constant shortening and lengthening velocities and parametrised them in terms of three variables: cycle frequency, FTS (fraction of the cycle time spent shortening) and length excursion (see Figure 4.1). Askew & Marsh (1997; 1998) examined three distinct FTS values: 0.25, 0.50 and 0.75. They found that AMPO was consistently highest for FTS = 0.75 at all tested cycle frequencies at the optimal length excursion. In a recent study (Reuvers et al., 2025), we further examined the effect of SSC parameters on the maximally attainable AMPO by combining experiments on rat *m. gastrocnemius medialis* with predictions using a Hill-type MTC model. Our results showed a near-perfect correlation between model-based predictions and experimental results ($r^2 > 0.98$). The effect of cycle frequency, FTS and length excursion on the maximally attainable AMPO was then further examined using the Hill-type MTC model for a broad range of SSC parameters. These model-based predictions showed that the optimal FTS ranged 0.84-0.88 across a wide range of cycle frequencies and length excursions.

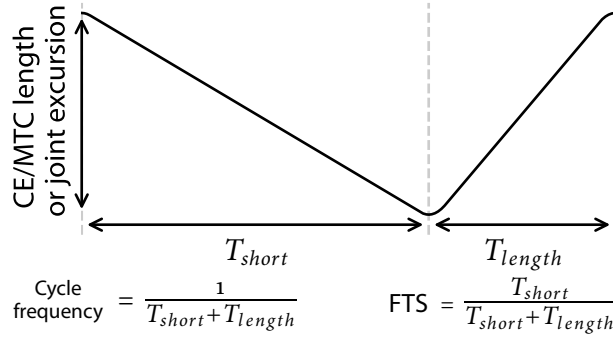


Figure 4.1: Parameterisation of stretch-shortening cycles. T_{short} and T_{length} denote the shortening and lengthening durations. FTS denotes the fraction of the cycle time spent shortening. In the example shown, the shortening duration is 65% of the cycle duration (i.e. $\text{FTS} = 0.65$).

Although animal studies have provided valuable insights, it remains unknown how these findings translate to human muscle. For example, is the optimal FTS for human muscle also in the order of 0.85? Although similar experiments can, in principle, be designed for human muscle, inherent differences exist between experiments involving human participants and experiments on isolated animal muscles. First, MTC properties can be accurately estimated in animal experiments based on dedicated experiments (see Reuvers and Kistemaker, 2025) that are challenging in human experiments (i.e. ‘quick-release’ experiments). Second, CE/MTC length is under full control in animal experiments, whereas in human movement it can only be imposed at the joint level. As a result, the exact CE/MTC length over time is uncertain in human experiments due to inter-individual variation in, for example, tendon compliance and moment arms. Third, stimulation is also under full control of the experimenter in animal experiments, but not in human experiments. Despite these differences, it is feasible to design human experiments in which imposed joint movements are systematically varied, with participants aiming to maximise AMPO. Similarly to animal experiments where CE/MTC length is under full control, an experimental design should be used in which joint angle over time is fully imposed, such that the behaviour of the participants does not have any influence on the kinematics (i.e. knee joint angle over time). Such an approach allows investigation of the effects of SSC parameters – cycle frequency, FTS, and joint excursion – on

maximally attainable AMPO. Surprisingly, to our knowledge such studies have not yet been conducted for human muscle – not even for sinusoidal SSCs. Consequently, the effect of SSC parameters on the maximally attainable AMPO in human muscle is currently unknown.

In this study, we examine the influence of SSC parameters on the maximally attainable AMPO of human *m. quadriceps femoris* during short-duration, all-out (sprint) conditions. Under such conditions, performance is determined by the mechanical properties of the MTC. Hill-type MTC models have been shown to accurately predict mechanical behaviour under these conditions (e.g. Ingen Schenau et al., 1988; Lemaire et al., 2016; Reuvers et al., 2025; Sandercock and Heckman, 1997; Wakeling and Johnston, 1999; Zandwijk et al., 1996). We therefore combined model predictions with *in vivo* knee dynamometer experiments. In both the simulations and the experiments, knee joint angle over time was fully imposed; in other words, the knee joint kinetics did not influence the knee joint movement. This allowed us to examine the direct influence of SSC parameters on the maximally attainable AMPO. The model was used to predict the maximally attainable AMPO across a wide range of imposed knee joint movements, from which nine insightful experimental conditions were selected. In the experiment, knee joint movements were fully imposed by a knee dynamometer, while participants aimed to maximise AMPO. Measured AMPO matched the model predictions reasonably well, confirming that the Hill-type MTC model reliably captures the influence of SSC parameters on the maximally attainable AMPO. This justified using the model to systematically examine the overall influence of imposed knee joint movement on the maximally attainable AMPO of human *m. quadriceps femoris*.

4.2 Methods

In this study, we examined the influence of imposed SSC parameters – cycle frequency, FTS, and joint excursion – on the maximally attainable AMPO of human *m. quadriceps femoris*. Experiments were combined with simulations and optimisations using a Hill-type MTC model. Based on model predictions, we selected one set of conditions expected to yield similar AMPO despite substantial variations in knee joint movements, and another set expected to yield large differences in AMPO. In the experiment, knee joint movements were fully imposed by a dynamometer. Participants were instructed to maximise AMPO and, for this purpose, received visual feedback on cumulative mechanical work throughout each cycle. Measured AMPO was compared with model predictions. After confirming that measurements closely matched predictions, the model was used to investigate the overall influence of SSC parameters on the maximally attainable AMPO of human *m. quadriceps femoris*.

4.2.1 Experiment

4.2.1.1 Participants and ethical approval

Approval of the local ethics committee was obtained before the start of the *in vivo* experiment (VCWE-2016-135). Four male and two female healthy participants (22.8 ± 2.2 years, mean $\pm$ SD), who all signed informed consent, participated in the experiment.

4.2.1.2 Measurement setup

A kinematically driven knee dynamometer (custom made, Vrije Universiteit, Amsterdam, the Netherlands, see Ruiter et al., 2006) was used to fully impose periodic knee joint angle over time. This dynamometer was capable of imposing large knee joint accelerations and was sufficiently stiff such that participant behaviour (i.e. knee joint moment changes) did not influence the resulting knee joint movement (see Results). During the experiment, participants sat in the dynamometer with the trunk, hip, and thigh firmly strapped. The foot was secured to the dynamometer footplate to fix the ankle joint. The fixed joint and segment angles were as follows:

the hip angle (between trunk and thigh, 0 rad = fully extended) was about 1.2 rad, the ankle angle (between shank and foot, 0 rad = fully extended) was about 1.2 rad and the thigh angle (relative to the horizontal) was about 3.3 rad (see Figure 4.2). The knee joint angle was defined such that 0 rad corresponded to a fully extended knee, while knee joint angle increased with flexion (see Figure 4.2). Participants performed the experiment with their preferred leg. The lower leg was attached to the sliding carriage of the dynamometer just proximal to the lateral malleolus (see Figure 4.2). To align the knee joint with the rotation axis of the dynamometer, the lateral epicondyle was initially used as a proxy for the knee joint axis while the participant generated a substantial knee extension moment. Next, the participant actively extended and flexed their knee over the experimental joint angle range. During this movement, the travel of the sliding carriage was visually inspected, and the knee joint alignment was adjusted to minimise carriage displacement. Net knee joint moment and knee joint angle were measured with a torque sensor (custom made, Vrije Universiteit, Amsterdam, the Netherlands) and an angle sensor (custom made, Vrije Universiteit, Amsterdam, the Netherlands) at a sample rate of 1000 Hz.

Surface EMG of the *m. vastus medialis*, taken as the reference muscle for *m. quadriceps femoris*, was measured during the AMPO measurements. In addition, EMG of *m. biceps femoris*, taken as the reference muscle for the hamstring muscle group, was measured to test for undesired co-contraction. Bipolar EMG electrodes (Ambu, Ballerup, Denmark) were placed at an interelectrode distance of 20mm, at 4/5 of the line originating from the anterior spina iliaca superior to the joint space in front of the anterior border of the medial ligament and at 1/2 of the line originating from the ischial tuberosity to the lateral epicondyle of the tibia for *m. vastus medialis* and *m. biceps femoris* respectively (SENIAM guidelines, Hermens et al., 2000). EMG recordings were amplified with a pre-amplifier (9200 EMG, Goodvaerts Instruments, Kortenhoeve, the Netherlands) and collected at a sample rate of 4000 Hz.

To synchronise the dynamometer and EMG measurement system, a trigger pulse generated by the dynamometer system was sent to the EMG measurement system at the start of each measurement. In addition, knee joint angle was recorded

simultaneously by both systems. After acquisition, the datasets were aligned based on the trigger pulse. To verify correct synchronisation, the knee joint angle signals recorded by the two systems were compared. The signals overlapped after alignment (i.e., maximal cross-correlation occurred at a time shift of 0), confirming successful time synchronisation between the dynamometer and EMG recordings. The EMG signals were downsampled from 4000 Hz to 1000 Hz, being the sampling rate of the dynamometer system. Before downsampling, a low-pass anti-aliasing filter with a cut-off frequency of 500 Hz (dynamometer sample rate divided by 2) was applied to prevent aliasing artifacts. All subsequent analyses were performed on the synchronised and downsampled datasets.

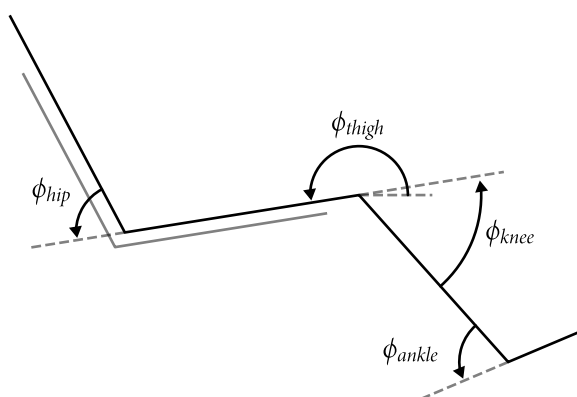


Figure 4.2: The experimental setup.

4.2.1.3 Isometric measurements

The maximally attainable AMPO of participants depends on their maximal isometric force. Consequently, for a meaningful comparison between the magnitude of measured AMPO and predicted maximally attainable AMPO, as well as for meaningful comparisons between participants, we needed an accurate estimate of each participant's maximal isometric force. To obtain this estimate, we measured the isometric knee joint moment-angle relationship for each participant individually. Maximal isometric net knee joint moment was determined at nine different knee joint angles equidistantly distributed over a range of knee joint angles between 0.2

rad and 1.8 rad. The order of these measurements was assigned pseudo-randomly such that they were equally distributed across participants. Participants were instructed to maximise their net knee joint moment with their *m. quadriceps femoris* at every knee joint angle for a period of about five seconds. Participants received real-time visual feedback about their net knee joint moment on a monitor which was placed in front of them and were verbally encouraged by the experimenter. There was a 3-minute rest between each isometric measurement. At the end of the experiment, the isometric measurements for the first two knee joint angles were repeated to assess fatigue.

4.2.1.4 AMPO measurements

Experimental conditions for the AMPO measurements were selected based on simulations using a Hill-type MTC model. Due to limits on knee angular acceleration, only a subset of combinations of cycle frequency, FTS and knee joint excursion were experimentally feasible (for details, see Appendix). Nine experimental conditions that differed in cycle frequency and FTS at a knee joint excursion of 1.6 rad were used in the experiment (see Figure 4.3). One subset of five experimental conditions was predicted to yield identical maximally attainable AMPO despite substantial variations in cycle frequency and FTS (0.35A-E; Figure 4.3). A second subset of five experimental conditions was predicted to yield substantial differences in maximally attainable AMPO due to variations in cycle frequency and FTS (0.20, 0.25, 0.30, 0.35C and 0.40; Figure 4.3). One experimental condition was shared between both subsets, resulting in nine unique experimental conditions.

In each experimental condition, 15 consecutive cycles of the respective knee joint movement were imposed; the participants were instructed to remain fully relaxed during cycles 1-5 and cycles 11-15 and to maximise AMPO with their *m. quadriceps femoris* during cycles 6-10. Throughout all cycles, participants were instructed to relax their hamstring muscle group to prevent it from producing negative and positive mechanical work during knee extension and flexion.

The periodic knee joint angle over time was fully imposed by the knee dynamometer and therefore periodic lengthening and shortening of the MTC was fully imposed.

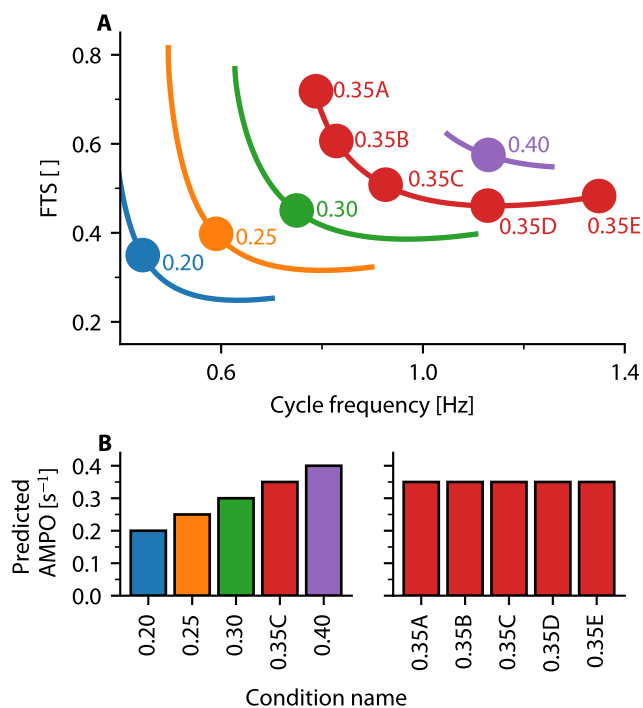


Figure 4.3: Predicted maximally attainable AMPO and experimental conditions. A) Contour lines depict model-based predictions of maximally attainable AMPO as a function of cycle frequency and FTS for a knee joint excursion of 1.6 rad. Contour lines are only shown for experimentally feasible conditions (due to limits set on knee joint acceleration; see Appendix). Dots indicate the nine conditions selected to be evaluated experimentally. B) Barplots of the same conditions: left, subset predicted to differ in maximally attainable normalised AMPO (i.e. 0.20, 0.25, 0.30, 0.35C and 0.40); right, subset predicted to yield identical maximally attainable normalised AMPO (0.35A-E).

Consequently, participants could only influence measured AMPO through their muscle activation. For the predictions derived with the Hill-type MTC model, muscle activation was optimised for AMPO (for details, see Methods: Musculoskeletal model). Thus, for a meaningful comparison between measured AMPO and the predicted maximally attainable AMPO, it was crucial that participants chose a (near-)optimal muscle activation of their *m. quadriceps femoris*. Therefore, each participant practised the nine conditions twice during both practice days (day 1 and 2) of the experiment. Additionally, all conditions were performed once at the end of day 3, at which day the isometric measurements took place. Thus, in total, participants practised each of the nine conditions five times, equalling 25 active cycles. During both practice and the actual AMPO measurements, participants received real-time visual feedback about their cumulative work throughout each cycle on a monitor and were verbally encouraged by the experimenter to maximise it.

During the AMPO measurements day (day 4), EMG measurements of *m. quadriceps femoris* and the hamstring muscle group were taken. At the start and at the end of the AMPO measurement, participants performed isometric contractions at a knee joint angle of 1.0 rad: twice for *m. quadriceps femoris* and twice for the hamstring muscle group in order to record EMG signals during maximal voluntary contractions (EMG^{MVC}) of both muscles, which were later used to normalise EMG activity during the AMPO measurements (see Methods: Data analysis). Before the actual measurement, the participants again practised with the protocol by performing the four most extreme conditions of the experiment (i.e. 0.20C, 0.35A, 0.35E and 0.40C). To prevent fatigue during the actual measurements, participants were instructed to perform these conditions sub-maximally. After this, participants performed the nine different conditions, from which the first two were repeated at the end to check for muscle fatigue. As with the isometric measurements, the order of the AMPO conditions was assigned pseudo-randomly such that they were equally distributed across participants. There was a 3-minute rest between each condition.

4.2.2 Musculoskeletal model

4.2.2.1 Description

A planar model consisting of an upper and lower leg connected by a frictionless hinge joint was used to relate knee joint angle to muscle-tendon-complex (MTC) length and to relate MTC force to knee joint moment (for details, see Appendix). The musculoskeletal model did not contain any knee flexors, as participants were instructed to maximise AMPO using their knee extensors only. A single Hill-type MTC model was used to represent the mechanical behaviour of all knee extensors in the human leg. The formulation of the Hill-type muscle-tendon complex model has been previously described in full detail (Reuvers and Kistemaker, 2025; Soest and Bobbert, 1993). Briefly, the Hill-type MTC model included a contractile element (CE) with a parallel elastic element (PEE) in parallel and a serial elastic element (SEE) in series with the CE. PEE and SEE were modelled as purely passive elastic structures, with their force non-linearly dependent on their length. CE force depended on CE length, CE velocity and active state. Active state, defined as the relative Ca^{2+} concentration bound to troponin C (Ebashi and Endo, 1968), depended on free Ca^{2+} concentration and CE length (Hatze, 1981, pp 31-32; Kistemaker et al., 2005). The free Ca^{2+} concentration in turn, depended on neural drive (*STIM*) by first-order dynamics. *STIM* and the imposed knee joint angle over time were the model's independent inputs.

4.2.2.2 Predictions of maximally attainable AMPO

To derive predictions of the maximally attainable AMPO, we used the musculoskeletal model with parameter values obtained from literature (see Table 4.A.1). As such, we derived generic predictions without tuning those to the muscle parameters of the participants in our experiment. Using the musculoskeletal model, we imposed periodic knee joint movements in which knee joint angle was centred around 1.0 rad. We set the knee angular velocities to constant values that could differ between flexion and extension. We examined a large number of knee joint movements over a range of cycle frequencies (0.4-3 Hz, in 0.2 Hz steps), FTS values (0.10-0.90, in 0.05 steps) and knee joint excursions (0.8-2.0 rad, in 0.1 rad steps). Muscle activation

onset time (i.e. the time instance at which *STIM* switches from 0 to its maximal value) was chosen at the start of MTC shortening for each imposed knee joint movement. Muscle activation duration (i.e. the time period during which *STIM* remains at its maximal value before returning to 0) was optimised for AMPO using a Nelder-Mead Simplex method (Lagarias et al., 1998). To ensure periodic behaviour, simulations were run for at least five cycles and until the difference in MTC force at the beginning and end of the cycle was less than $0.1\% F_{CE}^{max}$. AMPO was calculated for the last cycle.

4.2.3 Data analysis

The isometric measurements were used to estimate the maximal isometric force (F_{CE}^{max}) for each participant individually. The maximal net knee joint moment was determined for every measured knee joint angle and was defined as the highest 500 ms average after correcting for gravitational effects. The moment arm was assumed to be equal for all participants (i.e. moment arm = 4.2 cm, see Soest and Casius (2000)). The isometric knee joint moment-angle relationship of the musculoskeletal model depends on three parameters: F_{CE}^{max} , L_{CE}^{opt} (CE optimum length), and L_{SEE}^0 (SEE slack length). These parameters were estimated individually for each participant by minimising the sum of squared differences between the measured isometric net knee joint moment and those predicted by the musculoskeletal model. This optimisation was performed using a Nelder-Mead Simplex method (Lagarias et al., 1998).

EMG signals measured during the AMPO measurements were bandpass filtered using a bidirectional first order Butterworth filter with cut-off frequencies of 20 and 500 Hz (Butterworth, 1930). The EMG envelope was determined as the absolute value of the Hilbert transformed EMG signal (Hilbert, 1904). EMG^{MVC} values of *m. vastus medialis* and *m. biceps femoris* were defined as the maximal average value of the EMG envelope over 500 ms during the isometric contractions at a knee joint angle of 1.0 rad. Measured EMG signals of *m. vastus medialis* and *m. biceps femoris* during the AMPO measurements were smoothed by a moving average filter with a window of 100 ms and were normalised by their EMG^{MVC} value. The first and

last 0.1 seconds of knee extension were not taken into account for the calculation of the average EMG values, as EMG values increased and decreased during these time intervals. Therefore, average EMG values reported were calculated between a knee joint angle of 0.45 and 1.55 rad.

The knee joint moment measured by the dynamometer during the AMPO measurements represented the combined contributions of gravity, inertia (from both the participant's lower leg+foot and that of the dynamometer), passive structures, and active muscle force. To estimate the contribution of gravity, inertia and passive structures, participants were instructed to remain fully relaxed during the first and last five cycles of each experimental condition. The knee joint moment recorded during these passive cycles was assumed to reflect only the net non-muscular contribution. To isolate the muscular contribution, the average knee joint moment from passive cycles 2, 3, 4, 12, 13, and 14 was calculated and subtracted from the measured knee joint moment. The resulting moment, attributed to the moment resulting from muscle force only, was called the 'muscular' knee joint moment (M_{mus}). AMPO per individual cycle was calculated by integrating M_{mus} over knee joint angle using a trapezoid method, and dividing the result by the cycle duration.

Measured AMPO and predicted maximally attainable AMPO were normalised in two different ways. First, we assessed whether the magnitude of measured AMPO could be adequately predicted. The predicted maximally attainable AMPO scales linearly with maximal isometric force (F_{CE}^{max}) and also depends on CE optimum length (L_{CE}^{opt}). The sensitivity for L_{CE}^{opt} was, however, low: a 10% reduction in L_{CE}^{opt} resulted in only a $3.7 \pm 0.3\%$ decrease in AMPO across all experimental conditions. We used the product of the generic model parameter value of L_{CE}^{opt} (see Table 4.A.1) and each participant's individually estimated F_{CE}^{max} to normalise measured AMPO. The predicted maximally attainable AMPO was normalised to the generic model parameter values of L_{CE}^{opt} and F_{CE}^{max} . Second, we evaluated how well differences in AMPO between experimental conditions were predicted. For each subject, the mean (μ) and standard deviation (σ) of measured AMPO across all experimental

conditions were calculated, and individual condition values were transformed into z-scores for each condition (denoted by the subscript i):

$$z_i = \frac{AMPO_i - \mu}{\sigma} \quad (4.1)$$

This procedure removed the influence of all potential errors affecting the magnitude of measured AMPO and isolates the relative changes of AMPO. Model-predicted maximally attainable AMPO values were similarly transformed into z-scores, allowing a direct comparison between the measured and predicted z-scores. Importantly, regardless of the normalisation method, whether by the product of L_{CE}^{opt} and F_{CE}^{max} or by z-scores, measured AMPO was assessed relative to the maximally attainable AMPO calculated from the generic model parameters. Thus, normalised model predictions were based on generic model parameters obtained from literature.

4.3 Results

4.3.1 AMPO-scaling factors were successfully obtained

All six participants were able to execute the isometric measurements at all imposed knee joint angles. To test for fatigue, the first two knee joint angles were repeated at the end of the isometric measurements. The maximal knee joint moment was $6.6 \pm 5.9\%$ lower for the last two measurements compared to the first two measurements, averaged over all participants and the two repeated measurements. The normalised root mean square difference between measured and predicted isometric net knee joint moments was $5.3 \pm 1.6\%$ on average. Variance in the estimated values of L_{CE}^{opt} and L_{SEE}^o between participants was low (Table 4.1). A representative example of the measured and modelled maximal isometric knee joint moment as a function of knee joint angle is depicted in Figure 4.4.

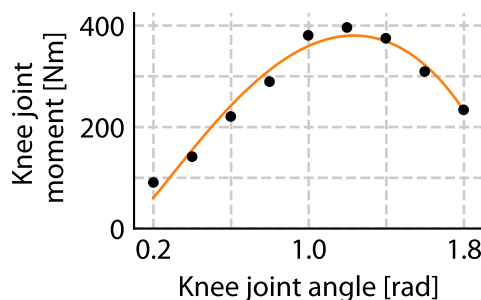


Figure 4.4: Representative example of maximal isometric knee joint moment as a function of knee joint angle, shown for participant 5. Maximal knee joint moment was measured (black dots) and used to estimate L_{SEE}^o , L_{CE}^{opt} and F_{CE}^{max} . The resulting maximal isometric knee joint moment of the musculoskeletal model is depicted by the orange line.

Table 4.1: Estimated parameter values for each individual participant.

Participant	F_{CE}^{max} [N]	L_{CE}^{opt} [cm]	L_{SEE}^o [cm]	RMSD [%]
1	4400	8.6	17	7.3
2	5100	9.9	15	6.9
3	5600	8.7	17	5.8
4	6800	8.6	16	4.1
5	9200	7.4	18	4.7
6	8500	8.8	16	2.9
AVG $\pm$ SD	6600 $\pm$ 1700	8.7 $\pm$ 0.7	17 $\pm$ 1	5.3 $\pm$ 1.6

F_{CE}^{max} , L_{CE}^{opt} and L_{SEE}^o were estimated from the isometric measurements for each participant individually. The root mean square difference (RMSD) was normalised by the individual maximal predicted knee joint moment, which is the product of F_{CE}^{max} and the moment arm of *m. quadriceps femoris*.

4.3.2 Knee joint movements were successfully imposed

For a meaningful comparison between the measured and predicted maximally attainable AMPO, it was crucial that the knee dynamometer adequately imposed the desired knee joint movement of the experimental conditions. This to ensure that both measured and predicted AMPO were based on (near-)identical knee joint movements. Averaged across all participants, conditions and cycles, the sum of squared differences between the desired and measured knee joint angle was only

$(7 \pm 12) \cdot 10^{-3}$ rad. As a result, knee extension lasted only 0.6 ± 0.5 ms shorter than desired, and knee flexion lasted 0.6 ± 0.5 ms shorter than desired. In addition, the difference between maximum and minimum knee joint angle in the experiment was $(0.7 \pm 0.2) \cdot 10^{-3}$ rad higher than desired, averaged across all participants, conditions and cycles. Overall, the dynamometer successfully imposed the desired knee joint movements both when participants were passive and when they aimed to maximise AMPO.

4.3.3 Timing of *m. quadriceps femoris* activation was adequate

EMG values during knee extension averaged over all conditions and participants were $77.7 \pm 31.9\%$ of EMG^{MVC} for *m. quadriceps femoris* and $16.7 \pm 12.2\%$ of EMG^{MVC} for the hamstring muscle group, between a knee joint angle of 0.45 and 1.55 rad. There was substantial interindividual variance in EMG values for both muscles, while variance in EMG values between conditions within each participant was small. EMG values of the hamstring muscle group increased just before the transition from flexion to extension and followed a similar time course as the EMG of *m. quadriceps femoris* (see Figure 4.5). This was reflected by a correlation coefficient of 0.76 ± 0.18 between EMG values of *m. quadriceps femoris* and of the hamstring muscle group, suggesting possible crosstalk between the two muscle groups. During flexion, EMG values of the hamstring muscle group were close to zero for all participants and conditions. EMG values of *m. quadriceps femoris* increased just before the transition from flexion to extension and were close to zero around the transition from extension to flexion for all participants and conditions (see Figure 4.5). Consequently, M_{mus} rose around the transition from flexion to extension and was close to zero around the transition from extension to flexion (Figure 4.6 and Figure 4.7). Except for occasional delays in the activation onset of *m. quadriceps femoris* during the first active cycle (cycle 6), M_{mus} exhibited little variance between cycles. In sum, participants were capable of finding an activation of their *m. quadriceps femoris* that appeared to be (near-)optimal.

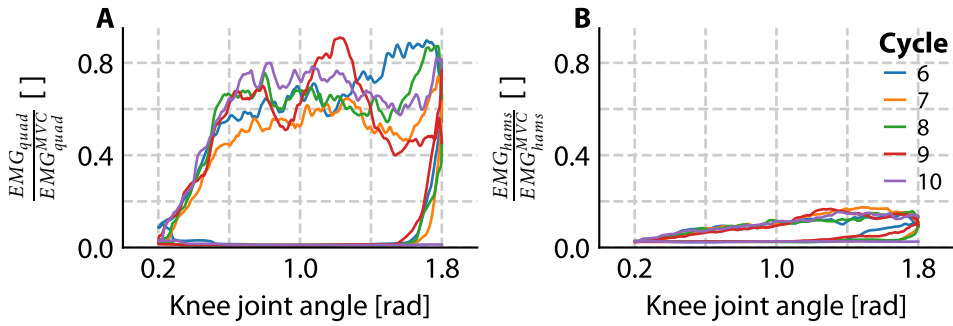


Figure 4.5: Representative example of measured EMG, shown for participant 5 under condition 0.35C. A) EMG of *m. vastus medialis* normalised by its EMG^{MVC} value as a function of knee joint angle. EMG values increased just before the transition from flexion to extension and were close to zero during flexion, indicating that the timing of *m. quadriceps femoris* activation of participants was adequate. B) EMG of the *m. biceps femoris* normalised by its EMG^{MVC} value as a function of knee joint angle. EMG values increased just before the transition from flexion to extension, following a similar time course as the EMG values of *m. vastus medialis*, suggesting possible co-contraction. During flexion, EMG values were close to zero, indicating participants adequately relaxed their the hamstring muscle group during flexion.

4.3.4 Mechanical work was achieved almost exclusively during knee extension

Participants were instructed to maximise mechanical work per cycle using their *m. quadriceps femoris*, while keeping their hamstring muscle group completely relaxed at all times. Average net knee joint moment during the passive cycles was subtracted from net knee joint moment during the active cycles to correct for knee joint moment due to inertia, gravity and passive structures (see Figure 4.6). The resulting knee joint moment (M_{mus}) was assumed to be predominantly the result of muscle force and was, as expected, substantially higher than zero during knee extension for all participants and conditions. Due to activation and deactivation delays, M_{mus} may be above zero just before the transition from flexion to extension and immediately after the transition from extension to flexion, in order to maximise AMPO (see also Figure 4.8). Yet, M_{mus} should be ideally zero during the middle part of knee flexion. For the active cycle the absolute value of M_{mus} was 3.0 ± 3.0 Nm between a knee joint angle of 0.45 and 1.55 rad (corresponding to 100 ms after and before the transitions). This minimal M_{mus} was a result of low activity of both

m. quadriceps femoris ($2.8 \pm 2.7\%$ of EMG^{MVC}) and hamstring muscle group ($4.1 \pm 4.0\%$ of EMG^{MVC}) during the same time interval. We therefore concluded that participants effectively relaxed their muscles during knee flexion, such that the net mechanical work was almost exclusively the result of positive mechanical work delivered during knee extension (see Table 4.2).

Table 4.2: Positive and negative mechanical work during knee extension and flexion as a percentage of the net mechanical work per cycle.

	<i>Condition</i>								
	0.20	0.25	0.30	0.35C	0.40	0.35A	0.35B	0.35D	0.35E
W_{ext}^+	102.3 ± 2.5	102.2 ± 2.0	103.4 ± 3.2	103.3 ± 4.3	104.2 ± 3.5	104.5 ± 4.0	102.5 ± 3.6	101.9 ± 4.2	100.5 ± 2.7
W_{flex}^+	1.3 $\pm$ 1.2	0.9 $\pm$ 0.9	1.1 $\pm$ 1.2	0.9 $\pm$ 0.7	1.4 $\pm$ 1.0	1.6 $\pm$ 1.5	1.7 $\pm$ 1.2	2.6 $\pm$ 1.9	3.3 $\pm$ 1.8
W_{ext}^-	-0.0 ± 0.1	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.1 ± 0.4
W_{flex}^-	-3.7 ± 2.2	-3.1 ± 2.0	-4.6 ± 3.3	-4.3 ± 4.1	-5.6 ± 3.4	-6.0 ± 3.6	-4.3 ± 3.0	-4.7 ± 4.4	-3.8 ± 2.0

W_{ext}^+ and W_{ext}^- represent positive and negative mechanical work during knee extension, respectively, whereas W_{flex}^+ and W_{flex}^- represent positive and negative mechanical work during knee flexion, respectively. These variables were calculated separately for periods in which instantaneous power was positive (positive work) or negative (negative work) during knee extension and flexion.

4.3.5 Measured AMPO closely matched predicted maximally attainable AMPO

To assess the reliability of AMPO measurements, we examined the variability of measured AMPO across cycles and repeated conditions. Averaged across each condition and for each participant, the second-highest measured AMPO was only $2.4 \pm 2.3\%$ lower than the highest value, indicating low variance in AMPO between cycles. Furthermore, in repeated conditions, the highest measured AMPO at the end of the experiment was just $2.3 \pm 4.1\%$ lower than at the start. Together, these results

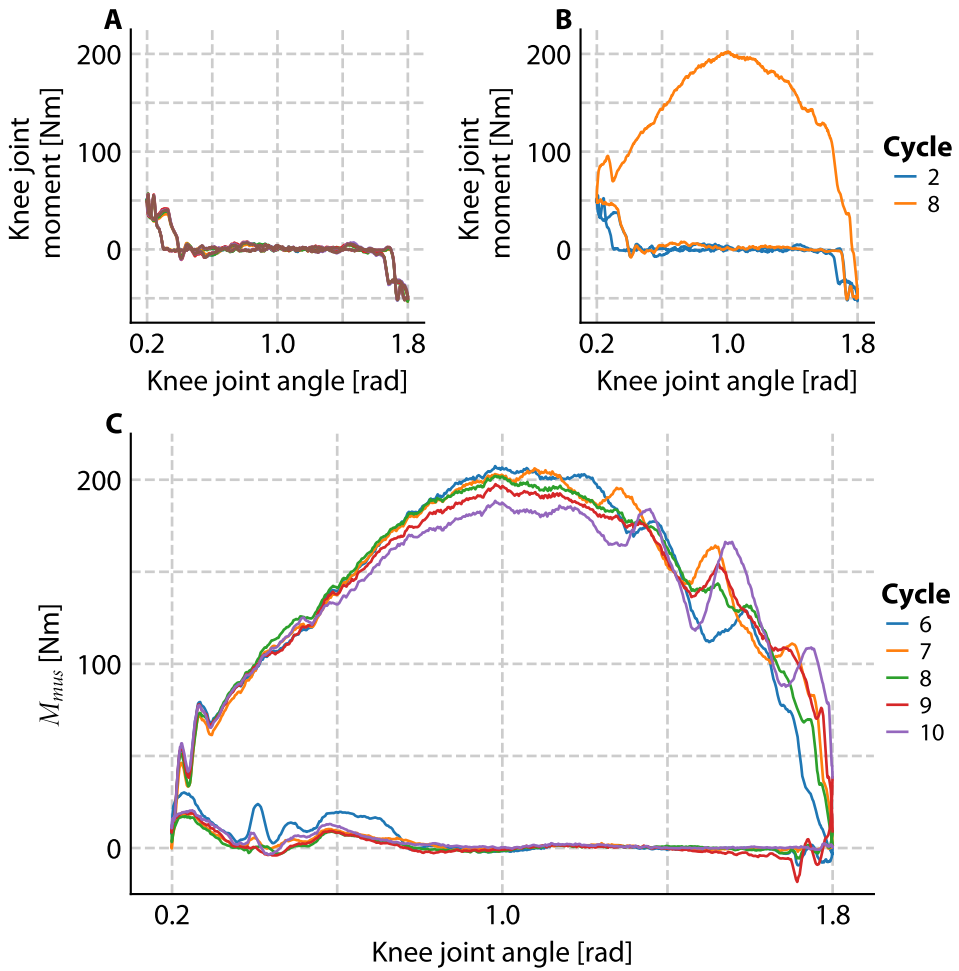


Figure 4.6: Representative example of mechanical behaviour, shown for participant 5 under condition 0.35C. A) Knee joint moment during passive cycles. Every individual cycle is represented by a different coloured line. B) Knee joint moment during (passive) cycle 2 (blue line) and during (active) cycle 8 (orange line). C) ‘Muscular’ knee joint moment (M_{mus}) as a function of knee joint angle. M_{mus} was calculated by subtracting average knee joint moment during the passive cycles from the knee joint moment during the active cycles (for details, see Methods). Every individual cycle is represented by a different coloured line. M_{mus} was close to zero during flexion, indicating that the net mechanical work was almost exclusively the result of mechanical work during extension.

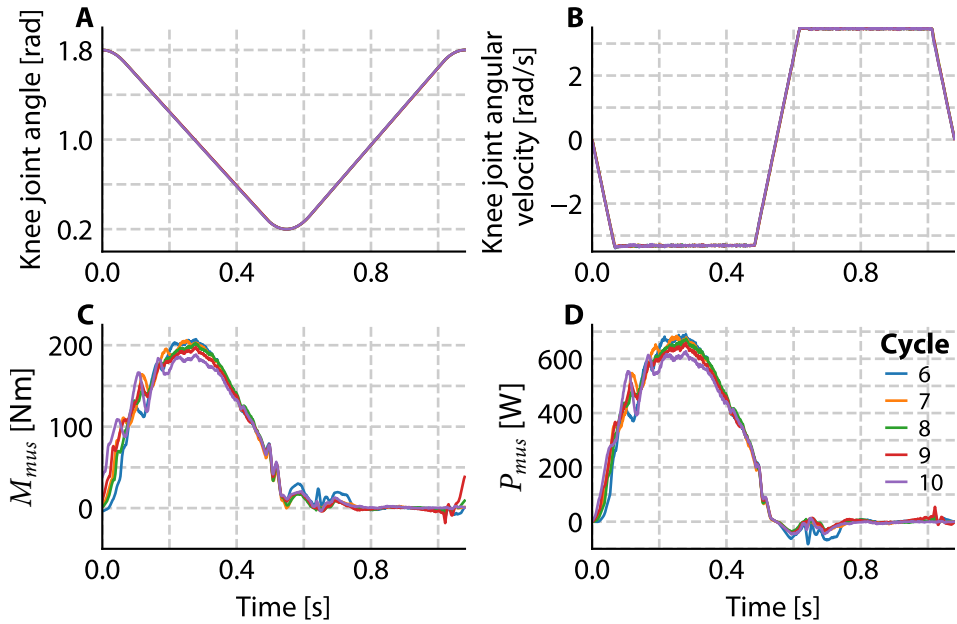


Figure 4.7: Representative example of mechanical behaviour, shown for participant 5 under condition 0.35C. Time series are only depicted for the active cycles, where time is set to zero at the start of knee extension for each cycle. Individual cycles are represented by a different coloured lines. A) Measured knee joint angle over time (note that these are indistinguishable between cycles). B) Measured knee joint angular velocity over time (indistinguishable between cycles). C) ‘Muscular’ knee joint moment (M_{mus}) over time, for cycles 6-10 (see Methods). M_{mus} was calculated by subtracting average knee joint moment during the passive cycles from the knee joint moment during the active cycles (see Methods). D) Instantaneous ‘muscular’ mechanical power (P_{mus}) over time, for cycles 6-10 (see Methods). P_{mus} was calculated as the product of M_{mus} and the measured knee joint angular velocity. Both M_{mus} and P_{mus} were close to zero during flexion, indicating that the net mechanical work was almost exclusively the result of mechanical work during extension.

indicate that AMPO measurements were consistent across cycles and throughout the experimental session.

Measured AMPO was compared against the predicted maximally attainable AMPO. First, AMPO was normalised by the generic value of L_{CE}^{opt} and participants' individually estimated F_{CE}^{max} (see Methods: Data analysis). As predicted, (1) measured AMPO differed substantially between conditions 0.20, 0.25, 0.30, 0.35 and 0.40 due to the imposed variations in cycle frequency and FTS and (2) measured AMPO was almost identical for conditions 0.35A-E, despite substantial variations in cycle frequency and FTS (Figure 4.9; Table 4.3). Overall, measured AMPO normalised by L_{CE}^{opt} and participants' individual F_{CE}^{max} correlated strongly with predicted maximally attainable AMPO ($r^2=0.75$). Second, AMPO was normalised by z-scores of measured and predicted maximally attainable AMPO. This revealed a very strong correlation between predictions and measurements ($r^2=0.95$). All in all, measured AMPO was remarkably close to the predicted maximally attainable AMPO.

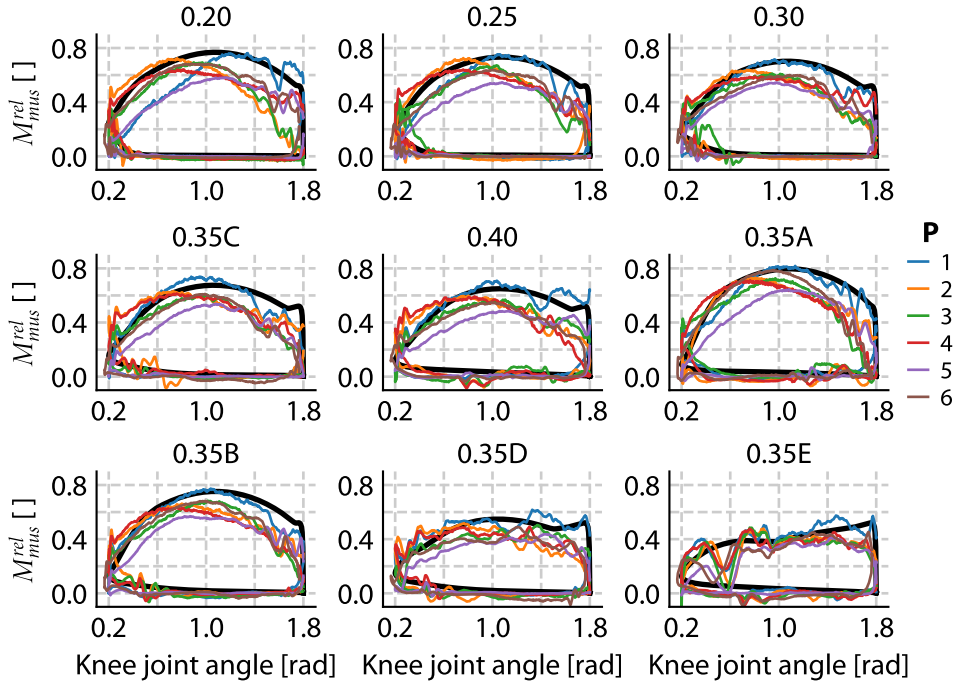


Figure 4.8: Experimentally estimated and model-predicted relative ‘muscular’ knee joint moment (M_{mus}^{rel}) for all conditions. M_{mus}^{rel} was calculated by dividing the ‘muscular’ knee joint moment by the maximal isometric knee joint moment of the individual participants (estimated from the isometric knee joint moment-angle relationship; see Figure 4.4) or by that of the model. The thick black line depicts the predicted M_{mus}^{rel} , while the thinner coloured lines depicts M_{mus}^{rel} for each individual participant (P). For each participant, M_{mus}^{rel} is shown for the cycle with highest measured AMPO.

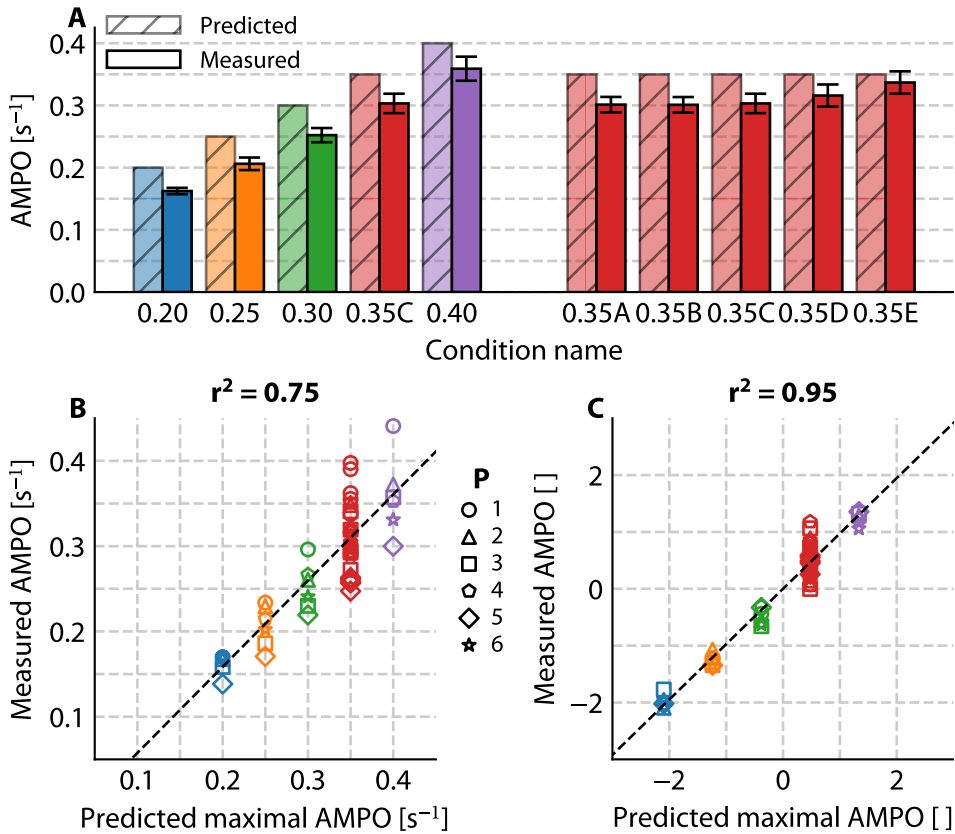


Figure 4.9: Measured AMPO versus predicted maximally attainable AMPO. A) Barplot showing predicted values with diagonal hatching and semi-transparent fill ($\alpha = 0.5$), and experimentally measured values fully filled. Error bars denote the standard error of the mean. B) Scatterplot of measured versus predicted maximally attainable AMPO normalised by L_{CE}^{opt} and F_{CE}^{max} , with F_{CE}^{max} estimated individually for each participant. C) Scatterplot of measured versus predicted maximally attainable AMPO normalised to z-scores. Participants (P) are distinguished by symbols.

Table 4.3: Overview of experimental conditions and measured AMPO per participant and condition.

	<i>Condition</i>								
	0.20	0.25	0.30	0.35C	0.40	0.35A	0.35B	0.35D	0.35E
<i>Movement characteristics</i>									
CF [Hz]	0.44	0.59	0.75	0.93	1.13	0.79	0.83	1.13	1.35
FTS []	0.35	0.40	0.45	0.51	0.58	0.72	0.61	0.46	0.48
T_{ext} [s]	0.79	0.67	0.60	0.55	0.51	0.91	0.73	0.41	0.36
T_{flex} [s]	1.46	1.02	0.73	0.53	0.38	0.36	0.48	0.48	0.38
<i>AMPO</i>									
P1 [s ⁻¹]	0.17	0.23	0.30	0.36	0.44	0.34	0.36	0.39	0.40
P2 [s ⁻¹]	0.17	0.23	0.26	0.32	0.37	0.32	0.30	0.32	0.35
P3 [s ⁻¹]	0.16	0.19	0.23	0.29	0.36	0.27	0.29	0.32	0.34
P4 [s ⁻¹]	0.17	0.21	0.27	0.30	0.35	0.30	0.29	0.30	0.35
P5 [s ⁻¹]	0.14	0.17	0.22	0.25	0.30	0.26	0.26	0.26	0.26
P6 [s ⁻¹]	0.17	0.20	0.24	0.29	0.33	0.31	0.30	0.30	0.32
AVG ±	0.16	0.21	0.25	0.30	0.36	0.30	0.30	0.32	0.34
SD [s ⁻¹]	±	±	±	±	±	±	±	±	±
	0.01	0.02	0.03	0.03	0.04	0.03	0.03	0.04	0.04
Predicted [s ⁻¹]	0.20	0.25	0.30	0.35	0.40	0.35	0.35	0.35	0.35

CF denotes cycle frequency, FTS denotes fraction of cycle time spent shortening, T_{ext} and T_{flex} denote the knee joint extension and flexion time per cycle. Measured AMPO was normalised to the product of the participant's individually estimated F_{CE}^{max} and the generic model parameter value of L_{CE}^{opt} , while predicted AMPO was normalised to the product of model parameters F_{CE}^{max} and L_{CE}^{opt} (see Methods).

4.3.6 Influence of SSC parameters on the maximally attainable AMPO

Having established that the influence of SSC parameters - cycle frequency, FTS and knee joint excursion - on the maximally attainable AMPO was accurately predicted by our Hill-type MTC model, we now examine the effects of the SSC parameters in detail. The predicted effect of cycle frequency and FTS at four distinct knee joint excursions are depicted in Figure 4.10). The optimal combination of SSC parameters was predicted to be at a cycle frequency of about 1.6 Hz, an FTS of about 0.80 and a knee joint excursion of about 2.0 rad. Because knee joint movements were centred around 1.0 rad, knee joint excursions beyond 2.0 rad would result in knee hyperextension and were therefore not physiologically meaningful. Nevertheless, model predictions showed that knee joint excursions above 2.0 rad did not further increase the maximally attainable AMPO.

Regarding the interrelationship between the SSC parameters, we found that the optimal FTS for AMPO remained relatively stable across the range of cycle frequencies and knee joint excursions studied, consistently ranging between 0.78 and 0.85. The largest variation in optimal FTS was observed at the smallest knee joint excursion studied (0.8 rad). By contrast, optimal cycle frequency showed substantial sensitivity to changes in FTS. Additionally, a strong interaction was found between cycle frequency and knee joint excursion: increasing one necessitates a decrease in the other to maximise AMPO.

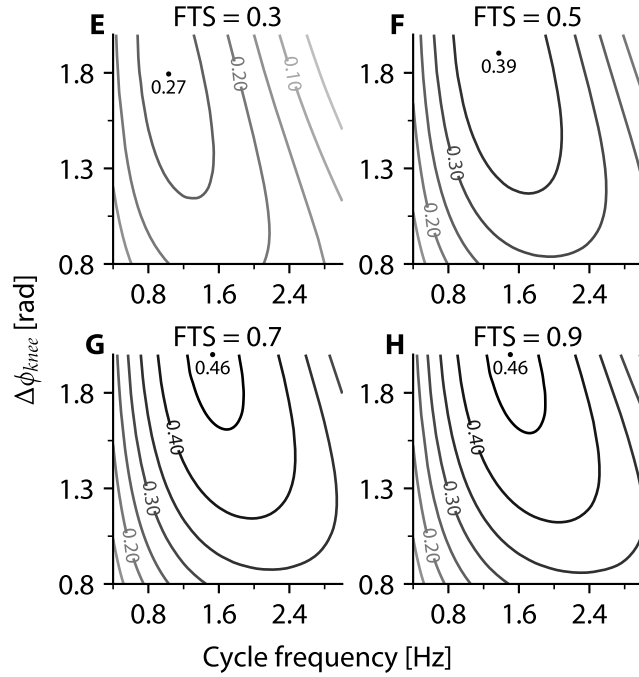
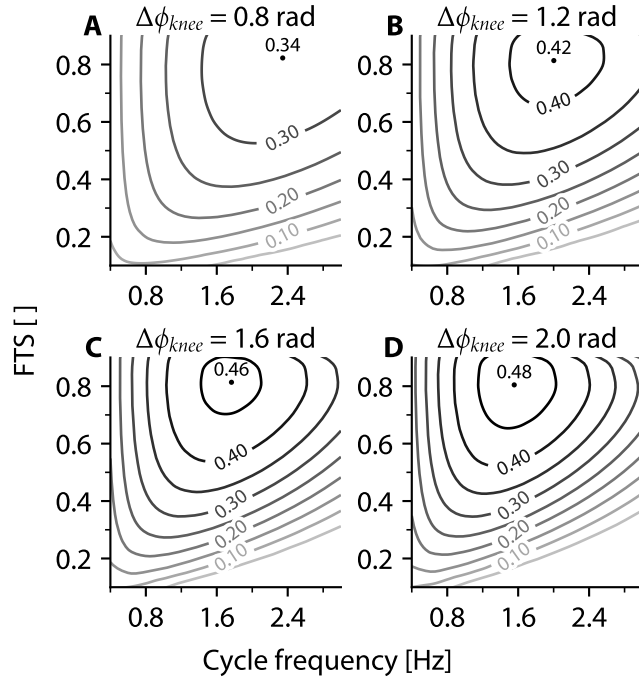


Figure 4.10: Predicted maximally attainable AMPO as a function of SSC parameters: cycle frequency, fraction of cycle time spent shortening (FTS), and knee joint excursion ($\Delta\phi_{\text{knee}}$). For each subplot, one SSC parameter was fixed, while the influence of the other two parameters on maximally attainable AMPO (normalised to L_{CE}^{opt} and F_{CE}^{max}) is depicted as contour lines. Dots indicate peak AMPO, corresponding to the optimal combination of the two varying SSC parameters; values are shown next to each dot. A–D) Predicted maximally attainable AMPO as a function of cycle frequency and FTS for four distinct knee joint excursions. E–H) Predicted maximally attainable AMPO as a function of cycle frequency and knee joint excursion for four distinct FTS values.

4.4 Discussion

In this study, we examined the influence of imposed knee joint movement on the maximally attainable AMPO of human *m. quadriceps femoris* by combining experiments with simulations and optimisations using a Hill-type MTC model. Our findings revealed a strong correlation between measured AMPO and predicted maximally attainable AMPO, for a wide variety of imposed knee joint movements. Model predictions revealed that the maximally attainable AMPO peaks at a cycle frequency of about 1.6 Hz, an FTS of about 0.80, and a knee joint excursion of about 2.0 rad. While the optimal knee joint excursion was predicted to decrease with increasing cycle frequency and vice versa, the optimal FTS was predicted to be largely unaffected by changes in cycle frequency and knee joint excursion.

4.4.1 Evaluation of model predictions against experimental data

Measured AMPO and predicted maximally attainable AMPO were strongly correlated, with measured AMPO being $87 \pm 10\%$ of the predicted values on average. The lower measured AMPO primarily resulted from M_{mus} values that were lower than values predicted for knee extension. It should be noted that, in theory, predicted maximally attainable AMPO should always exceed measured AMPO. For the model predictions, *m. quadriceps femoris* activation was set to be optimal in both timing and magnitude, and no hamstring co-contraction occurs. By contrast, participant muscle activation is inherently suboptimal.

Regarding *m. quadriceps femoris* activation, EMG of *m. quadriceps femoris* rose just at the transition from flexion to extension, causing M_{mus} to increase at this time instant. Thus, we consider the timing of *m. quadriceps femoris* of participants

to be adequate. However, EMG of *m. quadriceps femoris* did not exceed 100% of EMG^{MVC} for all conditions amongst five of the six participants. This differs from previous studies (e.g. Bobbert and Harlaar, 1993; Kellis and Baltzopoulos, 1998; Westing et al., 1991), which reported *m. quadriceps femoris* EMG values slightly exceeding 100% in knee dynamometer experiments. Hence, submaximal *m. quadriceps femoris* activation may partly explain the lower M_{mus} relative to predicted values.

Regarding the co-contraction of the hamstring muscle group, EMG of the hamstring muscle group during knee extension was $17.3 \pm 12.4\%$ of EMG^{MVC} , consistent with previous studies (e.g. Kellis and Baltzopoulos, 1998; Snow et al., 1993). When interpreting surface EMG as a proxy for muscle activation, potential crosstalk should be considered. In the context of this study, it is possible that both quadriceps and hamstrings activation are slightly overestimated due to crosstalk. However, the effect of EMG crosstalk on the net knee joint moment is probably smaller than suggested by EMG values alone, since the maximal isometric force of the hamstring muscle group is approximately 2.5 times lower than that of *m. quadriceps femoris* (e.g. Prietto and Caiozzo, 1989). Nevertheless, it is clear that co-contraction of hamstrings during knee extension may also partly explain the slight deficit in M_{mus} relative to predicted values.

Measured and predicted M_{mus} were normalised to the maximal isometric force (F_{CE}^{max} ; see Figure 4.8). F_{CE}^{max} was estimated individually for each participant from their measured isometric knee joint angle-moment relationship. In most cases, the actual participant's maximum measured isometric knee joint moment was higher than the maximal value of the model (see Figure 4.4), indicating that F_{CE}^{max} was not overestimated. Nevertheless, it should be noted that any inaccuracy in F_{CE}^{max} directly affects normalised M_{mus} . Aside from a simple normalisation of AMPO with F_{CE}^{max} , predictions were not individually tailored. Consequently, properties of participants' *m. quadriceps femoris* could be different from the properties used in the Hill-type MTC model. Measured force rise was generally slower than predicted (see Figure 4.8), suggesting that the model's activation dynamics may have been slightly too fast. Moreover, measured M_{mus} was consistently lower across various

knee joint angles, even where maximal activation was achieved in both experiment and model. The force–length relationship likely played only a minor role, because the isometric knee joint moment–angle relationship was reasonably similar for participants and the model. Instead, the force–velocity relationship appears to be the main contributor. In our model, the decline in knee extension moment with increasing angular velocity was at the lower end of experimentally observed values (for review, see Chen and Franklin, 2025). Overall, the slight deficit in M_{mus} relative to predicted values may partly be explained by differences between the generic model properties and participants' *m. quadriceps femoris* properties. Obviously, all other muscle properties implemented in the model may not exactly match those of (or between) participants' *m. quadriceps femoris*. For example, the moment arm (assumed to be constant in our model) directly influences M_{mus} , yet it was also used to estimate F_{CE}^{max} ; thus, any over- or underestimation would be compensated during normalisation. Additionally, the moment arm determines MTC velocity during knee extension, thereby affecting M_{mus} mainly through its influence on CE shortening velocity. Lastly, SEE compliance influences CE dynamics; the additional dynamics resulting from this CE-SEE interaction is captured in our model. Under the present experimental conditions, discrepancies between CE and SEE length and velocity are predicted to be small, as MTC excursions (~6.7 cm) were substantially larger than SEE excursions (~0.6 cm). This suggests that CE–SEE interaction, and therefore any mismatch between modelled and participant-specific SEE compliance, had only a minor influence on M_{mus} . We expect that, with individually tailored models, differences between experimental and predicted M_{mus} would have been even smaller than differences reported in this study.

We emphasise that our aim was not to precisely predict the magnitude of M_{mus} over time, but rather to examine the influence of SSC parameters on the maximally attainable AMPO. Thus, it is promising that we predict the magnitude of the maximally attainable AMPO remarkably well using a relatively simple Hill-type MTC model with generic parameter values obtained from literature, despite the aforementioned methodological limitations. Moreover, when assessing the influence of SSC parameters on the maximally attainable AMPO, the relative change in AMPO

is more informative than its (average) magnitude. In this context, it is particularly encouraging that AMPO normalised to z-scores revealed a very strong correlation between predictions and measurements ($r^2 = 0.95$). This finding aligns with previous studies demonstrating that Hill-type MTC models reliably predict mechanical behaviour, especially during concentric contractions (e.g. Ingen Schenau et al., 1988; Lemaire et al., 2016; Sandercock and Heckman, 1997; Wakeling and Johnston, 1999; Zandwijk et al., 1996). Concentric contractions primarily determine maximally attainable AMPO. All in all, we attribute the small discrepancy between measured and predicted maximally attainable AMPO to (a combination of) several factors, such as suboptimal *m. quadriceps femoris* activation, hamstring co-contraction, and generic model assumptions. Nevertheless, predictions aligned remarkably well with measurements, demonstrating the validity of the predicted influence of SSC parameters on the maximally attainable AMPO of human *m. quadriceps femoris*.

4.4.2 Results agree well with available literature

In the first part of the current study, L_{SEE}^o , L_{CE}^{opt} and F_{CE}^{max} were estimated for each participant individually based on isometric measurements at nine different knee joint angles. The measured net knee joint moment-angle relationship of participants was in line with literature (e.g. Anderson et al., 2007; Eijden et al., 2008; Murray et al., 1980). The estimation of the muscle model parameters was deemed successful as the root mean square error between the measured isometric knee joint moment and the musculoskeletal predicted isometric knee joint moment was low ($5.3 \pm 1.6\%$ of F_{CE}^{max} on average). Additionally, values of L_{SEE}^o and L_{CE}^{opt} corresponded well with values obtained from cadavers (e.g. Friederich and Brand, 1990; Ward et al., 2009; Wickiewicz et al., 1983) and values used in other modelling studies (e.g. Anderson and Pandy, 1999; Soest and Bobbert, 1993). Overall, we conclude that the values of L_{SEE}^o , L_{CE}^{opt} and F_{CE}^{max} were successfully estimated.

In the second part of the current study, we examined the maximally attainable AMPO for a large set of imposed knee joint movements. As our study is the first to investigate this for human muscle, the only available data for comparison are those of *in vitro* or *in situ* studies on animal muscle. Whereas we influenced muscle

length indirectly by imposing knee joint excursions, those studies directly imposed muscle length excursions. Those studies on animal muscle show that an increase in cycle frequency leads to a decrease in the AMPO-optimal length excursion, muscle stimulation duration and mechanical work per cycle (e.g. Askew and Marsh, 1997; James et al., 1995; Josephson, 1985; Reuvers et al., 2025). Our findings for human *m. quadriceps femoris* align with these observations. Additionally, in accordance with Askew and Marsh (1997) and Reuvers et al. (2025), we found that the optimal (length) excursion increases with increasing FTS and decreasing cycle frequency.

Regarding the values of cycle frequency, FTS and muscle fibre length excursion per se, it should be noted that these values substantially differ due to differences in muscle properties (e.g. maximal shortening velocity, excitation dynamics etc.). Accordingly, the optimal cycle frequency of about 1.6 Hz that was observed for human *m. quadriceps femoris* contrasts with about 4 Hz for mouse *m. soleus* (Askew and Marsh, 1997), about 9 Hz for mouse *m. extensor digitorum longus* (Askew and Marsh, 1997), and about 3.5 Hz for rat *m. gastrocnemius medialis* (Reuvers et al., 2025). Similarly, optimal muscle fibre length excursion, expressed as a percentage of the optimum fibre length, varies across muscles: 22% for mouse *m. soleus* (Askew and Marsh, 1997), 16% for mouse *m. extensor digitorum longus* (Askew and Marsh, 1997), and 58% for rat *m. gastrocnemius medialis* (Reuvers et al., 2025). All of these values are considerably lower than the 85% reported here for human *m. quadriceps femoris*. Interestingly, the optimal FTS of 0.80 for human *m. quadriceps femoris* is comparable to the range of 0.80-0.90 observed in animal muscle (Askew and Marsh, 1997; Reuvers et al., 2025). The optimal FTS is predominantly a trade-off between (the negative effect of) the force-velocity relationship on the one hand and (the negative effect of) the excitation dynamics on the other hand (for details, see Reuvers et al., 2025). Both factors depend on muscle fibre type; their negative effect on AMPO increases with slower muscle fibre types. Consequently, the optimal FTS is expected to be relatively robust for variations in muscle fibre type. Therefore, it is promising that the optimal FTS in our study aligns with those observed in animal muscle. Overall, our findings are consistent with the available literature, which strengthens our confidence in the validity of the results.

4.4.3 Experimental constraints and physiological relevance

In extensive pilot experiments, we explored a range of knee joint movements with different angular velocities and accelerations. Based on these pilot experiments and safety considerations, we limited knee joint acceleration to 50 rad/s^2 , thereby also limiting the (maximum achievable) knee joint angular velocity. As a result, only a subset of combinations of SSC parameters — cycle frequency, FTS, and knee joint excursion — could be tested experimentally. Importantly, this does not imply that higher velocities or accelerations are physiologically infeasible for human knee joint movements. In fact, substantially higher values are reached during various motor activities (for review, see Grimmer et al., 2020). For example, during sprint cycling, knee joint angular velocities reach $5\text{--}10 \text{ rad/s}$, and angular accelerations exceed 100 rad/s^2 . During sprint running these values are even higher, with knee joint angular velocities reaching about 20 rad/s and angular accelerations up to 200 rad/s^2 . Therefore, the joint movements that were not feasible in our experiment are nonetheless physiologically meaningful.

4.4.4 Physiological implications for human movement

AMPO in our study was normalised to the product of F_{CE}^{max} and L_{CE}^{opt} . This allowed comparison across participants, but also allows us to extend our findings to muscles of similar fibre types. For typical values of $L_{CE}^{opt}=9.3 \text{ cm}$ and $F_{CE}^{max}=5250 \text{ N}$ for *mm. vastii*, and $L_{CE}^{opt}=8.1 \text{ cm}$ and $F_{CE}^{max}=1750 \text{ N}$ for *m. rectus femoris* (Soest and Casius, 2000), peak AMPO for *m. quadriceps femoris* is predicted to be approximately 300 W per leg at the optimal combination of SSC parameters. This value may be used as a benchmark in assessing maximum AMPO of human *m. quadriceps femoris* AMPO in any short-duration task that involves periodic motion.

In our study, we measured AMPO of the human *m. quadriceps femoris* during fully imposed knee joint movements. This setup allowed us to directly control the kinematics of the stretch-shortening cycle (SSC) and to ensure that knee joint movements were identical across participants. We show that SSC parameters substantially influence maximally attainable AMPO (see Figure 4.10). For example, increasing FTS from 0.5 to its optimal value of about 0.8 is predicted to improve

AMPO by about 25% at the optimal cycle frequency, with smaller effects at lower frequencies and larger effects at higher frequencies. The resulting insights into the influence of SSC parameters on maximally attainable AMPO can be used to analyse any SSC of any muscle or muscle group, irrespective of whether this SSC arises from fully imposed movements or from unconstrained/self-generated movements. For any SSC, we can now readily estimate the maximal AMPO at the observed kinematics for any muscle (group), and relate that to the maximally attainable AMPO of that muscle (group).

In most real-world situations, movements are not fully constrained and typically involve several mechanical and kinematic degrees of freedom. In such cases, movements arise from muscle forces and mechanical interactions with the environment, which in turn determine the SSCs of the muscles involved. This means that, for example, during an all-out sprint cycling task, the coordination problem is to find muscle activations such that the total AMPO of all muscles combined is maximised over a full cycle. This inevitably leads to a trade-off between the contributions of individual muscles to the total AMPO. Our results provide not only insight into the mechanics of such a trade-off, but may also help in designing equipment that interacts with the musculoskeletal system to improve AMPO. For example, our results indicate that muscles with a larger volume should spend more time shortening than lengthening compared to muscles with a smaller volume in order to maximise the total AMPO. In many periodic sports, however, the shortening duration of *m. quadriceps femoris* does not exceed the lengthening duration. For instance, FTS ranges from 0.3–0.5 in rowing (Bechard et al., 2009; Ettema et al., 2022), is approximately 0.3 in cross-country skiing (Nilsson et al., 2004; Pellegrini et al., 2014), and is around 0.5 in cycling (Bobbert et al., 2020). Interestingly, in a clever experiment, Martin et al. (2002) showed that during single-leg cycling, the maximal AMPO can be improved by about 4% when the downstroke duration is approximately 10% longer than the upstroke duration. This longer downstroke duration allows the large leg muscles to spend more time shortening than lengthening, leading to an increase in their AMPO that more than offsets the decrease in AMPO of other (smaller) muscles that unavoidably spend less time shortening

than lengthening. This example highlight that redesign of equipment that allows muscles to operate closer to their optimal SSC parameters, has the potential to result in improved performance.

4.4.5 Conclusion

In this study, we examined how SSC parameters — cycle frequency, FTS, and knee joint excursion — influence the maximally attainable AMPO of human *m. quadriceps femoris* during short-duration, all-out (sprint) conditions. Because only a subset of SSC parameters can be tested experimentally, we combined experiments with musculoskeletal modelling. Differences in maximally attainable AMPO were adequately predicted by our Hill-type MTC model, justifying its use to examine the effect of SSC parameters on maximally attainable AMPO in detail. Model predictions showed a substantial influence of SSC parameters on the maximally attainable AMPO. Specifically, predictions revealed a strong interaction between cycle frequency and knee joint excursion: increasing one necessitates a decrease in the other to maximise AMPO. Even more interestingly, in order to maximise AMPO, *m. quadriceps femoris* should spend about 80% of the cycle duration while shortening, independent of cycle frequency and/or knee joint excursion.

4.A Appendix

4.A.1 Selection of experimental conditions

With our musculoskeletal model we predicted the maximally attainable AMPO for knee joint movements with different constant knee angular velocities during flexion and extension. Thus, the knee joint angular velocity changed instantaneously at the switch from flexion to extension (and vice versa), and the knee joint angular acceleration at that instant was infinitely high/low. Clearly, infinite knee joint accelerations cannot be achieved in an experimental setup. Instead, we chose to impose a constant knee angular acceleration until a predefined knee joint angular velocity was achieved — determined by the combination of cycle frequency, FTS, knee joint angle excursion and constant knee angular acceleration — in order to achieve the highest cycle frequency and the greatest difference between maximal

and minimal FTS. In extensive pilot tests, we determined that 50 rad/s^2 was the highest knee angular acceleration that participants were comfortable with and was used as an upper limit for the experiment. Accordingly, the experimental conditions were selected from the knee joint movement that were experimentally feasible (see Figure 4.3).

Due to the limit on maximal knee joint acceleration, the imposed knee joint movement in the experiment were slightly different from those with a constant knee angular velocity (i.e. with no limit on knee angular acceleration; see Figure 4.A.1). As a result, the maximally attainable AMPO differed slightly between those two for the same set of SSC parameters. Therefore, we derived predictions of the maximally attainable AMPO for knee joint movement with a limit of 50 rad/s^2 on the knee angular acceleration (see Figure 4.3).

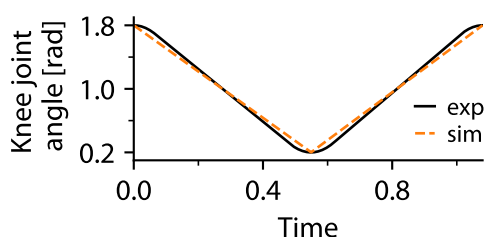


Figure 4.A.1: Comparison between knee joint movements during the experiment and simulations. In the experiment, knee joint movements were limited to an angular acceleration of 50 rad/s^2 . In contrast, in the simulations, knee joint movements were used with different constant angular velocities for flexion and extension in order to simulate a broad range knee joint movements.

Among the experimentally feasible knee joint movements, we observed that the influence of cycle frequency and FTS was greatest at highest knee joint excursion. Therefore, we chose to use a knee joint excursion of 1.6 rad , which was the maximum range of motion of the knee dynamometer. Based on the resulting mapping from cycle frequency and FTS at a knee joint excursion of 1.6 rad (see contour lines in Figure 4.3), we selected two sets of five experimental conditions. One condition was shared between both sets, resulting in nine unique experimental conditions in total. The first set of five conditions was chosen such that the maximally attainable AMPO was predicted to be identical, despite substantial variations in cycle frequency and FTS (0.35A-E; Figure 4.3). The second set of conditions was chosen such that the

maximally attainable AMPO was predicted to differ substantially due to the imposed variations in cycle frequency and FTS (0.20, 0.25, 0.30, 0.35A-E and 0.40; Figure 4.3).

4.A.2 Musculoskeletal model

In both our predictions, as well as in the experiments, the knee joint angle over time was fully imposed. Based on the imposed knee joint angle, we calculated the muscle–tendon complex (MTC) length over time based on Grieve et al. (1978):

$$L_{MTC} = A_0 + A_1 \cdot \phi_{joint} \quad (4.2)$$

A_0 denotes the muscle-tendon-complex length at a knee joint angle of 0 rad (full extension), with knee joint angle increasing with flexion. ϕ_{joint} denotes the knee joint angle. A_1 defines the change in MTC length based on knee joint angle. MTC mechanical work can be calculated both by $\int F_{MTC} dL_{MTC}$ and $\int M_{MTC} d\phi_{joint} = \int F_{MTC} \cdot r_{MTC} d\phi_{joint}$. An expression for the moment arm of the muscle-tendon-complex around the knee joint (r_{MTC}) can then be derived by substituting these equations into Equation 4.2:

$$r_{MTC} = \frac{dL_{MTC}}{d\phi_{joint}} = A_1 \quad (4.3)$$

Table 4.A.1: Key muscle parameter values.

Description	Symbol	Unit	Value	Reference
<i>Musculoskeletal geometry</i>				
MTC length at $\phi_{knee} = 0$	A_0	cm	21.3	Ref. 1.
Linear term of MTC- ϕ_{knee} relation	A_1	cm/rad	4.20	Ref. 1
Hill constant	a^{rel}	-	0.41	Ref. 1
<i>Contraction dynamics</i>				
Hill constant	b^{rel}	1/s	5.2	Ref. 1
Maximum isometric CE force	F_{CE}^{max}	N	5250	Ref. 1
PEE stiffness scaling parameter	k_{PEE}	N/m ²	$1.19 \cdot 10^7$	Ref. 2
SEE stiffness scaling parameter	k_{SEE}	N/m ²	$1.28 \cdot 10^8$	Ref. 1
CE optimum length	L_{CE}^{opt}	cm	9.3	Ref. 1
PEE slack length	L_{PEE}^0	cm	13.0	Ref. 2
SEE slack length	L_{SEE}^0	cm	16.0	Ref. 1
<i>Excitation dynamics</i>				
Sets CE force-length relation width	w	-	0.56	Ref. 1
Sets steepness γ - q relation	a_{act}	-	-4.59	Ref. 3
Sets pCa^{2+} level at which $q = 0.5$	$b_{act,1}$	-	5.17	Ref. 3
Sets pCa^{2+} level at which $q = 0.5$	$b_{act,2}$	-	1.08	Ref. 3
Sets pCa^{2+} level at which $q = 0.5$	$b_{act,3}$	-	-0.19	Ref. 3
Relates γ to pCa^{2+}	kCa	mol/L	$8.00 \cdot 10^{-6}$	Ref. 4
Activation time constant	τ_{act}	ms	88.9	Ref. 3
De-activation time constant	τ_{deact}	ms	88.9	Ref. 3

Parameter values are depicted only for those key to predict the maximally attainable AMPO.

Ref. 1: Soest and Casius (2000), Ref. 2: Soest and Bobbert (1993), Ref. 3: Hatze (1981) and Ref. 4:

Kistemaker et al. (2005)

Data availability

All data, code, and materials used in this study are openly available:

- **GitHub repository:** All raw data, processed data, and analysis code are hosted on GitHub at <https://github.com/edwinreuvers/hq-ampo>.
- **Reproducible analysis website:** Full analysis pipeline — including data, analysis code, and figure/table generation — is available at <https://edwinreuvers.github.io/publications/hq-ampo>.

Acknowledgments

The authors thank all participants for their sustained and dedicated effort during the experiment.

Grants

This work was funded by The Dutch Research Council (NWO) [21728 to D.A.K.].

Disclosures

The authors declare no competing or financial interests.

Author contributions

E.D.H.M.R., A.J.K.v.S., and D.A.K. conceived and designed research; E.D.H.M.R. performed experiments; E.D.H.M.R. analyzed data; E.D.H.M.R., A.J.K.v.S., and D.A.K. interpreted results of experiments; E.D.H.M.R. prepared figures; E.D.H.M.R. drafted manuscript; E.D.H.M.R., A.J.K.v.S., and D.A.K. edited and revised manuscript; E.D.H.M.R., A.J.K.v.S., and D.A.K. approved final version of manuscript.

**Limiting radial pedal forces greatly
reduces maximal power output and
efficiency in sprint cycling; an optimal
control study**

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Journal of Applied Physiology, 134(4), 980-991

<https://doi.org/10.1152/jappphysiol.00733.2021>

Abstract

A cyclist's performance depends critically on the generated average mechanical power output (AMPO). The instantaneous mechanical power output equals the product of crank angular velocity, crank length, and the tangential pedal force. Radial pedal forces do not contribute to mechanical power. It has been suggested that radial pedal forces arise from suboptimal pedaling technique and that limiting these would increase AMPO and efficiency. Here, we presented an optimal control musculoskeletal model of a cyclist (consisting of five segments driven by nine Hill-type muscle-tendon units) to predict maximal AMPO during sprint cycling at different levels of allowed radial pedal forces. Our findings showed that limiting radial pedal forces has a detrimental effect on maximal AMPO; it dropped from 1,115 W without a limit on radial forces to 528 W when no radial forces were allowed (both at 110 rpm). We explained that avoiding radial pedal forces causes ineffective use of muscles: muscles deliver less positive power and have a higher muscle power dissipation ratio (average mechanical power dissipated per unit of average positive power delivered). We concluded that radial pedal forces are an unavoidable by-product when optimizing for maximal AMPO and that limiting these leads to a performance decrease.

5.1 Introduction

The question has often been raised in the literature that pedaling technique in cycling leads to maximal performance (e.g., see Bini et al., 2013; Blake et al., 2012; Korff et al., 2007; Leirdal and Ettema, 2011; Mornieux et al., 2006; Mornieux et al., 2008; Theurel et al., 2012). Performance of a cyclist depends critically on the average mechanical power output (AMPO) generated by the cyclist and the average mechanical power lost to the environment (Ingen Schenau and Cavanagh, 1990). To start with the obvious, mechanical power is produced by the force on the pedals. This pedal force does not influence the (typical) circular pedal path that is a consequence of the design of the bike (i.e., a stiff crank that revolves around the bottom bracket). Therefore, the cyclist has the freedom to choose the direction of the pedal force without influencing the pedal path. However, the direction of the pedal force does influence the crank power. The instantaneous mechanical power delivered to the crank by the cyclist equals the product of crank angular velocity and the moment produced by the cyclist on the crank. This moment, in turn, is the product of crank length and the so-called “effective force”; the component of the force from the cyclist’s pedal (F_p) on the crank that is tangential to the crank motion (F_{tan} , see Figure 5.1). The radial component of the pedal force (F_{rad}) is known as the “ineffective force” (e.g., see Bini et al., 2013; Blake et al., 2012; Leirdal and Ettema, 2011) or “wasted force” (Faria, 1992), because it does not contribute to power output.

The fact that the radial component does not contribute to the mechanical power output has led Lafortune and Cavanagh (1983) to introduce the index of effectiveness (IE): the magnitude of F_{tan} and the magnitude of F_p are each integrated over a complete crank cycle, and IE is calculated as the ratio of the two integrals, also known as the “motive efficiency” (Böhm et al., 2008; Davis and Hull, 1981) or “force effectiveness” (e.g., see Leirdal and Ettema, 2011). In all-out sprint cycling, the value of IE has a maximum of ~ 0.8 at low pedaling rates and drops to values below 0.3 at high pedaling rates (Dorel et al., 2010). In a 1-h endurance time trial on an ergometer, IE values range from ~ 0.6 to ~ 0.8 among top cyclists (Coyle et al., 1991). From the finding that IE over a full cycle is less than 1, it follows that the pedal force is not purely tangential over the full crank cycle. In fact, it has been shown that

substantial radial pedal forces occur, especially during the down stroke and around bottom-dead-center (Beelen and Sargeant, 1991; Dorel et al., 2010). Various authors have taken the value IE to be an indicator of pedaling technique (Dorel et al., 2009a; Dorel et al., 2009b; Jobson et al., 2012; Korff et al., 2007; Leirdal and Ettema, 2011; Mornieux et al., 2008; Rossato, 2008; Zameziati et al., 2006) and thus implicitly assumed that the radial pedal forces during cycling arise from suboptimal pedaling technique. Based on this assumption, it has been proposed that improving IE is beneficial for mechanical power output (Bini et al., 2013) and gross efficiency in endurance cycling (Patterson and Moreno (1990)). In fact, commercially available power meters (e.g., Pioneer Pedaling Monitor System) have the option to provide riders with IE and supporting material states that cyclists should train such to increase IE to arrive at a better cycling technique/higher cycling efficiency.

Why would a cyclist produce metabolically costly muscle forces to generate radial pedal forces that do not contribute to mechanical power output? Is human (elite) cycling indeed suboptimal, or are radial pedal forces an unavoidable by-product when maximizing AMPO? In the first case, avoiding radial pedal forces may increase maximal attainable AMPO. In the second case, it is expected that limiting or avoiding radial pedal forces will decrease maximal attainable AMPO. In general, if a cyclist's goal is to optimize the controls (i.e., muscle activations over time) to deliver maximal AMPO during a full cycle, first principles of optimal control theory tells us that the addition of a constraint on the controls (i.e., to avoid radial pedal forces) can never lead to an increase in AMPO. Furthermore, several studies indeed indicate that limiting radial pedal forces is not a good idea. For example, Kautz and Hull (1993) decomposed pedal forces in cycling and showed that radial pedal forces were partly caused by outwardly directed inertial forces, which increased with cycle frequency, and by gravity. The authors argued that eliminating nonmuscular radial force contributions would require muscles to produce opposite radial pedal forces, which is metabolically costly. In line with this argument, Höchtl et al. (2010) showed that radial pedal forces also occur if muscle activation in a forward simulation model of cycling is optimized for minimal metabolic energy

consumption at a given AMPO, causing them to conclude that a certain amount of radial pedal force is needed for a pedaling technique that is optimal for efficiency.

Although it is true that inertial forces and gravity contribute to radial pedal forces, there is also a (much more substantial) muscular contribution to radial pedal forces (Kautz and Hull, 1993). Onasch and Herzog (2024) studied isometric situations on a bicycle and as such cancelled these inertial forces. They showed for different crank angles that the maximal isometric “effective” tangential pedal force was smaller when subjects were required to direct the pedal force perpendicular to the crank than when they were free to choose the force direction. Their explanation was as follows. Orienting the force on the pedal in a specific direction requires a specific muscle synergy. If the task is to produce a maximal tangential pedal force without constraint on the direction of the pedal force, subjects are free to choose the optimal synergy that yields the largest tangential force; in that case the corresponding direction of the pedal force is an irrelevant outcome (Doorenbosch et al., 1997). Yet, if the direction of the pedal force to be produced is constrained, subjects are forced to choose a synergy different from the optimal one; the corresponding maximal tangential force component of that synergy will be smaller than in the unconstrained condition. This result indicates that the radial pedal forces in cycling may occur as a “by-product” for maximal tangential pedal force. An analysis of cycling as done by Onasch and Herzog (2024) provides valuable insight in the maximal static tangential pedal force, but does not necessarily translate to actual sprint cycling: in sprint cycling the task is to maximize the AMPO and this is not the same as maximizing the tangential pedal force at each pedal position (see Section 5.4). Furthermore, a static analysis by definition cannot answer the question why limiting or avoiding radial pedal forces would lead to a reduced maximal attainable AMPO and by how much it would be reduced.

In this study, we present an optimal control musculoskeletal model of a human cyclist to predict maximal AMPO during sprint cycling at different levels of allowed radial pedal forces. In previous simulation studies on sprint cycling, we used so-called “bang-bang” control (meaning that muscles were either switched fully on or off), which resulted in a straightforward optimization problem that can be readily

tackled with conventional shooting optimization techniques (Bobbert et al., 2016a; Bobbert et al., 2020; Soest and Casius, 2000). In the present study, we will impose constraints to limit radial pedal forces, which greatly increase the complexity of the optimization problem (e.g., “bang-bang” control is no longer optimal). To allow for continuous change in muscle activation, we applied techniques from the optimal control framework [e.g., direct collocation and a sparse nonlinear optimizer; see Refs. Bobbert et al. (2016b); Kistemaker et al. (2013); Kistemaker et al. (2010); Bogert et al. (2011)]. For the interested reader on applying the optimal control framework to musculoskeletal models, we refer to the excellent work of the group of van den Bogert (Ackermann and Bogert, 2010; Bogert et al., 2011; Van den Bogert and Ackermann, 2009) and of De Groote (De Groote et al., 2016; Falisse et al., 2019). Using our optimal control model of a human cyclist, we will show and explain that constraining pedal forces to avoid radial pedal forces—although leading to a higher IE—will lead to a substantially lowered maximal attainable AMPO and increased negative muscle power in sprint cycling.

5.2 Methods

5.2.1 Musculoskeletal model

We used a two-dimensional musculoskeletal model (see Figure 5.1) similar to models used in studying vertical jumping Zandwijk et al. (2000) and sprint cycling (Bobbert et al., 2020; Soest and Casius, 2000). In short, the skeletal model consisted of five rigid segments connected in frictionless hinge joints. These segments represented the crank (0.17 m), foot (from ankle to crank: 0.165 m), lower leg (0.458 m), upper leg (0.485 m), and the “head-arms-trunk” (HAT; 0.82 m). The position of the hip and angle of HAT were fixed in space. The crank angle (ϕ_{cr}) was defined to be zero when the crank was upright (top-dead-center) and crank angular velocity was defined positive when rotating clockwise (see Figure 5.1). The crank’s origin was fixed in space and the crank angular velocity was fixed to have a constant value. Setting the crank velocity to a fixed value results in isokinetic cycling and is mechanically similar to having an infinite “effective inertia” [i.e., the inertia “felt” on the crank that depends on (rotational) inertia of the bike components and gear ratio; see Fregly et

al. (2000)]. Since maximal AMPO is typically delivered at high speeds and thus involving high gear ratios, effective inertia of a sprint cyclist is very high and hence cycling is well approximated by a fixed angular velocity (see Soest and Casius, 2000). The advantage of having a fixed crank angular velocity is that it greatly simplifies the optimization problem, as the two legs are mechanically decoupled (the torques delivered by one leg do not influence the accelerations of the other) and there is no need to model all the resistive and inertial forces acting on the cyclists (see Fregly et al., 2000). As such, only the muscle activations of one leg need to be optimized; the total AMPO is then simply twice that of one leg. Equations of motion for the skeletal model plus crank were derived using a Newton–Euler approach as described in Casius et al. (2004). The position of the hip relative to the crank's origin was optimized for maximal AMPO (see later).

The model was driven by nine Hill-type muscle tendon units representing m. iliopsoas (ILI), m. gluteus maximus (GLU), mm. vastii (m. vastus medialis + m. vastus lateralis + m. vastus intermedius; VAS), m. rectus femoris (REC), the bi-articular hamstring (m. biceps femoris caput longum, m. semimembranosus, m. semitendinosus; HAMb), the mono-articular hamstring (m. biceps femoris caput breve), m. gastrocnemius (GAS), m. soleus (SOL), and m. tibialis anterior (TIB). Every muscle tendon complex consisted of a contractile element (CE) and a series elastic element (SE). In contrast to previous work, we removed the parallel elastic element (PE) as we found that it was not delivering force in the optimal cycling motions but did slow down convergence to the optimal solutions. Activation dynamics, describing the relation between neural input to the muscle (STIM) and active state (q), here defined as the relative amount of Ca^{2+} bound to troponinC (Ebashi and Endo, 1968), was modeled according to Hatze Kistemaker et al. (2005). This model of activation dynamics first describes the dynamical relationship between the relative amount of intracellular Ca^{2+} (γ_{rel}) and STIM, the normalized α -motoneuron firing frequency. The model then nonlinearly relates active state (q) to γ and the length on the CE (l_{CE}). This last step accounts for the l_{CE} dependent $[\text{Ca}^{2+}]$ sensitivity of muscle fibers (see Kistemaker et al., 2005). Contraction dynamics describes how the muscle delivers force depending on q , l_{CE} , and CE contraction velocity (v_{CE})

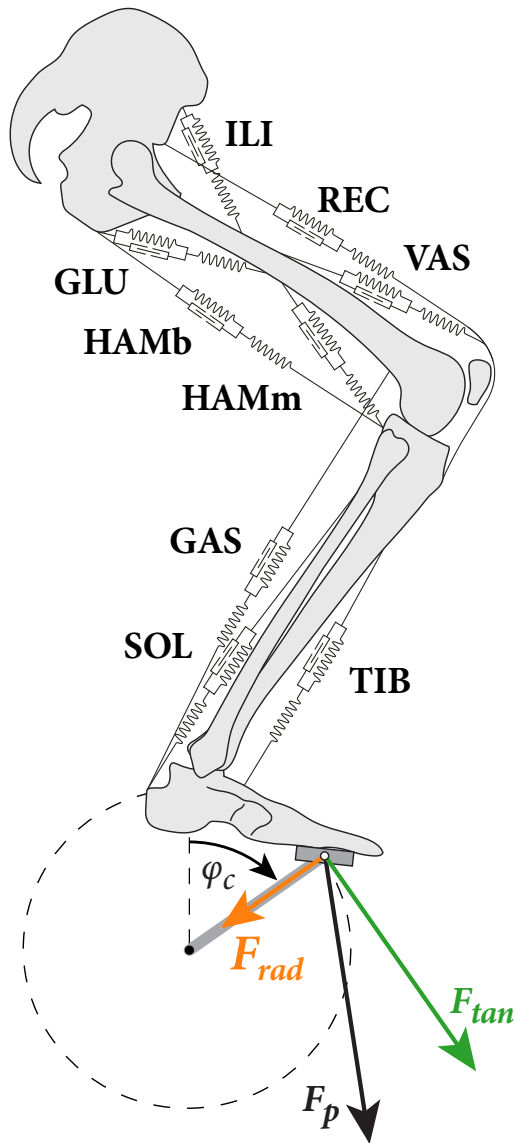


Figure 5.1: Schematic representation of the musculoskeletal model used to simulate cycling. The skeleton and crank were modeled as rigid segments connected in frictionless hinge joints and actuated by nine Hill-type muscle-tendon complexes. Pedal forces (F_p ; black arrow) are defined as the force from the pedal on the crank. Also shown are the radial (F_{rad} ; orange arrow) and tangential (F_{tan} ; green arrow) pedal force component. GAS, m. gastrocnemius; GLU, m. gluteus maximus; HAMb, bi-articular hamstring; HAMm, mono-articular hamstring; ILI, m. iliopsoas; REC, m. rectus femoris; SOL, m. soleus; TIB, m. tibialis anterior; VAS, mm. vastii.

and was modeled in two steps. First, we used a parabola to describe the maximal isometric force-length relationship of CE. This force is multiplied with q , and thus together with the activation dynamics describes the muscle activation-dependent isometric force. Second, CE force was nonlinearly dependent on v_{CE} . The eccentric force was modeled using a hyperbola. The concentric part was described using a classical Hill curve for which the maximal shortening speed was made dependent on q and l_{CE} . For a detailed overview of all modeled steps and all nonmuscle-specific parameters, please refer to our previous papers Kistemaker et al. (2007).

All muscle-specific parameters were the same as used in our previous studies on sprint cycling Soest and Casius (2000), apart from the hamstrings that were split in a mono- and bi-articular part (parameters based on Ward et al., 2009) and are shown in Table 5.1. To make the muscle model suitable for a direct collocation approach (see later), the force-length-velocity relationship and activation dynamics were slightly reformulated to a continuous and differentiable form (see also Kistemaker et al., 2014). The musculoskeletal model as a whole had 26 states (9 l_{CE} , 9 γ_{rel} , 4 segment angles, and 4 segment angular velocities). The trunk was implicitly modeled with a fixed angle and zero angular velocity. The independent inputs to the model were the crank angular velocity (set to a fixed value) and $STIM$.

5.2.2 Constrained Optimization

The goal of the optimization was to find the optimal muscle activations over time [$STIM(t)$] that maximized the average mechanical power output during a full cycle (AMPO), for different levels of allowed radial pedal forces. The optimization criterion for our (constrained) optimization problem was to maximize the total amount of mechanical work done on the crank (W_{cr}) over a full crank cycle with a particular cycle frequency (cf):

$$W_{cr} = \int_0^{1/cf} M_{cr} \cdot \dot{\phi}_{cr} dt = \int_0^{1/cf} F_{tan} \cdot l_{cr} \cdot \dot{\phi}_{cr} dt$$

Here, M_{cr} represents the torque around the bottom bracket on the crank, $\phi_{cr} = 2\pi \cdot cf$ represents the crank angular velocity and is positive when the crank rotates

Table 5.1: Muscle-specific parameters

Muscle	F_{max} [N]	$l_{CE,opt}$ [m]	$l_{SEE,o}$ [m]	α_o [m]	α_{1A} [m]	α_{1K} [m]	α_{1H} [m]	α_{2A} [m]	α_{2K} [m]
TIB	1200	0.087	0.317	0.448	0.037	0	0	0	0
SOL	3000	0.055	0.246	0.230	-0.0627	0	0	-0.0081	0
GAS	1500	0.055	0.382	0.394	-0.0627	-0.0186	0	-0.0081	0.0015
VAS	5250	0.093	0.160	0.231	0	0.042	0	0	0
REC	1750	0.081	0.340	0.418	0	0.042	0.035	0	0
GLU	2750	0.200	0.150	0.255	0	0	-0.062	0	0
HAMb	1925	0.104	0.370	0.410	0	-0.026	-0.077	0	0
HAMm	275	0.110	0.200	0.339	0	-0.026	0	0	0
ILI	4000	0.102	0.115	0.248	0	0	0.05	0	0

Muscle-specific parameters. F_{max} = maximal isometric contractile element (CE) force, $l_{CE,opt}$ = optimum CE length and $l_{SEE,o}$ = series elastic element slack length. Parameters define the muscle-tendon complex length (IMTC) as a function of the ankle (α_A), knee (α_K), and hip (α_H) joint angle. The muscle-tendon complex length for each muscle as a function of the joint angles is given by a sum of second order polynomials with the polynomial constants provided in Table ???. For example, $l_{MTC_HAMb} = \alpha_o + \alpha_{1A} \cdot \phi_A + \alpha_{2A} \cdot \phi_A^2 + \alpha_{1K} \cdot \phi_K + \alpha_{2K} \cdot \phi_K^2 + \alpha_{1H} \cdot \phi_H + \alpha_{2H} \cdot \phi_H^2$. Since α_{1A} , α_{2A} , and α_{2K} are zero for this muscle, and that α_{2H} is zero for all muscles (and not given in Table 1), this simplifies to: $l_{MTC_HAMb} = \alpha_o + \alpha_{1K} \cdot \phi_K + \alpha_{1H} \cdot \phi_H$. The moment arm (arm) of a particular muscle around a particular joint is found by taking the partial derivative of l_{MTC} with respect to that joint. For example, $arm_{HAMb} = \alpha_{1K}$ (note that since $\alpha_{2K} = 0$, the moment arm of the bi-articular hamstring (HAMb) around the knee is constant, and not dependent on the knee joint angle [this in contrast to, for example, the moment arm of m. gastrocnemius (GAS)]. TIB, m. tibialis anterior; SOL, m. soleus; GAS, m. gastrocnemius; VAS, mm. Vastii; REC, m. rectus femoris; GLU, m. gluteus maximus; HAMb, bi-articular hamstrings; HAMm, mono-articular hamstrings; ILI, m. Iliopsoas.

clockwise (see Figure 5.1). Alternatively, W_{cr} can be expressed in terms of F_{tan} and the length of the crank (l_{cr}). AMPO equals $W_{cr} \cdot cf$.

$STIM(t)$ for the nine modeled muscles of one leg were optimized to maximize W_{cr} . We only needed to optimize $STIM(t)$ for one leg as during isokinetic cycling legs are mechanically decoupled and we assumed symmetry between the legs. In the remainder of this article, we will provide values of the left and right leg muscles together, unless specifically stated otherwise. To ensure fully periodic behavior, we added constraints on all 9 $STIM(t)$ and 26 states of the musculoskeletal model $[x(t)]$ such that $[STIM(0) \ x(0)] = [STIM(1/cf) \ x(1/cf)]$. Finally, to investigate the influence of IE on maximal AMPO, we defined seven conditions by constraining the instantaneous radial pedal force at different values:

$$|F_{rad}(t)| \leq F_{rad_max}$$

with $F_{rad}(t)$ the absolute of the radial pedal force and $F_{rad,max}$ set to either 0, 100, 200, 300, 400, 500, or ∞ N. When $F_{rad,max}$ was set to 0 N, no radial pedal forces were allowed; hence IE is one. When, for example, $F_{rad,max}$ was set to 200 N, maximal allowed instantaneous radial pedal force was 200 N, and $F_{rad,max}$ when set to ∞ N there was no limit on F_{rad} .

To identify optimal $STIM(t)$, we used a direct collocation approach (Betts, 2010). To do so, all states and inputs were parameterized and all differential equations of the musculoskeletal model (activation, contraction, and skeletal dynamics) were discretized at every collocation point (Bogert et al., 2011; De Groote et al., 2016; Kistemaker et al., 2013; Kistemaker et al., 2014). These discretized differential equations, together with the constraint equations and optimization criterion described earlier, form a large-scale nonlinear programming problem, which was solved using sparse nonlinear optimal controller (SNOPT; TOMLAB Optimization, Pullman, WA). The derivatives and second derivatives of the constraints and cost function were computed analytically using PROPT (TOMLAB Optimization, Pullman, WA).

To check if we used enough collocation points to adequately discretize the musculoskeletal dynamics, we performed a forward simulation with the results obtained from the optimal control model. It was found that the states obtained

from a forward simulation and those obtained from our optimal control model were nearly identical when using 50 collocation points (on average ~1 collocation point per 10 ms).

To avoid the possibilities of arriving at (getting stuck in) a local minimum, each condition was optimized ten times, with each optimization starting from a random initial guess for all parameterized inputs and states. For each condition, we selected the solution leading to the highest AMPO.

5.2.3 Optimal Cycling Frequency and Saddle Position

Before investigating the influence of limiting radial pedal forces, we first identified the optimal cf and saddle position for unconstrained cycling (with $F_{rad,max} = \infty$). To do so, we added three additional optimization parameters to the optimal control problem described earlier, being cf itself and the x - and y -position of the hip with respect to the position of the bottom bracket. We then found the optimal cf and the hip position simultaneously with $STIM(t)$ that maximized AMPO. For all other optimizations, we fixed cf and hip position to these values.

5.2.4 Index of Effectiveness

In the literature, there seems to be a lack of consensus over the exact definition of IE; the index of effectiveness (cf. Bini et al., 2013; Coyle et al., 1991; Lafortune and Cavanagh, 1983; Mornieux et al., 2006; Mornieux et al., 2008; and Zameziati et al., 2006). Moreover, sometimes IE is defined unclearly or even in a mathematically unsound way. Although different definitions (e.g., using F_{tan} or the absolute of F_{tan}) may lead to (slightly) different values, we found that for this study these differences only marginally affected our solutions. Here, we have chosen to define IE along the lines of Lafortune and Cavanagh (1983): IE is the ratio of the magnitude of F_{tan} and the magnitude of F_p both integrated over crank angle (ϕ_{cr}) for a complete cycle:

$$IE = \frac{\int_0^{2\pi} |F_{tan}| d\phi_{cr}}{\int_0^{2\pi} |F_p| d\phi_{cr}}$$

F_{tan} was defined positive when it pointed in the direction of pedal movement and F_{rad} was defined positive when it pointed (inward) to the crank origin (see also Figure 5.1).

5.2.5 PDR; Average Muscle Mechanical Power Dissipation Ratio

To assess and compare the contribution of all muscles to the average positive and negative mechanical power delivered at different values of AMPO, we define average mechanical muscle power dissipation ratio (PDR); the ratio of average negative mechanical power dissipated ($AMPO^-$) per unit of average positive power delivered ($AMPO^+$) in one full cycle:

$$PDR = \frac{AMPO^-}{AMPO^+} = \frac{cf \cdot \sum_{n=1}^9 \int_0^{1/cf} P_{MTC_n}^- dt}{cf \cdot \sum_{n=1}^9 \int_0^{1/cf} P_{MTC_n}^+ dt} = \frac{cf \cdot \sum_{n=1}^9 \int_0^{1/cf} [F_{MTC_n} \cdot v_{MTC_n}]^- dt}{cf \cdot \sum_{n=1}^9 \int_0^{1/cf} [F_{MTC_n} \cdot v_{MTC_n}]^+ dt}$$

P_{MTC_n} is the instantaneous mechanical power delivered by the n th muscle-tendon complex and equals the product of the muscle-tendon complex force (F_{MTC_n}) and contraction velocity (v_{MTC_n}). $P_{MTC_n}^-$ denotes the negative instantaneous mechanical power of muscle n ; it equaled P_{MTC_n} when it was ≤ 0 and was set to 0 otherwise. Likewise, $P_{MTC_n}^+$ denotes the positive instantaneous mechanical power of muscle n .

5.3 Results

5.3.1 Optimal Saddle Position and Cycle Frequency

We first identified the optimal cf and hip position that yielded the highest AMPO. The identified optimal pedaling rate for unconstrained cycling was very close to 110 rpm. The results in terms of kinematics and AMPO were close to that found in Soest and Casius (2000), and close to the pedaling rate observed experimentally in sprint cycling (Vandewalle et al., 1987). For this article, cf was fixed to 1.83 Hz (equaling to cycle frequency of 110 rpm). The hip position was set to the optimal hip position for unconstrained cycling, which was $[-0.055 \ 0.87]$ m (effective seat

tube angle of 1.63 rad with respect to positive x-axis) relative to the origin of the crank (bottom bracket) and was close to that used by Soest and Casius (2000).

5.3.2 Convergence of Optimal Control Solutions

We were able to successfully identify the optimal $STIM(t)$ for maximal AMPO for all levels of constrained radial forces investigated. Furthermore, we were successful in constraining radial pedal forces at different levels of maximally allowed radial pedal forces. For example, the maximal value of radial force in the condition in which no radial pedal forces were allowed ($F_{rad_max} = 0$) was only $\sim 2 \cdot 10^{-9}$ N (rms = $\sim 5 \cdot 10^{-10}$ N). Visualizations of the cycling model for $F_{rad_max} = \infty$ and $F_{rad_max} = 0$ can be found online (Supplemental Material; see <https://doi.org/10.6084/m9.figshare.21711020>).

5.3.3 Comparison with Literature

The unconstrained ($F_{rad_max} = \infty$) AMPO was found to be 1,115 W, which agrees with values reported in the literature on sprint cycling (see Table 5.2). In addition, the identified maximal AMPO is very close to that found by Soest and Casius (2000). Note that this is not trivial, since they used a different approach in which they used “bang-bang” control (muscles were allowed to switched fully on or off once during each crank cycle) and in which they added the constraints on periodicity (see Section 5.2) as penalty terms in their optimization criterion. Reassuringly, in our optimal control solution for the unconstrained condition, we found that the muscle activation pattern was similar to earlier work of our group: muscles were either fully turned on or turned off (see also Bobbert et al., 2016a).

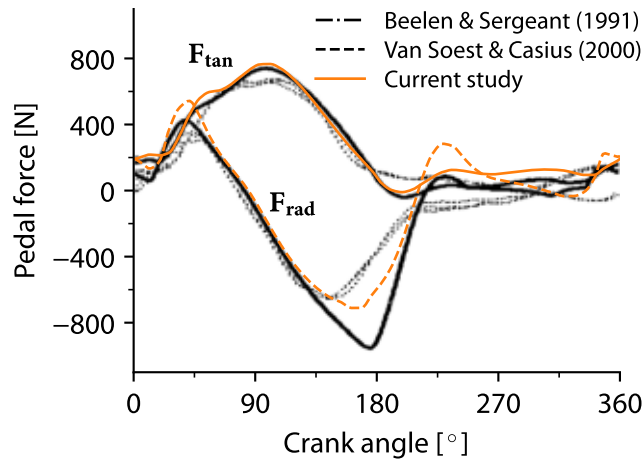


Figure 5.2: Tangential (F_{tan}) and radial (F_{rad}) component of the pedal force as a function of crank angle. Crank angle is equal to that defined in Figure 5.1 (0° is top-dead center). Black dotted lines are the experimentally observed force histories of three full crank cycles of a typical subject cycling at 120 rpm (Beelen and Sargeant, 1991). Solid black lines are the pedal force histories obtained by Soest and Casius (2000) and the orange lines are the pedal force histories of our optimal control model cycling at 110 rpm.

Table 5.2: Reported values for two-leg AMPO in the literature on sprint cycling

Source	AMPO
Dorel et al. (2010)	1250 ± 117 ($n=14$) W
Vandewalle et al. (1987) ^a	1150 ± 127 ($n=7$) W
Elmer et al. (2011) ^b	1126 ± 38 ($n=11$) W
McDaniel et al. (2014) ^b	1098 ± 34 ($n=10$) W
Martin & Brown (2009) ^b	1080 ± 62 ($n=13$) W
Van Soest & Casius (2004)	1076 W
Current study	1115 W

AMPO, average mechanical power output. ^a Vandewalle et al. reported on many different types of athletes; the value displayed here is for track cyclists (the other types of athletes scored similar values).

^b These studies only reported power output for one leg; right-left symmetry has been assumed to calculate the AMPO for two legs.

Figure 5.2 shows the optimal unconstrained ($F_{rad_max} = \infty$) radial and tangential pedal forces as a function of crank angle found in this study, as well as those found by Soest and Casius (2000) and those obtained experimentally by Beelen and Sargeant (1991). In general, the tangential pedal forces profiles look very similar. As with the study of Soest and Casius (2000), the radial pedal forces of the model are substantially higher around bottom-dead-center (crank angle = 180°) than experimental values. These higher radial pedal forces occur when tangential pedal forces are very small and thus when little power is delivered. As argued by Soest and Casius (2000), it may be that subjects avoid high radial pedal forces when mechanical power output is low. Be that as it may, we found that the radial pedal forces at bottom-dead-center were also influenced by the positioning of the hip, as illustrated by the differences in peak radial force between this study (note that we optimized the hip position for maximal AMPO for $F_{rad_max} = \infty$; see Section 5.2) and that of Soest and Casius (2000). In addition, we found that peak radial pedal forces were influenced by the values of the maximal isometric forces of the muscles modeled (data not shown). Since the values of the muscle properties originate from elite jumpers (Soest and Bobbert, 1993; Soest et al., 1993), it is very well possible that the differences in radial pedal forces between our model and those of elite cyclists are partly caused by differences in muscle properties (see also Bobbert et al., 2016a). Nevertheless, we believe that our model is capable of adequately capturing the important features of real cycling (visualizations of the model can be found in Supplemental Data).

5.3.4 Influence of Limiting Radial Pedal Forces on Maximal Attainable AMPO

In Figure 5.3, AMPO is plotted as a function of maximally allowed radial pedal force. As expected (see Section 5.1), the highest value of AMPO was found when the radial force was unconstrained ($F_{rad_max} = \infty$): 1,115 W. Also in line with our expectations is the clear reduction in AMPO with a reduction in the allowed radial force. When no radial pedal forces were allowed, maximal AMPO was reduced by as much as 53%, to a value of 528 W. In Figure 5.3, we also plotted the IE calculated

from our optimal solutions; clearly the highest IE does not coincide with the highest attainable AMPO. This furthermore indicates that the radial pedal forces are, as suggested in Section 5.1, unavoidable “by-products” when maximizing AMPO.

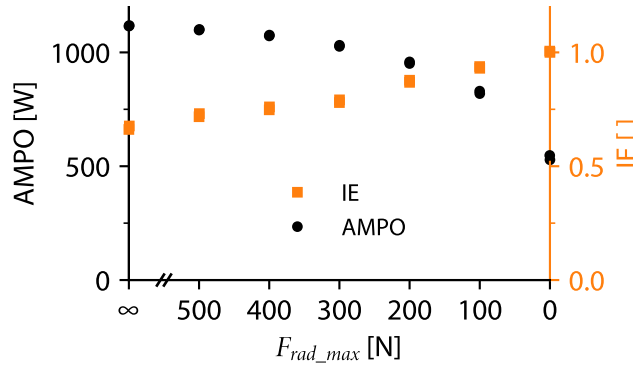


Figure 5.3: Maximal attainable average mechanical power output (AMPO) and accompanying index of effectiveness (IE) as a function of the maximal allowed “ineffective” radial pedal force.

5.3.5 Influence of Limiting Radial Pedal Forces on Maximal Tangential Pedal Forces

To understand why reducing the maximal allowed radial pedal force reduces the maximal attainable AMPO, we first looked at the pedal forces. In Figure 5.4, the optimal pedal force as a function of crank angle is plotted at fixed intervals for all levels of F_{tan_max} investigated. Clearly, the magnitude of the total pedal force (length of the arrows) increased monotonically with increased allowed radial pedal forces. This is even more apparent when plotting time histories of F_{rad} and F_{tan} as a function of crank angle for different levels of F_{rad_max} (Figure 5.5). When no radial pedal forces were allowed, pedal forces were obviously always tangential (i.e., $F_{rad} = 0$), but apparently this constraint comes with a substantial cost: tangential pedal forces are much lower, leading to a reduced AMPO. This leads to the obvious question: why is AMPO reduced when limiting radial pedal forces? Here, we will identify the muscles responsible for loss in power of our model, and in Section 5.4 we will try to unravel the underlying mechanism.

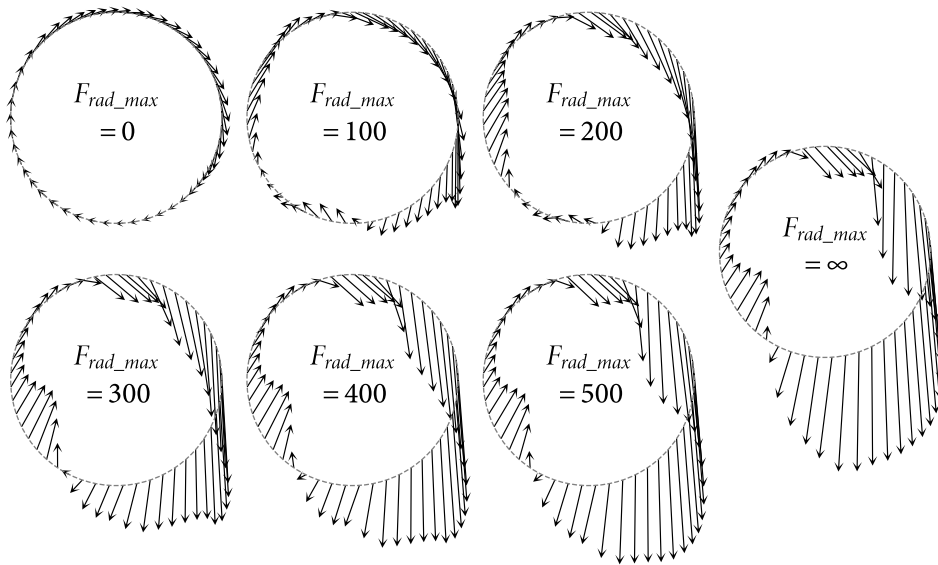


Figure 5.4: Pedal force as a function of crank angle, for all seven conditions. Pedal forces were drawn as vectors at fixed crank angle intervals.

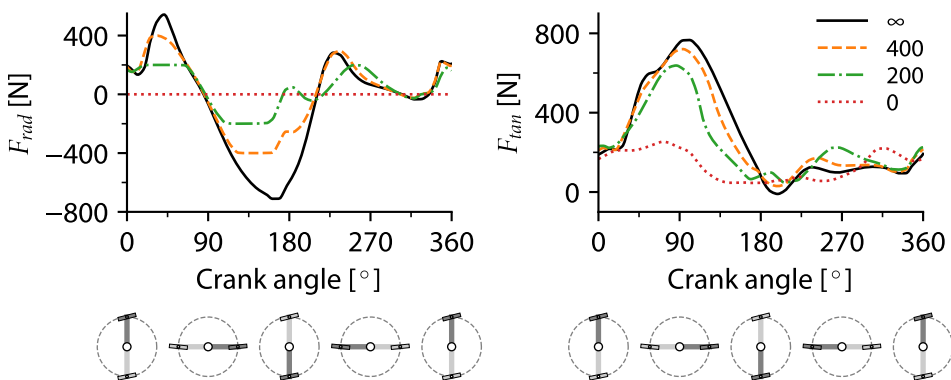


Figure 5.5: Tangential (F_{tan}) and radial (F_{rad}) component of the pedal force as a function of crank angle for four different conditions.

5.3.6 Influence of Limiting Radial Pedal Forces on Muscle Power

In Figure 5.6, we show for each modeled muscle (summed over both legs) the net mechanical work, total positive work (W^+) and the total negative work (W^-) done during one full crank cycle (note that AMPO equals net mechanical work delivered during one crank revolution multiplied by cycle frequency). In addition, we show for each muscle (summed over both legs) the instantaneous power delivered as a function of crank angle during one full crank revolution. This is shown for both the unconstrained ($F_{rad_max} = \infty$) and the fully constrained situation ($F_{rad_max} = 0$). Three clear observations about loss in power production can be made when closely observing muscle power. First, the biggest difference in average power production is caused by a reduction in power of mm. vasti: these muscles deliver ~195 W less per cycle when avoiding radial pedal forces. Second, the muscles spanning the ankle joint are not capable of delivering power when avoiding radial pedal forces. In fact, instead of delivering positive power at a rate of 185 W, the muscles spanning the ankle joint deliver -21 W in the fully constrained condition; note that delivering negative power means that they dissipated power. The third main reason for the loss in AMPO in the maximally constrained situation is a reduction of net power of m. iliopsoas. Even though m. iliopsoas delivers, perhaps counter-intuitively, more positive power when radial pedal forces are limited (by 25 W), it also delivers much more negative power (by -150 W); hence the contribution of m. iliopsoas to AMPO drops by ~125 W. In summary, constraining radial pedal forces causes mm. vastii to deliver much less positive power, causes plantar flexors to stop producing any power, and causes m. iliopsoas to dissipate more power. From the individual muscle contributions to AMPO, we calculated the average muscle mechanical power dissipation ratio (see Section 5.2). For the unconstrained condition, PDR was -0.20 and increased to -0.45 in the fully constrained condition.

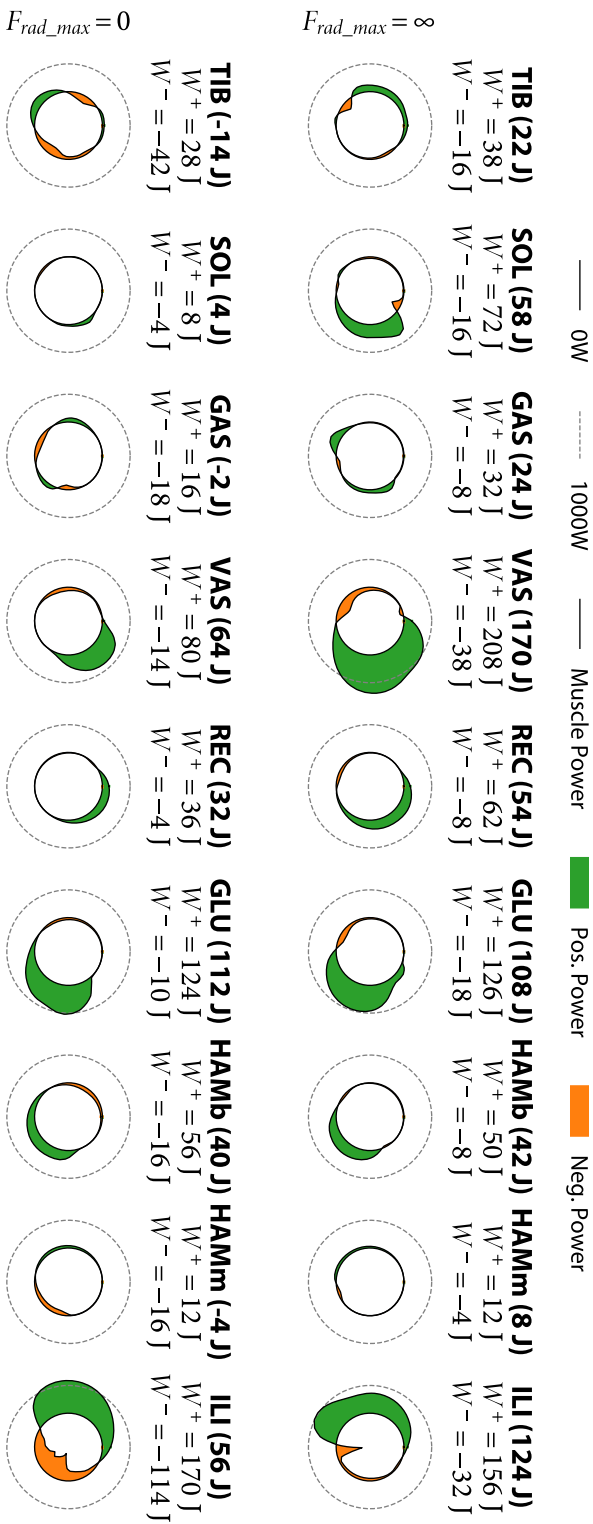


Figure 5-6: Instantaneous mechanical (muscle tendon complex) power delivered by each of the modeled muscles as a function of crank angle. Top row: unconstrained ($F_{rad_max} = \infty$) cycling. Bottom row: fully constrained ($F_{rad_max} = 0$). Outer dashed circle was drawn at 1,000 W and the inner solid line circle at 0 W. Green denotes when muscles deliver positive power and orange when delivering negative power. The total work, total positive work (W^+) and total negative work (W^-) done during one full cycle are given for each muscle separately. Note that the green and orange area scales (nonlinearly) with positive and negative work. Furthermore, note that average mechanical power output (AMPO) equals the sum of all work done by each muscle multiplied by the cycle frequency (≈ 1.83 Hz).

5.4 Discussion

In this study, we presented an optimal control model of a cyclist and showed that radial pedal forces in cycling occur as a “by-product” for maximal tangential pedal force, which means that reducing radial pedal forces reduces the maximal average mechanical power output (AMPO). When completely avoiding radial pedal forces (leading to a “perfect” IE of 1), maximal AMPO during one cycle drops by more than 50% compared with the unconstrained condition (leading to an IE of 0.66). In the next section, we will first attempt to explain the mechanisms responsible for the decrease in AMPO when avoiding radial pedal forces. Then we will argue that sprint cycling efficiency will also decrease when avoiding radial pedal forces.

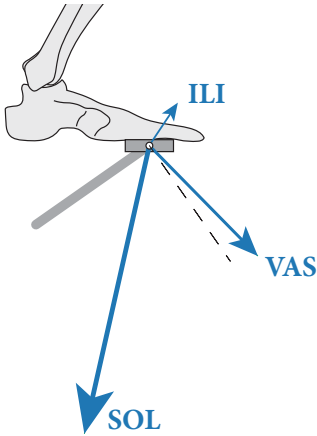
5.4.1 Why Does AMPO Decrease When Avoiding Radial Pedal Forces?

In Figure 5.6, we showed the power delivered and work done by the individual muscles during a full cycle, both for unconstrained cycling and constrained cycling when completely avoiding radial pedal forces. It is clear that in the latter case, muscles not only deliver less positive power, but also deliver more negative power (and thus dissipate energy). To get an intuitive idea of why this is the case, we will look at the at the optimal isometric (static) contributions of three muscles (mm. vastii, m. iliopsoas, and m. soleus) to the pedal force (green arrows; Figure 5.7) for a crank angle of $\sim 55^\circ$ (see appendix for the calculation of the force contributions). As explained in appendix, in general, the contribution of a unit of force of a muscle to the pedal force decreases with increasing distance from joint(s) that the muscle spans to the pedal. Thus, during cycling, 1 N delivered with an ankle plantar flexor leads to a much higher pedal force than 1 N delivered by a hip flexor (in fact, up to two orders of magnitude). Without a constraint on radial pedal forces ($F_{rad_max} = \infty$), finding the optimal combination of isometric muscle forces for maximal F_{tan} is straightforward: just activate maximally all muscles that have a positive F_{tan} component; in this case both m. soleus as well as mm. vastii (Figure 5.7B). This leads to a large F_{tan} (blue arrow) and a substantial F_{rad} (red arrow). When avoiding radial pedal forces, the optimal solution for maximal static F_{tan} is far less straightforward (Figure 5.7C). The mm. vastii are activated maximally and the outward radial component of

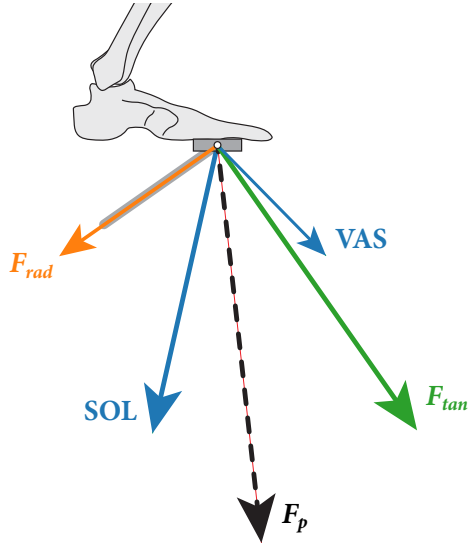
the generated pedal force is compensated by an inward radial component of m. soleus. However, since m. soleus has a much larger inward radial component than the outward F_{rad} component of mm. vastii [this is augmented as the ankle joint (spanned by the m. soleus) is closer to the pedal than the knee joint (spanned by the mm. vastii); see appendix], the activation of m. soleus is very limited. The activation of m. soleus can be increased if m. iliopsoas is activated, such that m. iliopsoas compensates for the inward F_{rad} component of m. soleus. This is beneficial as the positive F_{tan} component of the m. soleus is larger than the negative F_{tan} component of the m. iliopsoas. In the optimal isometric solution ($F_{tan_max} = 0$), m. soleus is submaximally activated, whereas m. iliopsoas—which negatively contributes F_{tan} —needs to be fully activated to avoid radial pedal forces. Consistent with the experimental findings of Onasch and Herzog (2024), our isometric muscle force optimization example shows that avoiding radial pedal forces leads to a decrease in activity of muscles that can contribute positively to F_{tan} . In addition, it shows that avoiding radial pedal forces may lead to an increase in activity of muscles that contribute negatively to F_{tan} .

Although a static/isometric analysis yields insight, it does not tell the full story. Several important factors play a role that complicate understanding the optimal muscle activation for maximal AMPO. First, activation dynamics causes a delay between the change in muscle activation and corresponding change in muscle force. In previous research, we have shown that activation dynamics plays an important role in explaining the optimal pedaling frequency (Soest and Casius, 2000) and the linear relationship between pedal force and crank angular velocity (Bobbert et al., 2016a). Here, we show that activation dynamics augments the reduction in muscle power when avoiding radial pedal forces. For example, this is the explanation of the vast reduction in power (by 195 W) of mm. vastii. From about a crank angle of 135° to a crank angle of 180° , mm. vastii and m. gluteus maximus produce a large outward radial force. This radial force becomes very large mostly due to the mm. vastii (a muscle crossing the knee joint has a greater effect on the pedal force than a muscle crossing the hip joint; see appendix). Only the bi-articular hamstring can compensate for this outward radial force while producing positive power, but is

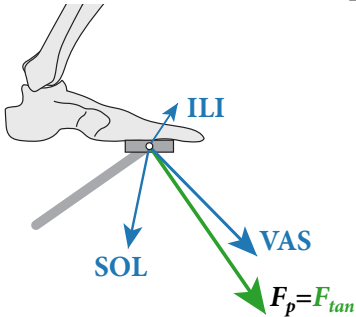
A) Maximal static contribution



B) Optimal static solution $F_{tan_max} = \infty$



C) Optimal static solution $F_{tan_max} = 0$



D) Optimal dynamic solution F_{tan_max}

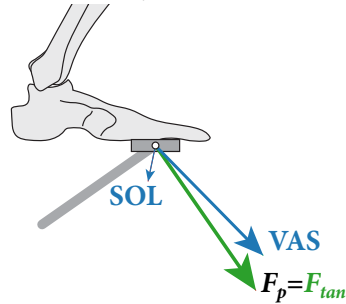


Figure 5.7: Illustration of the maximal static F_{tan} from three muscles (note that the length of the vectors are not drawn to size; see Section 5.A). A: maximal pedal force contribution of three muscles (blue arrows). Note that the contribution of a muscle to the pedal force diminishes with increasing distance from the pedal to the joint(s) that muscle spans (see also Section 5.A). B: optimal muscle force distribution for maximal F_{tan} (green arrow) when F_{rad} is unconstrained ($F_{rad_max} = \infty$). mm. vastii (VAS) and m. soleus (SOL) can be activated maximally. The pedal force (F_p ; black dashed arrow) has a substantial radial component (F_{rad} ; orange arrow) C: optimal muscle force distribution for maximal F_{tan} when F_{rad} is fully constrained ($F_{rad_max} = 0$). VAS is maximally activated. m. iliopsoas (ILI) cannot contribute to a positive F_{tan} but is maximally activated to compensate for submaximally activated SOL. Here, F_p equals F_{tan} since F_{rad} is constrained to be zero. D: optimal “quasi-static” solution; when taking into account capability of delivering positive instantaneous muscle power, m iliopsoas is no longer activated, which further reduces the activation of the m. soleus.

much smaller than the mm. vastii. To avoid radial forces, the mm. vastii is switched off much earlier even though it can contribute greatly to power production. Due to activation dynamics, shutting this muscle group off needs to be done ahead in time. The result is that not only the mm. vastii are generating less instantaneous power, they are also doing so over a much smaller range in crank angles (cf. Figure 5.6 VAS 1st and 2nd row). Since the m. gluteus maximus is crossing the hip joint, its effect on the radial pedal force is much smaller compared with that of the mm. vastii. The bi-articular hamstring can be fully activated to compensate for most of the outward radial forces caused by the m. gluteus maximus. In fact, the bi-articular hamstring is the only muscle that is not affected by the constraint on the pedal force (it actually delivers a little bit more power when $F_{rad} = 0$, but this is due to small differences in cycling kinematics). Around 135° , the inward radial force of the bi-articular hamstring is too small. The m. gluteus maximus needs to be shut off, but, due to activation dynamics this is relatively slow (and thus would cause a great decrease in positive power) and the optimal solution is to both decrease the activity of the m. gluteus maximus and to increase its antagonist the m. iliopsoas, causing a large increase in negative AMPO (from ~ -60 to -210 W cf. Figure 5.6 ILI 1st and 2nd row).

A second complicating factor is that there is no direct relation between instantaneous crank power and instantaneous muscle power. As discussed earlier, in Figure 5.7 we plotted the contribution of the muscle forces to the pedal forces. When looking at the power delivered by the muscles, such a static analysis is not insightful. It is true that over a full cycle average mechanical crank power equals average mechanical muscle power, because the movement is periodic. However, in general, the instantaneous crank power (P_{crank}) is not equal to instantaneous muscle (-tendon complex) power (P_{MTC}). P_{MTC} equals P_{crank} plus the rate of change in kinetic and potential energy [P_{K+P} ; Ingen Schenau and Cavanagh (1990), Chen (2006)]. Figure 5.8 shows these power terms as a function of crank angle for the optimal unconstrained solution. We found that the peak rate of change in kinetic energy can be as large as -550 W (independent of a constraint on radial

forces). Thus, maximizing instantaneous crank power would yield instantaneous joint accelerations that are inconsistent with a cycling motion.

A third complicating factor is that although muscle force contributions to the pedal force are insightful for understanding how to coordinate muscles to avoid radial pedal forces, they are not insightful when it comes to understanding how (instantaneous) muscle power relates to crank power. The reason is that there is no direct relationship between the instantaneous power delivered by a muscle and the direction/magnitude of its contribution to the pedal force (to explain this in detail goes beyond the scope of this article, but see Chen, 2006). For example, this means that a muscle contributing to a large tangential pedal force may not produce any power at the muscle level, but simply “transfers” power from other muscles Zajac et al. (2002). In the end, the power delivered by a muscle force is fully determined by the angular velocity of the joint(s) the muscle spans. As an example, we look again at Figure 5.7C. In this figure, we plotted the optimal isometric muscle forces for maximal tangential force while avoiding radial force ($F_{rad_max} = 0$). We see that m. iliopsoas is substantially activated in order to compensate for the radial pedal force component of the m. soleus. However, during cycling, at this crank angle, the hip joint is extending/retro flexing, which would lead to a large instantaneous negative muscle power of m. iliopsoas. This is obviously not optimal when considering that the task is to maximize average mechanical power output over a full cycle (in the optimal control solution, this muscle is indeed not activated around this crank angle; see Figure 5.6). When allowing only the use of muscles that deliver positive instantaneous muscle power, m. iliopsoas is no longer activated. Since m. iliopsoas was required to compensate for the radial pedal force component of the m. soleus, the consequence is that the m. soleus activation is reduced to almost zero (Figure 5.7D). This is in fact a general “problem” for the muscles spanning the ankle joint during a full cycle. The ankle is close to the pedal and therefore the muscles spanning the ankle joint have a large contribution to the (radial) pedal force. Thus, when avoiding radial pedal forces, ankle spanning muscles can hardly be activated. This is the reason why m. soleus, m. gastrocnemius. and m. tibialis anterior are not

able to contribute to AMPO when no radial pedal forces are allowed, leading to a 170 W reduction in AMPO (see Figure 5.6).

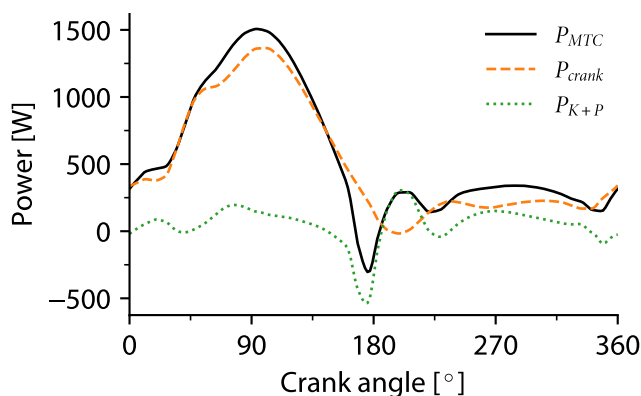


Figure 5.8: Time histories of the instantaneous mechanical power terms of our bike model as a function of crank angle for the optimal unconstraint solution. P_{MTC} = muscle-tendon complex power, P_{crank} = mechanical power delivered on the crank, and P_{K+P} = sum of the rate of change of kinetic and potential energy. Given the definitions of directions of forces, the power equation equals $\sum P = P_{MTC} - P_{crank} = P_{K+P}$ (see Ingen Schenau and Cavanagh (1990))

5.4.2 IE and Efficiency

One may argue (see e.g., Leirdal and Ettema, 2011) that since radial pedal forces are ineffective (not producing power) and are the consequence of active muscles that use metabolic energy, avoiding radial pedal forces may lead to more efficient cycling (i.e., less metabolic power needed for a given value of AMPO). Although this argument may sound reasonable at first glance, it is incorrect. First, the theoretical argument in Section 5.1 also holds for efficiency: if the goal is to maximize efficiency (at a given fixed value of AMPO), then adding a constraint (to avoid radial pedal forces) cannot lead to a better efficiency. Second, we can see from our analyses that avoiding radial pedal forces not only greatly diminished maximal AMPO, it also greatly increased the dissipation of power. From Figure 5.6, we calculated the PDR: the ratio of negative to positive AMPO. This ratio allows us to compare the muscle power dissipation between conditions, as it is independent of the net AMPO delivered. In the unconstrained condition, PDR is about -0.2, which means that for every W of positive mechanical power delivered, muscles dissipate 0.2 W. PDR

increases when limiting radial pedal forces and more than doubles to a value of -0.45 when completely avoiding radial pedal forces. From this increased PDR, it follows that the net amount of negative muscle power for a given amount of positive muscle power increases. This increase will result in a reduction in metabolic efficiency during sprint cycling while avoiding radial forces.

5.4.3 Conclusions

Using an optimal control model of a cyclist, we showed and explained that the radial pedal forces that occur during cycling are unavoidable when maximizing average mechanical power output during a full cycle. It is true that the “ineffective” radial pedal forces do not contribute to the power delivered by the cyclist to the crank. However, the idea that avoiding these radial pedal forces will lead to more efficient pedaling is incorrect. In fact, the opposite is true: avoiding radial pedal forces causes ineffective use of muscles that not only leads to a decrease in maximal AMPO in sprint cycling, but also to a decrease in pedaling efficiency.

5.A Appendix

5.A.1 Contribution of Muscle Forces to Pedal Forces

To analyze differences in outcome for different levels of constrained radial pedal forces, we calculated the (potential) contribution of each muscle to the pedal force, along the lines of the work of Zajac and coworkers (Zajac, 2002; Zajac et al., 2002) described our differential equations of the skeletal dynamics (see Methods section, *Musculoskeletal Model*) as follows:

$$A \cdot [\vec{F}_j \ \vec{\phi} \ \vec{F}_{con} \ M_{crank} \ \vec{F}_{pedal}]^T = \vec{b}$$

here A is a matrix describing the system dynamics pertaining to all unknown joint reaction forces ($\vec{F}_j$), joint angular accelerations ($\vec{\phi}$, constraint forces ($\vec{F}_{con}$; forces constraining the fixed position of the hip and bottom bracket), constraint moment (M_{crank} ; moment enforcing the constant crank angular velocity), and pedal forces ($\vec{F}_{pedal}$). $\vec{b}$ is a vector pertaining to the gravitational and Coriolis forces and,

importantly, the sum of the muscle moments around each joint. We structured our differential equations by identifying several blocks:

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \cdot \begin{bmatrix} \vec{x} \\ \vec{F}_{pedal} \end{bmatrix} = \begin{bmatrix} \vec{b}_1 \\ \vec{b}_2 \end{bmatrix}$$

Here, $\vec{x}$ are all unknowns except $\vec{F}_{pedal}$. We then used some basic general linear algebra manipulations to eliminate all unknowns other than $\vec{F}_{pedal}$:

$$\vec{F}_{pedal} = \left[A_4 - A_3 A_1^{-1} \cdot A_2 \right]^{-1} \cdot (\vec{b}_2 - A_3 A_1^{-1} \vec{b}_1)$$

This allowed us to calculate the pedal forces for any given state of the system (i.e., joint angles and angular velocities) and any net joint torque. Moreover, for any point along the cycle, we were able to use the states and joint torques obtained from our optimization and express the pedal forces as a function of net joint moments around the hip (M_{hip}), knee (M_{knee}), and ankle (M_{ankle}) and a term that represents the inertial component of the pedal forces ($\vec{F}_{dyn}$):

$$\vec{F}_{pedal} = \begin{bmatrix} a_x \\ a_y \end{bmatrix} M_{hip} + \begin{bmatrix} b_x \\ b_y \end{bmatrix} M_{knee} + \begin{bmatrix} c_x \\ c_y \end{bmatrix} M_{ankle} + \vec{F}_{dyn}$$

Since the net joint moment is simply the sum of all muscle joint torques, pedal force can be expressed as a sum of eight muscle contributions:

$$\vec{F}_{pedal} = \sum_{i=1}^8 \left(\begin{bmatrix} a_x \\ a_y \end{bmatrix} M_{hip,i} + \begin{bmatrix} b_x \\ b_y \end{bmatrix} M_{knee,i} + \begin{bmatrix} c_x \\ c_y \end{bmatrix} M_{ankle,i} \right) + \vec{F}_{dyn}$$

Here, $M_{hip,i}$ means the hip moment of the i th muscle, etc. Finally, by taking the summations apart one can calculate the contribution of any muscle to the pedal force ($F_{pedal,i}$):

$$\vec{F}_{pedal,i} = \begin{bmatrix} a_x \\ a_y \end{bmatrix} M_{hip,i} + \begin{bmatrix} b_x \\ b_y \end{bmatrix} M_{knee,i} + \begin{bmatrix} c_x \\ c_y \end{bmatrix} M_{ankle,i}$$

To give an example, for Figure 5.7 we calculated the directions of the contributions of the several muscles at a crank angle of 55° (0° being top-dead center):

$$\vec{F}_{pedal,i} = \begin{bmatrix} 0.02 \\ 0.03 \end{bmatrix} M_{hip,i} + \begin{bmatrix} 1.66 \\ -1.43 \end{bmatrix} M_{knee,i} + \begin{bmatrix} 2.17 \\ 5.95 \end{bmatrix} M_{ankle,i}$$

From this transfer function from joint torque to pedal force it can be appreciated that the contribution of a joint torque (and hence the muscle force producing that torque) to the pedal force greatly diminishes with increasing distance to the origin of the pedal force. In fact, for this angle, contribution of hip muscle to the pedal force is about two orders of magnitude smaller than the ankle muscles. The (normalized) direction of the force delivered by the m. soleus is $\begin{bmatrix} -0.34 & -0.94 \end{bmatrix}^T$ (note that the m. soleus delivers a negative torque around the ankle joint), that of the m. vastus is $\begin{bmatrix} 0.76 & -0.65 \end{bmatrix}^T$ and that of the m. iliopsoas is $\begin{bmatrix} 0.55 & 0.83 \end{bmatrix}^T$.

Data availability

Data will be made available upon reasonable request.

Supplemental data

Supplemental Material: <https://doi.org/10.6084/m9.figshare.21711020>

Acknowledgments

The authors would like to thank Gijs van der Woerd for his help in preparing this study and analyzing model outcomes and thank Koen Lemaire for insightful discussions.

Disclosures

No conflicts of interest, financial or otherwise, are declared by the authors.

Author contributions

D.A.K., R.M.T., E.D.H.M.R., and M.F.B. conceived and designed research; D.A.K. and R.M.T. analyzed data; D.A.K. and M.F.B. interpreted results of experiments; D.A.K. and R.M.T. prepared figures; D.A.K. drafted manuscript; D.A.K., R.M.T., E.D.H.M.R., and M.F.B. edited and revised manuscript; D.A.K., R.M.T., E.D.H.M.R., and M.F.B. approved final version of manuscript.

6

General discussion

In locomotor activities such as cycling, rowing, speed skating, and running, performance is largely determined by the average mechanical power output (AMPO) produced by muscles over successive stretch–shortening cycles (SSCs). In each periodic SSC, a muscle lengthens from its initial length and subsequently shortens back to that same length. The primary aim of this thesis was to examine how variations in the parameters defining SSCs—cycle frequency, FTS, and length excursion¹—influence the maximally attainable AMPO of muscle over one full periodic cycle. In addition, this thesis investigated how constraints affect the maximally attainable AMPO. The results presented in the preceding chapters, together with several preliminary findings not included in this thesis, give rise to a number of broader points of discussion. We introduce these topics below and then elaborate on each of them in separate sections.

The first section reflects on the methodological transition from highly controlled *in situ* rat experiments (Chapter 3) to less controlled human *in vivo* experiments (Chapter 4). Although Chapter 3 and Chapter 4 addressed similar questions, the experimental paradigms differed substantially in terms of the control of MTC length, the control of stimulation, the specificity of the measured output, and the

¹For a graphical definition of the SSC parameters, see Figure 3.2 or Figure 4.1.

degree of model individualisation. We discuss these differences to clarify why strong agreement between model predictions and experimental measurements in rat *in situ* experiments does not automatically imply that strong agreement is to be expected in human *in vivo* experiments.

The second section examines how sensitive the AMPO-optimal SSC behaviour is to muscle properties. Since estimating muscle properties inevitably involves errors and because properties also vary between individuals, it is important to assess how robust the model predictions are to variations in muscle properties. Ideally, the predicted optimal behaviour would remain largely unchanged across a realistic range of muscle property values. To explore this, a sensitivity analysis is conducted in which individual muscle property was systematically varied, complemented by additional analysis investigating the influence of muscle fibre type.

The third section addresses the sensitivity of AMPO to variations in three SSC parameters: cycle frequency, FTS and length excursion. Typically, we visualised their influence with contour plots. A question that arises at this point is: how sensitive is the maximally attainable AMPO to variations in these three SSC parameters? We explain why it is difficult to make general statements about the sensitivity to individual SSC parameters.

The fourth section extends the analysis from isolated muscles to the musculoskeletal level. In most motor tasks, the objective is not to maximise AMPO of a single muscle, but rather to maximise AMPO summed over multiple muscles. Muscles spanning the same joint cannot all operate at their individual optimum simultaneously. Consequently, the joint kinematics that maximise summed AMPO inherently reflect a trade-off. We discuss this trade-off using a simple model of an agonist-antagonist pair of muscles and a more complex model of the human lower extremity.

The fifth section considers the difference between fully imposed motion and motion that arises from the interaction between the musculoskeletal system and the environment. In this light, we compare the theoretical upper limit for AMPO of the human lower extremity with the maximally attainable AMPO predicted during isokinetic sprint cycling. We discuss the mechanisms underlying this difference.

The final section discusses the broader implications of this thesis. We explain how the present approach can be used to estimate the maximally attainable AMPO for observed SSCs in various motor tasks, and how this may help to interpret behaviour observed in these tasks. We then discuss how changing the mechanical interaction between the musculoskeletal system and the environment may allow muscles to operate closer to their AMPO-optimal SSCs. These insights provide general design principles for sport and rehabilitation equipment aimed at improving AMPO.

6.1 From *in situ* to *in vivo* experiments

In Chapter 3 and Chapter 4, experiments were combined with simulations and optimisations using a Hill-type muscle-tendon complex (MTC) model to investigate the influence of parameters defining SSCs—cycle frequency, FTS, and length excursion—on the maximally attainable AMPO. These studies involved two fundamentally different experimental paradigms: an isolated rat hindlimb muscle preparation *in situ* and a human knee extensor experiment performed *in vivo*. Although both approaches aimed to examine the influence of SSC parameters on the maximally attainable AMPO, the transition from the highly controlled *in situ* experiment to the substantially less controlled *in vivo* experiment introduced several important sources of uncertainty.

In the rat experiment, *m. gastrocnemius medialis* was isolated from surrounding tissue, while all nerve branches except the branch innervating *m. gastrocnemius medialis* were severed to ensure that contributions from surrounding muscles to the measured force were eliminated. This was confirmed by simultaneously measuring forces of *m. plantaris* and *m. gastrocnemius lateralis*. Consequently, the measured force directly reflected the mechanical behaviour of *m. gastrocnemius medialis*. By contrast, in the human experiment, the measured knee joint moment reflected the combined action of multiple muscles spanning the knee joint. During knee extension, EMG measurements indicated low levels of both agonist and antagonist activity, and the resulting net knee joint moment was close to zero. During knee flexion, however, EMG recordings suggested substantial antagonist co-activation, with activity reaching approximately 17% of activity during a maximal voluntary

contraction. Consequently, the experimentally measured net knee joint moment no longer represented the isolated behaviour of a single MTC, which complicated direct comparison between model predictions and measurements.

A second important distinction concerns control of MTC length over time, $L_{MTC}(t)$. In the rat experiments, MTC was directly connected to the motor and thus under full experimental control. In the human experiments, however, the muscle remained embedded within the musculoskeletal system. Consequently, the motion could not be imposed directly at the MTC level, but only indirectly by imposing knee joint angle over time. Due to inter-individual differences in moment arms and musculoskeletal geometry, identical knee joint kinematics do not correspond to identical $L_{MTC}(t)$ across participants. Therefore, the actual $L_{MTC}(t)$ of the muscle group of interest in the experiment may have differed from the $L_{MTC}(t)$ assumed in the model.

Substantial differences also existed in the control of muscle stimulation over time, $STIM(t)$. In the rat experiments, stimulation was imposed directly through controlled nerve stimulation. This allowed stimulation timing and intensity to be optimised for each experimental condition, whilst they were also highly reproducible between trials. Moreover, the experimentally applied stimulation pattern was used as input to the model, making the experimental and simulated stimulation conditions directly comparable. In the human experiments, however, stimulation resulted from voluntary neural drive. Participants were instructed to maximise mechanical work per cycle and received visual feedback on their performance. After familiarisation, the resulting stimulation over time (assessed using EMG) and the resulting net joint moment appeared to be close to optimal. Nevertheless, voluntary stimulation is inherently variable and is unlikely to be perfectly optimal. By contrast, in our model $STIM(t)$ was optimised for maximally attainable AMPO without any variability between cycles. Therefore, unlike in the rat experiments, stimulation over time differed between the human measurements and the corresponding model simulations.

A final important difference concerns the degree of model individualisation. In the rat experiments, key muscle properties were estimated separately for each speci-

men and incorporated into specimen-specific models. This allowed us to compare experimental measurements with highly individualised model predictions. In the human experiments, direct estimation of many key muscle properties was not feasible. Instead, models were individualised only by scaling maximal isometric force for each participant, while other muscle properties remained generic. Consequently, the model predictions in the human experiment were less tailored to individual participants than in the rat experiment, increasing the amount of uncertainty in the comparison between model predictions and experimental measurements.

As described above, the transition from highly controlled *in situ* rat experiments to less controlled *in vivo* human experiments introduces several (additional) sources of uncertainty that complicate direct comparisons between experimental measurements and model predictions. For example, in the human experiments, the contribution of antagonist co-contraction to the measured net joint moment, and thus to measured AMPO, may differ between conditions. Similarly, sub-optimal voluntary activation may be different between conditions, such that deviations between experimental measurements and model predictions (where stimulation over time was optimal) may vary between experimental conditions. These factors therefore contribute primarily to differences *between* conditions. Against this background, the set of five experimental conditions in which maximally attainable AMPO was predicted to be similar across conditions represents a particularly challenging test of the model. It is therefore a highly encouraging result that experimentally measured AMPO across these conditions were reasonably similar, remaining within $4\pm 4\%$ of the mean. All in all, despite these additional sources of uncertainty, model predictions still showed very strong agreement with experimental measurements in both paradigms. This suggests that the overall influence of these additional sources of uncertainty was small and, more generally, that the mechanisms captured by the Hill-type MTC model are robust across levels of physiological complexity.

6.2 Sensitivity of AMPO and AMPO-optimal SSC parameters to muscle properties

In this thesis, we found that the AMPO-optimal SSC parameters differ substantially between rat *m. gastrocnemius medialis* and human *m. quadriceps femoris* (see Chapter 3 and Chapter 4). Specifically, for rat *m. gastrocnemius medialis*, AMPO peaked at a cycle frequency of approximately 3.5 Hz and an optimal CE length excursion of about 58% L_{CE}^{opt} , whereas for human *m. quadriceps femoris* these values were approximately 1.6 Hz and 80%, respectively. In addition, Askew and Marsh (1997) reported corresponding values of approximately 4.1 Hz and 22% for mouse *m. soleus*, and 9.4 Hz and 16% for mouse *m. extensor digitorum longus*. Together, these findings indicate that the optimal cycle frequency and muscle length excursion strongly depend on muscle properties. By contrast, the optimal FTS was found to be relatively constant, ranging between approximately 0.80 and 0.85 across all muscles mentioned above. This raises a general question: to what extent do the optimal SSC parameters depend on specific muscle properties?

Table 6.1: Results of the sensitivity analysis of maximally attainable AMPO and AMPO-optimal SSC parameters to muscle properties.

	AMPO	CF	FTS	ΔL_{CE}
<i>Excitation dynamics</i>				
a_{act}	-3.8	-5.0	-0.0	2.1
m	-4.1	-6.0	-0.5	2.0
q_0	-0.1	0.1	0.0	0.0
<i>Contraction dynamics</i>				
a^{rel}	2.2	1.6	-0.7	1.0
b^{rel}	-5.4	-3.5	1.2	-1.6
F_{asympt}	0.9	1.6	0.8	-0.4
r_{as}	0.6	0.5	0.7	0.4
r_{slope}	0.0	0.0	-0.0	-0.0
w	-3.5	4.5	0.1	-7.7
AVG $\pm$ STD	2.3 $\pm$ 1.8	2.6 $\pm$ 2.2	0.7 $\pm$ 0.5	1.8 $\pm$ 2.3

A sensitivity analysis was performed for different muscle model parameters. Each parameter was varied by $\pm 10\%$ relative to its standard values. Changes in maximally attainable AMPO, optimal cycle frequency (CF), FTS and CE length excursion (ΔL_{CE}) are all expressed relatively to the values obtained with the standard model parameters.

To investigate the sensitivity of the optimal SSC parameters and the maximally attainable AMPO to muscle properties, we performed a model-based sensitivity analysis using muscle properties of human *m. quadriceps femoris*. Muscle property values were individually increased and decreased by 10% relative to their standard values. For each changed muscle property, we optimised CE length over time for maximal AMPO using Optimal Control techniques (for details, see S3 of Chapter 3). Detailed results and mechanistic explanations for how individual parameter changes affected AMPO and the AMPO-optimal SSC parameters are provided in Section 6.5. Here, we provide only a general overview of the effects on maximally attainable AMPO, optimal cycle frequency, optimal FTS, and optimal CE length excursion. In addition, the influence of muscle fibre type was investigated separately.

In relative terms, the effects of individual muscle properties on maximally attainable AMPO and the AMPO-optimal SSC parameters were smaller than the imposed parameter variations. The optimal FTS proved to be highly robust, showing a maximal deviation of only 1.2% across all tested variations of 10%. The optimal cycle frequency, CE length excursion and maximally attainable AMPO showed somewhat greater sensitivity to changes in muscle properties, although the overall effects remained modest (less than 3% on average). Muscle properties related to excitation dynamics, the CE force–length relationship, and the CE force–velocity relationship appeared to contribute in a broadly comparable manner; however, their relative importance varied across outcome measures considered (i.e. maximally attainable AMPO, optimal cycle frequency and optimal CE length excursion). All in all, while muscle properties do affect the optimal SSC parameters and the maximally attainable AMPO itself, their impact remained limited in magnitude.

To examine the effect of muscle fibre type, we performed additional optimisations in which the activation and deactivation time constants were assumed to scale proportionally with maximal shortening velocity, consistent with the observations of Close (1972). Specifically, when the activation and deactivation time constants are scaled by a factor C , maximal shortening velocity (v_{CE}^{max}) decreases by a factor $1/C$. Under this assumption, both AMPO and optimal cycle frequency scale linearly with $1/C$, whereas the optimal FTS and CE length excursion remain unchanged.

This can be explained by two observations. First, starting from the same initial intramyofilament calcium concentration, it takes C times longer to reach a given intramyofilament calcium concentration in the scaled system, assuming non-steady-state conditions. Second, CE force at shortening velocity v_{CE} in the unscaled system equals CE force at a shortening velocity v_{CE}/C in the scaled system. Consequently, when cycle frequency is also scaled by $1/C$, the active state as a function of normalised time (defined as time divided by the cycle duration) is the same for the scaled and unscaled system. Furthermore, the CE force–velocity effects remain the same, provided that the CE length over normalised time is preserved. Because the CE force–length relationship is independent of muscle fibre type, this results in similar CE length and CE force as a function of normalised time throughout the cycle. Therefore, changes in muscle fibre type primarily affect the optimal cycle frequency and AMPO.

In general, the sensitivity analysis demonstrated that variations in individual muscle properties had only limited influence on the shape of the optimal CE length over time (data not shown) as well as the optimal SSC parameters. This is an important finding, because it suggests that the predicted optimal SSC behaviour is relatively robust to variations in muscle properties, which reflect both inter-individual variability and uncertainties in property estimation, as well as to the muscle characteristics represented in the model. In other words, although specific muscle property values affect the maximally attainable AMPO and the exact values of the optimal SSC parameters, they appear to have relatively little influence on the overall optimal behaviour for maximising AMPO.

6.3 Sensitivity of AMPO to SSC parameters

In Chapter 3 and Chapter 4, the influence of cycle frequency, FTS, and length excursion on AMPO was visualised using contour plots. These plots are useful because they show the influence of SSC parameters on the maximally attainable AMPO. However, making a general statement about the sensitivity of AMPO to SSC parameters using these contour plots is not straightforward.

One possible approach to investigate this sensitivity is to define a region around optimum in which AMPO remains above a certain percentage of its peak value, for example 95%. This allows one to describe how much cycle frequency, FTS, or length excursion can be changed before AMPO drops below 95% of its peak value. However, such an analysis raises a fundamental problem: changes in different SSC parameters are not directly comparable, because the parameters have different units and scales. For example, it is not obvious what change in cycle frequency (expressed in Hz) is equivalent to a given change in FTS (which is unitless) or to a given change in MTC/CE length excursion (expressed in mm).

A second reason is that the parameterisation itself is arbitrary. In this thesis, SSCs were parameterised using cycle frequency, FTS, and MTC/CE length excursion. However, the same SSC could also be described using many other combination of parameters. For example, FTS could be replaced by absolute shortening duration, T_{short} . The sensitivity of AMPO to FTS would then differ from the sensitivity of AMPO to T_{short} , although the sensitivity of AMPO to the underlying change of the SSC obviously does not change. At this point in time, we consider the contour plots as presented in Chapter 3 and Chapter 4 to be the most informative way to show the influence of SSC parameters on the maximally attainable AMPO.

6.4 Trade-offs at the musculoskeletal level

Results of the analysis of the influence of SSC parameters on the maximally attainable AMPO for an individual muscle, as presented in Chapter 3 and Chapter 4, provides a major improvement of our understanding of muscle function. Yet, as most motor tasks involve multiple muscles, the objective is generally not to maximise AMPO of a single muscle, but rather to maximise AMPO summed over all muscles involved. Because each joint is typically spanned by several muscles, the SSC that maximises AMPO for a single muscle cannot be realised in all muscles. This is particularly evident for FTS, as the values that maximises AMPO for a single muscle (~ 0.80 – 0.85) cannot occur in both agonistic and antagonistic muscles spanning the same joint. Consequently, the joint kinematics that maximise summed AMPO represent a trade-off among all muscles spanning that joint, and due to the presence of bi-

articular muscles in fact over multiple muscles spanning multiple joints. The optimal kinematics therefore arise from this trade-off between all muscles involved and depend on properties of all muscles involved, such as maximal isometric force and muscle fibre length. Due to the trade-off, the SSC parameters of individual muscles are expected to shift towards more intermediate values, i.e. closer to an FTS of 0.5, when optimising the summed AMPO across all muscles, especially when agonist and antagonist muscles have similar properties.

To investigate this issue, we first considered a simplified system in which two theoretical human muscles spanned a single joint: one agonist and one antagonist. The fully imposed joint angle over time was optimised such that summed AMPO of both muscles was maximal. For simplicity, the two muscles were assigned identical properties, except for maximal isometric CE force: agonist maximal isometric CE force was kept constant, whereas antagonist maximal isometric CE force was systematically varied. This was done such that the ratio between antagonist and agonist maximal isometric CE force ($F_{CE,anta}/F_{CE,ago}$) ranged from 0 to 1.

As expected, when agonist and antagonist maximal isometric CE forces were equal (i.e. $F_{CE,anta}/F_{CE,ago} = 1$), the optimal joint kinematics resulted in an FTS of 0.5 for both muscles. And, not surprisingly, when antagonist maximal isometric CE force was zero (i.e. $F_{CE,anta}/F_{CE,ago} = 0$), optimal joint kinematics caused the agonist FTS to converge to the value previously identified for an isolated muscle, approximately 0.80. Also as expected, maximal summed AMPO (normalised to “muscle volume”) increased with decreasing $F_{CE,anta}/F_{CE,ago}$.

More interesting is the relationship between $F_{CE,anta}/F_{CE,ago}$ and both the optimal FTS and maximal summed AMPO. Reductions in $F_{CE,anta}/F_{CE,ago}$ at lower relative values had a substantially larger effect than equivalent reductions at higher relative values. For example, decreasing $F_{CE,anta}/F_{CE,ago}$ from 1.0 to 0.6 had only a minor influence on maximal summed AMPO, whereas the effect was substantially larger when decreasing $F_{CE,anta}/F_{CE,ago}$ from 0.4 to 0 (Figure 6.1). Similarly, optimal FTS (for the agonist) increased only slightly when $F_{CE,anta}/F_{CE,ago}$ was close to 1, but increased more strongly towards the isolated-muscle optimum as antagonist maximal isometric CE force approached zero.

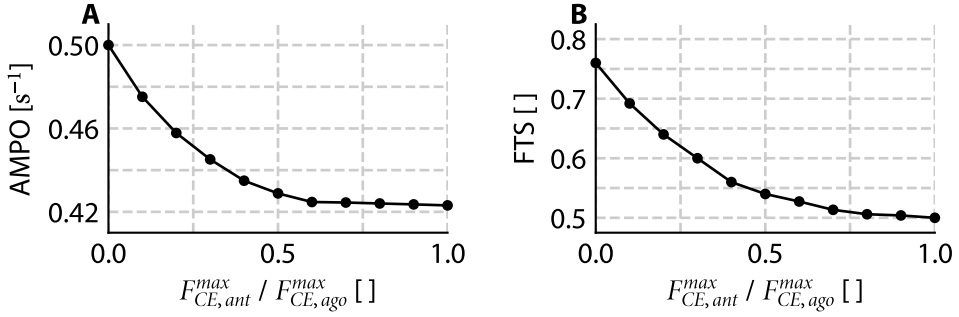


Figure 6.1: Maximally attainable AMPO and optimal FTS for agonistic and antagonistic muscle spanning the same joint. The ratio between antagonist and agonist maximal isometric CE force ($F_{CE,anta}/F_{CE,ago}$) was varied from 0 to 1 and the fully imposed joint angle together with stimulation over time was optimised such that summed AMPO of both muscles was maximal. The maximally attainable AMPO was scaled to the sum of the product of CE optimum length and CE maximal isometric force of both muscles to account for differences in “total muscle volume” induced by varying $F_{CE,anta}/F_{CE,ago}$. The optimal FTS for maximally attainable AMPO was given for the agonist. Note that FTS of antagonist equals 1-FTS of the agonist.

Similar principles likely apply to differences in muscle fibre length between agonistic and antagonistic muscles. Specifically, the optimal FTS of the muscle with longer fibres are expected to shift towards higher values than that of the muscle with shorter fibres. In addition, the optimal joint excursion is expected to shift towards ranges that are more favourable, in terms of AMPO, for the muscle with longer fibres. While such analysis is insightful, real musculoskeletal systems involve multiple agonists and antagonists, which may have properties with competing or even opposing effects on the optimal solution (e.g. agonists with long muscle fibres and low maximal isometric CE force versus antagonists with short muscle fibres and high maximal isometric CE force). A more detailed exploration of such trade-offs is beyond the scope of this general discussion and is an interesting direction for future work. Overall, these considerations imply that, if the goal is to maximise summed AMPO, muscles capable of attaining relatively high AMPO (e.g. due to long muscle fibres and/or high maximal isometric force) should operate closer to their individual optimum than muscles with a lower attainable AMPO.

To quantify how this trade-off between muscles manifests itself at the level of the complete lower extremity, we previously predicted the maximal AMPO that could

be produced by the human lower extremity (Reuvers et al., 2019). The optimisation problem was to find the fully imposed hip, knee, and ankle joint angles together with muscle stimulations over time for maximal AMPO, with the constraint that the solution was fully periodic. The maximal AMPO produced by the lower extremity was approximately 15% lower than the theoretical maximum AMPO that would be achieved if all muscles could simultaneously operate at their individual optimum (i.e. if each muscle underwent its individually optimal SSC with optimal muscle stimulation). In line with the trade-offs at the joint level considered earlier, the joint kinematics that maximised summed AMPO were such that they were more favourable for muscle capable of attaining relatively high AMPO (see Figure 6.2).

6.5 From imposed motion to coordinated action

The maximal AMPO predicted for the lower extremity in Reuvers et al. (2019) defines a theoretical upper limit for the human lower extremity, i.e. the maximal AMPO achievable when optimal hip, knee, and ankle joint angles over time together are imposed, accompanied by optimal muscle stimulation. When this value is compared with the maximal AMPO predicted during isokinetic sprint cycling using the same musculoskeletal model (see Chapter 4), it becomes apparent that the theoretical upper limit is approximately 65% higher than the maximal value attained during isokinetic sprint cycling. Two main factors explain this difference.

The first reason for this difference is that the joint kinematics were fully imposed in Reuvers et al. (2019). This means that muscle forces did not influence the kinematics and, consequently, the MTC lengths of the muscles involved. As a result, the optimal muscle stimulation was simply the set of stimulations that maximised the AMPO of each individual muscle. During isokinetic sprint cycling, however, the kinematics are not imposed, and muscle forces therefore do influence the kinematics. This means that the force produced by one muscle affects the MTC lengths of all muscles. Consequently, the optimal muscle stimulation is not simply the set of stimulations that maximised the AMPO of each individual muscle, but rather the solution to a coordination problem involving all muscles. To illustrate this, consider a strong knee extensor muscle (e.g. *mm. vastii*) and relatively weak ankle plantarflexors

(e.g. m. soleus and m. gastrocnemius). While maximal AMPO of *mm. vastii* would require maximal stimulation (at least during the beginning of shortening), this may induce substantial ankle dorsiflexion that cannot be counteracted by the relatively weak plantarflexors. To prevent substantial ankle dorsiflexion, muscle stimulation of *mm. vastii* cannot be maximal, or should cease well before lengthening starts (for details, see Van Soest et al., 2005). Taken together, when kinematics are not fully imposed, optimal muscle stimulation requires a compromise between the optimal operating conditions of all muscles involved. By contrast, when kinematics are imposed, this coordination problem is eliminated, allowing muscle stimulations to be tuned to those that maximise AMPO for each individual muscles. Consequently, higher overall power outputs can, in principle, be achieved under optimally chosen imposed kinematics than when kinematics emerge from the coordinated action of all muscles. This is consistent with the first principle of optimal control theory, which states that introducing additional constraints cannot increase the optimal value of the objective function. Since this is a universal principle, it applies not only to the situation described above, but also to *any* constraints imposed on the task itself. A clear example is provided in Chapter 5, where we showed that adding a task constraint—namely limiting radial pedal force—led to a substantial reduction in performance (in this case, more than 50% reduction in maximally attainable AMPO when no radial forces were allowed compared to the condition in which there was no constraint on radial pedal force).

The second reason for the difference between maximally attainable AMPO during isokinetic sprint cycling and the theoretical upper limit on AMPO is that isokinetic sprint cycling imposes a kinematic constraint. During isokinetic cycle, the pedal is constrained to follow a circular trajectory at constant crank angular velocity (i.e. with a constant pedal velocity magnitude). This kinematic constraint imposes a strong restriction on the possible joint excursions and thereby on the SSCs of the individual muscles.

To quantify the influence of these two factors separately, we performed additional optimisations to identify how hip, knee, and ankle joint angles over time should be imposed to maximise AMPO under the constraint that the “ball of the feet”

(~MTP₅) followed a circular trajectory with a constant velocity magnitude. This mimicked the trajectory of the pedal during normal cycling. The results showed that maximal AMPO summed across all muscles of the lower extremity increased by about 40% compared to normal cycling, yet still 20% lower than when all joint angles were imposed without the circular toe-trajectory constraint (Figure 6.3). These findings provide initial insight into the extent to which constraints limit maximal performance during periodic motor tasks.

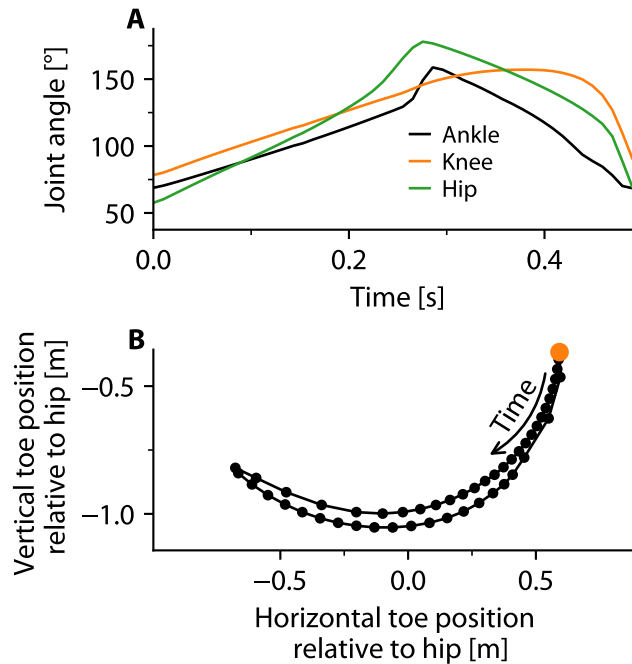


Figure 6.2: Optimal joint angles over time and resulting toe trajectory for the optimisation in which all imposed joint angles over time were optimised. A) Optimal joint angles over time. All joint angles increase with (plantar)flexion. For the hip and knee, a joint angle of 0° corresponds to maximal extension. For the ankle, a joint angle of 90° corresponds to the anatomical standing position, with the foot flat on the ground and the shank perpendicular to the foot. B) The resulting toe trajectory (relative to hip position). Dots are equidistant in time (0.005s) and the solid orange dot in B corresponds to $t=0$ in A.

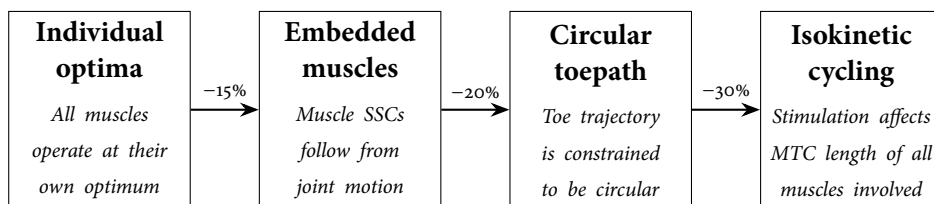


Figure 6.3: Comparison of the sum of the maximally attainable AMPO summed over all muscles in the human lower extremity for different periodic motions. Maximal AMPO is predicted to decrease with each additional constraint. First MTC length depends on joint angle and therefore MTC lengths are not independent. Second, MTP₅ follows a prescribed circular trajectory with a constant velocity magnitude. Third, muscle stimulation influences the kinematics and therefore MTC length over time of all muscles involved.

6.6 General insight obtained from this thesis

In Chapter 3 and Chapter 4, the direct influence of SSC parameters on maximally attainable AMPO was examined. In both studies, MTC length over time was imposed, either directly in Chapter 3 or indirectly via knee joint angle in Chapter 4. Consequently, the net joint moments resulting from muscle force did not influence the MTC length over time. This approach allowed us to isolate the direct effects of SSC parameters on maximally attainable AMPO, which was desirable given the specific aims of these studies. Because AMPO depends on MTC length and stimulation over time, the maximally attainable AMPO can, in principle, be determined for any SSC, irrespective of whether that SSC is externally imposed or emerges from the interaction between the musculoskeletal system and the environment. Therefore, for any observed SSC, it is possible to estimate the maximally attainable AMPO associated with the observed kinematics of a given muscle or muscle group.

Such an analysis — together with the consideration of trade-offs at the musculoskeletal level — may help to improve our understanding of muscle behaviour during a wide range of motor tasks. For example, during flight, birds actively adjust wing kinematics by folding and unfolding their wings such that the shortening duration of *m. pectoralis major* exceeds its lengthening duration (e.g., Askew and Marsh, 2001; Biewener et al., 1998; Earls, 2000; Ellerby and Askew, 2007; Jackson and Dial, 2011; Spedding, 1987; Williamson et al., 2001). According to the present findings, such kinematics are favourable for maximising AMPO of the muscles contributing to propelling flight.

Whereas birds can actively adjust the relative shortening and lengthening durations of *m. pectoralis major*, the extent to which SSC parameters can be adjusted is more limited in many other motor tasks. For example, during running, leg extensor muscles typically spend more time lengthening than shortening, as ground contact time is shorter than flight time. The present findings may help identify which changes in running kinematics, such as changes in stride length or stride frequency, could increase AMPO through their effects on SSC parameters. However, the SSC parameter combination that maximises AMPO may not be attainable during running because of task-specific mechanical constraints. This may also apply to many other motor tasks in which the FTS of the major power-producing muscles is well below the isolated-muscle optimum.

For many motor tasks, the only way to substantially alter SSC parameters is to change the mechanical interaction between the musculoskeletal system and the environment. This can be achieved through equipment, which can constrain or change movement kinematics. For example, in rowing, the resistance during the stroke and recovery phases could in principle be adjusted to alter muscle shortening and lengthening durations. More broadly, such (re)design principles could be applied to any equipment, thereby enabling the SSCs of the involved muscles to operate closer to their optimum for AMPO.

When (re)designing such equipment, the present results provide a useful starting point. As argued throughout this thesis, optimal equipment design depends on both the trade-off between muscles and the mechanical interaction of the musculoskeletal system with the environment. In this regard, the present findings suggest a clear general principle: equipment should be designed such that the muscles capable of producing relatively high AMPO spend more time shortening than lengthening. The kinematics that maximise depend on the properties of the muscles involved (e.g. fibre length and maximal isometric force). An interesting implication of this principle is that when fewer muscles are active, the remaining muscles may be able to operate closer to their individual optimum. This may be particularly relevant in clinical populations such as individuals with spinal cord injury, where only a limited number of muscles can be voluntarily recruited or electrically stimulated.

Such populations may therefore benefit substantially from equipment designed specifically to improve AMPO.

Overall, this thesis shows that AMPO is strongly influenced by MTC length over time and that the SSC that maximises AMPO for a single muscle is characterised by a shortening duration that exceeds the lengthening duration. At the same time, this optimum cannot generally be realised when muscles operate within a musculoskeletal system, because the optimal joint kinematics represent a compromise between multiple muscles and might be further influenced by (additional) constraints. The challenge for future research is therefore to determine how equipment can be designed such that the most important power-producing muscles operate as close as possible to their AMPO-optimal SSCs.

6.S Supplementary material

6.S.1 Sensitivity of AMPO and AMPO-optimal SSC parameters to muscle properties

In this supplementary material, detailed results and mechanistic explanations are provided on how individual parameter changes affect AMPO and the AMPO-optimal SSC parameters, which are considered the outcome measures. To assess their sensitivity to muscle properties, each contractile element (CE) property was varied by 10% relative to its standard value. The sensitivity of the outcome measures to changes in muscle properties is discussed separately for the concentric CE force–velocity relationship, the eccentric CE force–velocity relationship, the CE force–length relationship, and excitation dynamics.

Regarding the concentric CE force–velocity, the parameters a^{rel} and b^{rel} determine the maximum CE shortening velocity at $F_{CE}^{isom,rel} = 1$ (i.e. $v_{CE}^{rel,max} = -b^{rel}/a^{rel}$). Whereas b^{rel} only influences $v_{CE}^{rel,max}$, a^{rel} also changes the curvature of the concentric CE force–velocity relationship. Increasing b^{rel} or decreasing a^{rel} increased maximal AMPO and the optimal cycle frequency, because the negative effect of the CE force–velocity relationship on maximal AMPO became smaller. At higher cycle frequencies, the absolute time for CE lengthening decreases. To prevent substantial negative mechanical work during lengthening, the optimal FTS

decreased when b^{rel} increased and/or when a^{rel} decreased. A somewhat counter-intuitive finding was that increasing b^{rel} had a larger effect on the optimal cycle frequency, FTS and maximal AMPO itself than decreasing a^{rel} , despite the fact that a 10% decrease in a^{rel} results in a larger increase in $v_{CE}^{rel,max}$. This can be explained by the fact that a^{rel} changes the curvature of the force–velocity relationship, whereas b^{rel} does not. At relative CE shortening velocities (i.e. CE shortening velocity divided by CE optimum length) below approximately -12.2 s^{-1} , the CE force at a given CE shortening velocity is higher for a 10% increase in b^{rel} than for a 10% decrease in a^{rel} . Consequently, increasing b^{rel} had a larger effect than decreasing a^{rel} .

Next, we investigated the sensitivity of the outcome measures of parameters F_{asympt} , r_{as} and r_{slope} . Increasing all these values resulted in a higher CE force at a given CE lengthening velocity (and CE length and active state). As a result, maximal AMPO decreased slightly for higher values of these parameters. The optimal cycle frequency and FTS also decreased slightly (with a maximum change of 1.7%; see Table 6.1), which reduced CE lengthening velocity and thereby limited excessive negative mechanical work during the lengthening phase. The effect on the optimal CE length excursion was even smaller. These small effects can be explained by the fact that maximally attainable AMPO is primarily determined during concentric contractions, making it relatively insensitive to the eccentric CE force–velocity relationship.

With regards to the CE force–length relationship, we investigated the effect of the parameter w that determines the width of the CE force–length relationship. Increasing w resulted in a higher maximal AMPO and a lower optimal cycle frequency. However, changing w had little effect on the optimal FTS, supporting the idea that the optimal FTS is determined primarily by the interaction between excitation dynamics and the concentric CE force–velocity relationship.

Regarding the excitation dynamics, we investigated the sensitivity of the outcome measures to several model parameters. First, we found that the parameter defining the minimal active state (q_0) had no influence on either AMPO or the optimal SSC parameters. This is because this value is very small and, under the simulated

conditions, the calcium concentration between the myofilaments (γ) never reaches zero. Consequently, changing the minimal value of γ also had almost no effect on the outcome measures. Next, we investigated the effect of the speed of the calcium dynamics by varying parameter m . Increasing m increases both activation and deactivation speed, which in turn increased maximal AMPO, the optimal cycle frequency, and the optimal FTS. Subsequently, the effect of a_{act} , which determines the steepness of the $\gamma - q$ relationship, was examined. Increasing a_{act} results in faster activation and deactivation dynamics for $q > 0.5$, but slower activation and deactivation dynamics for $q < 0.5$. In addition, increasing a_{act} increases the active state (q) at $\gamma = 1$. In other words, at the same value of γ , higher values of a_{act} result in higher q when $q > 0.5$, but lower q when $q < 0.5$. Despite these mixed effects, the overall result was that the optimal cycle frequency and FTS increase with increasing a_{act} . This suggests that the faster deactivation at low activation levels (i.e. $q < 0.5$) outweighs the slower deactivation dynamics at high activation levels (i.e. $q > 0.5$).

The parameters b_{act}^1 , b_{act}^2 and b_{act}^3 determine the value of γ at $q = 0.5$ (i.e. the ratio between speed of activation and deactivation) for a given L_{CE}^{rel} . The dependency of q on b_{act}^1 , b_{act}^2 and b_{act}^3 is highly non-linear and therefore changing primarily the parameter values of b_{act}^1 , but also those of b_{act}^2 , has dramatic consequences for the excitation dynamics. Changing the ratio between speed of activation and deactivation will therefore be discussed separately.

For the excitation dynamics of human m. quadriceps femoris, the difference in activation and deactivation speed is fully caused by the relationship between γ and q . Since the calcium saturation point (the point where $q = 0.5$) is at a low value of γ , the activation dynamics are much faster than the deactivation dynamics. For illustration, the active state increases from 0.1 to 0.8 in 13 ms when $STIM$ is 1, whereas it takes 133 ms to decrease from 0.8 to 0.1 when $STIM$ is 0 (for the AMPO-optimum $L_{CE}^{rel}(t)$ and $STIM(t)$ at a cycle frequency of 1.6 Hz). As activation dynamics are very fast, it was found that CE motion and maximal AMPO hardly changed when making activation dynamics instantaneous ($q = STIM$ for $STIM > q$). Hence, it was concluded that the influence of activation dynamics on CE behaviour is negligible compared to the influence of deactivation dynamics and therefore any change in parameter values

governing the (de)activation dynamics almost completely influences CE behaviour by its influence on the deactivation dynamics.

The ratio between activation and deactivation dynamics was altered by changing the parameter value of b_{act}^1 such that γ at the calcium saturation point (CSP) became 25% lower ($b_{act}^1 = 5.30$ and $a_{act} = -4.05$; CSP 0.75) and 25% higher ($b_{act}^1 = 5.07$ and $a_{act} = -5.09$; CSP 1.25) for $L_{CE}^{rel} = 1$ compared to the original formulation and by changing a_{act} accordingly such that $q(\gamma = 1)$ was unaltered (see Figure 6.S.1). This resulted in faster activation dynamics and slower deactivation dynamics for CSP 0.75, while it resulted in slower activation dynamics and faster deactivation dynamics for CSP 1.25 compared to the original parameter values. Maximal AMPO was found to be 0.41 s^{-1} and 0.50 s^{-1} at the optimal cycle frequency of 1.57 and 1.67 Hz with an optimal FTS of 0.75 and 0.77 for CSP 0.75 and CSP 1.25, respectively.

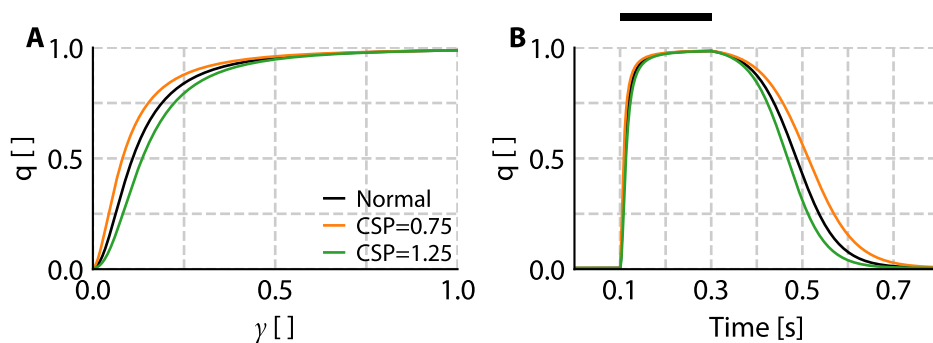


Figure 6.S.1: Influence of the calcium saturation point on the excitation dynamics. The parameters b_{act}^1 and a_{act} were changed such that the calcium saturation point (CSP) was at a 25% lower value of γ (CSP 0.75) and at a 25% higher value of γ (CSP 1.25) compared to the original model for $L_{CE}^{rel} = 1$. A) Effect on the relation between γ and active state B) The effect on active state over time. CE stimulation is maximal during the period indicated by the black bar and off elsewhere.

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Summary

When we walk, run, cycle, or propel a wheelchair, our muscles act as the motors of the biological system, driving the limbs in repetitive, cyclic motions that generate movement. During each cycle, muscles repeatedly lengthen (eccentric contraction) and shorten (concentric contraction), a pattern known as the stretch–shortening cycle (SSC). Muscles deliver negative mechanical power when producing force while lengthening, whereas they deliver positive mechanical power when producing force while shortening. Across a movement cycle, these phases result in negative and positive mechanical work, the sum of which determines the net mechanical work produced by the muscles over one full cycle. Dividing this net mechanical work by the cycle duration yields the average mechanical power output (AMPO), a key determinant during self-propelled high-speed motor tasks. Yet, despite its importance, surprisingly little was known about how muscles should operate in order to maximise AMPO in a particular movement task and, related, how much AMPO we can maximally generate with our muscles.

Put in its simplest form, the main question addressed in this thesis is: What muscle length over time and corresponding muscle activation over time maximises AMPO? In addition, there are several related questions such as: What is the influence of muscle length excursion and cycle frequency on AMPO? Do these parameters interact? Trying to answer these questions using experiments alone provides limited knowledge as only a small set of all possible SSCs can be investigated within a measurement session: Which SSCs should be selected to gain as much insight as possible? In this thesis, we addressed these questions using experimental techniques in combination with simulation and optimisations of mathematical muscle–tendon complex models in order to gain fundamental insight into the power-producing capabilities of muscles.

In **Chapter 2** and **Chapter 3**, we focused on the muscle–tendon complex (MTC) of the rat hindlimb. The advantage of working on an isolated MTC preparation was that we had full control over the MTC length and activation over time. Thus, we could readily test the influence of various SSC parameters (e.g. cycle frequency and length excursion) on the maximally attainable AMPO. Nevertheless, as mentioned earlier, only a limited set of SSCs can be tested in a single measurement session. To address this, we complemented experimental work with predictions derived using Hill-type MTC models. Obviously, model predictions are only valuable if they are sufficiently accurate. Our approach was to evaluate model predictions against a set of insightful experimental conditions. If the model predictions proved sufficiently accurate, it would justify to use the model to further explore the SSC parameter space beyond what could be experimentally tested. Importantly, AMPO is influenced by muscle properties, such as maximal isometric force and muscle fibre length as well as all other properties of the contraction and excitation dynamics. Thus, to reduce inter-individual variance in predictions due to differences in MTC properties, we chose to estimate the properties of each individual MTC based on commonly used experimental protocols. This raises an important question: Can these properties be estimated with sufficient accuracy using these protocols?

Chapter 2 presents a theoretical study on the accuracy of commonly used experimental protocols for estimating key MTC properties in isolated MTC preparations, such as quick-release and step-ramp experiments. We simulated these protocols with a Hill-type MTC model based on parameter values obtained from literature. We then estimated the parameters from the simulated data and compared them with their *a priori* values. In quick-release experiments, MTC first contracts isometrically (i.e. at a constant length), after which a quick change in length is imposed (hence the name: quick-release). A common assumption underlying the quick-release experiments is that the release is so fast that muscle fibres do not shorten during the quick-release. Consequently, under this assumption, all length change is taken up by the series elastic element (SEE) in series with the muscle fibres. SEE stiffness can then be estimated from its force and length changes measured before and after the release. However, we found that this assumption is invalid: muscle fibres do

shorten even within quick-releases of only 10 ms. Consequently, SEE stiffness is substantially underestimated (by about 35%) under this commonly used assumption. This underestimation in SEE stiffness subsequently propagates to the estimation of other MTC properties, with estimations of excitation dynamics properties being affected most strongly (up to about 20%). We developed an improved method that accounts for the shortening of muscle fibres during quick-releases. This improved method yielded parameter estimates that were close to their *a priori* values (within 3%). This improved method was then used in **Chapter 3**.

In **Chapter 3**, we combined *in situ* experiments with simulations and optimisations using a Hill-type MTC model to investigate the influence of various SSCs on the maximally attainable AMPO of rat *m. gastrocnemius medialis*. For each individual specimen, key MTC properties were estimated individually using the methodology developed in **Chapter 2**. This enabled a direct comparison between experimentally measured and model-predicted AMPO for each specimen. We found near-perfect agreement between experimentally measured and model-predicted AMPO ($r^2 > 0.98$), which then justified the use of the Hill-type MTC model to systematically explore how variations SSCs influence the maximally attainable AMPO. We predicted the maximally attainable AMPO for a large set of SSCs, parametrised by cycle frequency, FTS (the fraction of the cycle time shortening; e.g. an FTS of 0.65 means that 65% of the cycle time is spent shortening) and MTC length excursion. AMPO was predicted to peak at a cycle frequency of 3.5 Hz, an FTS of 0.85 and an MTC length excursion of 8 mm. Notably, cycle frequency and MTC length excursion showed a strong interaction: increasing one necessitated a decrease in the other in order to maximise AMPO. By contrast, the optimal FTS remained remarkably constant across all tested combinations of cycle frequency and MTC length excursion. **Chapter 3** provides an extensive overview of how SSC parameters influence the maximally attainable AMPO of rat *m. gastrocnemius medialis*.

Given the success of the approach used in **Chapter 3**, we applied a similar approach in **Chapter 4** to investigate the effect of SSC parameters on the maximally attainable AMPO of the human *m. quadriceps femoris in vivo*. However, the

approach differed in several key aspects. First, instead of individually tailored models for each participant, we employed a Hill-type model with generic MTC properties derived from literature to investigate the influence of SSC parameters on the maximally attainable AMPO. Second, in the experimental setup, knee joint movements were fully imposed using a knee dynamometer, while participants were instructed to maximise AMPO. Thus, muscle activation was not directly controlled but resulted from voluntary neural drive. Third, outcome measures were based on measured net knee joint moment, which reflects the combined contribution of all muscles spanning the joint rather than that of *m. quadriceps femoris* alone. Taken together, the *in vivo* human experiments were less controlled than the *in situ* rat experiments described in **Chapter 3**, which introduced several (additional) sources of uncertainty. The expectation was that this would reduce the agreement between model predictions and experimental results. Model predictions were compared with experimental data using two complementary scaling approaches. When scaling the experimental data only to the maximal isometric force of each participant, we observed good agreement between predictions and measurements ($r^2 = 0.75$). When investigating differences between conditions (expressed in units of standard deviation), the differences between model predictions and experimental data decreased, resulting in stronger agreement ($r^2 = 0.95$). These results confirmed that predictions derived with a Hill-type MTC model with generic MTC properties were valid for investigating the influence of SSC parameters on the maximally attainable AMPO of human *m. quadriceps femoris in vivo*. These predictions revealed that the maximally attainable AMPO peaks at a cycle frequency of about 1.6 Hz, an FTS of about 0.80, and a knee joint excursion of about 2.0 rad (centred around a knee joint angle of 1.0 rad, which is about the knee joint angle yielding maximal isometric force). Similar to rat *m. gastrocnemius medialis*, we found that cycle frequency and knee joint excursion strongly interacted, while the optimal FTS was again predicted to be largely unaffected by changes in cycle frequency and knee joint excursion. **Chapter 4** provides an extensive overview of how SSC parameters influence the maximally attainable AMPO of human *m. quadriceps femoris in vivo*.

In **Chapter 5**, we investigated how varying constraints on radial pedal forces affect the maximally attainable AMPO during isokinetic sprint cycling. For this purpose, we used a musculoskeletal model of the human lower limb and optimised muscle activation over time to maximise AMPO summed over all muscles under different constraints on allowed radial pedal forces. Maximal AMPO decreased when lower radial pedal forces were allowed. Specifically, the maximally attainable AMPO was reduced by 53% when no radial forces were allowed compared to the condition in which there was no constraint on radial pedal forces. This reduction in maximally attainable AMPO was not only due to a decrease in positive mechanical power production by muscles, but also because some muscles substantially increased their negative mechanical power in order to satisfy the constraint on radial pedal force. Taken together, when radial pedal forces are constrained, this imposes a constraint on the resulting behaviour that must be satisfied through an appropriate choice of muscle activation over time. Consequently, muscle activation can no longer be solely optimised for the performance criterion, thereby reducing maximally attainable performance. This holds in general: requiring the resulting behaviour to satisfy additional constraints, which must be enforced through the choice of control inputs, can never improve performance; it can only leave it unchanged (if the constraint does not affect the optimal solution) or reduce it. Another example of this general principle is discussed in **Chapter 6**.

In **Chapter 6**, we discussed the broader implications of the findings presented in this thesis. First, we reflected on the transition from highly controlled *in situ* experiments on rat muscle to less controlled *in vivo* experiments on human muscle. Despite reduced experimental control and the use of generic rather than individualised model parameters, the Hill-type MTC models still captured the influence of SSC parameters on the maximally attainable AMPO *in vivo*. Second, we showed that both AMPO-optimal SSC parameters and the maximally attainable AMPO are relatively robust to variations in MTC properties. Third, we explained why no general statement can be made about the sensitivity of maximally attainable AMPO to individual SSC parameters. Fourth, we explained that the AMPO-optimal SSC-parameters of individual muscles cannot be realised in a musculoskeletal

system, in which individual joints are spanned by multiple muscles. As a result, the maximally attainable AMPO summed over all muscles represents a trade-off between agonistic and antagonistic muscles. Consequently, the FTS-values that maximise summed AMPO for SSCs of individual muscles are expected to shift towards more intermediate values, i.e. closer to 0.5, especially when agonistic and antagonistic muscles have similar properties. Fifth, as explained before, in many motor tasks, motion emerges from interaction between the musculoskeletal system and the environment. Consequently, muscle activation of an individual muscle affects the MTC length over time of all muscles involved. We explained why this limits performance in the context of sprint cycling. Finally, we discussed how insights obtained in this thesis may help improve performance in motor tasks involving equipment, such as cycling and rowing. Specifically, movement tasks and equipment could potentially be designed such that the main power-producing muscles operate closer to their AMPO-optimal SSCs. Overall, **Chapter 6** shows that maximising AMPO in a motor task requires consideration of both the intrinsic power-producing capacity of muscle and the constraints imposed by the task.

In short, in this thesis, we showed how SSC parameters — cycle frequency, FTS, and length excursion — influence maximally attainable AMPO. Specifically, we showed that cycle frequency and length excursion strongly interact in order to maximise AMPO, both in rat and human muscle. By contrast, the optimal FTS for maximising AMPO was relatively robust against changes in the other SSC parameters and was about 0.80–0.85 in both rat and human muscle. Since multiple muscles span the same joint, this isolated-muscle optimum cannot be realised for all muscles within a musculoskeletal system. In addition, additional constraints might further reduce the maximally attainable AMPO. The challenge for future research is therefore to determine how equipment can be designed such that the most important power-producing muscles operate as close as possible to their AMPO-optimal SSCs.

Samenvatting

Wanneer we wandelen, hardlopen, fietsen of een rolstoel voortbewegen, functioneren onze spieren als de motoren van mens en dier. Zij drijven de ledematen aan in herhaalde, cyclische bewegingen, waardoor het lichaam zich kan voortbewegen. Tijdens elke cyclus verlengen spieren, wat een eccentriche contractie wordt genoemd, om daarna weer te verkorten, wat een concentrische contractie wordt genoemd. In het Engels staat dit patroon bekend als de *stretch-shortening cycle* (SSC). Spieren leveren negatief mechanisch vermogen wanneer zij kracht produceren terwijl zij verlengen, en positief mechanisch vermogen wanneer zij kracht produceren terwijl zij verkorten. Over een volledige bewegingscyclus leiden deze fasen tot negatieve en positieve mechanische arbeid, waarvan de som bepaalt hoeveel netto mechanische arbeid de spieren produceren over een volledige cyclus. Door deze netto mechanische arbeid te delen door de duur van de cyclus, wordt het gemiddelde mechanische vermogen verkregen (Engels: *average mechanical power output*; AMPO). AMPO is een belangrijke bepalende factor tijdens zelfaangedreven, bewegingsactiviteiten met hoge snelheid, zoals sprintfietsen. Ondanks het belang van AMPO was echter verrassend weinig bekend over hoe spieren zouden moeten functioneren om de prestatie in een specifieke bewegingstaak te maximaliseren en hoeveel AMPO wij maximaal met onze spieren kunnen genereren.

De hoofdvraag in de meest eenvoudige vorm is: Welke spierlengte als functie van tijd en bijbehorende activatie als functie van tijd maximaliseren AMPO? Daarnaast zijn er verschillende gerelateerde vragen, zoals: Wat is de invloed van de maximale lengteverandering van de spier (d.w.z. het verschil tussen de maximale en minimale spierlengte tijdens een cyclus) en de cyclusfrequentie op AMPO? En beïnvloeden deze parameters elkaar? Het uitsluitend experimenteel beantwoorden van deze vragen levert beperkte kennis op, omdat slechts een klein deel van alle mogelijke

SSC's kan worden onderzocht binnen een meetsessie. Welke SSCs zou men dienen te kiezen om zoveel mogelijk inzicht te verkrijgen? In deze dissertatie hebben we deze vragen onderzocht door experimentele technieken te combineren met simulaties en optimalisaties met wiskundige spiermodellen, met als doel fundamenteel inzicht te verkrijgen in hoe verschillende SSCs het mechanisch vermogen van spieren beïnvloeden.

In **Hoofdstuk 2** en **Hoofdstuk 3** richten we ons op het spier-peescomplex (MTC; muscle-tendon complex) van de achterpoot van de rat. Het voordeel van werken met een geïsoleerd MTC-preparaat was dat we volledige controle hadden over de lengte en activatie van MTC als functie van tijd. Daardoor konden we de invloed van SSC-parameters (bijvoorbeeld cyclusfrequentie en maximale lengteverandering) op AMPO onderzoeken. Echter, zoals hierboven aangegeven kan in een experiment slechts een beperkt aantal condities worden onderzocht. Daarom vulden we het experimentele werk aan met voorspellingen op basis van Hill-type MTC-modellen. Modelvoorspellingen zijn uiteraard alleen waardevol als ze voldoende accuraat zijn. Onze aanpak was om modelvoorspellingen te evalueren aan de hand van een set zorgvuldig gekozen experimentele condities. Indien de modelvoorspellingen voldoende accuraat bleken, zou het een valide aanpak zijn om het model te gebruiken om de SSC-parameterruimte verder te verkennen buiten wat experimenteel getest kon worden. Belangrijk is dat AMPO wordt beïnvloed door eigenschappen van een individueel MTC, zoals de maximale isometrische kracht en de spiervezellengte, evenals alle overige eigenschappen van de contractie- en activatiedynamica. Om interindividuele variatie in voorspellingen te verminderen, kozen we ervoor om de eigenschappen van elk individueel MTC te schatten op basis van veelgebruikte experimentele protocollen. Dit roept een belangrijke vraag op: Kunnen deze parameters met deze protocollen voldoende nauwkeurig worden geschat?

Hoofdstuk 2 bevat een theoretische studie naar de nauwkeurigheid van veelgebruikte experimentele protocollen voor het schatten van belangrijke MTC-eigenschappen in geïsoleerde MTC-preparaten, zoals *quick-release*- en *step-ramp*-experimenten. We simuleerden deze protocollen met behulp van een Hill-type MTC-model met parameterwaarden gekozen uit de literatuur. Vervolgens

werden de parameters geschat uit de gesimuleerde data en vergeleken met hun vooraf gekozen waarden. In quick-release-experimenten contraheert MTC eerst isometrisch, dus bij constante lengte, waarna een snelle lengteverandering wordt opgelegd; vandaar de naam *quick-release*. Een veelgebruikte aanname bij deze *quick-release*-experimenten is dat de *release* zo snel verloopt dat spiervezels tijdens de *release* niet verkorten. Onder deze aanname wordt alle lengteverandering toegeschreven aan het *series elastic element* (SEE) in serie met de spiervezels. De SEE stijfheid kan dan worden geschat uit de kracht- en lengteveranderingen die vóór en na de release worden gemeten. Wij vonden echter dat deze aanname ongeldig is: zelfs tijdens *quick-releases* van slechts 10 ms verkorten spiervezels. Hierdoor wordt de SEE stijfheid met de gebruikelijke aanname aanzienlijk onderschat, met ongeveer 35%. Deze onderschatting van SEE stijfheid werkt vervolgens door in de schatting van andere MTC-eigenschappen, waarbij met name de schattingen van eigenschappen van de activatiedynamica sterk werden beïnvloed, tot ongeveer 20%. We ontwikkelden een verbeterde methode die rekening houdt met de verkorting van spiervezels tijdens *quick-releases*. Deze verbeterde methode leverde parameterschattingen op die dicht bij de vooraf bekende waarden lagen, binnen 3%. Deze verbeterde methode werd vervolgens toegepast in **Hoofdstuk 3**.

In **Hoofdstuk 3** combineerden we *in situ* experimenten met simulaties en optimalisaties op basis van een Hill-type MTC-model om de invloed van verschillende SSC-parameters op de maximaal haalbare AMPO van rat *m. gastrocnemius medialis* te onderzoeken. Voor elk preparaat werden belangrijke MTC-eigenschappen individueel geschat met behulp van de methode die in **Hoofdstuk 2** werd ontwikkeld. Dit maakte een directe vergelijking mogelijk tussen de experimenteel gemeten en door het model voorspelde AMPO voor elk individueel preparaat. De overeenkomst tussen beide bleek vrijwel perfect ($r^2 > 0,98$). Deze sterke overeenkomst rechtvaardigde vervolgens het gebruik van het Hill-type MTC-model om systematisch te onderzoeken hoe variaties in SSCs parameters de maximaal haalbare AMPO beïnvloeden. We voorspelden de maximaal haalbare AMPO voor een grote reeks SSC's, met als gemanipuleerde parameters cyclusfrequentie, MTC-lengteverandering en FTS (de fractie van de cyclustijd waarin verkorting plaatsvindt; een FTS van 0,65 betekent

bijvoorbeeld dat 65% van de cyclustijd aan verkorting wordt besteed). Er werd voorspeld dat maximale AMPO optrad bij een cyclusfrequentie van 3,5 Hz, een MTC-lengteverandering van 8 mm, en een FTS van 0,85. Cyclusfrequentie en MTC-lengteverandering interacteerden sterk met elkaar: een toename van de een vereiste een afname van de ander om AMPO te maximaliseren. Daarentegen bleef de optimale FTS opmerkelijk constant over alle geteste combinaties van cyclusfrequentie en MTC-lengteverandering. **Hoofdstuk 3** biedt een uitgebreid overzicht van hoe SSC-parameters de maximaal haalbare AMPO van *m gastrocnemius medialis* van de rat beïnvloeden.

Aangezien de aanpak gebruikt in **Hoofdstuk 3** succesvol was, pasten we in **Hoofdstuk 4** een vergelijkbare methode toe om het effect van SSC-parameters op de maximaal haalbare AMPO van menselijke *m. quadriceps femoris in vivo* te onderzoeken. De aanpak verschilde echter in een paar belangrijke aspecten. Ten eerste werd, in plaats van individueel afgestemde modellen voor elke deelnemer, een Hill-type model met generieke MTC-eigenschappen uit de literatuur gebruikt om de invloed van SSC-parameters op de maximaal haalbare AMPO te onderzoeken. Ten tweede werden in de experimentele opzet de kniegewrichtsbewegingen volledig opgelegd met behulp van een kniedynamometer, terwijl deelnemers werden geïnstrueerd om de AMPO te maximaliseren. Hierdoor werd spieractivatie niet direct gecontroleerd, maar bepaald door de vrijwillige neurale aansturing van de deelnemers. Ten derde waren de uitkomstmaten gebaseerd op het gemeten netto kniegewrichtsmoment, dat de gecombineerde bijdrage van alle spieren die het gewricht overspannen weerspiegelt in plaats van uitsluitend die van *m. quadriceps femoris*. Samengevat waren de *in vivo* humane experimenten minder gecontroleerd dan de *in situ* ratexperimenten beschreven in **Hoofdstuk 3**, wat leidde tot een aantal (additionele) bronnen van onzekerheid. De verwachting was dat dit ten koste zou gaan van de overeenkomst tussen modelvoorspelling en experimentele resultaten. Modelvoorspellingen werden vergeleken met experimentele data met behulp van twee verschillende methoden van schaling. Wanneer de experimentele data alleen werden geschaald naar de maximale isometrische kracht van elke deelnemer, vonden we een goede overeenkomst tussen voorspellingen en metingen ($r^2 = 0,75$). Wanneer

verschillen tussen condities werden uitgedrukt in eenheden van standaarddeviatie, namen de verschillen tussen modelvoorspellingen en experimentele data af, wat resulteerde in een sterkere overeenkomst ($r^2 = 0,95$). Deze resultaten lieten zien dat voorspellingen op basis van een Hill-type MTC-model met generieke MTC-eigenschappen valide waren voor het onderzoeken van de invloed van verschillende SSC-parameters op de maximaal haalbare AMPO van menselijke *m. quadriceps femoris in vivo*. De voorspelling was dat de maximaal haalbare AMPO piekt bij een cyclusfrequentie van ongeveer 1,6 Hz, een FTS van ongeveer 0,80 en een kniegewrichtsuitslag van ongeveer 2,0 rad (gecentreerd rond een kniegewrichtshoek van 1,0 rad, wat overeenkomt met de kniegewrichtshoek waarbij de maximale isometrische kracht wordt geproduceerd). Net als bij *m. gastrocnemius medialis* van de rat vonden we dat cyclusfrequentie en kniegewrichtsuitslag sterk met elkaar interacteerden. Daarnaast werd opnieuw voorspeld dat de optimale FTS grotendeels ongevoelig was voor veranderingen in cyclusfrequentie en kniegewrichtsuitslag. **Hoofdstuk 4** biedt een uitgebreid overzicht van hoe SSC-parameters de maximaal haalbare AMPO van menselijke *m. quadriceps femoris in vivo* beïnvloeden.

In **Hoofdstuk 5** onderzochten we de invloed van het beperken van radiale pedaalkrachten op sprintfietsprestatie. Hiervoor gebruikten we een spierskelet model van de menselijke onderste extremiteit en optimaliseerden we spieractivatie als functie van tijd om de som van AMPO over alle spieren te maximaliseren bij verschillende niveaus van toegestane radiale pedaalkrachten. Zoals verwacht nam de maximale AMPO af wanneer lagere radiale pedaalkrachten werden toegestaan. Wanneer geen radiale pedaalkrachten werden toegestaan nam de maximaal haalbare AMPO af met 53% ten opzichte van de conditie waarin er geen beperking was op radiale pedaalkrachten. De afname in maximaal haalbare AMPO werd niet alleen veroorzaakt doordat spieren minder positief mechanisch vermogen produceerden, maar ook doordat sommige spieren aanzienlijk meer negatief mechanisch vermogen gingen produceerden om te voldoen aan de beperking van de toegestane pedaalkracht. Wanneer radiale pedaalkrachten worden beperkt, wordt er een extra beperking opgelegd aan het gedrag dat het spierskeletstelsel. Omdat spieractivatie als functie van tijd de enige input is, betekent dit dat er een beperking wordt opgelegd

aan spieractivatie als functie van tijd. In het algemeen geldt dat wanneer er een (extra) beperking wordt opgelegd aan het gedrag van een systeem, de prestatie nooit beter kan worden. Het kan alleen gelijk blijven (als de beperking geen effect heeft op de optimale oplossing) of slechter worden. Een ander voorbeeld van dit idee wordt besproken in **Hoofdstuk 6**.

In **Hoofdstuk 6** bespraken we de bredere implicaties van de bevindingen uit deze dissertatie. Ten eerste reflecteerden we op de overgang van sterk gecontroleerde *in situ* experimenten in rattenspieren naar minder gecontroleerde *in vivo* experimenten in menselijke spieren. Ondanks de verminderde experimentele controle en het achterwege laten van modelindividualisatie konden Hill-type MTC-modellen nog steeds de belangrijkste effecten van SSC-parameters op de maximaal haalbare AMPO *in vivo* voorspellen. Ten tweede lieten we zien dat de AMPO-optimale SSC-parameters en de maximaal haalbare AMPO relatief robuust zijn voor variatie in MTC-eigenschappen. Ten derde legden we uit waarom er geen algemene uitspraak kan worden gedaan over de gevoeligheid van maximaal haalbare AMPO voor individuele SSC-parameters. Ten vierde hebben we uitgelegd dat de AMPO-geoptimaliseerde SSC-parameters van een individuele spier niet realiseerbaar zijn in een spierskeletstelsel, waarin individuele gewrichten door meerdere spieren worden overspannen. Als gevolg hiervan vormt de maximaal haalbare AMPO, gesommeerd over alle spieren, een compromis tussen agonistische en antagonistische spieren. Daarom wordt verwacht dat de FTS-waarden die de gesommeerde AMPO maximaliseren voor SSC's van individuele spieren verschuiven naar lagere waarden (d.w.z. dicht bij 0,5), met name wanneer agonistische en antagonistische spieren vergelijkbare eigenschappen hebben. Ten vijfde, zoals eerder uitgelegd, ontstaat beweging bij veel motorische taken uit de interactie tussen het spierskeletstelsel systeem en de omgeving. Daardoor beïnvloedt de activatie van een individuele spier de MTC-lengte als functie van tijd van alle betrokken spieren. We hebben uitgelegd waarom dit de prestaties beperkt, in de context van sprintfietsen. Ten slotte hebben we besproken hoe de inzichten die in deze dissertatie zijn verkregen kunnen bijdragen aan het verbeteren van de prestatie in motorische taken waarbij apparatuur wordt gebruikt, zoals fietsen en roeien. Bewegingstaken en apparatuur zouden zo

ontworpen dienen te worden dat de belangrijkste vermogensproducerende spieren dichter bij hun AMPO-optimale SSC's functioneren. Over het geheel genomen laat **Hoofdstuk 6** zien dat het maximaliseren van AMPO in een motorische taak vereist dat rekening wordt gehouden met zowel de intrinsieke vermogensproducerende capaciteit van spieren als de beperkingen van de taak.

Kortom, in deze dissertatie onderzochten we hoe SSC-parameters — cyclusfrequentie, MTC-lengteverandering en FTS — de maximaal haalbare AMPO beïnvloeden. Specifiek lieten we zien dat cyclusfrequentie en lengteverandering sterk met elkaar interacteerden voor het maximaliseren van AMPO, zowel in een rattenspier als in een menselijke spier. Daarentegen was de optimale FTS voor het maximaliseren van AMPO relatief robuust tegen veranderingen in de andere SSC-parameters en bedroeg deze ongeveer 0,80–0,85 in zowel de rattenspier als de menselijke spier. Omdat meerdere spieren werken rondom hetzelfde gewricht, kan het optimum voor een enkele spier niet gerealiseerd worden voor alle spieren in een spierskeletstelsel. Bovendien zijn er in veel motorische taken extra beperkingen die de maximaal haalbare AMPO verder limiteren. De uitdaging voor toekomstig onderzoek is om te bepalen hoe apparatuur zo ontworpen kan worden dat de belangrijkste vermogensproducerende spieren zo dicht mogelijk bij hun AMPO-optimale SSC's functioneren.

Acknowledgements

Zo'n 13 jaar geleden liep ik de Vrije Universiteit binnen als (naïeve) bachelorstudent Bewegingswetenschappen. Dit was het begin van een traject dat uiteindelijk heel geleidelijk overging in een research master en een baan als docent, waarbij er (een klein beetje) ruimte was om onderzoek te doen. Hoewel het traject nog steeds, weer geheel geleidelijk, doorloopt, biedt het afronden van mijn proefschrift een mooi moment om even stil te staan bij de inspirerende mensen, steunpilaren en kritische denkers die mij de afgelopen 13 jaar hebben omringd en ieder op hun eigen wijze hebben bijgedragen aan de totstandkoming van dit proefschrift. Waarvoor dank!

Dinant, wij leerden elkaar op de tennisbaan kennen in mijn eerste studiejaar. Jij had geen idee dat ik al voor de minor BWSB had gekozen, ik had geen idee dat jij een van de docenten was in deze minor. Jouw aanstekelijke enthousiasme van toen heb je nog steeds. Daarmee enthousiasmeer jij niet alleen mij, maar ook vele anderen om dingen écht te begrijpen. Onze intensievere samenwerking begon tijdens mijn research master. In deze begeleiding heb ik altijd vertrouwen en waardering gevoeld. We kunnen heerlijk met elkaar samenwerken, op een natuurlijke manier en zonder dat daar veel expliciete afspraken voor nodig zijn. Juist die vanzelfsprekendheid heb ik altijd erg gewaardeerd. In de loop van de jaren is naast de fijne samenwerking ook een vriendschappelijke band ontstaan. Dank voor alle support, het vertrouwen en de fijne samenwerking.

Knoek, ik herinner me nog dat jij in het eerste college van S3 de universiteit omschreef als een "sociale werkplek voor hoger opgeleiden". Ik wist niet wat ik hoorde, maar inmiddels weet ik wel beter. Jouw begeleiding, zowel tijdens het miniproject bij S3 als tijdens mijn BOP, heeft mij geïnspireerd bij het onderzoek dat in dit proefschrift wordt beschreven. Tijdens onze besprekingen is jouw vermogen om terug te gaan naar de basis tot een kunst verheven: een probleem strak

formuleren en van daaruit verder redeneren. Juist bij ogenschijnlijk ingewikkelde (mechanische) problemen blijkt die aanpak vaak essentieel. Door de jaren heen ben ik er ook achter gekomen dat je je op allerlei manieren inzet voor anderen, zonder daar ook maar enige ruchtbaarheid aan te geven. Dat waardeer ik in je. Dank voor je inspiratie, je scherpe blik en de manier waarop je in het leven staat.

Maarten, voor mij is de cirkel rond nu jij als mijn promotor optreedt. Niet alleen bestaat daarmee het voltallige begeleidingsteam uit de *Biofysica-crew*, maar bovenal uit mensen die een belangrijke bijdrage hebben geleverd aan mijn ontwikkeling. Jouw luisterend oor en de steun die je me na het overlijden van mijn moeder hebt gegeven, hebben ontzettend veel voor mij betekend. Dat zal ik nooit vergeten. Jij was (en bent) altijd oprecht geïnteresseerd in hoe het nou écht met mensen gaat. Die oprechte aandacht en empathie zijn eigenschappen die ik bijzonder in jou waardeer. Zowel op persoonlijk vlak als inhoudelijk ben je altijd eerlijk en rechte doorzee. Jouw kritische en scherpe feedback raakt altijd de kern, met een substantiële verbetering van het “schrijfsel” als resultaat. Dank voor je steun, je oprechte aandacht en voor het feit dat je altijd zegt waar het op staat.

Mijn eerste ervaring als docent deed ik op bij *Meten van Fysische Grootheden*, waar ik als student-assistent aan de slag mocht. **Rob**, jij gaf mij het vertrouwen om deze rol op mij te nemen en later, toen ik nog masterstudent was, ook de verantwoordelijkheid voor de coördinatie en inhoudelijke kant van de practica. Deze werkzaamheden hebben niet alleen bijgedragen aan mijn ontwikkeling als docent, maar hebben mij ook de mogelijkheid gegeven om mijn onderzoek voort te zetten en dit proefschrift te schrijven. Ik ben je dan ook bijzonder dankbaar voor de kansen die je mij hebt gegeven en voor het vertrouwen dat je daarbij steeds in mij hebt gehad.

Bij de uitvoering van de experimenten voor dit proefschrift heb ik veel waardevolle hulp en ondersteuning gekregen. **Huub**, hartelijk dank dat je mij de mogelijkheid hebt geboden om gebruik te maken van jouw opstelling. **Wendy**, jouw expertise, precisie en chirurgische vaardigheid bij het prepareren van de spieren hebben geleid tot data van uitmuntende kwaliteit bij dit experiment, waarvoor ik je hartelijk wil bedanken. Dank gaat ook uit naar alle leden van de **TOD**,

zowel voor de prettige samenwerking als voor jullie vlotte hulp bij vragen en/of problemen. **Peter**, in het bijzonder bedankt voor alle ondersteuning bij zowel de kniedynamometeropstelling als de opstelling in het Myologisch laboratorium. Dank voor je geduld en behulpzaamheid, vooral wanneer ik weer met lastige vragen kwam over hoe die opstelling nou écht precies werkte.

Door de jaren heen heb ik het geluk gehad om met veel verschillende **stafleden** samen te werken aan uiteenlopende cursussen. Al deze samenwerkingen hebben mij op hun eigen manier veel geleerd en zijn stuk voor stuk waardevolle ervaringen geweest. Daarnaast ben ik dankbaar voor de vele **studenten** met wie ik de afgelopen jaren heb mogen samenwerken. In het bijzonder wil ik ook de studenten bedanken die ik tijdens hun BOP-traject heb mogen begeleiden. Het was mooi om van dichtbij mee te mogen kijken naar hun ontwikkeling en daar zelf ook veel van te leren.

Het laatste jaar van mijn promotietraject – toen ik eindelijk ook officieel als promovendus was aangesteld – bracht ik steeds meer tijd door met mijn **collega's (promovendi)**. Kamer A621, *heuj!* Met jullie was het altijd gezellig, zowel op de kamer als bij het koffiepunt. In het algemeen heb ik een fantastische tijd gehad, met vele leuke activiteiten zoals zaalvoetbal (BW Bikkels!), beachvolleybal, ECR-weekenden, en meer. Bovenal koester ik de persoonlijke betrokkenheid die ik heb ervaren. Dank aan alle collega-promovendi voor de fijne tijd; jullie aanwezigheid maakte het afmaken van dit proefschrift een stuk aangenamer.

Tijdens de studie Bewegingswetenschappen heb ik met verschillende mensen mogen optrekken. **Dax, Pieter, Thomas**, jullie waren mijn eerste vrienden in het grote Amsterdam. Zowel voor, tijdens als na college was het altijd gezellig. *Woensdag stampopotdag!* **Jaap**, pas tijdens de minor BWSB leerde ik je kennen, en kwam ik erachter dat je niet alleen een superslimme, maar ook een superlieve gast bent. Ik ben blij dat we elkaar later weer op de A12 zijn tegengekomen. **Tim**, wij vormden samen de BW-nerds van ons jaar, met als klap op de vuurpijl een onwijs gaaf BOP-project, waar ik met veel plezier op terugkijk. Het lijkt me leuk om in de toekomst nog eens samen een project op te pakken. **Guus**, met jou kan ik alles delen. Ik waardeer onze goede en vertrouwelijke gesprekken en de mooie vriendschap die daaruit is ontstaan. **Koen**, door één ogenschijnlijk simpele vraag zorgde je er niet

alleen voor dat ik binnen twee maanden vegetariër was, maar bovenal dat ik oog kreeg voor het grotere plaatje. Dank voor alle interessante en gezellige gesprekken.

Het idee is dat wetenschap bijdraagt aan een betere wereld. De impact van een proefschrift blijft echter vaak bescheiden. Wat is de bijdrage van meer kennis over de werking van spieren in een tijd van grote maatschappelijke vraagstukken? In dat kader zijn er mensen die dagelijks strijden voor veel belangrijkere zaken zoals de toekomst van onze planeet, de rechten van anderen en sociale rechtvaardigheid. Via deze weg wil ik mijn dankbaarheid en waardering uitspreken voor de **activisten** die zich onverminderd inzetten voor rechtvaardigheid. Op lokaal niveau heb ik het genoeg gehad om met vele geweldige mensen samen te werken. Aan alle activisten van XR & SR: jullie doen goed! Mijn dank gaat ook uit naar alle vrijwilligers van **De Sering**. Jullie creëren een fantastische plek waar iedereen samen kan komen: *All good things begin with a bowl of soup*. Ik ben bijzonder dankbaar voor het beschikbaar stellen van deze plek voor het diner ter gelegenheid van mijn promotie. Voor mij was er geen betere locatie denkbaar.

Pap en mam, bedankt voor al jullie ondersteuning. Jullie hebben, ieder op jullie eigen manier, gezorgd voor een omgeving waarin wij als kinderen konden floreren. Jullie gaven ons de ruimte om te ontdekken, te leren en onze eigen interesses te ontwikkelen, maar ook om zorgeloos te studeren. Daar ben ik jullie ontzettend dankbaar voor. **Mama**, jij stond altijd voor ons klaar. Ik herinner mij vele gezellige kilometers van en naar tennistraining en toernooien. Studeren heb je vooral meegemaakt als een noodzakelijk kwaad en ik denk niet dat je ooit zou hebben verwacht dat ik een academisch proefschrift zou schrijven. Toch weet ik zeker dat je ongelooflijk trots zou zijn geweest. Ik had zó graag gewild dat je had kunnen zien hoe ver ik het heb geschopt op verschillende vlakken. **Papa**, hoewel jij jezelf wat meer op de achtergrond positioneerde, had je altijd goed in de smiezen hoe alles ging. Die rol nam je op je voor zowel mama als voor ons. Qua persoonlijkheid lijken wij toch aardig op elkaar. Iets wat ik vroeger niet graag wilde toegeven, maar wat ik nu koester. Voor mij ben je echt een vaderfiguur, maar bovenal een “sparringspartner” met wie ik zowel inhoudelijk als persoonlijk graag van gedachten wissel. **Claudia**, jouw strijdbaarheid in moeilijke jaren is niet alleen een voorbeeld

geweest voor mij, maar ook voor velen om je heen. Ik ben zo blij dat je nu het prachtige gezin hebt waarvan je altijd droomde. Ik vind het geweldig om te zien hoe je daarin floreert; de rest van het gezin heeft enorm veel geluk met zo'n fantastische moeder! **Gwen**, ik ben zo blij dat wij samen zijn. Ik vind het fijn hoe makkelijk en natuurlijk alles tussen ons gaat. We kunnen samen lachen, uren kletsen, maar ook heerlijk een avondje spelletjes spelen. Inmiddels hebben we daar een nieuwe hobby aan toegevoegd in de vorm van kamperen op mooie plekjes. Ik kijk uit naar alle avonturen die daar nog uit voortkomen. Jouw positiviteit, enthousiasme en humor brengen mij verlichting die ik erg waardeer. Daarmee heb je niet alleen een belangrijke bijdrage geleverd aan dit proefschrift, maar ook aan mijn levensgeluk.

Gihigugma ko ikaw!

